

Guidelines for Carbon Accounting and Emission Reduction In the Urban Water Sector

Organized by the China Urban Water Association

Edited and translated by Xiaodi Hao and Ranbin Liu



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Contents

List of Contributors	ix
Foreword.....	xi
Preface	xiii
Acknowledgements.....	xv
 Chapter 1 General principles	 1
1.1 Purposes	1
1.2 Application scope	1
 Chapter 2 Urban water sectors and their carbon emissions	 3
2.1 Urban water sectors	3
2.2 Carbon emissions of urban water sectors	4
2.2.1 Greenhouse gases and global warming potential	4
2.2.2 Reporting boundaries and emission activities.....	4
 Chapter 3 Carbon accounting principles and methodologies	 11
3.1 Basic principles	11
3.2 Accounting procedures	11
3.2.1 General introduction.....	11
3.2.2 Water companies	12
3.2.3 Water sector associations.....	13
3.2.4 Water departments.....	13
3.3 Accounting methodologies	13
3.3.1 General introduction.....	13
3.3.2 Water companies	15
3.3.3 Water sector associations.....	21
3.3.4 Water departments.....	21
 Chapter 4 Planning and construction.....	 23
4.1 Overview	23

4.2 Direct emissions of fossil fuel	23
4.3 Indirect emissions from electricity consumption	24
4.4 Indirect emissions from material consumption	24
4.5 Indirect emissions from transportation	24
4.6 Overall estimation as a whole	25
4.6.1 Estimation based on water volume handled	25
4.6.2 Estimation based on investment	31
Chapter 5 Operation and maintenance	35
5.1 Overview	35
5.2 General provisions	35
5.2.1 Direct emissions of fossil fuel	35
5.2.2 Indirect emissions from electricity consumption	35
5.2.3 Indirect emissions of material consumption	36
5.2.4 Indirect emissions of transportation	36
5.3 Water supply systems	37
5.3.1 Accounting boundary	37
5.3.2 Water abstraction facilities	40
5.3.3 Water treatment plants	40
5.3.4 Desalination plants	41
5.3.5 Water distribution network	42
5.3.6 Long-distance water transfer	42
5.4 Wastewater management systems	43
5.4.1 Accounting boundary	43
5.4.2 Wastewater collection networks	46
5.4.3 Wastewater treatment plants	50
5.4.4 Sludge treatment and disposal	57
5.5 Water reclamation systems	61
5.5.1 Accounting boundary	61
5.5.2 Advanced treatment plants	63
5.5.3 Water distribution network	63
5.6 Stormwater systems	63
5.6.1 Accounting boundary	63
5.6.2 Stormwater sewers	66
5.6.3 Stormwater control facilities	67
Chapter 6 Asset replacement and demolition	71
6.1 Overview	71
6.2 Direct emissions of fossil fuel	71

Contents

6.3 Indirect emissions from electricity consumption	72
6.4 Indirect emissions of transportation	72
6.5 Carbon offset via material recovery	73
Chapter 7 Carbon reduction pathways in urban water sectors	75
7.1 Overview	75
7.2 Synergy of subsystems' carbon emission reduction in urban water sectors	81
7.3 Planning and construction and asset replacement and demolition	84
7.3.1 Overview	84
7.3.2 Carbon reduction pathways	88
7.3.3 Carbon replacement pathways.....	90
7.4 Water supply and water reclamation systems.....	90
7.4.1 Overview	90
7.4.2 Carbon reduction pathways	94
7.4.3 Carbon replacement pathways.....	94
7.5 Wastewater management systems	95
7.5.1 Overview	95
7.5.2 Carbon reduction pathways	100
7.5.3 Carbon replacement pathways.....	102
7.6 Stormwater systems	103
7.6.1 Overview	103
7.6.2 Carbon reduction pathways	107
7.6.3 Carbon replacement pathways.....	108
7.6.4 Carbon removal pathways	109
Chapter 8 Data acquisition and management	111
8.1 Overview	111
8.2 Activity data acquisition.....	111
8.3 Emission factors acquisition	122
8.3.1 General emission factors	122
8.3.2 CH ₄ and N ₂ O emission factors in wastewater management systems	124
Chapter 9 Result interpretation and reporting	131
9.1 Result interpretation	131
9.1.1 Key information analyzed	132
9.1.2 Analyzing methodologies	133
9.2 Reporting protocol	137
9.2.1 Information of reporting entity	137
9.2.2 Data sources	137
9.2.3 Emission factors	137

Guidelines for Carbon Accounting and Emission Reduction in the Urban Water Sector

9.2.4 Carbon emissions	137
Appendix A Terms and symbols	139
Appendix B Emission factors	143
Appendix C Construction inventory and integrated emission factor formulation	159
Appendix D Carbon accounting: case studies	195
Appendix E Carbon emission reduction potential of typical technologies	221
Appendix F Reporting protocol	231
References.....	237

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Foreword



In September 2020, at the 75th session of the United Nations General Assembly, President Xi Jinping solemnly pledged that China will strive to peak carbon dioxide emissions before 2030 and achieve carbon neutrality before 2060, i.e. dual carbon goals. This major strategic decision was based on our sense of responsibility in building a community with a shared future for mankind and our own need to secure sustainable development. The Central Committee of the Communist Party of China and the State Council released a document titled "*Working Guidance for Carbon Dioxide Peaking and Carbon Neutrality in Full and Faithful Implementation of the New Development Philosophy*". Together with the "*Action Plan for Carbon Dioxide Peaking Before 2030*" and "*Implementation Plan for Synergizing the Reduction of Pollution and Carbon Emission*", China formulated a "1+N" policy framework to guide these dual carbon goals. From the perspective of top-level design, the dual carbon goals have been broken down one by one, and directions have been established for the development and transformation of all industries and carbon emission reduction actions.

As an important part of urban infrastructure, ensuring the normal life of residents and the healthy development of the economy, urban water sectors are a key factor in the sustainable development of Chinese cities. Since the implementation of reform and opening up, China's urban water supply has continued to grow and develop, with the construction and service capabilities of water supply and drainage facilities having improved year by year, and the production capacity and utilization of reclaimed water reaching a record high. In response to the extreme rainfall brought about by climate change, China is also actively promoting "sponge cities" and improving the planning and construction of urban waterlogging prevention and control facilities. However, China's urban water sectors are also facing the huge challenges of being labeled as large energy consumers. According to preliminary statistics and estimates, the power consumption of the urban water supply system alone accounts for about 1.5% of the whole state's power consumption. Moreover, due to the low efficiency of urban sewage treatment systems, effluent quality is only able to meet the discharging standard at the cost of the consumption of energy and chemicals. Consequently, the above activities cause greatly increased greenhouse gas emissions. With the continuous urbanization of China leading to urban water supply and drainage capacities increasing year by year, coupled with increasing stress on the ecological environment, it is foreseeable that the energy consumption and carbon emissions of the urban water sector will inevitably increase unless effective steps are taken swiftly. Under the pressure of dual carbon goals, the urban water

Guidelines for Carbon Accounting and Emission Reduction in the Urban Water Sector

sectors are facing enormous challenges in terms of technological innovation and upgrading. However, these challenges may also provide opportunities. Driven by the multiple goals of improving the ecology of the urban water sector, environment, resources and safety, the urban water sector must reduce pollution and carbon emissions, leverage their synergy to achieve low-carbon and green, and set itself on a more sustainable course.

Recently, seven ministries of the PRC, including the Ministry of Ecological Environment, jointly issued the “*Implementation Plan for Synergizing the Reduction of Pollution and Carbon Emission*”. In it, they proposed to “carry out carbon emission measurement for urban sewage treatment and resource utilization, and optimize the energy consumption and carbon emission management of sewage treatment facilities”. This shows that China's reduction in carbon accounting and emissions has gradually evolved from heavy industries with high carbon emissions to routine life and service businesses. Cities and towns have been included in the struggle to reduce pollution and carbon emissions. Therefore, the urban water sector cannot be excluded; indeed, it is imperative that China takes swift action to reduce carbon emissions and achieve low-carbon development. Carbon accounting is the first step for the industry to carry out low-carbon development and clarify the entanglement between carbon emission and profits, i.e., high-carbon coupled with a deficit, low-carbon reversing deficit, and carbon negative leading to profits.

There is a wide range of complex carbon emission activities in the urban water sector, and there is no general agreement on a standard definition of their accounting boundaries. In particular, the unclear identification of greenhouse gas emission activities leads to omission, overcalculation, and miscalculation, which is not conducive to the recognition and consensus on carbon emission in the urban water sectors, as well as to identifying the key points to reduce carbon emissions. Based on this, the China Urban Water Association organized the compilation of the *Guidelines for Carbon Accounting and Emission Reduction in the Urban Water Sector* (hereinafter referred to as the *Guidelines*). This *Guidelines* focuses on clarifying the boundaries and methods of carbon accounting of urban water sectors under the dual carbon goals, analyzing and categorizing carbon emission paths and strategies. It is believed that the release of the *Guidelines* will definitely promote the practice of the dual carbon goals action in the urban water sectors, help to meet the responsibilities and obligations of the industry and make contributions to the development of China as a green and low-carbon system.



Linwei Zhang

President of China Urban Water Association

Preface

China's striving to achieve carbon peaking by 2030 and carbon neutrality by 2060 are critical strategic decisions made by the Central Committee of the Communist Party of China. They are related to the sustainable development of China as well as to the building of a Community of Shared Future for humanity. They also demonstrate China's firm determination to actively respond to climate change and take the high-quality development path of prioritizing the ecology, green transformation, and low carbon emissions. The Central Committee of the Communist Party of China and the State Council successively issued “*Working Guidance for Carbon Dioxide Peaking and Carbon Neutrality in Full and Faithful Implementation of the New Development Philosophy*”, “*Action Plan for Carbon Dioxide Peaking Before 2030*”, and “*Implementation Plan for Synergizing the Reduction of Pollution and Carbon Emission*”. These documents will help China achieve a comprehensive green transformation of economic and social development through the dual carbon goals by indicating the direction to take and providing a blueprint for successful development. To ensure the realization of these dual carbon goals, “establishing a unified and standardized carbon emission accounting system” has been listed as an important task. The “*Action Plan for Carbon Dioxide Peaking Before 2030*” clearly highlights that “strengthening the capacity-building of carbon emission statistical accounting, deepening the research on accounting methods, accelerating the establishment of a unified and standardized carbon emission statistical accounting system, supporting industries and enterprises to carry out carbon emission and algorithm methodology research according to their characteristics, and establishing the carbon emission measurement system”.

The urban water sector is the basic foundation of daily life for urban residents, and the lifeline to maintaining urban running. It also intersects with many other areas of the national economy, which makes carbon emission activities complex. In recent years, a great deal of research has been carried out on the carbon emissions of urban water sectors, providing important support for the development of low-carbon practices. To align with the “*Working Guidance for Carbon Dioxide Peaking and Carbon Neutrality in Full and Faithful Implementation of the New Development Philosophy*” and the “*Action Plan for Carbon Dioxide Peaking Before 2030*”, take a leading role in industry associations, and accelerate the establishment of a unified and standardized carbon emission accounting system, the China Urban Water Association organized the preparation of the *Guidelines for Carbon Accounting and Emission Reduction in the Urban Water Sectors* (hereinafter referred to as the *Guidelines*) to guide the sectors in carrying out carbon emission reduction.

The *Guidelines* consists of 10 chapters, including general principles, urban water sectors and carbon emissions, carbon accounting principles and methodologies, planning and construction, operation and maintenance, asset replacement and demolition, carbon emission reduction pathways of urban water

Guidelines for Carbon Accounting and Emission Reduction in the Urban Water Sector

sectors, data acquisition and management, interpretation of result and reporting, and the appendices. The *Guidelines* regulates the carbon emission accounting methods in a whole life cycle by dividing the urban water sectors into four subsectors: water supply sector, sewage sector, reclaimed water sector, and rainwater management sector. The *Guidelines* begins by clarifying the carbon emission activities of each subsector, then unifies the carbon emission accounting boundary, gives transparent carbon emission factors and finally unifies the carbon emission accounting and reporting template. To guide carbon emission reduction in the urban water sector, the *Guidelines* proposes strategies for carbon emission reduction in urban water systems in the following five areas: source control, process optimization, technology upgrades, renewable energy, and carbon sinks. Overall, the *Guidelines* highlights the following features:

Scientificity. The accounting principles, accounting framework, and accounting methods follow the established greenhouse gas accounting system, i.e., “*Greenhouse Gas Protocol*” and “*Greenhouse gases—Part 1: Specification with guidance at the organization level for quantification and reporting of greenhouse gas emissions and removals* (ISO 14064-1:2018)”, reflecting the scientific consensus.

Consistency. Carbon emission factors come from governmental open-access data and/or scientific literature, particularly local documents, and reflect the transparency of the accounting process and the accuracy of the accounting results.

Modularization. The *Guidelines* provides accounting methods in terms of four subsectors – the water supply sector, the sewage sector, the reclaimed water sector, and the rainwater management sector – as well as three life cycle stages (planning and construction, operation and maintenance, and asset replacement and demolition) with a total of 12 modules. These can be flexibly applied to carbon emission accounting for different purposes.

Comprehensiveness. The *Guidelines* covers all carbon emission activities throughout the life cycle of the urban water sectors. It also provides carbon emission reduction pathways of different subsectors, which can be used for carbon emission accounting by different operating entities as well as for carbon emission reduction pathway analysis.

Hierarchy. The *Guidelines* provides various accounting methods and emission factors based on the availability of data, which can be flexibly selected according to accounting purposes.

Guidance. The *Guidelines* standardizes the carbon emission accounting process, outlines the emission reduction technology and strategy of each subsector, unifies the carbon emission accounting report, and illustrates the guidance role of the industry association on the development of the water sector.

The compilation of the *Guidelines* was led by the Beijing University of Civil Engineering and Architecture, together with Beijing Capital Eco-Environmental Protection Group Co., Ltd., Beijing Urban Drainage Group Co., Ltd., CSCEC SCIMEE Sci. and Tech. Co., Ltd., Central and Southern China Municipal Engineering Design and Research Institute Co., Ltd., and Harbin Institute of Technology. In the preparation of the *Guidelines*, the China Urban Water Association held several expert consultation meetings to solicit opinions from various parties. On this basis, the first technical guidelines for carbon accounting and emission reduction pathways of the urban water sector were outlined. If any further strategies or information are developed or deemed noteworthy for the purposes of correcting or enhancing the *Guidelines*, please do not hesitate to contact us so that the compilation group may update and improve their information.

Guidelines' editing team
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Chapter 1

General principles

1.1 PURPOSES

The objectives of these *Guidelines* are to clarify carbon (or greenhouse gas, GHG) accounting boundaries, carbon emission activities and carbon emission categories in the urban water sector. They also serve to standardize data acquisition and carbon accounting methodologies; to help navigate accounting implementation in a scientific manner; and to regulate carbon emission reporting. In addition, the *Guidelines* summarize the strategies and actions that can achieve carbon reduction in the water sector; promote the role of carbon accounting and reduction in leveraging sustainable management of the urban water sector; and strengthen top-level design to guide the move towards low carbon development of the water sector.

1.2 APPLICATION SCOPE

Guidelines is applicable to the carbon accounting and reporting of urban water sectors in China, and can also be referred to when formulating and implementing carbon reduction plans and actions, which can include:

- Guiding the water utilities to recognize and clarify their boundaries, activities and categories of carbon emissions, and serving as a basic framework for formulating comprehensive policies and/or plans for carbon management;
- Instructing urban water sectors to implement carbon accounting over the course of a life cycle (planning and construction, operation and maintenance, and asset replacement and demolition) and/or for specific emission activities;
- Guiding the urban water sectors to analyze carbon accounting results and, based on these results, carrying out top-level design of carbon reduction actions, program development, strategy implementation, evaluation feedback, and interdepartmental collaboration.

Chapter 2

Urban water sectors and their carbon emissions

2.1 URBAN WATER SECTORS

These *Guidelines* aim at urban water sectors for carbon accounting focusing on four main systems – water supply systems, wastewater management systems, water reclamation systems and stormwater management systems as shown in Figure 2.1.

- **Water supply systems** consist of five modules: water abstraction, water treatment plants, seawater desalination plants, water distribution networks, and long-distance water conveyance.
- **Wastewater management systems** consists of three modules, wastewater collection networks (including septic tanks), wastewater treatment plants, and sludge treatment and disposal facilities.
- **Water reclamation systems** consists of two modules: advanced treatment plants and water distribution networks.
- **Stormwater systems** consists of two modules of stormwater sewers and stormwater control facilities that store, infiltrate, or reuse stormwater and reduce flows to sewer systems or to surface waters.

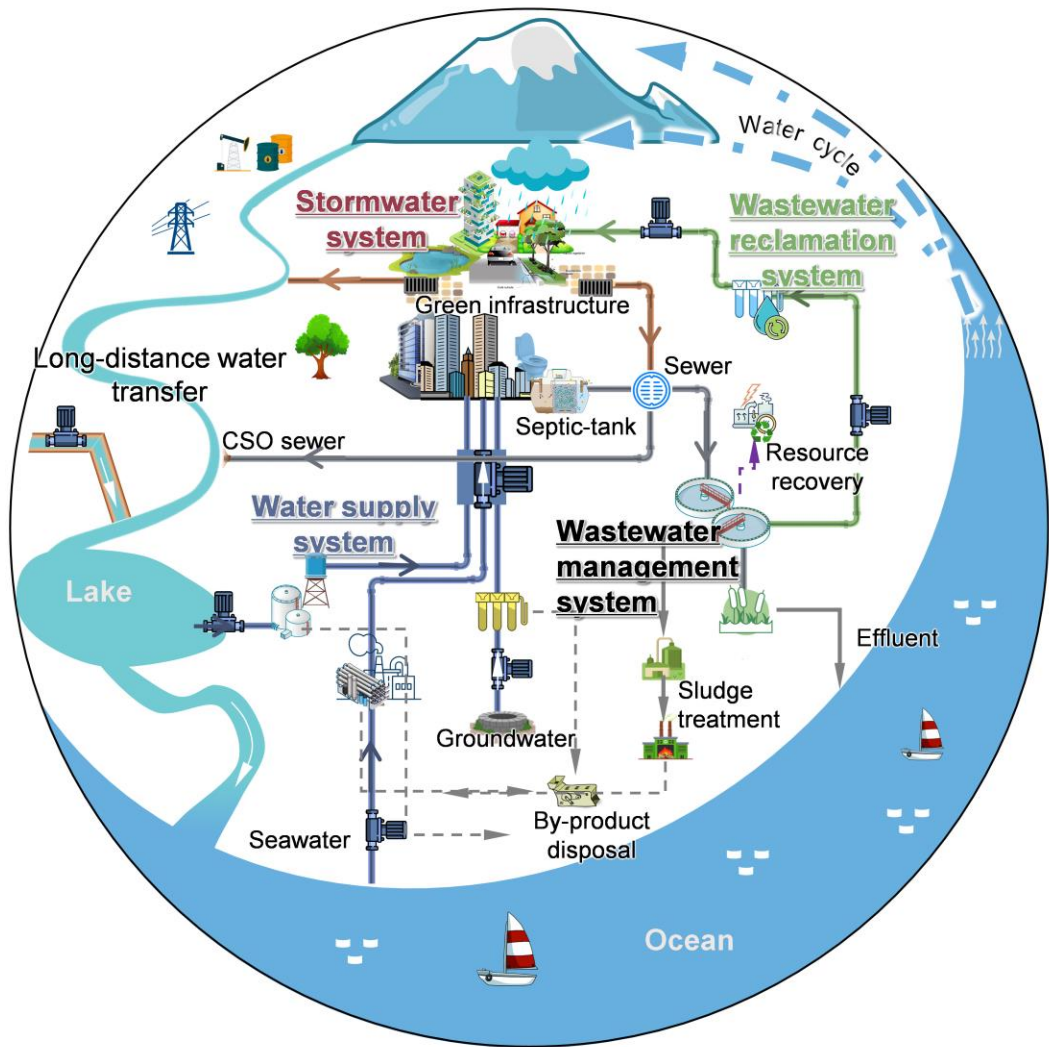


Figure 2.1 Schematic diagram of urban water sectors targeted in these *Guidelines*.

2.2 CARBON EMISSIONS OF URBAN WATER SECTORS

2.2.1 Greenhouse gases and global warming potential

The types of greenhouse gases (GHG) considered in *Guidelines* include carbon dioxide (CO₂), methane (CH₄), and nitrous oxide (N₂O). According to the *International standard of ISO14064-1: 2018* (ISO, 2018), GHG emission accounting is thought to utilize Global Warming Potential (GWP) with 100-year times span as shown in the reference standard (Table 2.1).

Table 2.1 Global warming potential (GWP) values relative to CO₂ (IPCC, 2013).

GHG	GWP 100
CH ₄	28
N ₂ O	265

2.2.2 Reporting boundaries and emission activities

The water sectors should establish and document their accounting boundaries (reporting boundaries) to ensure the accuracy and representativeness of carbon accounting results. This is also important for result

interpretation and comparison analysis. Besides, accounting boundaries consist of multiple dimensions which is also a detrimental factor for the emission quantification results. Thus, in this *Guidelines*, time range (temporal dimension) and the interaction boundary between system and nature (spatial dimension) are the most basic two dimensions within which the carbon emission accounting method of urban water sectors are developed.

In terms of time range, carbon emission occurs in association with the life cycle of the construction, operation, and demolition of structures and buildings in the water sectors. From the perspective of carbon reduction, a great deal of work can be done at every stage of the life cycle in order to achieve carbon reduction. Thus, the carbon accounting and reduction in the urban water sector is supposed to consider its whole life cycle *from cradle to grave* (Figure 2.2), namely: 1) planning and construction, the whole process preceding the formal operation of the facility; 2) operation and maintenance, the entire process from the formal operation to end-of-run; 3) asset replacement and demolition, the entire process of using the facility for other purposes or completely removing it after the end of its operational lifespan. In terms of the spatial boundary, carbon accounting in urban water sectors includes not only GHG emission within the organizational boundary of the urban water sector, but also related material flow and other activities, such as energy input, effluent discharge and sludge disposal, all of which are addressed below in detail.

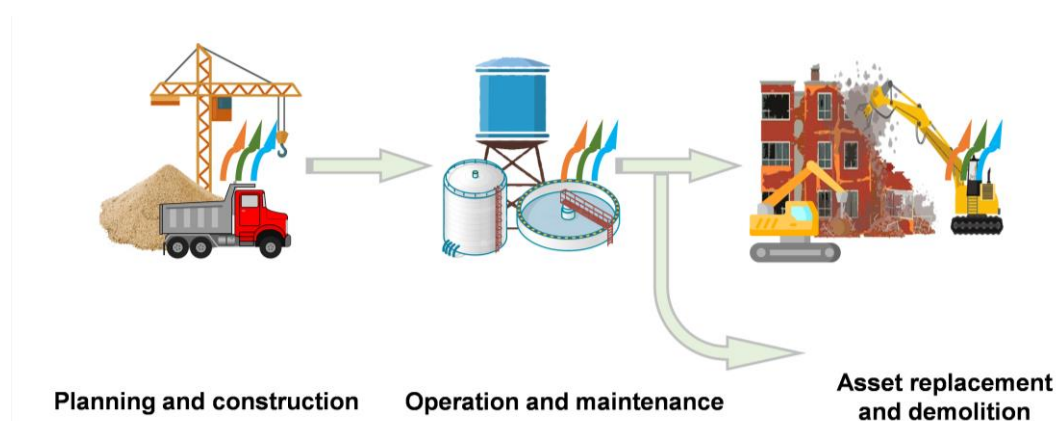


Figure 2.2 Temporal boundary of carbon emission accounting in urban water sector.

To ensure the results are easy to read and interpret, the reporting boundaries are in line with the international standards of the *Greenhouse Gas Protocol* (WBCSD/WRI, 2004) and *Greenhouse gas - Part 1: Specification with guidance at the organization level for quantification and reporting of greenhouse gas emissions and removals* (ISO 14064-1: 2018) (ISO, 2018). In the *Greenhouse Gas Protocol*, the GHG emissions are classified into three scopes (Table 2.2): **Scope 1**: direct GHG emissions produced at facilities and/or assets owned by the water sectors; **Scope 2**: indirect GHG emissions incurred by an energy provider when producing the energy purchased by the water sectors; **Scope 3**: other indirect GHG emissions not attributable to energy, including emissions generated by the productions of goods, transportation. By contrast, ISO 14064-1: 2018 standard divides three scopes into six categories, which are: 1) direct GHG emissions and removals; 2) indirect GHG emissions from imported energy; 3) indirect GHG emissions from transportation; 4) indirect GHG emissions from products used by the organization; 5) indirect GHG emissions associated with the use of products from the organizations; 6) indirect GHG emissions from other sources. As summarized in Table 2.2, these two kinds of classifications are

consistent.

Next, according to the above-mentioned classification methods, the GHG emission activities of urban water sectors are summarized into 5 groups (Table 2.2):

- Emission from the consumption of fossil fuels within the organizational boundaries and emission in the production of the imported energy in a whole life cycle;
- Emissions from the production of goods consumed by the water sectors in the whole life cycle, such as building materials and chemicals;
- Emissions from the transportation for building materials, chemicals, byproducts exported, and building waste in asset replacement and demolition;
- Emissions from the biological processes occurring in wastewater collection, treatment or sludge treatment and disposal and in CSO events and effluent discharge;
- Carbon credits from resources recovery, renewables, and vegetation sequestrations.

It is noteworthy that several sources of GHG emissions in the urban water sector specified in this *Guidelines* are explicitly excluded, including:

Biogenic CO₂

According to different sources of organic matter, the CO₂ can be divided into biogenic carbon (BC) and fossil carbon (FC) (Isaksen, 2012). From the perspective of the carbon cycle, if they can be traced to the organic matter generated by plant photosynthesis, they are referred to as biogenic CO₂. Because they come from the atmospheric carbon pool, which they also eventually return to, and because they are in the earth's short-term carbon cycle. Therefore, biogenic CO₂ does not result in a net increase of the total amount of CO₂ in the atmosphere. On the other hand, if the organic matter is ultimately traced back to fossil fuels or their converted products, the produced CO₂ is defined as fossil CO₂ (or non-biogenic carbon). since it belongs to the long-term carbon cycle represented by fossil fuels. In terms of wastewater management system, a small fraction of carbon comes from fossil sources (i.e., some components of cosmetics and skin care products). Therefore, the fossil (or non-biogenic) CO₂ is counted in the accounting boundaries. It should be emphasized that the CH₄ produced by incomplete mineralization of biogenic carbon should also be included in carbon emission accounting.

GHG emissions from industrial wastewater treatment facilities before discharge into urban sewage systems

Industrial wastewater, is required to meet quality standards before being discharged into urban wastewater collection systems as indicated by the *Water Quality Standards for Discharge of Sewage into Urban Sewers* (GB/T 31962-2015) (GB/T 31962-2015, 2015). Treatment facilities are generally built and operated by industries themselves and not included in the organizational boundaries of the water sectors in this *Guidelines*. Besides, industrial wastewater is different from domestic wastewater in terms of both composition and treatment processes. The carbon emission accounting methods used for domestic wastewater cannot be adopted directly for industrial wastewater. Therefore, the carbon emissions from industrial wastewater treatment facilities are not included in this *Guidelines*.

Secondary water supply stations

In the water supply system, water distribution networks are designed to convey enough water volume and pressure for consumers by pumps. In some cases, the pressure provided by the municipal level public water supply cannot meet the demand of some high-rising buildings. Thus, a secondary water supply station is often adopted with booster pumps which lead to an increase of energy consumption and GHG emissions. At present, the secondary water supply station is generally charged by certified properties. Overall, the GHG emissions of secondary water supply stations are excluded in this *Guidelines*. It should be noted that if the jurisdiction of the secondary water supply stations changes, their inclusion in the reporting boundaries of urban water sectors will need to be reconsidered.

Table 2.2 Summary of GHG emission activities of the urban water sector.

Scope (GHG Protocol)	Category (ISO 14064-1: 2018)	Water supply system	Wastewater management system	Water reclamation system	Stormwater system
Direct GHG emissions from activities within the management boundary of the accounting entity	Direct GHG emissions or carbon sinks	—	GHG emissions caused by oil, coal, and gas energy consumed on site at each stage of the full life cycle Non-biogenic CO ₂ , CH ₄ and N ₂ O produced by biochemical reactions in sewage collection, in-plant treatment or sludge in-plant treatment Carbon compensation in the form of resource and energy recovery	—	GHG emissions from stormwater wetland units Green stormwater facility carbon sink
Indirect GHG emissions due to electricity, steam, heating/cooling sources	Indirect GHG emissions - electricity and heat consumption	—	GHG emissions caused by electricity and thermal energy consumed on site at each stage of the life cycle	—	—
Other indirect GHG emissions caused by the activities of the accounting subject but outside the management	Indirect GHG emissions - transport Indirect GHG emissions - material inputs and services Indirect GHG emissions - asset and By-Product disposal ¹	—	GHG emissions at each stage of the life cycle due to the materials used and the means of transportation used for off-site disposal of waste GHG emissions from purchased and consumed materials at each stage of the life cycle GHG emissions generated during off-site disposal of muddy water or sludge generated during water quality treatment	—	—

Continued

Urban water sectors and their carbon emission

Scope (GHG Protocol)	Category (ISO 14064-1: 2018)	Water supply system	Wastewater management system	Water reclamation system	Stormwater system
boundary of the accounting subject, such as purchased consumables, waste disposal	Indirect GHG emissions - other	—	CO ₂ , CH ₄ and N ₂ O produced by combined system overflow (CSO) pollution and effluent discharged into the receiving water body biochemically	—	—
		—	—	—	—

1. It mainly covers the GHG emissions caused by the reaction of carbon and nitrogen contained in the by-products themselves when these by-products produced by each system (for example, sludge) are transported to the outside of the management boundary for disposal. However, emissions from normal activities of other operating entities mainly engaged in waste treatment are not included. For example, the residual sludge of a sewage treatment plant is transported to a landfill after in-plant treatment (stabilized dehydration, etc.), and the carbon biochemical reaction contained in the generated CH₄ should be counted but the GHG generated by normal operations such as landfills are not counted.

Chapter 3

Carbon accounting principles and methodologies

3.1 BASIC PRINCIPLES

The application of principles is the key to ensure true and fair of accounting, as well as being important in interpreting and analyzing the results. The principles which are the basis for making an accounting plan and implementing accounting practices which are (ISO, 2018):

- **Correlation:** All activities related to greenhouse gas (GHG) emissions in the urban water sector should be included in the accounting boundary to ensure that the accounting results truly represent the level of GHG emissions.
- **Integrity:** All emission activities within the reporting boundaries should be accounted for and any exemptions should be explicitly stated and explained.
- **Consistency:** The accounting schemes, boundaries, and methods in an inventory period should be consistent, and any revisions and adjustments that may affect the accuracy of results should be clearly recorded and marked.
- **Transparency:** Data sources, accounting equations and factors, and any assumptions made must be clearly introduced and displayed.
- **Accuracy:** Measures should be taken to avoid foundational and/or subjective errors in accounting practices, to reduce uncertainty, and to ensure accuracy.

3.2 ACCOUNTING PROCEDURES

3.2.1 General introduction

The operation and management of the urban water sectors involve multiple entities, including the water companies responsible for the operation and maintenance and the water departments responsible for the regulations and supervision. In addition, water sector associations also play a critical role in grouping companies, data collection and sharing, formulating group standards, and ensuring the healthy development of the urban water sector. The sector associations not only play the role in collaboration between the water companies, but also form a bridge between water companies and the water departments. In terms of their roles in carbon accounting of water sectors, these three entities vary a lot in accounting boundaries, accounting purposes, and accuracy requirements. As depicted in Figure 3.1, water companies are the smallest unit implementing carbon accounting independently. In some cases, the accounting boundaries adopted by different water companies are probably not consistent, which will induce difficulties for the results interpretation and comparative analysis. Herein, water sector

associations and water departments could do more by defining the accounting boundary and guiding the accounting practice. Then, the accounting results of water companies can be collected and easy to compile by water sector associations to present the GHG emission of the whole water sectors whereby water departments then could formulate relevant policy more scientifically.

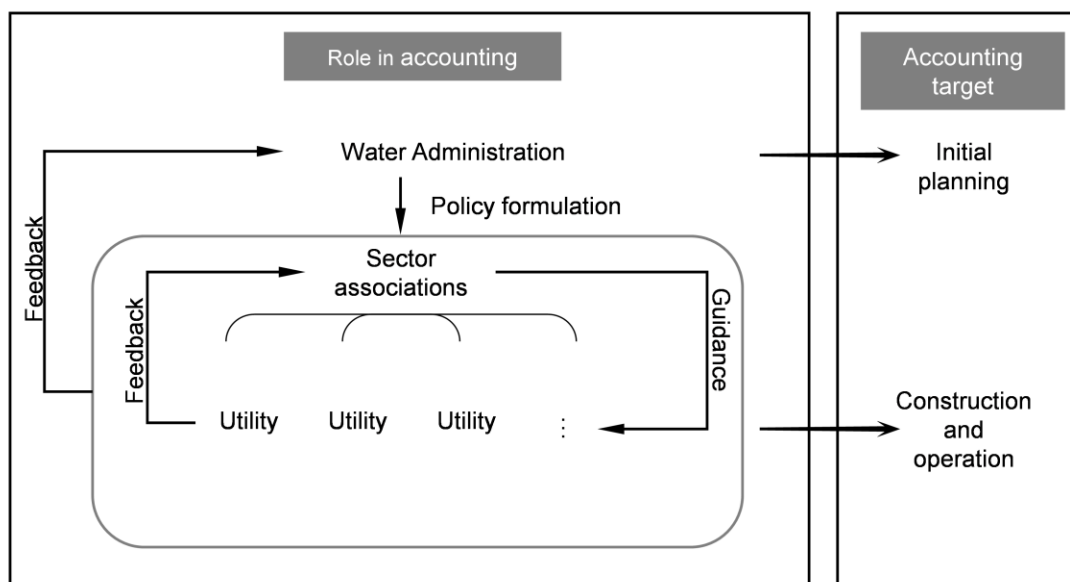


Figure 3.1 Correlation and roles of water utilities, sector associations, and water departments in water sector carbon accounting.

3.2.2 Water companies

Water companies are the executors of carbon accounting of their owned facilities as they have the direct access to the emission activity data. The key issues for the water companies are to ensure the data quality, scientific determination of accounting boundaries, and rationality of the accounting methods as discussed above. Thereinto, sector associations and/or water departments could play roles and provide the basic frameworks and guidance. Overall, to make the accounting results more accurate and convincing, water companies are recommended to follow a standardized procedure as summarized below.

- Determine the accounting boundary and clarify the emission activities and categories of GHG emissions;
- Determine the accounting methods in terms of each emission activity and the required accounting accuracy;
- Collect and obtain the required activity data and emission factors, and do on-site monitoring if necessary;
- Implement carbon accounting, summarize and organize the accounting results, and calculate the total carbon emissions and carbon emission intensity at the company level;
- Compile and archive carbon accounting reports;
- Report the results to the sector associations or water departments.

3.2.3 Water sector associations

The roles of the water sector associations in carbon accounting lie in two aspects. One is to instruct water companies to implement accounting to avoid interpretation bias. The other is to summarize and analyze the results submitted by water companies whereby the water sector associations can calculate the total emission and intensity at the sector level. To ensure the quality and accuracy of accounting results from water companies, water sector associations should take their role as follows:

- Coordinate the members of water companies to implement carbon accounting;
- **Frame** a unified accounting boundary, accounting method, emission factors, etc.;
- Guide members to obtain activity data and to implement accounting
- **Summarize** and analyze the accounting results from members to calculate total carbon emission and emission intensity at sector level (See Section 3.3.3);
- Compile and archive carbon accounting reports;
- Share the results with the water departments.

3.2.4 Water departments

Water departments are taking the responsibility of guiding the water companies to provide qualified water and/or wastewater services to residents. Besides, it is also the duty of water departments to promote the low carbon development of water companies. As such, there are also two roles water departments could play in carbon accounting of water sectors. On the one hand, water departments could guide water companies to proceed carbon accounting as sector association does (See Section 3.2.3). On the other hand, water departments also do carbon accounting by themselves at a city-level to support governments to do water sector planning. In most cases, the later scenario is in the planning and decision making stage, and the data necessary for carbon accounting should be provided by water companies, such as average activity data and/or emission factors. Overall, water companies can implement the second scenario with the following steps:

- Map the water flow and balance of the water sector under various planning schemes, and obtain the necessary emission activity data;
- Obtain and summarize the emission factors for carbon accounting based on the results provided by water sector associations or water companies;
- Implement carbon accounting and support planning decision making (See Section 3.3.4);
- Compile and archive carbon accounting reports;

3.3 ACCOUNTING METHODOLOGIES

3.3.1 General introduction

As stated above, the obtaining of emission activity data is vitally important to the accuracy of carbon accounting. In these *Guidelines*, a variety of activity data and sources are considered. If the direct activity data (such as electricity or fuel consumption) can be obtained, carbon accounting can be implemented by multiplying the activity data and the emission factors (Figure 3.2, Method 1). However, it may be challenging to obtain direct activity data in some cases, though some other related activity data can be obtained which can then be used to obtain the direct activity data and so to complete the carbon accounting—namely, Method 2 and Method 3 (Figure 3.2). It is worth noting that the accuracy of three

methods differs, and that Method 3 is considered *good practice* while Method 1 is the most accurate method and should be given priority.

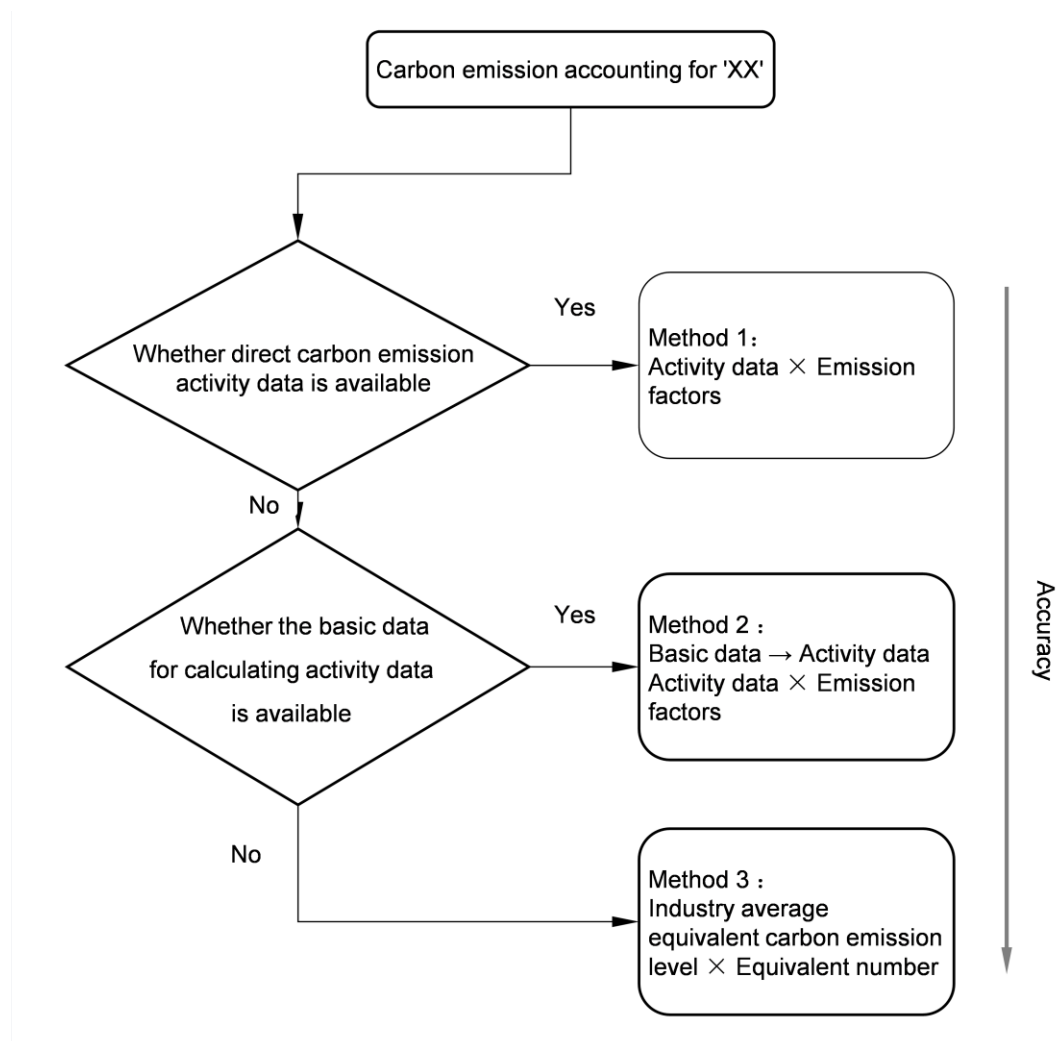


Figure 3.2 Schematic diagram of carbon emission accounting methods and their accounting accuracy levels.

The accounting results of GHG emissions are presented as CO₂ equivalent (CO₂ -eq) in these *Guidelines*. For carbon accounting in the planning and construction and asset replacement and demolition stages, the total carbon emissions are calculated. In the operation and maintenance stage, carbon emission intensity is calculated. In terms of carbon emission intensity, the water supply system, wastewater management system, and water reclamation system are calculated based on the water volume handled in the inventory year (plants are based on the treatment capacity and pipe networks are based on water volume conveyed). Regarding the stormwater management system, emission intensity is calculated based on the volume of stormwater handled by the management facilities.

The carbon emission and carbon emission intensity can be converted between each other as shown in Equation (3.1-3.2):

$$CE = CES \times Q \times 365 \quad (3.1)$$

$$CES = \frac{CE}{Q \times 365} \quad (3.2)$$

Where:

- CE — Total carbon emissions in the inventory year, kg CO₂-eq
- CES — Carbon emission intensity in the inventory year, kg CO₂-eq/m³
- Q — Average daily treated water volume, m³/d, water supply and sewage treatment plants are measured by water quality to the standard, water distribution networks and sewage pipelines are measured by total water conveyance volume and rainwater sectors are measured by conveyance and management water volume
- 365 — 365 days/year

3.3.2 Water companies

The emission activities of water sectors and the corresponding accounting method index are summarized and depicted in Figure 3.3~Figure 3.7.

Guidelines for Carbon Accounting and Emission Reduction in the Urban Water Sector

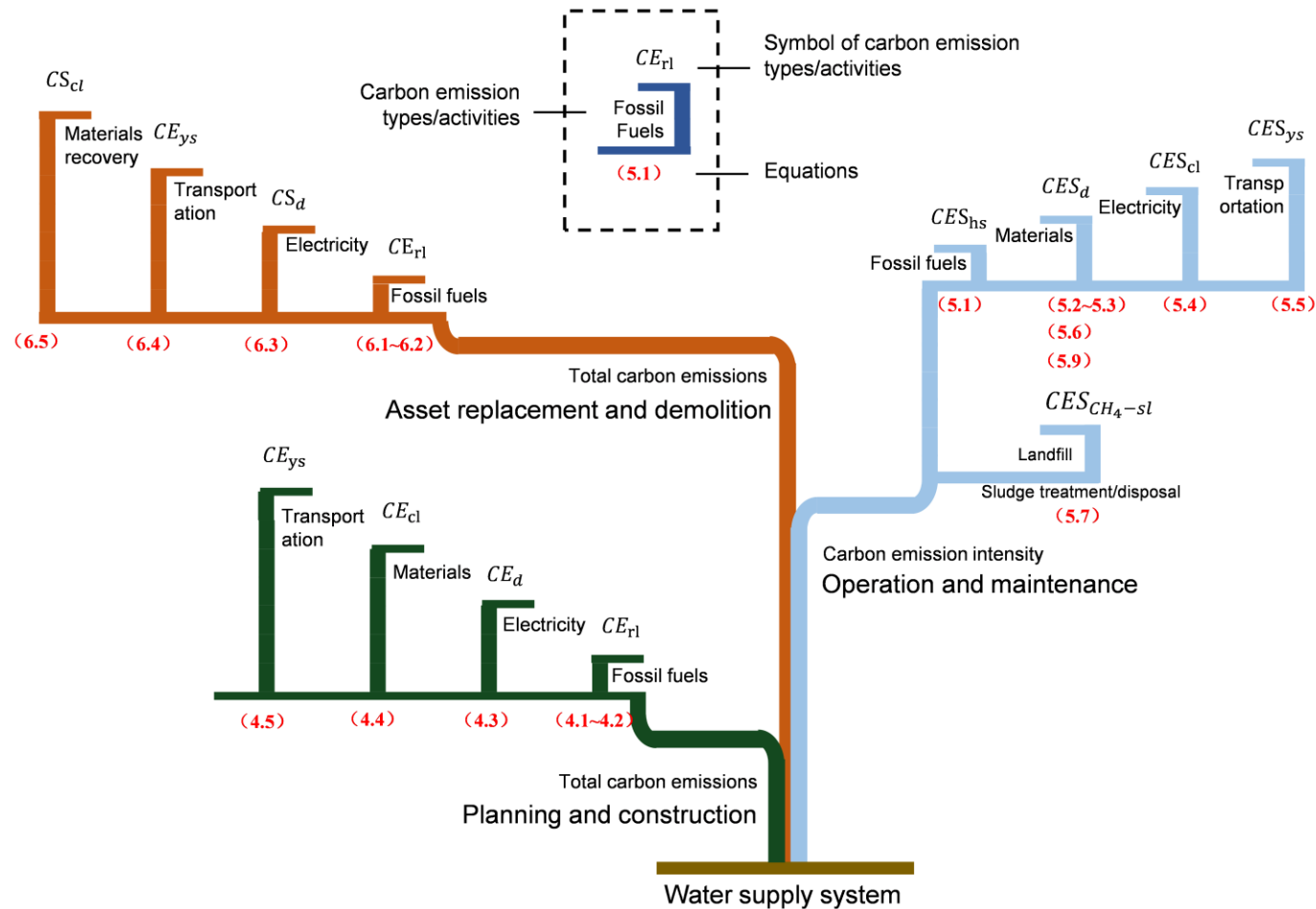


Figure 3.3 Relationship tree and accounting index tree of carbon emission activities in the whole life cycle of water supply system.

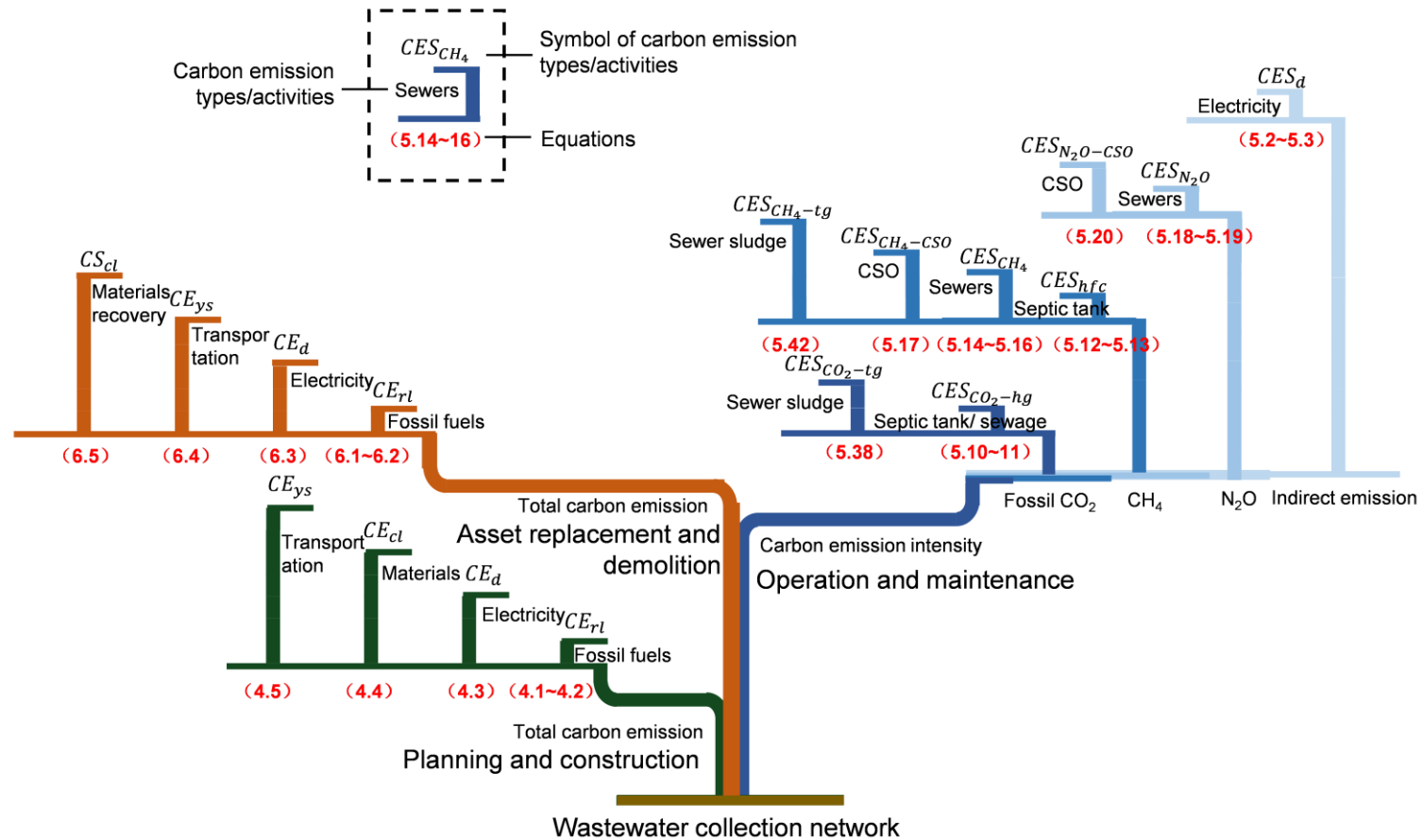


Figure 3.4 Relationship tree and accounting index tree of carbon emission activities in the whole life cycle of wastewater collection networks.

Guidelines for Carbon Accounting and Emission Reduction in the Urban Water Sector

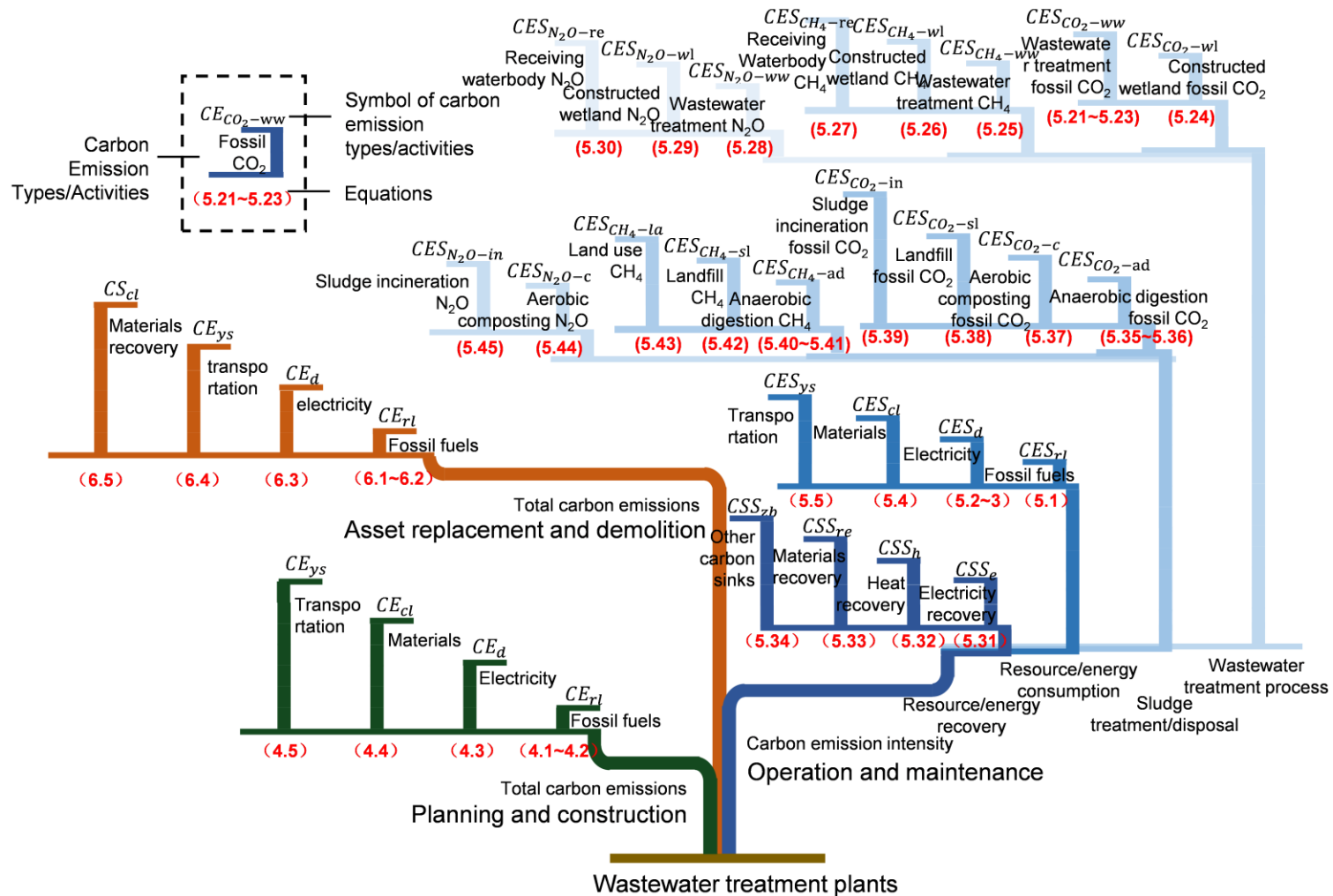


Figure 3.5 Relationship tree and accounting index tree of carbon emission activities in the whole life cycle of wastewater treatment plants.

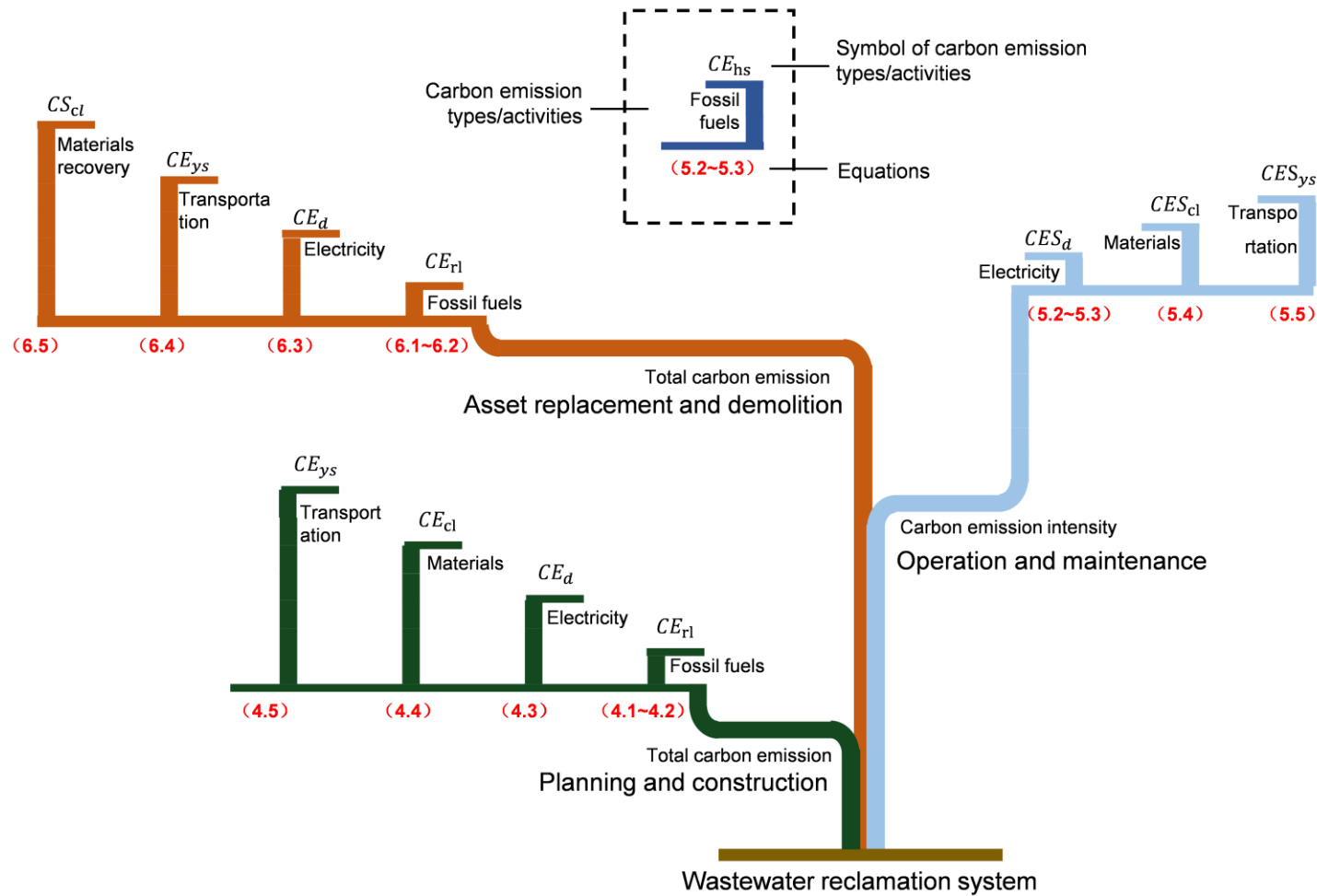


Figure 3.6 Relationship tree and accounting index tree of carbon emission activities in the whole life cycle of wastewater reclamation system.

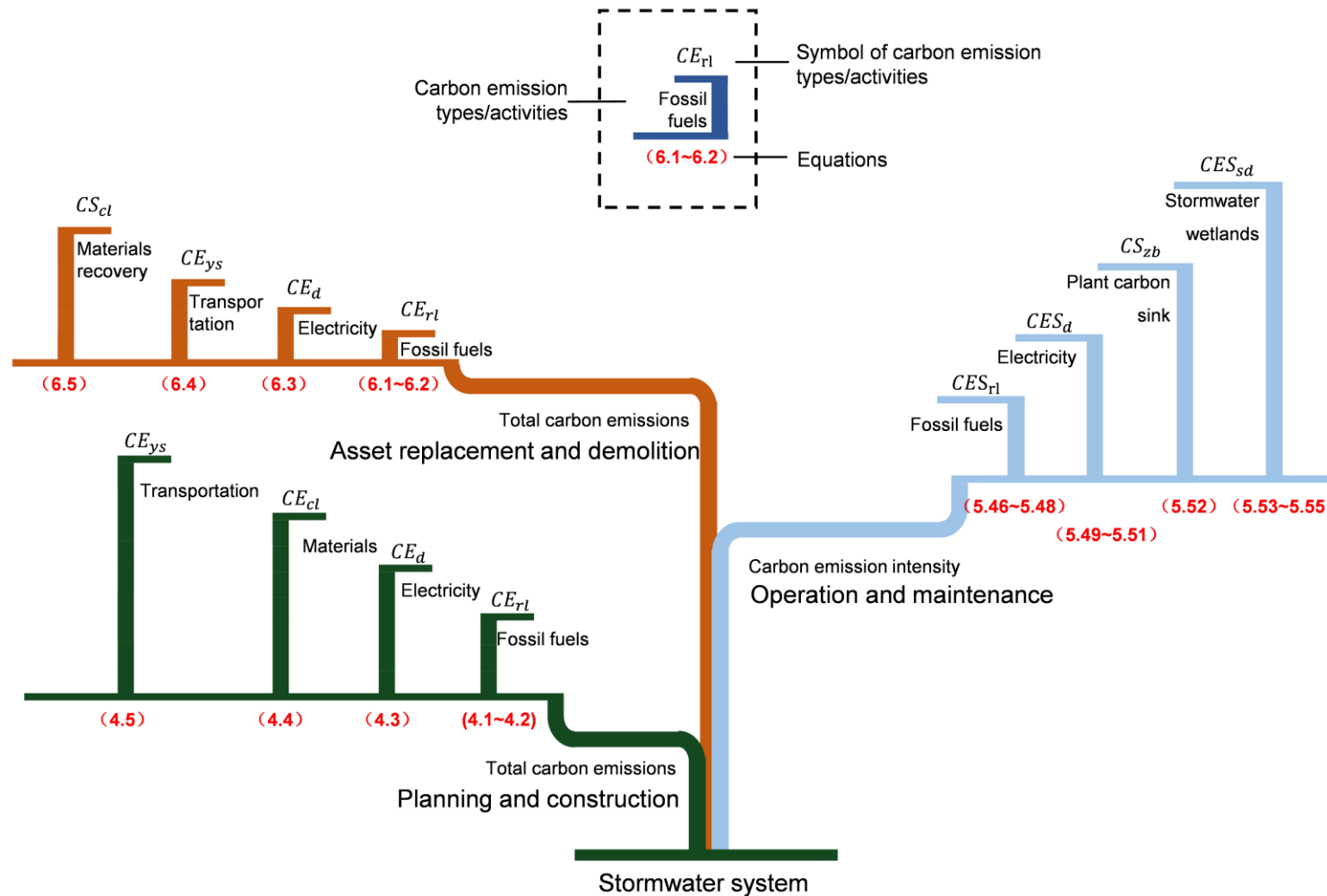


Figure 3.7 Relationship tree and accounting index tree of carbon emission activities in the whole life cycle of stormwater management system.

3.3.3 Water sector associations

As stated above, water sector associations collect the accounting results from water companies and provide intellectual support, e.g., emission factors, for water sector planning. In terms of the carbon emission intensity, it is highly related to the facility scale, treatment method (such as the main biological treatment unit of sewage treatment plant), influent and effluent water quality, etc. Thus, when calculating the carbon emission intensity factor, it is recommended to consider the above factors if possible. If the results cannot be grouped due to a lack of data, one emission factor can be calculated for each system following Equation (3.3) and (3.4).

$$\overline{CES} = \frac{\sum_{i=1}^n CE_i}{\sum_{i=1}^n Q_i \times 365} \quad (3.3)$$

$$\overline{CES} = \frac{\sum_{i=1}^n (CES_i \times Q_i)}{\sum_{i=1}^n Q_i} \quad (3.4)$$

Where:

- $\overline{CES}$ — System weighted average carbon emission intensity, kg CO₂-eq/m³
- CE_i — The first i operating company, kg CO₂-eq
- CES_i — Of the first i operating company, kg CO₂-eq/m³
- Q_i — The first i operating enterprise, m³/d, the reference water quality and volume of water supply and sewage treatment plants, the reference water transfer volume of water supply pipe network and sewage pipes, and the reference transfer and management water volume of rainwater system
- n — The group l runs the business together

The pipe network is recommended to be classified according to material and pipe diameter. For water treatment plants and wastewater treatment plants, it is recommended to group the results to calculate specific emission factor as follows:

- **Classification of water treatment plants:** scale (<50,000 m³/d; 5~100,000 m³/d; >100,000 m³/d) and treatment methods (conventional method; pretreatment + conventional; pretreatment + conventional + advanced treatment).
- **Classification of wastewater treatment plants:** scale (<10,000 m³/d; 1~100,000 m³/d; >100,000 m³/d) and process (AAO; AO; oxidation ditch; SBR; common aeration tank).

3.3.4 Water departments

The accounting work taken by water departments is different from that of water companies and water sector associations. In terms of accounting procedures, it can be divided into two steps. The first step is to establish the water flow and balance among different systems within the organizational boundaries. The second step is to conduct carbon emission accounting based on the water flow and balance and the emission factors provided by water sector associations. If the carbon emission accounting results of all water companies within the boundaries are available, the total carbon emission is calculated based on Equation (3.5). In addition, another total carbon emission can be obtained with water balance based on Equation (3.6) which can be used to check the integrity of carbon accounting by the water companies. For some planning areas, if it is difficult to obtain the accounting results from water companies directly, the total carbon emission can only be calculated according to Equation (3.6).

$$CE = \sum_{i=1}^n CE_i \quad (3.5)$$

Where:

- CE — Total carbon emissions of the city, kg CO₂-eq
 CE_i — The first i enterprise or group, kg CO₂-eq

$$CE = \sum_{i=1}^n CES_i \times Q_i \quad (3.6)$$

Where:

- CE — Total annual carbon emissions of the city, kg CO₂-eq
 CES_i — The first i water system, kg CO₂-eq/m³
 Q_i — The first i water affairs system, m³, water supply and sewage treatment plants are measured by the water quality to the standard, the water transmission and distribution network and sewage pipelines are measured by the total transfer water volume, and the rainwater system is measured by the transfer and management water volume

Chapter 4

Planning and construction

4.1 OVERVIEW

The main emission activities in the planning and construction stage of water sectors include: i) direct carbon emission generated by burning fossil fuels; ii) indirect carbon emission from the generation of imported electricity; iii) indirect carbon emission from the production of the construction materials consumed; iv) indirect carbon emission from the transportation of various goods. As the activities in planning and construction for different systems are consistent, the accounting equations provided in this chapter are applicable to any systems of urban water sectors.

According to the completeness and accessibility of the material and energy inventory of the planning and construction, two accounting methods are provided for emission activities in this chapter: i) if the inventory can be obtained, the activity data and corresponding emission factors (EFs) are used for accounting (Sections 4.2 to 4.4), which has a relatively high accuracy and should be given priority; ii) if it is difficult to obtain the inventory, the total carbon emission can be roughly estimated according to the processing capacity or investment of the facilities by leveraging the prefabricated calculation diagram and/or the quick reference tables (Section 4.5).

4.2 DIRECT EMISSIONS OF FOSSIL FUEL

The direct emission of fossil fuel mainly refers to carbon emissions caused by the on-site consumption of fossil fuels (coal, oil, natural gas and their derivative fuels) for construction machinery. According to the available construction inventory, carbon emission of fossil fuel consumption can be calculated using the following two methods: i) if the types of fossil fuels and the corresponding consumption can be obtained, Method 1 is recommended for calculating the carbon emissions (Equation (4.1)), and the accuracy of the results is relatively high; ii) if the details of fossil fuel consumption are not available, it can be calculated according to the number of machine shifts (Method 2, Equation (4.2)), though the accuracy of the result is quite low.

Method 1:

$$CE_{rl} = \sum_{i=1}^n (M_{rl,i} \times EF_{rl,i}) \quad (4.1)$$

Where:

- CE_{rl} — Carbon emissions of fossil fuel, kg CO₂-eq
- $M_{rl,i}$ — Total amount of the fossil fuel i consumed, kg or m³
- $EF_{rl,i}$ — The EFs of fossil fuel i , kg CO₂-eq/kg or kg CO₂-eq/m³. See Appendix B.1
- n — Total number of fossil fuels used

Method 2:

$$CE_{rl} = \sum_{i=1}^n (T_i \times S_i \times EF_{rl,i}) \quad (4.2)$$

Where:

- CE_{rl} — Carbon emissions of fossil fuel, kg CO₂-eq
- T_i — The number of shifts of machine i
- S_i — The fossil fuel consumption per shift of machine i . See *Building Carbon Emission Calculation Standard (GB/T 51366-2019)*
- $EF_{rl,i}$ — The EFs of fossil fuel for machine i , kg CO₂-eq/kg or kg CO₂-eq/m³. See Appendix B.1
- n — The number of machine types used

4.3 INDIRECT EMISSIONS FROM ELECTRICITY CONSUMPTION

The electricity consumption in the construction period also contributed a significant portion of the total greenhouse gas (GHG) emission which is calculated as shown in Equation (4.3).

$$CE_d = E_d \times EF_d \quad (4.3)$$

Where:

- CE_d — Carbon emission of electricity consumption, kg CO₂-eq
- E_d — Total consumption of electricity, kWh
- EF_d — The EFs specific for the imported electricity, kg CO₂-eq/kWh. See Appendix B.2

4.4 INDIRECT EMISSIONS FROM MATERIAL CONSUMPTION

The indirect emission from the production processes of various materials used in the planning and construction stage can be calculated as shown in Equation (4.4).

$$CE_{cl} = \sum_{i=1}^n (M_{cl,i} \times EF_{cl,i}) \quad (4.4)$$

Where:

- CE_{cl} — Carbon emission in the production process of consumed materials, kg CO₂-eq
- $M_{cl,i}$ — The amount of material used by type i , t or m³
- $EF_{cl,i}$ — The EFs of material i , kg CO₂-eq/m³ or kg CO₂-eq/t. See Appendix B.3
- n — Total number of construction materials used

4.5 INDIRECT EMISSIONS FROM TRANSPORTATION

The construction materials and/or services consumed during the planning and construction stage require transportation which will generate a certain amount of GHG emissions. The accounting can refer to Equation (4.5).

$$CE_{ys} = \sum_{i=1, j=1}^{n, l} (M_{ys, i, j} \times L_{ys, i, j} \times EF_{ys, j}) \quad (4.5)$$

Where:

- CE_{ys} — Carbon emission of material transportation, kg CO₂-eq
- $M_{ys, i, j}$ — The total amount of materials transported by the transportation type j in the procurement i , t
- $L_{ys, i, j}$ — The distance of materials transported by the transportation j in the procurement i , km
- $EF_{ys, j}$ — The EFs of transportation j , kg CO₂-eq/(t·km). See Appendix B.4
- n — The number of procurement activities
- l — The number of transportations in one procurement activity

4.6 OVERALL ESTIMATION AS A WHOLE

4.6.1 Estimation based on water volume handled

In some cases, the detailed inventory about materials and/or energy consumption is not available and the above methods are no longer applicable. Alternatively, the GHG emission of the construction of a specific facility could be quantified by regarding the facility's emission as a whole. Thereinto, the emission activities are the amounts of water the facility handles, i.e., treatment capacity for water treatment plants, wastewater treatment plants, and wastewater reclamation plants; water volume conveyed by pipe networks. Then, the corresponding emission intensities were formulated by summarizing and analyzing dozens of case studies as shown in Table 4.1~4.4 (See Appendix C.1-C.5 for details). Overall, the methods for different facilities refer to Equation (4.6~4.9) as listed below.

$$CE_{js} = CES \times Q \quad (4.6)$$

Where:

- CE_{js} — Carbon emission of a water treatment plant or a wastewater treatment plant, t CO₂-eq
- CES — Emission intensities, t CO₂-eq/(10,000 m³/d). See Table 4.2 and Table 4.4
- Q — Treatment capacity, m³/d

$$CE_{js} = CES \times L \quad (4.7)$$

Where:

- CE_{js} — Carbon emission of pipe networks, t CO₂-eq
- CES — Emission intensities, t CO₂-eq/km. See Table 4-1 and Table 4-3
- L — Pipe length with different diameters, km

$$CE_{gh} = \sum_{i=1, j=1}^{n, l} (CES_i \times S_i + CES_j \times L) \quad (4.8)$$

Where:

- CE_{gh} — Carbon emissions of stormwater management system, 10⁴ t CO₂-eq
- CES_i — Emission intensities of facility i , kg CO₂-eq/m². See Appendix C.5

- S_i — The footprint of green facility i , m^2
 CES_j — Emission intensity of stormwater sewer j , $kg\ CO_2\text{-eq}/m^2$
 L — The length of the stormwater sewer, m. If the length of stormwater sewer is not available, Equation (4.9) can be applied
 n — The number of facilities types used
 l — The number of rainwater pipes types used

$$L = S \times D \times 10^3 \quad (4.9)$$

Where:

- S — Area of organizational boundary, km^2
 D — Density of stormwater sewer, km/km^2

Table 4.1 Summary of EFs of pipes by materials and diameters for water distribution network.

Diameter	Emission factor (t CO ₂ -eq/km) ¹	
	Ductile iron pipes	Steel pipe
DN 300	105.1	181.3
DN 400	155.4	228.6
DN 500	214.0	287.2
DN 600	280.7	345.4
DN 700	356.1	403.4
DN 800	440.9	461.9
DN 900	532.3	519.9
DN 1000	630.5	578.1
DN 1200	856.0	694.4

1. The depth of cover is 0.7 m.

Table 4.2 Summary of EFs of water treatment plants by different processes and treatment capacity.

Processes	Emission factor (t CO ₂ -eq/10,000 m ³)		
	< 50,000 m ³ / d	50,000 m ³ / d-100,000 m ³ / d	≥ 100,000 m ³ / d
Conventional treatment	1,209.5	1,070.8	833.1
Pretreatment + conventional treatment	1,445.9	1,300.7	1,011.1
Pretreatment + conventional treatment+ advanced treatment	1,942.8	1,785.5	1,437.5
Water distribution pump station	389.6	339.4	266.8

Table 4.3 Summary of carbon emission factors of wastewater collections pipes by materials and diameters for drainage pipelines.

Reinforced concrete pipes		HDPE pipes	
Diameter	EFs ¹ (t CO ₂ -eq/km)	Diameter	EFs ¹ (t CO ₂ -eq/km)
<i>dn</i> 600	52.4	<i>DN</i> 600	17.9
<i>dn</i> 800	86.7	<i>DN</i> 800	20.8
<i>dn</i> 1000	126.7	<i>DN</i> 1000	24.4
<i>dn</i> 1200	178.8	<i>DN</i> 1200	29.9
<i>dn</i> 1400	263.5	<i>DN</i> 1400	36.1
<i>dn</i> 1600	346.4	<i>DN</i> 1600	44.1
<i>dn</i> 1800	414.9	<i>DN</i> 1800	49.6
<i>dn</i> 2000	536.5	<i>DN</i> 2000	55.1

1. The depth of cover is 0.7 m.

Table 4.4 Summary of carbon emission factors of wastewater treatment plants by different processes and treatment capacity.

Processes	Emission factor (t CO ₂ -eq/10,000 m ³)		
	< 10,000 m ³ /d	10,000 m ³ /d-100,000 m ³ /d	≥ 100,000 m ³ /d
Wastewater treatment plant meeting Discharge Standard Class I-B (excluding sludge digestion)	3,275.0	2,239.9	1,575.3
Wastewater treatment plant meeting Discharge Standard Class I-B (including sludge digestion)	4,226.4	2,543.8	2,022.9
Wastewater treatment plant meeting Discharge Standard Class I-A (no sludge digestion)	4,748.9	2,919.8	2,293.1
Wastewater treatment plant meeting Discharge Standard Class I-A (including sludge digestion)	5,657.7	2,503	2,750.8
Upgrading and transforming to a wastewater treatment plant meeting Discharge Standard Class I-A (excluding sludge digestion)	1,602.3	1,037.6	794.1

Upgrading and transforming

to a wastewater treatment

plant meeting Discharge

2 421.2

1,369.5

1,054.7

Standard Class quasi-IV

(excluding sludge digestion)

1. The influent COD is assumed to be 350 mg/L (BOD=170 mg/L).

Carbon emissions can also be directly determined through the carbon emission calculation diagram of planning and construction (Figure 4.1 ~ Figure 4.4) which are visualized from Table 4.1~4.4.

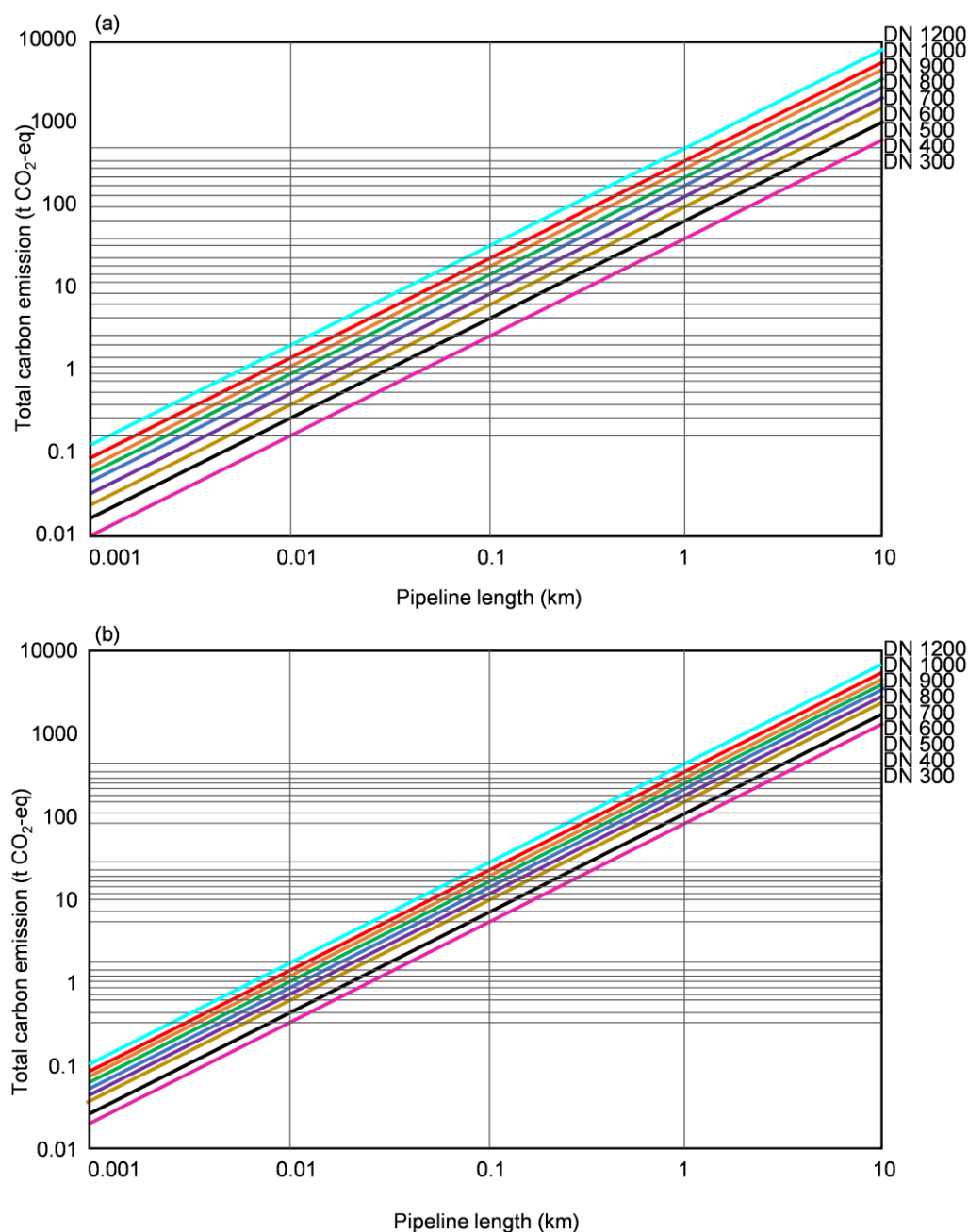


Figure 4.1 Carbon emission calculation diagram of water distribution network construction with different materials (a. Ductile iron pipes; b. Steel pipes) (with a soil cover depth of 0.7m).

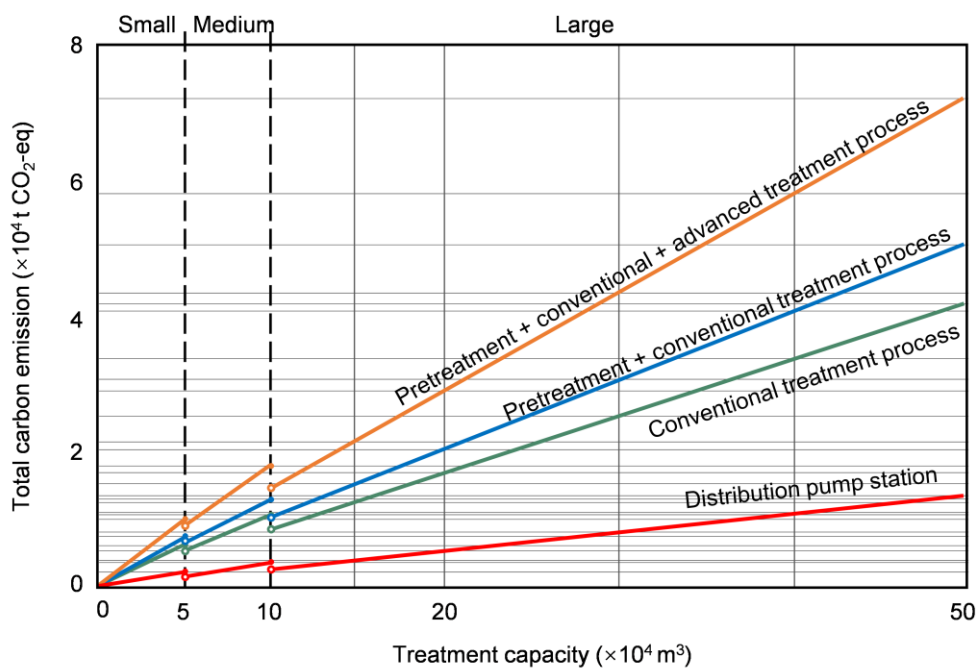


Figure 4.2 Carbon emission calculation diagram for construction of water treatment plants and pumping stations.

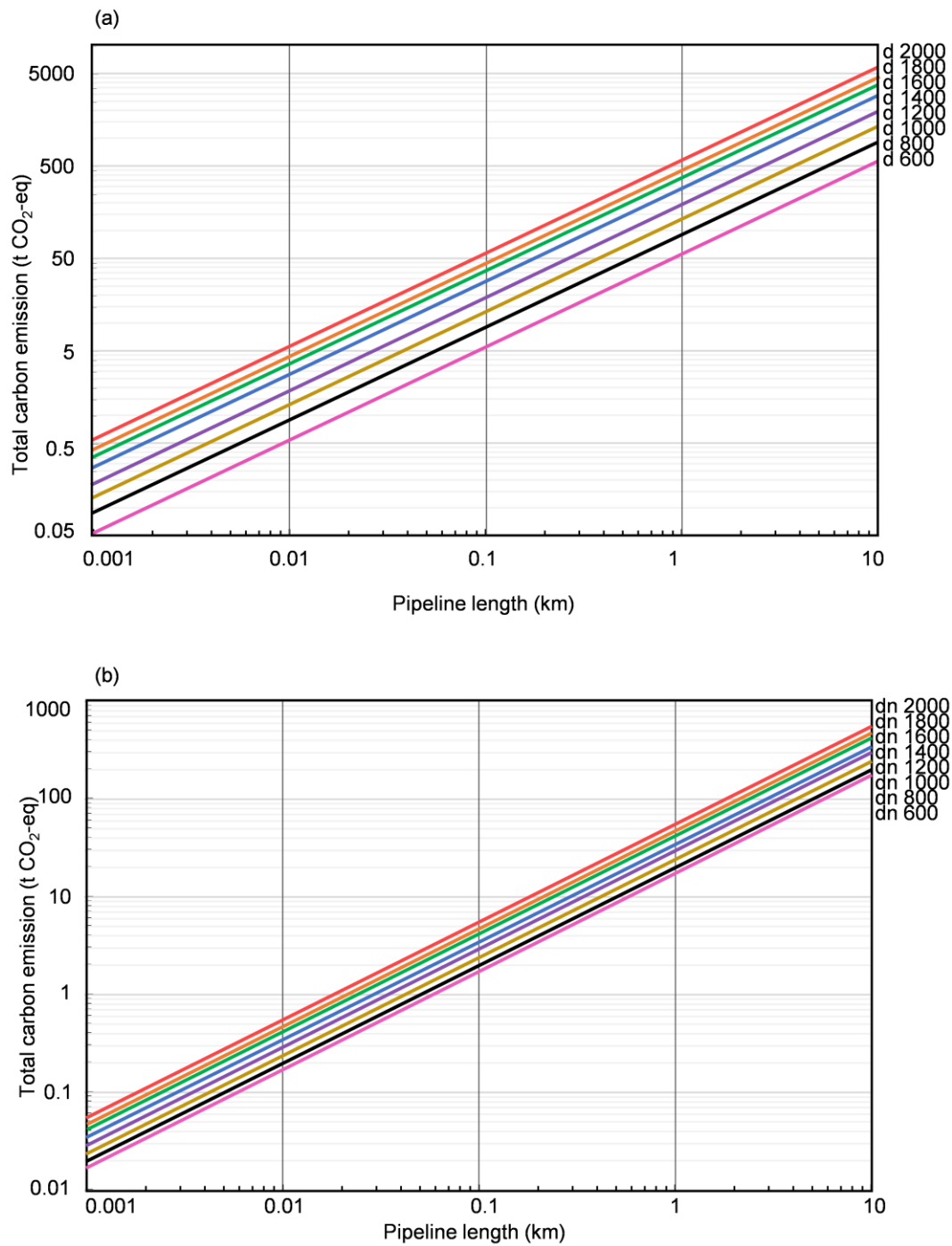


Figure 4.3 Carbon emission calculation diagram of sewage pipe construction with different materials (a. Reinforced concrete pipes; b. HDPE pipes) (with a soil cover depth of 0.7m).

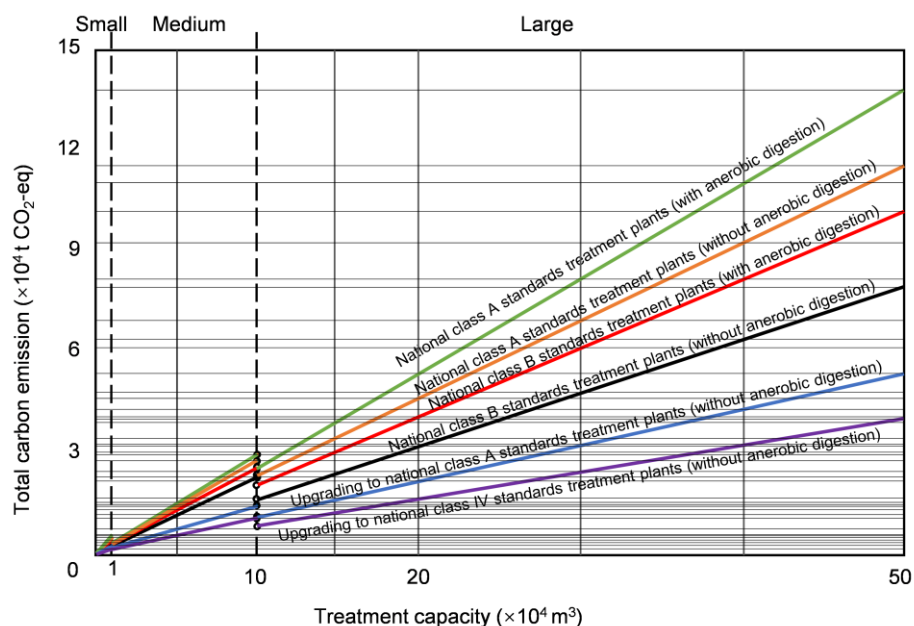


Figure 4.4 Calculation diagram of carbon emission in construction of wastewater treatment plant with different effluent discharge standards.

4.6.2 Estimation based on investment

In view of the capital cost embodied in the materials used in construction, carbon emissions can also be roughly calculated according to investment as shown in Equation (4.10):

$$CE_{js} = CES_{tz} \times TZ \quad (4.10)$$

Where:

- CE_{js} — Carbon emissions of planning and construction, 10^4 t CO_2 -eq
- CES_{tz} — Carbon emission intensity, 10^4 t CO_2 -eq/ 10^4 CNY. See Table 4.5~4.9
- TZ — The total investment, 10^4 CNY

Table 4.5 Summary of investment carbon emission factors of water distribution pipes by different materials and diameters.

Diameter	Emission factor (t CO_2 -eq/ 10^4 CNY)	
	Ductile Iron Pipes	Steel pipe
DN 300	1.33	1.88
DN 400	1.54	1.99
DN 500	1.60	1.90
DN 600	2.12	2.40
DN 700	2.42	2.68
DN 800	2.91	2.98
DN 900	3.13	2.91
DN 1000	3.40	2.98
DN 1200	4.05	3.16

Table 4.6 Summary of investment carbon emission factors of water treatment plants by different treatment processes and capacities.

Processes	Emission factor (t CO ₂ -eq/10 ⁴ CNY)		
	< 50,000 m ³ /d	50,000 m ³ /d-100,000 m ³ /d	≥ 100,000 m ³ /d
Conventional treatment	1.03	1.02	1.00
Pretreatment + conventional treatment	1.03	1.04	1.01
Pretreatment + conventional treatment+ advanced treatment	1.00	1.01	1.00
Water distribution pump station	1.00	0.98	0.99

Table 4.7 Summary of investment carbon emission factors of water treatment plants by different treatment processes and capacities.

Reinforced concrete pipes		HDPE pipe	
Diameter	EFs (t CO ₂ -eq/ 10 ⁴ CNY)	Diameter	EFs (t CO ₂ -eq/10 ⁴ CNY)
DN 600	3.1	DN 600	0.5
DN 800	4.3	DN 800	0.5
DN 1000	5.2	DN 1000	0.5
DN 1200	6.1	DN 1200	0.5
DN 1400	8.1	DN 1400	0.5
DN 1600	8.8	DN 1600	0.5
DN 1800	9.2	DN 1800	0.5
DN 2000	10.7	DN 2000	0.5

Table 4.8 Summary of investment EFs of wastewater treatment plants by different effluent standards and treatment capacity.

Processes	EFs (t CO ₂ -eq/ 10 ⁴ CNY)		
	< 10,000 m ³ /d	10,000 m ³ /d-100,000 m ³ /d	≥ 100,000 m ³ /d
Wastewater treatment plant meeting Discharge Standard Class I-B (excluding sludge digestion)	1.3	1.3	1.1
Wastewater treatment plant meeting Discharge Standard Class I-B (including sludge digestion)	1.3	1.2	1.2
Wastewater treatment plant meeting Discharge Standard Class I-A (no sludge digestion)	1.3	1.2	1.2

Processes	EFs (t CO ₂ -eq/ 10 ⁴ CNY)		
	< 10,000 m ³ /d	10,000 m ³ /d-100,000 m ³ /d	≥ 100,000 m ³ /d
Wastewater treatment plant meeting Discharge Standard Class I-A (including sludge digestion)	1.3	0.9	1.2
Upgrading and transforming to a wastewater treatment plant meeting Discharge Standard Class I-A (excluding sludge digestion)	1.3	1.2	1.2
Upgrading and transforming to a wastewater treatment plant meeting Discharge Standard Class quasi-IV (excluding sludge digestion)	1.3	1	1

Table 4.9 Summary of investment EFs of green facilities.

Stormwater control facilities		Total emission intensity (t CO ₂ -eq/10 ⁴ CNY)	
Grassed swales	Transfer type dry grass planting ditch	0.42	2.81
	Permeable dry grass ditch	0.99	6.59
	Wet grass ditch	1.83	12.22
Permeable pavement	Sidewalk	1.89	6.32
	Driving load ≤ 5 t	4.01	13.37
	Driving load = 5-8 t	5.20	17.34
	Driving load = 8-13 t	6.00	19.99
Bioretention facility	Bioretention zone	0.19	1.03
	Rain garden	0.55	2.92
	(taking purification rain garden as an example)		
	Simple ecological tree pool	2.07	11.06
	Purifying ecological tree pool	1.95	10.39
	High flower beds	2.87	15.31
	(taking the stagnant high flower beds as an example)		
Low elevation greenbelt		2.62	3.27
Green roof	Simple	2.85	8.56
	Garden style	6.58	19.74
Cistern		4.38	6.57
Pond wetland		0.14	0.22

Chapter 5

Operation and maintenance

5.1 OVERVIEW

The main emission activities of the urban water sector in operation and maintenance stage include: i) direct carbon emissions from biochemical processes, particularly in wastewater management system; ii) direct carbon emissions generated by burning fossil fuel; iii) indirect carbon emissions from the consumption of imported electricity; iv) indirect carbon emissions from the production of chemicals and/or materials consumed; v) indirect carbon emissions from the transportation of goods. It is worth noting that some facilities of the urban water sector can fix CO₂ and achieve GHG removals through plants sequestration. Besides, energy and resource recovery and importation also produce carbon credits, a factor which is also considered.

The accounting methods for some common emission activities among various systems are summarized in Section 5.2, and the other exclusive activities for specific system are presented in Sections 5.3~5.6, respectively.

5.2 GENERAL PROVISIONS

5.2.1 Direct emissions of fossil fuel

The direct emission of fossil fuels mainly refers to carbon emissions caused by the on-site consumption of fossil fuels (coal, oil, natural gas and their derivative fuels) by some mechanical pieces of equipment. This is calculated by Equation (5.1).

$$CES_{rl} = \sum_{i=1}^n (M_{rl,i} \times EF_{rl,i}) / Q \quad (5.1)$$

Where:

- CES_{rl} — Carbon emission intensity of fossil fuel, kg CO₂-eq/m³
- $M_{rl,i}$ — Total amount of fossil fuel i consumed in the inventory year, kg/a or m³/a
- $EF_{rl,i}$ — The emission factors (EFs) of fossil fuel i , kg CO₂-eq/kg or kg CO₂-eq/m³. See Appendix B.1
- Q — The volume of water handled by the targeted systems and/or facilities in the inventory year, m³/a. See Section 3.3.1
- n — Total number of fossil fuel used

5.2.2 Indirect emissions from electricity consumption

The calculation of carbon emissions caused by electricity consumption in the operation and maintenance stage can be implemented by the following methods: i) when electricity consumption data is available, Method 1 can be applied (Equation (5.2)); ii) in terms of pumps, if the electricity consumption data is

not available, Method 2 is recommended by leveraging the pump working parameters and lifting height information (Equation (5.3)).

Method 1:

$$CES_d = (E_d \times EF_d) / Q \quad (5.2)$$

Where:

- CES_d — Carbon emission intensity of electricity consumption, kg CO₂-eq/m³
- E_d — The total imported electricity consumption in the inventory year, kWh/a
- EF_d — The EFs specific for the imported electricity, kg CO₂-eq/kWh. See Appendix B
- Q — The volume of water handled by the targeted systems and/or facilities in the inventory year, m³/a. See Section 3.3.1

Method 2:

$$CES_d = \sum_{i=1}^n \left(\frac{g \times l \times \rho \times EF_d}{3.6 \times 10^6 \times \eta_i} \right) \quad (5.3)$$

Where:

- CES_d — Carbon emission intensity of electricity consumption, kg CO₂-eq/m³
- g — Gravity acceleration, 9.8 m/s²
- l — Elevation height, m
- ρ — Density of water, kg/m³. Default is 1 000 kg/m³
- EF_d — The EFs specific for the imported electricity, kg CO₂-eq/kWh. See Appendix B.2
- η_i — The working efficiency of the pump by type i
- n — Total number of pump types used

5.2.3 Indirect emissions of material consumption

Indirect emission from the production processes of various chemicals or materials used in the operation and maintenance stage can be calculated as shown in Equation (5.4).

$$CES_{cl} = \sum_{i=1}^n (M_{cl,i} \times EF_{cl,i}) / Q \quad (5.4)$$

Where:

- CES_{cl} — Carbon emission intensity of chemicals and/or materials consumption, kg CO₂-eq/m³
- $M_{cl,i}$ — The amount of material used by type i in the inventory year, kg/a
- $EF_{cl,i}$ — The EFs of materials or chemicals consumed by type i , kg CO₂-eq/kg. See Appendix B.5
- n — Total number of construction material types used
- Q — The volume of water handled by the targeted systems and/or facilities in the inventory year, m³/a. See Section 3.3.1

5.2.4 Indirect emissions of transportation

The construction materials and/or chemicals consumed during the planning and construction stage require transportations which will generate a certain amount of GHG emissions. The accounting can refer

to Equation (5.5).

$$CES_{ys} = \sum_{i=1, j=1}^{n, l} (M_{ys, i, j} \times L_{ys, i, j} \times EF_{ys, j}) / Q \quad (5.5)$$

Where:

- CES_{ys} — Carbon emission intensity of material or chemical transportation, kg CO₂-eq/m³
- $M_{ys, i, j}$ — The total amount of materials transported by the transportation of type j in the i th procurement in the inventory year, t/a
- $L_{ys, i, j}$ — The distance of materials or chemicals transported by the transportation of j in the i th procurement in the inventory year, km
- $EF_{ys, j}$ — The EFs of transportation of j , kg CO₂-eq/ (t·km). See Appendix B.4
- n — The number of procurement activities
- l — The number of transportation types in the i th procurement activity
- Q — The volume of water handled by the targeted systems and/or facilities in the inventory year, m³/a. See Section 3.3.1

5.3 WATER SUPPLY SYSTEMS

5.3.1 Accounting boundary

A water supply system is composed of water abstraction facilities, water treatment plants, and water distribution networks. These modules work together to serve residents with qualified and safe water in a sequent and synergic way. According to the type of water sources, water treatment facilities can be divided into conventional water treatment plants using surface water and groundwater as sources, as well as desalination plants that make use of seawater. In addition, long-distance water transfer can alleviate the shortage of water supply for some cities, and so has also been included as part of the water supply system in this *Guidelines*.

As shown in Figure 5.1, the organizational boundary of the water supply system covers all facilities from water source to users, including long-distance water transfer pipelines and pumping stations, water abstraction pipelines and pumping stations, water treatment plants, and water distribution pumps and networks. The specific carbon emission activities and emission categories are summarized in Table 5.1.



Figure 5.1 Schematic diagram of the organizational boundary of water supply systems.

Operation and maintenance

Table 5.1 Summary of carbon emission activities of water supply systems.

Scope (GHG protocol)	Category (ISO 14064-1: 2018)	Water abstraction facilities and distribution network	Conventional water treatment plant	Desalination plant	Long-distance water transfer
Direct GHG emissions produced at facilities and/or assets owned by water sectors	Direct GHG emissions and removals		Fossil fuel consumption		
Indirect GHG emissions incurred by an energy provider when producing energy purchased by water sectors	Indirect GHG emissions from imported energy	Electricity consumption by water abstraction and/or distribution pumps	Electricity consumption due to backwashing, mechanical mixer, etc.	Electricity consumption by vacuum pumps and filter presses	Electricity consumption for pumps
	Indirect GHG emissions from transportation	Transportation for materials, chemicals input, by-products output, etc.			
Other indirect GHG emissions not attributable to energy	Indirect GHG emissions from products used by organization	Material and chemical consumption	Material and chemical consumption	Material and chemicals consumption	Material consumption
	Indirect GHG emissions associated with the use of products from the organizations	-	Disposal of sludge	-	-
	Indirect GHG emissions from other sources	-	—	-	-

5.3.2 Water abstraction facilities

Water abstraction facilities are comprised of pipes and pumps which work together to collect raw water from a source, e.g., groundwater or surface water, and convey it to a distribution network or a treatment plant. The main GHG emission comes from electricity consumption by pumps of which the accounting method is shown in Equation (5.2) and Equation (5.3). When electricity consumption data are difficult to obtain, the following methods can be referred to, but the accuracy of the results is poor. The calculation is as shown in Equation (5.6).

$$CES_d = EI \times EF_d \quad (5.6)$$

Where:

- CES_d — Carbon emission intensity of electricity consumption, kg CO₂-eq/m³
- EI — Electricity intensity of abstracting groundwater or surface water in specific area, kWh/m³. See Table 5.2
- EF_d — The EFs specific for the imported electricity, kg CO₂-eq/kWh. See Appendix B.2

Table 5.2 Electricity intensity of water abstraction by region (Xiang, 2019).

Groundwater		Surface water	
Province	Electricity intensity kWh/m ³	Province	Electricity intensity kWh/m ³
Hebei	0.53	Jiangsu	0.36
Henan	0.30	Zhejiang	0.43
Shaanxi	0.64	Anhui	0.32
Shanxi	0.62	Fujian	0.40
Inner Mongolia	0.30	Jiangxi	0.37
Liaoning	0.21	Guangdong	0.41
Ningxia	0.27	Guangxi	0.34
Gansu	0.50	Hainan	0.41
Hubei	0.22	Chongqing	0.57
Hunan	0.40	Sichuan	0.30
Shandong	0.47	Guizhou	0.36
Beijing	0.44	Yunnan	0.45
Tianjin	0.66	Tibet	0.29
Jilin	0.35	Qinghai	0.52
Heilongjiang	0.43	Xinjiang	0.60
Shanghai	0.39		

0.2 (national average)

5.3.3 Water treatment plants

5.3.3.1 Indirect emission of electricity consumption

For carbon emission from imported electricity consumption for operation and maintenance of water treatment plants, refer to Equation (5.2).

5.3.3.2 Indirect emission of material consumption

The indirect carbon emission of the materials consumption for operation and maintenance of water treatment plants can refer to Equation (5.4).

5.3.3.3 Indirect emissions of transportation

The indirect carbon emission from the transportation of chemical import or sludge export in the water treatment plant can refer to Equation (5.5).

5.3.3.4 Water treatment sludge disposal

All water treatment plants produce waste/residue known as water treatment sludge (WTS) from the sedimentation process and/or backwashing of filtration. WTS is mainly composed of inorganic particles such as fine sand and sediment, along with very small amounts of organic components. After conditioning and dewatering, WTS is transported outside plants and disposed of in landfilling. Given the minor amounts of organic content in WTS, the GHG emission due to the organic degradation is usually not considered. If the organic content in WTS is relatively large, CH₄ can be produced by the anaerobic fermentation of organic matter which can be calculated by Equation (5.7).

$$CES_{CH_4-sl} = M_{ss} \times DOC \times DOC_f \times MCF \times F \times (1 - OX) \times 16/12/Q \times 28 \quad (5.7)$$

Where:

- CES_{CH_4-sl} — CH₄ emission intensity of WTS landfilling, kg CO₂-eq/m³
- M_{ss} — Total dry solid of WTS in the inventory year, kg TS/a
- DOC — Degradable organic carbon content, kg C/ kg TS
- DOC_f — Fraction of DOC decomposed, %. Default is 50%
- MCF — CH₄ correction factor. Default is 1.0
- F — Proportion of CH₄ in released gas, %. Default is 50%
- OX — The ratio of CH₄ being oxidized before escaping into atmosphere. Default is 0.1 (when well managed and covered with breathable material)) or 0 (when not handled properly) (IPCC,2019)
- 16/12 — Molecular mass ratio of CH₄ and C
- Q — The total amount of water treated in treatment plant in the inventory year, m³/a
- 28 — Global warming potential of CH₄, 28 kg CO₂-eq/kg CH₄

5.3.4 Desalination plants

5.3.4.1 Direct carbon emissions of fossil fuels

The direct emissions of fossil fuel mainly refer to carbon emission caused by the on-site consumption of fossil fuels (coal, oil, natural gas and their derivative fuels) to support, for example, the distillation process. Carbon emission intensity is calculated according to Equation (5.8).

$$CES_{rl} = \sum_{i=1}^n (A \times EF_{rl,i} \times R_i) \quad (5.8)$$

Where:

- CES_{rl} — Carbon emission intensity of fossil fuel consumption, kg CO₂-eq/m³

- A — Thermal energy intensity of desalination unit, GJ/m³, provided by the equipment manufacturer
- $EF_{r,i}$ — The EFs of fossil fuel by type i , kg CO₂-eq/GJ. See Appendix B.1
- R_i — The proportion of fuel by type i used
- n — total number of fossil fuel used

5.3.4.2 Indirect carbon emission of electricity consumption

The carbon emission from imported electricity consumption for operation and maintenance of desalination plants can refer to Equation (5.2).

5.3.4.3 Indirect carbon emission of material consumption

The indirect carbon emission of the materials consumption for operation and maintenance of desalination plants can refer to Equation (5.4).

5.3.4.4 Indirect carbon emission of transportation

The indirect carbon emission from the transportation of chemical or other goods import in the desalination plant can refer to Equation (5.5).

5.3.5 Water distribution network

The carbon emission from imported electricity consumption for operation and maintenance of water distribution network can refer to Equation (5.2) and (5.3).

5.3.6 Long-distance water transfer

Long-distance water transfer is the moving of water from a watershed with a surplus (donor basin) to a watershed suffering from a shortage (recipient basin). The water is transferred primarily to alleviate water scarcity in the recipient basin and travels long distances via complex pipeline and canal systems. The indirect carbon emissions from imported electricity consumption (mainly by pumps) can be calculated according to Equation (5.2) and (5.3). When the electricity consumption data are not available, the following methods can be referred to, on the one hand, the water lifting height and transfer distance can be used to estimate carbon emission intensity (Method 1, Equation (5.9)). On the other hand, a quick calculation diagram (Figure 5.2) is prepared for directly reading emission intensity based on water lifting height and transfer distance (Method 2). It should be noted that the entire water transfer line can be divided into several sections based on slope which then are calculated separately, and finally summed up.

Method 1:

$$CES_d = \sum_i^n \left(EI \times l_i + \frac{g \times h_i \times \rho}{3.6 \times 10^6 \times \eta_i} \right) \times EF_d \quad (5.9)$$

Where:

- CES_d — Carbon emission intensity of long-distance water transfer, kg CO₂-eq/m³
- EI — Energy intensity of long-distance water transfer, kWh/(m³·km). Default is 9.73×10^{-6} kWh/(m³·km) (Xiang, 2019)
- EF_d — The EFs specific for the imported electricity, kg CO₂-eq/kWh. See Appendix B-2
- l_i — The length of section i by dividing the entire transfer line into n sections, km

Operation and maintenance

- g — Gravity acceleration, 9.8 m/s²
 h_i — The lifting height of the section i , m
 ρ — The density of water, kg/m³. Default is 1,000 kg/m³
 η_i — The working efficiency of the pump unit in the section i . Default is 75%-80%

Method 2:

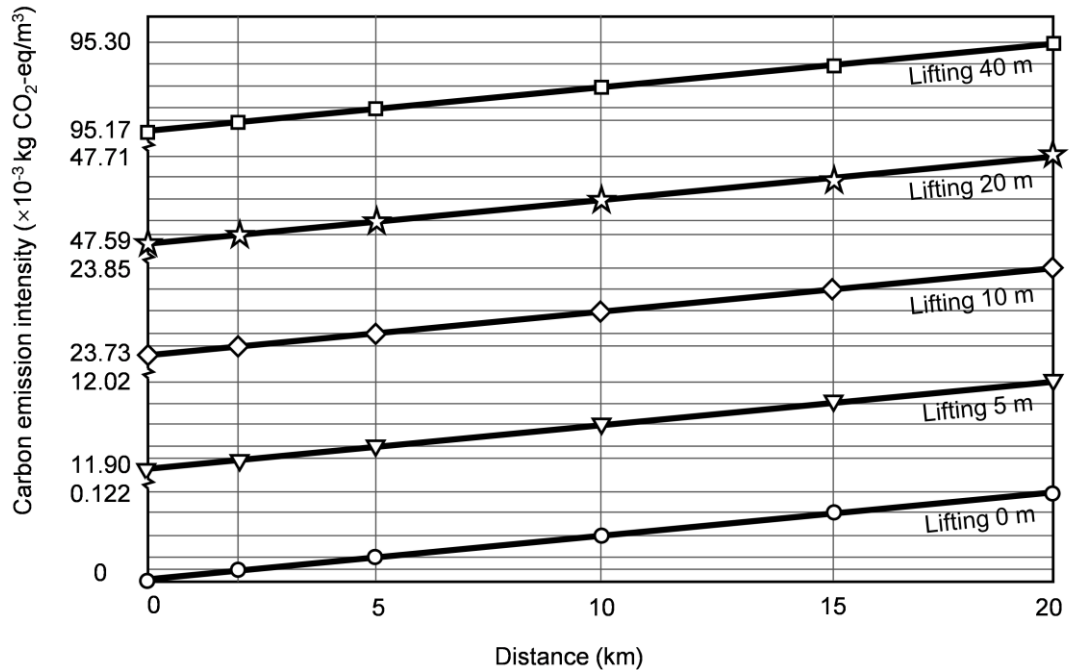


Figure 5.2 Quick calculation diagram of carbon emission intensity of long-distance water transfer.

5.4 WASTEWATER MANAGEMENT SYSTEMS

5.4.1 Accounting boundary

Wastewater management system refers to all the facilities for wastewater collection, conveyance, treatment, discharge, and associated by-product disposal. According to specific functions and working sequence, wastewater management system can be divided further into three modules: wastewater collection network, wastewater treatment plants, and sludge treatment and disposal facilities.

The carbon emission activities of wastewater management systems are complicated, with their own exclusive features because of the high concentration of organic matter/kjeldahl nitrogen loaded in wastewater. Facilities with organic matter/nitrogen flowing through them are all potential sites of GHG emissions and should be included in the accounting boundary. As depicted in Figure 5.3, the organizational boundary of the wastewater management system covers all treatment units from the beginning points of the community septic tank or wastewater being discharged into the collection network to the point of the treated wastewater being discharged into the receiving water body. The accounting boundary and emission activities are summarized in Table 5.3 in detail.

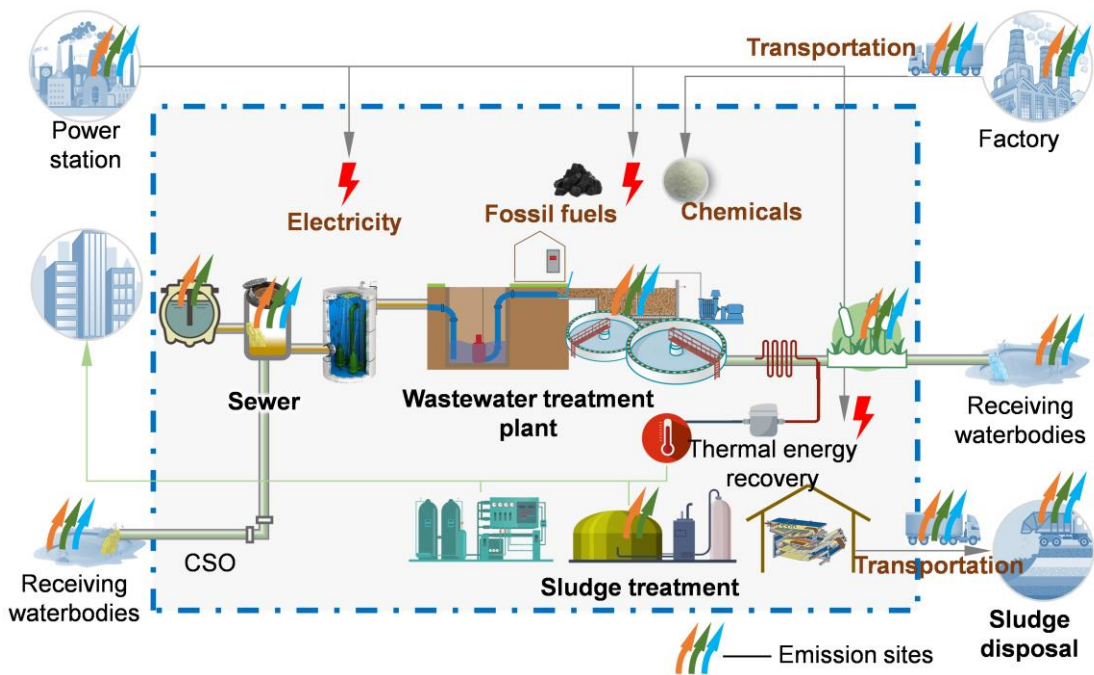


Figure 5.3 Schematic diagram of the organizational boundary of the wastewater management system.

Operation and maintenance

Table 5.3 Summary of the carbon emission activities of the wastewater management system.

Scope (GHG Protocol)	Category (ISO 14064-1: 2018)	Wastewater collection network	Wastewater treatment plant	Sludge treatment and disposal
Direct GHG emissions produced at facilities and/or assets owned by the water sectors	Direct GHG emissions and removals	Emissions from biochemical processes in septic tanks and drainage pipes	Emissions from biochemical processes in wastewater treatment unit Fossil fuel consumption	Emissions from biochemical processes in sludge treatment unit
Indirect GHG emissions incurred by an energy provider when producing the energy purchased by the water sectors	Indirect GHG emissions from imported energy	Electricity consumption by septic tank cleaning, pumping stations, etc.	Electricity consumption by mechanical equipment, such as water pumps, mixers, and aerators	Electricity consumption by pumps, heating, mixers, etc.
	Indirect GHG emissions from transportation	Transportation for various materials, chemicals input, and sludge output, etc.		
	Indirect GHG emissions from products used by organization	Material and/or chemical consumption, such as phosphorus removal agents, coagulants, disinfectants, etc.		
Other indirect GHG emissions not attributable to energy	Indirect GHG emissions associated with the use of products from the organizations	Emissions from biochemical processes due to combined sewer overflow (CSO)	—	Emissions from biochemical processes due to sludge disposal
	Indirect GHG emissions from other sources	—	—	—

5.4.2 Wastewater collection networks

5.4.2.1 Fossil CO₂ emissions

The organic matter in wastewater is partly attributed to the usage of chemical products, e.g., detergents, PPCPs, which have ingredients sourced from fossil fuels. As such, the produced CO₂ from the degradation of this part of organic matter should be classified as abiogenic and included in the accounting boundary.

5.4.2.1.1 Septic tanks/drainage systems

The quantification of fossil CO₂ from septic tanks/drainage systems is based on the proportion of fossil organic matter in the total organic matter in wastewater as well as total CO₂ produced. The quantification can be calculated as Equation (5.10) and (5.11) (Topark, 1995).

$$CES_{CO_2-hg} = FCF \times EF_{CO_2} \times COD \times \left(1 - \frac{1}{1 + \eta_T \times t}\right) \quad (5.10)$$

$$\eta_T = \eta_{20} \cdot \varepsilon^{T-20} \quad (5.11)$$

Where:

- CES_{CO_2-hg} — Emission intensity of fossil CO₂ from septic tanks/drainage system, kg CO₂-eq/m³
- FCF — Proportion of fossil organic matter in total organic matter in wastewater, %. 5 %~20% in general and 10 % recommended provided no data
- EF_{CO_2} — Maximum CO₂ producing capacity under anaerobic condition, kg CO₂/kg COD. Default is 1.47 (IPCC, 2006)
- COD — Total organically degradable matter on average in wastewater flowing to septic tanks/drainage system, kg COD/m³
- η_T — Degradation fraction of organic matter in septic tanks/drainage system, See Equation (5.11)
- t — Hydraulic retention time of septic tanks/drainage system, d
- η_{20} — Degradation fraction of organic matter at 20 °C. Default is 0.221
- ε — Correction factor. Default is 1.117
- T — The annual average temperature of the wastewater in septic tanks/drainage system, °C

5.4.2.2 CH₄ emissions

In the wastewater collection system, due to the existence of an anaerobic environment, CH₄ may be produced from the anaerobic degradation of organic matter in wastewater.

5.4.2.2.1 Septic tanks

Septic tanks generally operate in anaerobic environments, under which conditions organic matter is converted into CH₄. Two accounting methods are provided as follows: 1) when the influent COD concentration of the septic tank is available, it is recommended to apply the emission factor method (IPCC, 2019) (Method 1); or 2) the population equivalent method, which can be used to estimate CH₄ emission (Method 2) (Leverenz, 2011). Method 1 is more accurate than Method 2. The equations for two methods are listed as Equation (5.12-5.13).

Method 1:

$$CES_{hfc} = \sum_{i=1}^n [COD_i \times (1 - \frac{1}{1 + \eta_{T,i} t_i}) \times Q_{hfc,i}] \times EF_{CH_4} / Q_y \times 28 \quad (5.12)$$

Where:

- CES_{hfc} — Emission intensity of CH₄ from septic tanks, kg CO₂-eq/m³
- COD_i — Organically degradable matter on average in wastewater flowing to septic tank i, kg COD/m³
- $\eta_{T,i}$ — Degradation fraction of organic matter in septic tank i. See Equation (5.11)
- t_i — Hydraulic retention time of septic tank i, d
- $Q_{hfc,i}$ — Total wastewater influent volume of septic tank i in the inventory year, m³/a
- EF_{CH_4} — Emission factor, kg CH₄/kg COD. Default is 0.25 (IPCC,2019)
- Q_y — Total wastewater conveyance volume of the drainage system in the inventory year, m³/a
- n — The number of septic tanks in the accounting boundary
- 28 — Global warming potential of CH₄, 28 kg CO₂-eq/kg CH₄

Method 2:

$$CES_{hfc} = \frac{EF_{PE}}{Q_{PE} \times 365} \quad (5.13)$$

Where:

- CES_{hfc} — Emission intensity of CH₄ from septic tanks, kg CO₂-eq/m³
- EF_{PE} — CH₄ emission per capita for a typical septic tank, kg CO₂-eq/(ca·a). Default is 8.5 (Leverenz,2010)
- Q_{PE} — Wastewater production per capita, m³/(ca·a)

5.4.2.2.2 Wastewater sewers

Two accounting methods are provided as follows: 1) when the hydraulic parameters of each sewer section can be obtained, it is recommended to apply Method 1 (Willis, 2018) by calculating the carbon emission of each section respectively of which the accounting accuracy is high; 2) if the organic matter content of wastewater is available, Method 2 (Topark, 1995), based on EFs, can also be applied, although its accuracy is lower than method 1. The equations for the two methods are listed as Equation (5.14-5.16).

Method 1:

$$CES_{CH_4} = 0.419 \times 1.05^{(T-20)} \times \sum_{i=1}^n \left[\frac{Q_i^{-0.74}}{(3600 \times 24)^{0.26}} \times D_i^{0.28} \times S_i^{-0.138} \times L_i \right] \times 28 \quad (5.14)$$

Where:

- CES_{CH_4} — Emission intensity of CH₄ from sewer, kg CO₂-eq/m³
- T — Temperature of wastewater on average in sewer, °C
- Q_i — Wastewater flow in sewer of section i, m³/d
- D_i — The diameter of sewer section i, m

- S_i — The slope of sewer section i
- L_i — The length of sewer section i, m
- 28 — Global warming potential of CH₄, 28 kg CO₂-eq/kg CH₄
- n — The number of segments sewer is divided

Method 2:

$$CES_{CH_4} = EF_{CH_4} \times COD \times \left(1 - \frac{1}{1 + \eta_T \cdot t}\right) \times 28 \quad (5.15)$$

$$\eta_T = \eta_{20} \varepsilon^{T-20} \quad (5.16)$$

Where:

- CES_{CH_4} — Emission intensity of CH₄ from sewer, kg CO₂-eq/m³
- EF_{CH_4} — Emission factor, kg CH₄/kg COD. Default is 0.25 (IPCC,2006)
- COD — Total organically degradable matter on average in wastewater flowing to sewer, kg COD/m³
- η_T — Degradation fraction of organic matter in sewer. See Equation (5.16)
- t — Hydraulic retention time of sewer on average, d
- 28 — Global warming potential of CH₄, 28 kg CO₂-eq/kg CH₄
- η_{20} — Degradation fraction of organic matter at 20 °C. Default is 0.221
- ε — Correction factor, 1.117
- T — The annual average temperature of the wastewater in sewer, °C

5.4.2.2.3 Combined sewer overflow

Combined Sewer Overflow (CSO) is a term used to describe what happens when combined sewer systems become overwhelmed by excess stormwater and overflow into nearby streams and rivers. Since the overflowed wastewater is untreated, or only partially treated, the residual organic matter and/or nitrogen will induce the production of greenhouse gases in streams and rivers. The CH₄ emission intensity of CSO is calculated as Equation (5.17).

$$CES_{CH_4-CSO} = \sum_{i=1}^n (M_{O,i} \times Q_{CSO,i}) \times EF_{CH_4-CSO} \times 28 / Q_y \quad (5.17)$$

Where:

- CES_{CH_4-CSO} — Emission intensity of CH₄ from CSO events, kg CO₂-eq/m³
- $M_{O,i}$ — The organically degradable matter content of overflowed wastewater on average in event i, kg COD/m³ or kg BOD/m³
- $Q_{CSO,i}$ — The overflow volume of wastewater in event i, m³
- EF_{CH_4-CSO} — Emission factor of CH₄, kg CH₄/kg COD or kg CH₄/kg BOD. See Table 5.4
- 28 — Global warming potential of CH₄, 28 kg CO₂-eq/kg CH₄
- Q_y — Total wastewater conveyance volume of the drainage system in the inventory year, m³/a
- n — The number of CSO event in the inventory year

Table 5.4 Summary of CH₄ EFs of wastewater by discharge pathway (IPCC,2019).

Discharge pathway	CH ₄ EFs ²	
	kg CH ₄ /kg BOD ₅	kg CH ₄ /kg COD
Discharge to aquatic environments ¹	0.068 (0.0024 ~ 0.16)	0.028 (0.001 ~ 0.068)
Discharge to reservoirs, lakes, estuaries	0.114 (0.048 ~ 0.16)	0.048 (0.02 ~ 0.068)
Discharge to non-reservoirs, rivers, estuaries	0.021 (0.0024 ~ 0.036)	0.009 (0.001 ~ 0.015)

1. When there is no information about discharge pathway, this can be referred to.

2. For value provided a range, the value can be taken based on the water quality of receiving body. A lower value is used for high-quality water, and vice versa.

5.4.2.3 N₂O emissions

In wastewater collection system, the biological transformation of nitrogen compounds, i.e., nitrification and denitrification, can lead to the production of N₂O.

5.4.2.3.1 Wastewater sewers

Two accounting methods are provided as follows: 1) when the nitrogen content of wastewater flowing into sewer is available, it is recommended that the emission factor method be applied (Method 1) or 2) the population equivalent method can be used to estimate N₂O emission (Method 2). Method 1 is more accurate than Method 2. The equations for the two methods are listed as Equation (5.18) and Equation (5.19).

Method 1:

$$CES_{N_2O} = EF_{N_2O} \times (TN_o - TN_e) \times \frac{44}{28} \times 265 \quad (5.18)$$

Where:

- CES_{N_2O} — Emission intensity of N₂O from sewer, kg CO₂-eq/m³
- EF_{N_2O} — Emission factor, kg N₂O/kg N. Default is 0.005 (IPCC, 2006)
- TN_o — The average nitrogen concentration of wastewater at the beginning of sewer, kg N/m³
- TN_e — The average nitrogen concentration of wastewater at the end of sewer, kg N/m³
- 265 — Global warming potential of N₂O, 265 kg CO₂-eq/kg N₂O
- $\frac{44}{28}$ — The conversion of kg N₂O-N into kg N₂O

Method 2:

$$CES_{N_2O} = \frac{EF_{N_2O,PE}}{Q_{PE} \times 365} \times 265 \quad (5.19)$$

Where:

- CES_{N_2O} — Emission intensity of N₂O from sewer, kg CO₂-eq/m³
- $EF_{N_2O, PE}$ — N₂O emission per capita, kg N₂O/(ca·a). Default is $1.5 \sim 3.5 \times 10^{-3}$ (Elena, 2015)

- Q_{PE} — Wastewater production per capita, m³/(ca·a)
265 — Global warming potential of N₂O, 265 kg CO₂-eq/kg N₂O

5.4.2.3.2 CSO

The N₂O emission intensity of CSO is calculated as Equation (5.20).

$$CES_{N_2O-CSO} = \sum_{i=1}^n (TN_{O,i} \times Q_{CSO,i}) \times EF_{N_2O-CSO} \times \frac{44}{28} \times 265 / Q_y \quad (5.20)$$

Where:

- CES_{N_2O-CSO} — Emission intensity of N₂O from CSO events, kg CO₂-eq/m³
 $TN_{O,i}$ — The nitrogen content of overflowed wastewater on average in event i, kg N/m³
 $Q_{CSO,i}$ — The overflow volume of wastewater in event i, m³
 EF_{N_2O-CSO} — Emission factor of N₂O, kg N₂O-N/kg N. See Table 5.5
44/28 — The conversion of kg N₂O-N into kg N₂O
265 — Global warming potential of N₂O, 265 kg CO₂-eq/kg N₂O
 Q_y — Total wastewater conveyance volume of the drainage system in the inventory year, m³/a
 n — The number of CSO event in the inventory year

Table 5.5 Summary of N₂O EFs of wastewater by discharge pathway (IPCC, 2019).

Discharge water body	N ₂ O EF (kg N ₂ O-N/kg N) ²
Discharge to lakes, oceans and other natural water bodies ¹	0.005 (0.0005 ~0.075)
Discharge to natural water bodies with excess nutrients or oxygen deficiency	0.019 (0.0041 ~0.091)

1. When there is no information about discharge pathway, this can be referred to.

2. For value provided a range, the value can be taken based on the water quality of receiving body. A lower value is used for high-quality water, and vice versa.

5.4.2.4 Other emission activities

In wastewater collection networks, if there are some wastewater lifting pumping stations, the carbon emission of power consumption should be considered as well, and the accounting methods to refer to are Equation (5.2) and Equation (5.3).

5.4.3 Wastewater treatment plants

5.4.3.1 Fossil CO₂ emissions

5.4.3.1.1 Wastewater treatment

The biotreatment of wastewater is performed through three processes: endogenous decay, BOD oxidation, and nitrogen removal. In these three processes, CO₂ is produced and consumed. For example, in the process of biological ammonia removal (nitrification), CO₂ is consumed leading to a CO₂ credit. It should be noted that only fossil CO₂ is included in inventory. In addition, if external carbon source is dosed in the biotreatment of wastewater, the resultant CO₂ is included in fossil CO₂. The fossil CO₂ production is calculated as Equation (5.21) – (5.23).

$$CES_{ww} = MFCF \times \left\{ \left[1.1 \times (BOD_{5in} + BOD_{5ex} - BOD_{5eff}) \right. \right. \\ \times \left(1.47 - 1.42 \times \frac{0.67}{1 + K_d \times SRT} \right) \left. \right] + (1.947 \times HRT \times MLVSS \times K_d) \\ - 4.49 \quad (5.21)$$

$$\times \left[(TKN_{in} - TKN_{eff}) - (BOD_{5in} + BOD_{5ex} - BOD_{5eff}) \times \left(\frac{0.67}{1 + K_d \times SRT} \right) \times 0.124 \right] \times 10^{-3} \\ MFCF = \frac{FCF \times BOD_{5in} + BOD_{5ex}}{BOD_{5in} + BOD_{5ex}} \quad (5.22)$$

$$K_d = 0.05 \times 1.04^{T_b - 20} \quad (5.23)$$

Where:

CES_{ww}	—	Emission intensity of fossil CO ₂ in wastewater treatment plant, kg CO ₂ -eq/m ³
$MFCF$	—	Modified proportion of fossil organic matter in total organic matter in wastewater, %. See Equation (5.22)
1.1	—	Conversion factor of kg COD mineralization to CO ₂
BOD_{5in}	—	Average influent BOD ₅ concentration of wastewater treatment plant, mg BOD ₅ /L
BOD_{5ex}	—	Dosage quantity of external carbon source, mg BOD ₅ /L
BOD_{5eff}	—	Average effluent BOD ₅ concentration of wastewater treatment plant, mg BOD ₅ /L
1.47	—	Conversion factor of BOD ₅ to BOD
1.42	—	Conversion factor of kg biomass to kg BOD, kg BOD/kg MLVSS
0.67	—	Biomass yield coefficient, kg MLVSS/kg BOD ₅
K_d	—	Endogenous decay coefficient, d ⁻¹ . See Equation (5.23)
SRT	—	Sludge retention time, d
1.947	—	Conversion factor of decay of kg biomass to CO ₂ , kg CO ₂ /kg MLVSS
HRT	—	Hydraulic retention time, d
$MLVSS$	—	Concentration of mixed liquid volatile suspended solids, mg MLVSS/L
4.49	—	kg CO ₂ fixed by the nitrification process of kg ammonia nitrogen, kg CO ₂ /kg NH ₄ ⁺ -N
TKN_{in}	—	Average influent kjeldahl nitrogen concentration of wastewater treatment plants, mg N/L
TKN_{eff}	—	Average effluent kjeldahl nitrogen concentration of wastewater treatment plants, mg N/L
0.124	—	The amount of nitrogen in the biomass, kg N/kg MLVSS
FCF	—	Proportion of fossil organic matter in total organic matter in wastewater, %. 5%~20% in general and 10% recommended provided no data
T_b	—	Wastewater temperature, °C

5.4.3.1.2 Constructed wetlands

Constructed wetlands are treatment systems that use natural processes involving wetland vegetation, soils, and their associated microbial assemblages to improve water quality. They are generally applied for advanced treatment of the secondary effluent of wastewater treatment plants and inevitably, greenhouse gas is produced and emitted. Fossil CO₂ is calculated as Equation (5.24).

$$CES_{wl} = MFCF \times \left(\frac{1.1 \times BOD'_{5in}}{44 \times 10^3} - \frac{CES_{CH_4-wl}}{28 \times 16} \right) \times 44 \quad (5.24)$$

Where:

- CES_{wl} — Emission intensity of fossil CO₂ in constructed wetland, kg CO₂-eq/m³
- $MFCF$ — Modified proportion of fossil organic matter in total organic matter in wastewater, %. See Equation (5.22)
- 1.1 — Conversion factor of kg COD mineralization to CO₂
- BOD'_{5in} — Average influent BOD₅ concentration of constructed wetland, mg BOD₅/L
- 44 — The molecular mass of CO₂, 44 g/mol
- 28 — Global warming potential of CH₄, 28 kg CO₂-eq/kg CH₄
- 16 — The molecular mass of CH₄, 16 g/mol
- CES_{CH_4-wl} — Emission intensity of CH₄ in constructed wetland, kg CO₂-eq/m³. See Equation (5.26)

5.4.3.2 CH₄ emissions

5.4.3.2.1 Wastewater treatment

In a typical wastewater treatment plant, despite the engineered CH₄ production points such as sludge anaerobic digester, there are also many natural CH₄ production points due to the prevailing anaerobic conditions including screens, grit chambers, the primary clarifier, biological treatment units, secondary sedimentation tanks, and some advanced treatment units. The emission factor method generally used to calculate the CH₄ emission of the wastewater treatment plant is provided by the *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories* (IPCC, 2019). Meanwhile, a series of EFs are provided based on literature reviews. However, there were some errors in processing the referred data, which compromised the accuracy of EFs and the match to equation (see Appendix B.6 for details). Therefore, this guideline revises the equation and updates the associated emission factors. It should be noted that the equation provided by IPCC is more scientific (see Appendix B.6 for details) and is recommended once EFs have been modified and updated. This guideline recommends the Equation (5.25) (excluding the leakage of CH₄ from anaerobic digestion of sludge tanks):

$$CES_{CH_4-ww} = (BOD_{5in} \times EF_{CH_4-ww} - M_{CH_4-T}) \times 28 \times 10^{-3} \quad (5.25)$$

Where:

- CES_{CH_4-ww} — Emission intensity of CH₄ in wastewater treatment plant, kg CO₂-eq/m³
- BOD_{5in} — Average influent BOD₅ concentration of wastewater treatment plant, mg BOD₅/L
- EF_{CH_4-ww} — Emission factor, kg CH₄/kg BOD₅. See Table 5.6
- M_{CH_4-T} — Recovered or removed amount of CH₄, kg CH₄/m³
- 28 — Global warming potential of CH₄, 28 kg CO₂-eq/kg CH₄

Table 5.6 Summary of CH₄ emission factors by wastewater treatment process.

Treatment process	CH ₄ EFs (kg CH ₄ /kg BOD ₅)
Integrated Emission Factor	
Centralized, aerobic treatment plant ¹	0.0121 (0.000336 ~ 0.048) (General)
	0.0036 (0.000336 ~ 0.0177) (China specific)

Operation and maintenance

Anaerobic reactor (such as upflow anaerobic sludge blanket digestion) ²	0.48
Anaerobic deep lagoon	0.5
Anaerobic shallow lagoon	0.125
Aerated aerobic pond	0.018
Emission factors by different processes	
AAO	0.0142 (7) ³
AO	0.0083 (7)
Oxidation ditch	0.0096 (4)
SBR	0.0100 (3)
Aerobic tank	0.0152 (6)

1. See Appendix B.6 for details

2. Citation from IPCC (2019)

3. The numbers in brackets indicate the amount of reference data, see Appendix B.6 for details

5.4.3.2.2 Constructed wetland

CH₄ emission from constructed wetland is calculated as Equation (5.26).

$$CES_{CH_4-wl} = BOD'_{5in} \times EF_{CH_4-wl} \times 28 \times 10^{-3} \quad (5.26)$$

Where:

- CES_{CH_4-wl} — Emission intensity of CH₄ in constructed wetland, kg CO₂-eq/m³
 BOD'_{5in} — Average influent BOD₅ concentration of constructed wetland, mg BOD₅/L
 EF_{CH_4-wl} — Emission factor, kg CH₄/kg BOD₅. See Table 5.7
 28 — Global warming potential of CH₄, 28 kg CO₂-eq/kg CH₄

Table 5.7 Summary of CH₄ EFs by different type of constructed wetland (IPCC, 2019)

Type	CH ₄ EFs			
	kg CH ₄ /kg BOD ₅		kg CH ₄ /kg COD	
	Defaults	Range	Defaults	Range
Surface flow constructed wetland	0.24	0.048 ~0.42	0.1	0.02 ~0.175
Horizontal subsurface constructed wetland	0.06	0.042 ~0.078	0.025	0.0175 ~0.0325
Vertical subsurface constructed wetland	0.006	0.0024 ~0.0096	0.0025	0.001 ~0.004

5.4.3.2.3 Receiving water bodies

The effluent discharged to receiving water bodies still contains a small amount of organic matter which has the potential to produce CH₄. The emission factor method can be applied to calculate CH₄ emissions as shown in Equation (5.27).

$$CES_{CH_4-re} = BOD''_{5in} \times EF_{CH_4-re} \times 28 \times 10^{-3} \quad (5.27)$$

Where:

- CES_{CH_4-re} — Emission intensity of CH₄ in receiving water body, kg CO₂-eq/m³
 $BOD_5''_{in}$ — Average BOD₅ concentration of wastewater discharge to receiving water body, mg BOD₅/L
 EF_{CH_4-re} — Emission factor, kg CH₄/kg BOD₅. See Table 5.4;
28 — Global warming potential of CH₄, 28 kg CO₂-eq/kg CH₄

5.4.3.3 N₂O emissions

5.4.3.3.1 Wastewater treatment plants

In wastewater treatment plants, nitrogen removal is a common target and achieved by the biological nitrogen removal process. In this process, N₂O is produced from microbial mediated nitrification and denitrification which have been categorized into four pathways. The N₂O emission from wastewater treatment plants is calculated as Equation (5.28).

$$CES_{N_2O-ww} = (TN_{in} \times EF_{N_2O-ww} \times 44/28 \times 10^{-3} - M_{N_2O-T}) \times 265 \quad (5.28)$$

Where:

- CES_{N_2O-ww} — Emission intensity of N₂O in wastewater treatment plant, kg CO₂-eq/m³
 TN_{in} — Average influent kjeldahl nitrogen concentration of wastewater treatment plants, mg N/L
 EF_{N_2O-ww} — Emission factor, kg N₂O-N/kg N. See Table 5.8
 M_{N_2O-T} — Recovered or removed amount of N₂O, kg N₂O/m³
44/28 — The conversion of kg N₂O-N into kg N₂O
265 — Global warming potential of N₂O, kg CO₂-eq/kg N₂O

Table 5.8 Summary of N₂O EFs by wastewater treatment process

Treatment process	N ₂ O EFs (kg N ₂ O-N/kg N)
Integrated Emission Factor	
Centralized, aerobic treatment plant ¹	0.00 93 (General) 0.0106 (China specific)
Anaerobic reactor (such as upflow anaerobic sludge blanket digestion) ²	0
Anaerobic deep lagoon	0
Anaerobic shallow lagoon	0 ~0.001
Aerated aerobic pond ²	0 ~0.001
Emission factors by different processes	
AAO	0.00466 (9) ³
AO	0.00680 (27)
Oxidation ditch	0.00641 (13)
SBR	0.02020 (11)
Aeration tank	0.00166 (10)
Short-cut nitrification-ANAMMOX	0.02000 (1)
Aerobic Granular Sludge	0.00330 (1)

1. See Appendix B.7 for details

2. Citation from IPCC, 2019

3. The numbers in brackets indicate the amount of reference data, and see Appendix B.7 for details

5.4.3.3.2 Constructed wetland

N₂O emission from constructed wetland is calculated as Equation (5.29).

$$CES_{N_2O-wl} = TN'_{eff} \times EF_{N_2O-wl} \times 44/28 \times 265 \times 10^{-3} \quad (5.29)$$

Where:

- CES_{N_2O-wl} — Emission intensity of N₂O in constructed wetland, kg CO₂-eq/m³
- TN'_{eff} — Average influent nitrogen concentration of constructed wetland, mg N/L
- EF_{N_2O-wl} — Emission factor, kg N₂O-N/kg N. See Table 5.9
- 44/28 — The conversion of kg N₂O-N into kg N₂O
- 265 — Global warming potential of N₂O, kg CO₂-eq/kg N₂O

Table 5.9 Summary of N₂O EFs by different type of constructed wetland (IPCC, 2019)

Type	N ₂ O EFs (kg N ₂ O-N/kg N)
Surface flow constructed wetland	0.0013
Horizontal subsurface constructed wetland	0.0079
Vertical subsurface constructed wetland	0.00023

5.4.3.3.3 Receiving water body

Emission factor method can be applied to calculate N₂O emission as Equation (5.30).

$$CES_{N_2O-re} = TN''_{eff} \times EF_{N_2O-re} \times 44/28 \times 265 \times 10^{-3} \quad (5.30)$$

Where:

- CES_{N_2O-re} — Emission intensity of N₂O in receiving water body, kg CO₂-eq/m³
- TN''_{eff} — Average nitrogen concentration of wastewater discharge to receiving water body, mg N/L
- EF_{N_2O-re} — EFs, kg N₂O-N/kg N. See Table 5.5
- 44/28 — The conversion of kg N₂O-N into kg N₂O
- 265 — Global warming potential of N₂O, kg CO₂-eq/kg N₂O

5.4.3.4 Other emissions

The GHG emissions from fossil fuel, electricity, material/chemical consumption, and transportation for chemicals and/or sludge disposal are calculated according to Section 5.2.

5.4.3.5 Carbon reduction by resource/energy recovery

5.4.3.5.1 Carbon offset by energy recovery

Wastewater is rich in chemical and thermal energy which can be recovered by anaerobic digestion-CHP and heat pumps to offset carbon emissions of the wastewater management system. When the recovered energy is exported, the carbon offset should be considered and calculated with Equation (5.31) and (5.32).

$$CSS_e = E_{cs-d} \times EF_d / Q_{in} \quad (5.31)$$

Where:

- CSS_e — Carbon offset of electricity production, kg CO₂-eq/m³
- E_{cs-d} — The total electricity exported in the inventory year, kWh/a
- EF_d — The EFs of electricity replaced by the exported electricity, kg CO₂-eq/kWh. See Appendix B.2;
- Q_{in} — The total amount of wastewater treated in the inventory year, m³/a

$$CSS_h = E_{cs-h} \times EF_h / Q_{in} \quad (5.32)$$

Where:

- CSS_h — Carbon offset for heat recovery, kg CO₂-eq/m³
- E_{cs-h} — The total thermal energy exported in the inventory year, GJ/a
- EF_h — EFs of heat replaced by the exported heat, kg CO₂/GJ. See Appendix B.1
- Q_{in} — The total amount of wastewater treated in the inventory year, m³/a

5.4.3.5.2 Carbon offset from resource recovery

There are abundant resources in wastewater and many technologies have been proposed and practiced in full-scale projects to recover a variety of products. For example, phosphorus in wastewater can be recovered via struvite or vivianite, which are a type of fertilizer and a high value-added material, respectively. These products can be substituted for industrial products to induce carbon offset and can be calculated with Equation (5.33).

$$CSS_{re} = \sum_{i=1}^n M_{re,i} \times EF_{re,i} / Q_{in} \quad (5.33)$$

Where:

- CSS_{re} — Carbon offset of resource recovery, kg CO₂-eq/m³
- n — The number of product types recovered in the inventory year
- $M_{re,i}$ — The industrial product equivalent of the recovered product i in the inventory year, kg/kg
- $EF_{re,i}$ — EFs of the industrial product replaced by recovered product i, kg CO₂/kg
- Q_{in} — The total amount of wastewater treated in the inventory year, m³/a

5.4.3.5.3 Other carbon sink

When constructed wetlands are applied for advanced purification of effluent, the vegetation and/or plants can be carbon sinks. The resulting carbon credit can be calculated with Equation (5.34).

$$CSS_{zb} = EF_{zb} \times S_{zb} \quad (5.34)$$

Where:

- CSS_{zb} — Carbon sink of plant carbon sequestration, kg CO₂-eq
- EF_{zb} — Carbon fixation capacity of plants, kg CO₂-eq/m². See Appendix B.8
- S_{zb} — Plant coverage area, m²

5.4.4 Sludge Treatment and Disposal

5.4.4.1 Fossil CO₂ emission

5.4.4.1.1 Anaerobic digestion

The anaerobic digestion process is widely applied in wastewater treatment plant to stabilize and valorize waste sludge. In this process, CH₄-rich biogas is produced and fuels the combined heat and power (CHP) needed for electricity generation. It means the greenhouse gas emitted from anaerobic digestion is in the form of CO₂ released into the atmosphere. As mentioned above, part of the organic matter in influent is fossil organic matter and part of the CO₂ released will be abiogenic. The *Guidelines* provides two accounting methods for fossil CO₂ quantification. Method 1 is based on mass balance and method 2 assumes that the volume of biogas is available. The equations for the two methods are listed as shown in Equation (5.35-5.36):

Method 1:

$$CES_{ad} = MFCF \times \left[(1 - P) \times \frac{44 \times F_{CH_4}}{44 - 28 \times F_{CH_4}} + \frac{44 - 44 \times F_{CH_4}}{44 - 28 \times F_{CH_4}} \right] \times Q_{ss} \times (VSS_{o-ad} - VSS_{e-ad}) / Q_{in} \quad (5.35)$$

Where:

- CES_{ad} — emission intensity of fossil CO₂ in anaerobic digestion, kg CO₂-eq/m³
- $MFCF$ — Modified proportion of fossil organic matter in total organic matter in wastewater, %. See Equation (5.22)
- P — The proportion of biogas leakage, %. See Table 5.10
- 44 — the molecular mass of CO₂, 44 g/mol
- F_{CH_4} — Proportion of CH₄ in biogas, %
- 28 — Global warming potential of CH₄, 28 kg CO₂-eq/kg CH₄
- Q_{ss} — the volume of waste sludge entering anaerobic digester in the inventory year, m³/a
- VSS_{o-ad} — the amount of volatile suspended solid in the influent, kg VSS/m³
- VSS_{e-ad} — the amount of volatile suspended solid in the effluent, kg VSS/m³
- Q_{in} — the total amount of wastewater treated in the inventory year, m³/a

Method 2:

$$CES_{ad} = \frac{MFCF \times M_{zq}}{(1 - P) \times 22.4 \times Q_{in}} \times 44 \quad (5.36)$$

Where:

- CES_{CH_4-ad} — Emission intensity of fossil CO₂ in anaerobic digestion, kg CO₂-eq/m³
- $MFCF$ — Modified proportion of fossil organic matter in total organic matter in wastewater, %. See Equation (5.22)
- M_{zq} — The amount of biogas collected in the inventory year, m³/a
- F_{CH_4} — Proportion of CH₄ in biogas, %
- 22.4 — The volume of one mole of a gas at STP
- P — The proportion of biogas leakage, %. See Table 5.10
- Q_{in} — The total amount of wastewater treated in the inventory year, m³/a
- 44 — The molecular mass of CO₂, 44 g/mol

Table 5.10 Summary of biogas leakage ratio by different conditions of anaerobic digester (IPCC, 2019).

Quality of digester	Conditions for storage	Climate zone	
		Cool temperate	Warm temperate
high quality	High gastight	1.00%	1.00%
	Poor gastight	1.41%	1.41%
	average	1.99%	2.27%
low quality	High gastight	9.59%	9.59%
	Poor gastight	10.00%	10.00%
	average	10.58%	10.85%

5.4.4.1.2 Aerobic composting

In the process of aerobic composting of waste sludge, organic matter can be mineralized to CO₂ part of which is fossil CO₂. It can be calculated as Equation (5.37).

$$CES_c = MFCF \times M_{ss} \times DOC \times DOC_{f-c} \times 44/12/Q_{in} \quad (5.37)$$

Where:

- CES_c — Emission intensity of fossil CO₂ in aerobic composting, kg CO₂-eq/m³
- $MFCF$ — Modified proportion of fossil organic matter in total organic matter in wastewater, %. See Equation (5.22)
- M_{ss} — Total dry solid of waste sludge in the inventory year, kg TS/a
- DOC — Degradable organic carbon, kg C/kg TS
- DOC_{f-c} — Fraction of DOC that decompose, %
- $44/12$ — Molecular mass ratio of CO₂ and C
- Q_{in} — The total amount of wastewater treated in the inventory year, m³/a

5.4.4.1.3 Landfill

When sludge is disposed in a landfill, CO₂ and CH₄ can be produced. The equation for fossil CO₂ calculation is listed as Equation (5.38).

$$CES_{sl} = MFCF \times M_{ss} \times DOC \times DOC_f \times (1 - MCF \times F) \times 44/12/Q_{in} \quad (5.38)$$

Where:

- CES_{sl} — Emission intensity of fossil CO₂ in landfill, kg CO₂-eq/m³
- $MFCF$ — Modified proportion of fossil organic matter in total organic matter in wastewater, %. See Equation (5.22)
- M_{ss} — Total dry solid of waste sludge in the inventory year, kg TS/a
- DOC — Degradable organic carbon, kg C/ kg TS
- DOC_f — Fraction of DOC that decompose, %
- MCF — Correction factor. Default is 1
- F — Proportion of CH₄ in released gas, %. Default is 50%
- $44/12$ — Molecular mass ratio of CO₂ and C
- Q_{in} — The total amount of wastewater treated in the inventory year, m³/a

5.4.4.1.4 Sludge incineration

The equation for fossil CO₂ calculation from sludge incineration is listed as Equation (5.39).

$$CES_{in} = MFCF \times M_{ss} \times CF \times OF \times 44/12/Q_{in} \quad (5.39)$$

Where:

- CES_{in} — Emission intensity of fossil CO₂ in incineration, kg CO₂-eq/m³
- $MFCF$ — Modified proportion of fossil organic matter in total organic matter in wastewater, %. See Equation (5.22)
- M_{ss} — Total dry solid of waste sludge in the inventory year, kg TS/a
- CF — Total carbon content percentage, %
- OF — Default is 100% in a well-managed incinerator
- $44/12$ — Molecular mass ratio of CO₂ and C
- Q_{in} — The total amount of wastewater treated in the inventory year, m³/a

5.4.4.2 CH₄ emissions

5.4.4.2.1 Anaerobic digestion

In the anaerobic digestion process, a small quantity of CH₄ may escape the reactor due to the quality of gastight conditions. The *Guidelines* provides two accounting methods of accounting for the CH₄ leakage. Method 1 is based on mass balance and Method 2 assumes that the volume of biogas is available. The equations for the two methods are listed as Equation (5.40) and Equation (5.41).

Method 1:

$$CES_{CH_4-ad} = P \times \left[Q_{ss} \times (VSS_{o-ad} - VSS_{e-ad}) \times \frac{16 \times F_{CH_4}}{44 - 28 \times F_{CH_4}} \right] / Q_{in} \times 28 \quad (5.40)$$

Where:

- CES_{CH_4-ad} — Emission intensity of CH₄ leakage in anaerobic digestion, kg CO₂-eq/m³
- P — The proportion of biogas leakage, %. See Table 5.10
- Q_{ss} — The volume of waste sludge entering anaerobic digester in the inventory year, m³/a
- VSS_{o-ad} — The amount of volatile suspended solids in the influent, kg VSS/m³
- VSS_{e-ad} — The amount of volatile suspended solids in the effluent, kg VSS/m³
- 44 — The molecular mass of CO₂, 44 g/mol
- 28 — Global warming potential of CH₄, 28 kg CO₂-eq/kg CH₄
- 16 — The molecular mass of CH₄, 16 g/mol
- F_{CH_4} — Proportion of CH₄ in biogas, %
- Q_{in} — The total amount of wastewater treated in the inventory year, m³/a

Method 2:

$$CES_{CH_4-ad} = M_{zq} \times F_{CH_4} / 22.4 \times P / (1 - P) / Q_{in} \times 16 \times 28 \quad (5.41)$$

Where:

- CES_{CH_4-ad} — Emission intensity of CH₄ leakage in anaerobic digestion, kg CO₂-eq/m³
- M_{zq} — The amount of biogas collected in the inventory year, m³/a
- F_{CH_4} — Proportion of CH₄ in biogas, %
- 22.4 — The volume of one mole of a gas at STP
- P — The proportion of biogas leakage, %. See Table 5.10

- Q_{in} — The total amount of wastewater treated in the inventory year, m³/a
- 16 — The molecular mass of CH₄, 16 g/mol
- 28 — Global warming potential of CH₄, 28 kg CO₂-eq/kg CH₄

5.4.4.2.2 Landfill

The equation for CH₄ emission from landfill is listed as Equation (5.42).

$$CES_{CH_4-sl} = M_{ss} \times DOC \times DOC_f \times MCF \times F \times (1 - OX) \times 16/12/Q_{in} \times 28 \quad (5.42)$$

Where:

- CES_{CH_4-sl} — Emission intensity of CH₄ in landfill, kg CO₂-eq/m³
- M_{ss} — Total dry solid of waste sludge in the inventory year, kg TS/a
- DOC — Degradable organic carbon, kg C/ kg TS
- DOC_{f-c} — Fraction of DOC that decompose, %
- MCF — Correction factor. Default is 1
- F — Proportion of CH₄ in released gas, %. Default is 50%
- OX — The ratio of CH₄ being oxidized before escaping into the atmosphere. Default value (when well managed and covered with breathable material) is 0.1, or (when not handled properly) 0 (IPCC, 2019)
- 16/12 — Molecular mass ratio of CH₄ and C
- Q_{in} — The total amount of wastewater treated in the inventory year, m³/a
- 28 — Global warming potential of CH₄, 28 kg CO₂-eq/kg CH₄

5.4.4.2.3 Land application

Emissions of CH₄ can occur from the disposal of sludge to land and can be calculated by Equation (5.43).

$$CES_{CH_4-la} = EF_{CH_4-la} \times M_{ss}/Q_{in} \times 28 \quad (5.43)$$

Where:

- CES_{CH_4-la} — Emission intensity of CH₄ in land application, kg CO₂-eq/m³
- EF_{CH_4-la} — Emission factor, kg CH₄/kg TS. Default is 0.00318 (IPCC, 2019)
- M_{ss} — Total dry solid of waste sludge in the inventory year, kg TS/a
- Q_{in} — The total amount of wastewater treated in the inventory year, m³/a
- 28 — Global warming potential of CH₄, 28 kg CO₂-eq/kg CH₄

5.4.4.3 N₂O emissions

5.4.4.3.1 Aerobic composting

Emissions of N₂O can occur from the aerobic composting of sludge due to the biochemical transformation of nitrogen compounds and can be calculated by Equation (5.44).

$$CES_{N_2O-c} = M'_{ss} \times EF_{N_2O-c}/Q_{in} \times 265 \quad (5.44)$$

Where:

- CES_{N_2O-c} — Emission intensity of fossil CO₂ in aerobic composting, kg CO₂-eq/m³
- M'_{ss} — Total dry solid of waste sludge in the inventory year, kg TS/a
- EF_{N_2O-c} — EFs, kg N₂O/kg TS. Default is 0.2~1.6 × 10⁻³ kg N₂O/kg TS (IPCC, 2019)
- Q_{in} — The total amount of wastewater treated in the inventory year, m³/a
- 265 — Global warming potential of N₂O, kg CO₂-eq/kg N₂O

5.4.4.3.2 Sludge incineration

The calculation for N₂O production in the incineration process is listed as Equation (5.45).

$$CES_{N_2O-in} = M_{ss} \times EF_{N_2O-in} / Q_{in} \times 265 \times 10^{-3} \quad (5.45)$$

Where:

- CES_{N_2O-in} — Emission intensity of N₂O in incineration, kg CO₂-eq/m³
- M_{ss} — Total dry solid of waste sludge in the inventory year, kg TS/a
- EF_{N_2O-in} — EFs, kg N₂O/t TS. Default is 0.99 kg N₂O/t TS (IPCC,2019)
- Q_{in} — The total amount of wastewater treated in the inventory year, m³/a
- 265 — Global warming potential of N₂O, kg CO₂-eq/kg N₂O

5.5 WATER RECLAMATION SYSTEMS

5.5.1 Accounting boundary

The water reclamation system is a process of reclaiming and purifying wastewater into a reusable form. As shown in Figure 5.4, the organizational boundary of the water reclamation system covers all relevant facilities from the outlet of the wastewater treatment plant to the user, including treatment facilities and distribution network. The carbon emission activities and emission types are summarized in Table 5.11.

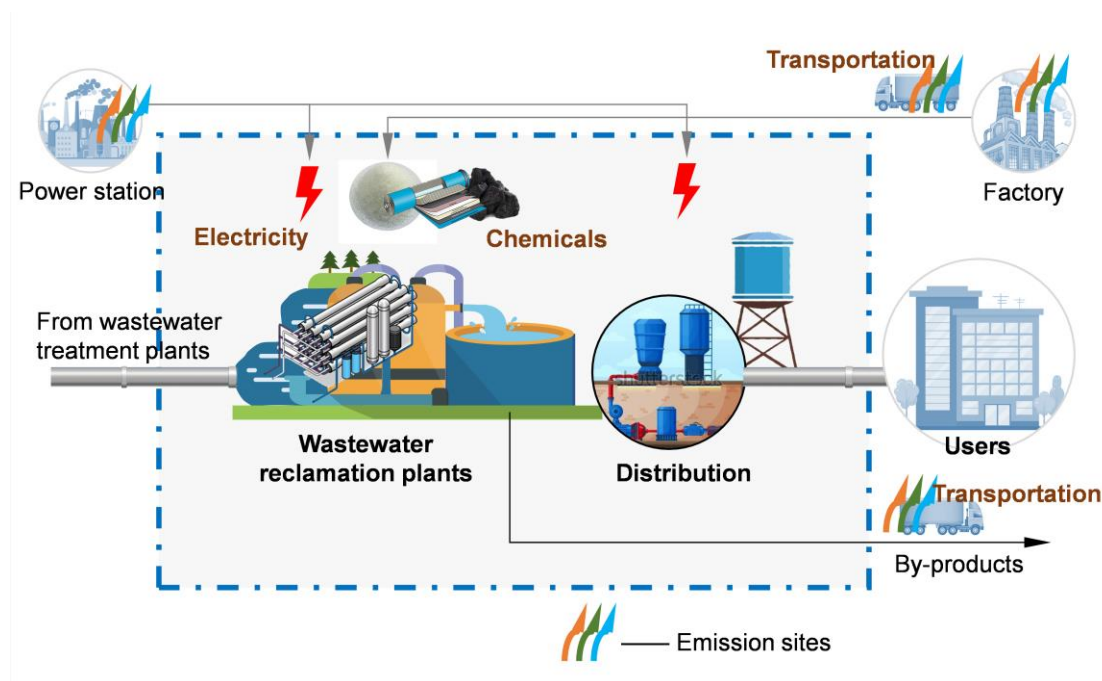


Figure 5.4 Schematic diagram of the organizational boundary of water reclamation systems.

Table 5.11 Carbon emission activities of water reclamation system.

Scope (carbon emission protocol)	Type (ISO 14064-1: 2018)	Advanced treatment plant	Water distribution network
Direct GHG emissions produced at facilities and/or assets owned by the water sectors	Direct GHG emissions and removals	Fossil fuel consumption	
Indirect GHG emissions incurred by an energy provider when producing the energy purchased by the water sectors	Indirect GHG emissions from imported energy	Electricity consumption by treatment units	Electricity consumption by distribution pumps
	Indirect GHG emissions from transportation	Transportation for materials, chemicals input, by-products output, etc.	
Other indirect GHG emissions not attributable to energy	Indirect GHG emissions from products used by organization	Material and chemical consumption	Material and chemical consumption
	Indirect GHG emissions associated with the use of products from the organizations	Disposal of sludge	-
	Indirect GHG emissions from other sources	-	-

5.5.2 Advanced treatment plants

Advanced treatment plants mainly receive secondary effluent of wastewater treatment plants and purify the effluent further for groundwater recharging, industrial and agricultural reuse, and landscape irrigation. Generally, a flexible combination of treatment processes can be exclusively designed according to the reuse purpose, including coagulation, precipitation, and filtration. In addition, advanced treatment units, such as membrane separation and advanced oxidation, can also be added to achieve high water quality standard. The carbon emission activities of water reclamation system are similar to those of water supply systems, primarily including electricity consumption, chemical usage, and transportation.

5.5.2.1 Indirect carbon emission of electricity consumption

The carbon emission from imported electricity consumption for advanced treatment plant can refer to Equation (5.2) and (5.3).

5.5.2.2 Indirect carbon emission of material consumption

The indirect carbon emission of the materials consumption for advanced treatment plant can refer to Equation (5.4).

5.5.2.3 Indirect carbon emission of transportation

The indirect carbon emission from the transportation of chemical import or sludge export in the advanced treatment plant can refer to Equation (5.5).

5.5.2.4 Sludge disposal

If the organic content in sludge produced from the advanced treatment processes is relatively large, CH₄ can be produced by the anaerobic fermentation of organic matter which can be calculated by Equation (5.7).

5.5.3 Water distribution network

The carbon emission from the consumption of imported electricity for various treatment processes can be calculated according to Equation (5.2) and (5.3).

5.6 STORMWATER SYSTEMS

5.6.1 Accounting boundary

The stormwater system is an urban construction model to absorb rain and prevent flooding. It is composed of a variety of facilities and modules taking the functions of stormwater infiltration, transmission, retention and detention, storage, pollution interception, purification and utilization, respectively. It can be divided into two modules, stormwater sewers and stormwater control facilities. Thereinto, stormwater sewers include conventional drainage pipes and planted ditches/infiltration pipes. Stormwater control facilities can be divided into five categories, including detention and retention facilities, harvesting and utilization facilities, detention and storage facilities, interception and purification facilities and other technical facilities.

As shown in Figure 5.5, the organizational boundary of carbon emission of the stormwater system covers all facility units from the discharge of rainwater source to the discharge into natural water bodies, including drainage pipes, pumping stations and other transfer facilities in stormwater pipe facilities, and green and gray infrastructures in stormwater control facilities. The specific carbon emission activities and emission categories are summarized in Table 5.12.

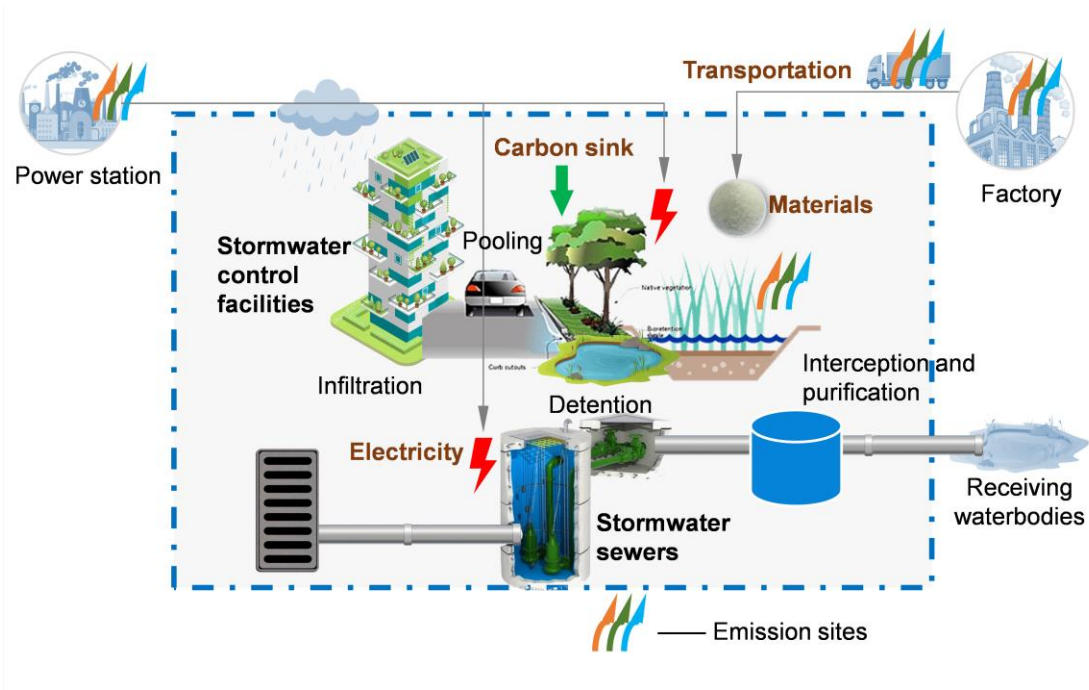


Figure 5.5 Schematic diagram of the carbon emission accounting boundary of the stormwater system.

Table 5.12 Summary of carbon emissions activities of stormwater system.

Scope (carbon emission accounting system)	Category (ISO 14064-1: 2018)	Stormwater sewers	Stormwater control facilities
Direct GHG emissions produced at facilities and/or assets owned by the water sectors	Direct GHG emissions and removals	Fossil fuel consumption	
Indirect GHG emissions incurred by an energy provider when producing the energy purchased by the water sectors	Indirect GHG emissions from imported energy	-	CH ₄ emission from stormwater wetland, storage pond, bioretention system. Besides, carbon removal by plant sequestration
Other indirect GHG emissions not attributable to energy	Indirect GHG emissions from transportation	Electric consumption by pumps or irrigation hydraulic giant	Electricity consumption by purification and utilization facilities
Other indirect greenhouse gas emissions caused by the activities of the accounting subject but outside its accounting boundary	Indirect GHG emissions from products used by organization	transportation for materials, chemicals input, by-products output, etc.	
	Indirect GHG emissions associated with the use of products from the organizations	Material consumption	
Direct GHG emissions produced at facilities and/or assets owned by the water sectors	Indirect GHG emissions from other sources	-	-
	Direct GHG emissions and removals	-	-

5.6.2 Stormwater sewers

5.6.2.1 Drainage pipes and ancillary structure

5.6.2.1.1 Direct carbon emission of fossil fuel

In stormwater drainage pipes and ancillary structures (pumping stations, etc.), some mechanical equipment consumes gasoline and diesel fossil fuels during operation, thus generating a certain amount of direct carbon emission. The calculation can refer to Equation (5.1). In terms of stormwater pumping stations, carbon emissions due to the consumption of fossil fuels can be calculated according to Equation (5.46) – (5.48).

$$CES_{rl} = W_{rl} \times EF_{rl} \times 10^{-9} \quad (5.46)$$

$$W_{rl} = \sum_{i=1}^n \frac{\rho g (H_{net} + H_{loss})}{\eta_i} \quad (5.47)$$

$$H_{loss} = 0.00124 \times \frac{v^2}{d^{1.33}} \times L \quad (5.48)$$

Where:

- CE_{rl} — Carbon emission intensity of fossil fuel consumption for water lifting or delivery, kg CO₂-eq/m³
- W_{rl} — Energy consumption of using oil pumps to transport stormwater to natural water bodies or sewage treatment plants, J/m³
- EF_{rl} — The EFs of fossil fuel, kg CO₂-eq/GJ. See Appendix B.1
- ρ — Rainwater density, kg/m³. Default is 1,000 kg/m³
- g — Acceleration of gravity. Default is 9.8 m/s²
- H_{net} — Elevation difference between the start and end points of water delivery, m
- H_{loss} — The hydraulic loss along the pipe network, m
- η_i — Unit efficiency of type i pump, %
- n — A total of pump units with different working efficiency are used
- v — Velocity, m/s
- d — Water pipeline diameter, m
- L — The length of the water pipeline, m

5.6.2.1.2 Indirect carbon emission of electricity consumption

The electricity consumption in the water pumping process and/or from landscape lighting also leads to carbon emission, which can be calculated by use of Method 1 (Equation (5.49)). In the absence of total electricity consumption data, carbon emissions can be calculated according to Method 2 (Equation (5.50) and (5.51)).

Method 1:

$$CE_d = EH \times EF_d / Q \quad (5.49)$$

Where:

- CE_d — Carbon emission intensity of the stormwater system purchased electricity, kg CO₂-eq/m³
- EH — The annual power consumption of pumping station system of the rainwater pipe network, kWh/a

- EF_d — EFs of electricity, kg CO₂-eq/kWh. See Appendix B.2
 Q — The total amount of rainwater transported during the year, m³/a

Method 2:

$$CE_d = E_d \times EF_d \quad (5.50)$$

$$E_d = \sum_{i=1}^n \frac{\rho g (H_{net} + H_{loss})}{3.6 \times 10^6 \times \eta_i} \quad (5.51)$$

Where:

- CE_d — Carbon emission intensity of electricity consumption for water fetching or water delivery, kg CO₂-eq/m³
 E_d — Energy consumption generated by using electric pumps to transport unit water to natural water bodies or sewage treatment plants, kWh/m³
 EF_d — EFs of electricity, kg CO₂-eq/kWh. See Appendix B.2
 $M_{rl,i}$ — Evaluate the total amount of fossil fuel i consumed during the year, kg/a or m³/a
 ρ — The EFs of fossil fuel i , kg CO₂-eq/kg or kg CO₂-eq/m³. See Appendix B.1
 g — Acceleration of gravity, 9.8 m/s²
 H_{net} — Elevation difference between the start and end points of water delivery, m
 H_{loss} — The hydraulic loss along the pipe network, m, which can be calculated by Equation (5.48)
 η_i — Unit efficiency of type i pump, %
 n — A total of pump units with different working efficiency are used

5.6.2.2 Other transfer facilities

5.6.2.2.1 Indirect carbon emission of electricity consumption

The calculation of carbon emissions from electricity consumption by other green conveyance facilities can refer to Equation (5.46) – (5.48).

5.6.2.2.2 Carbon removal by green infrastructures

Plants and soils in green infrastructures can sequester GHG, such as green roofs, rain gardens, low elevation greenbelt and other vegetation-containing facilities. The amount of carbon credit by vegetation is calculated by Equation (5.52).

$$CS_{zb} = EF_{zb} \times S_{zb} \quad (5.52)$$

Where:

- CS_{zb} — Carbon sink capacity of vegetation, kg CO₂-eq
 EF_{zb} — Vegetation carbon sequestration factor, kg CO₂-eq/m². See Appendix B.8
 S_{zb} — Vegetation area, m²

5.6.3 Stormwater control facilities

5.6.3.1 Detention and retention facilities

Carbon emissions from the operation and maintenance of detention and retention facilities include indirect emissions from imported electricity and carbon sinks of vegetation in green infrastructures which are calculated according to Section 5.6.2.2.

5.6.3.2 Harvesting and utilization facilities

Carbon emission from harvesting and utilization facilities includes indirect emissions from imported electricity which is calculated by Equation (5.2) and (5.3).

5.6.3.3 Detention and storage facilities

Carbon emissions from the detention and storage facilities include indirect emissions from imported electricity, carbon removal by vegetation in green infrastructures, and direct greenhouse gas emissions from stormwater wetlands which are calculated according to Section 5.6.2.2.

In addition, as an important facility for improving water quality in stormwater control system, stormwater wetland probably produces CH₄ and N₂O emissions. The carbon emission mechanism of stormwater wetlands are complex, and the emission rate and amount are affected by various factors such as stability, plants, soil background value, and water flow mode (Table 5.13). For simplicity, this *Guidelines* only considers the carbon emissions produced by wetlands due to receiving stormwater.

Table 5.13 Factors affecting CH₄ and N₂O emissions in stormwater wetland.

Factor/process	CH ₄	N ₂ O
Increase of temperature of water/soil/air	Increase in almost all cases	No relationship
Elevated moisture in the soil or filter material (increased soil pore water content)	Obvious increase	Reduction
Increased runoff of collected rainwater	Increase	Increase
Plants with aerated leaves	As the case may be	As the case may be
Fluctuating hydrological regime (intermittent loading)	Significant reduction	As the case may be
Groundwater level deepening in horizontal flow constructed wetlands	Reduction	Increase

5.6.3.3.1 CH₄ emission

CH₄ emission from stormwater wetlands is calculated by Equation (5.53) and (5.54).

$$CE_{CH_4} = \sum_{i=1}^n (TOW_i \times EF_{CH_4 i}) \times 28 \quad (5.53)$$

$$TOW_i = COD_i \times W_i \times 365 \quad (5.54)$$

Where:

- CE_{CH_4} — Annual carbon emission of CH₄ converted into CO₂ equivalent from stormwater wetlands, kg CO₂-eq/a
- TOW_i — Total COD of rainwater entering type i stormwater wetland each year, kg COD/a. Calculated by Equation (5.55)
- EF_{CH_4} — Emission factor of CH₄ in the stormwater wetland i , kg CH₄/kg COD. See Table 5.7
- i — Stormwater wetland type
- 28 — Global warming potential constant of CH₄, 28 kg CO₂-eq/kg CH₄

Operation and maintenance

- COD_i — The average COD concentration of rainwater entering type i stormwater wetland each year, kg COD/m³
- W_i — The daily stormwater capacity of type i stormwater wetland, m³/d
- i — Stormwater wetland type

5.6.3.3.2 N₂O emission

N₂O emission from stormwater wetland are calculated by Equation (5.55).

$$CE_{N_2O} = \sum_{i=1}^n (N_i \times EF_{N_2O} \times 44/28) \times 265 \quad (5.55)$$

Where:

- CE_{N_2O} — Annual carbon emissions of N₂O converted into CO₂ equivalent from stormwater wetlands, kg CO₂-eq/a
- N_i — Total nitrogen of rainwater entering type i stormwater wetland each year, kg N/a
- EF_{N_2O} — Emission factors of N₂O in the stormwater wetland i , kg N₂O-N/kg N. See Table 5.9
- i — Stormwater wetland type
- 265 — Global warming potential constant of N₂O, 265 kg CO₂-eq/kg N₂O
- 44/28 — The conversion of kg N₂O-N into kg N₂O

5.6.3.4 Interception and purification facilities

Carbon emissions from the interception and purification facilities include indirect emissions from imported electricity, and carbon removal from vegetation in green infrastructures which are calculated according to Section 5.6.2-5.6.3.

5.6.3.5 Other green facilities

The accounting method of carbon emissions from the other green infrastructures can refer to Section 5.6.2-5.6.3.

Chapter 6

Asset replacement and demolition

6.1 OVERVIEW

The main carbon emission activities of the urban water sector in the asset replacement and demolition stage include: i) direct carbon emissions generated by burning fossil fuels; ii) indirect carbon emissions from the consumption of imported electricity; iii) indirect carbon emissions from the transportation of building waste. In addition, building waste produced in the asset replacement and demolition stage can be recycled and reused (such as steel bars, etc.) which can then produce carbon credit and offset carbon emissions. As the activities in planning and construction for different systems are consistent, the accounting equations provided in this section are applicable to any system in the urban water sectors.

6.2 DIRECT EMISSIONS OF FOSSIL FUELS

The direct emissions of fossil fuels are mainly carbon emissions caused by the on-site consumption of fossil fuels (coal, oil, natural gas and their derivative fuels) by construction machinery. According to available construction inventory, carbon emissions of fossil fuel consumption can be calculated using the following two methods: i) if the types of fossil fuels and the corresponding consumption can be obtained, Method 1 is recommended for calculating carbon emissions (Equation (6.1), with relatively high accuracy of results; ii) if the details of fossil fuel consumption are not available, emissions can be calculated according to the number of machine shifts (Method 2, (Equation (6.2), though the accuracy of the results of this method are low.

Method 1:

$$CE_{rl} = \sum_i^n (M_{rl,i} \times EF_{rl,i}) \quad (6.1)$$

Where:

CE_{rl}	—	Carbon emissions of fossil fuel, kg CO ₂ -eq
$M_{rl,i}$	—	Total amount of the fossil fuel i consumed,, kg, m ³
$EF_{rl,i}$	—	The emission factor of fossil fuel i,, kg CO ₂ -eq/kg or kg CO ₂ -eq/m ³ . See Appendix B.1
n	—	Total number of fossil fuels used

Method 2:

$$CE_{rl} = \sum_i^n (T_i \times S_i \times EF_{rl,i}) \quad (6.2)$$

Where:

CE_{rl}	—	Carbon emissions of fossil fuel, kg CO ₂ -eq
T_i	—	The number of shifts of machine i
S_i	—	The fossil fuel consumption per shift of machine i . See <i>Building Carbon Emission Calculation Standard</i> (GB/T 51366-2019)
$EF_{rl,i}$	—	The emission factor of fossil fuel for machine i , kg CO ₂ -eq/kg or kg CO ₂ -eq/m ³ , see Appendix B.1
n	—	Total number of machine types used

6.3 INDIRECT EMISSIONS FROM ELECTRICITY CONSUMPTION

The electrical consumption in the asset replacement and demolition stage also contributes a significant portion of the total GHG emissions, and is calculated as shown in Equation (6.3).

$$CE_d = E_d \times EF_d \quad (6.3)$$

Where:

CE_d	—	Carbon emissions of electrical consumption, kg CO ₂ -eq
E_d	—	Total consumption of electricity, kW·h
EF_d	—	The emission factor specific for the imported electricity, kg CO ₂ -eq/kW·h. See Appendix B.2

6.4 INDIRECT EMISSIONS OF TRANSPORTATION

The building waste produced in the asset replacement and demolition stage requires transportation for disposal or recycling and reuse, which will generate a certain amount of GHG emissions. The accounting method can refer to Equation (6.4).

$$CE_{ys} = \sum_{i,j}^{n,l} (M_{ys,i,j} \times L_{ys,i,j} \times EF_{ys,j}) \quad (6.4)$$

Where:

CE_{ys}	—	Carbon emission of transportation for waste or other service, kg CO ₂ -eq
$M_{ys,i,j}$	—	The total amount of waste transported by the transportation type j in the activity i , t
$L_{ys,i,j}$	—	The distance of waste transported by the transportation j in the activity i , km
$EF_{ys,j}$	—	The emission factor of transportation j , kg CO ₂ -eq/ (t·km), see Appendix B.4
n	—	The number of transportation activities
l	—	The number of transportations in one activity

6.5 CARBON OFFSET VIA MATERIAL RECOVERY

The carbon credit by waste recycling and reuse can be calculated by Equation (6.5).

$$CS_{cl} = \sum_i^n (M_{cl,i} \times EF_{cl,i}) \quad (6.5)$$

Where:

- CS_{cl} — Carbon credit of waste recycling and reuse, kg CO₂-eq
- $M_{cl,i}$ — The industrial product equivalent of the recovered waste by type i, kg, m³
- $EF_{cl,i}$ — Emission factor of the industrial product replaced by recovered waste i, kg CO₂-eq/m³ or kg CO₂-eq/t, see Appendix B.3
- n — The number of waste types recovered

Chapter 7

Carbon reduction pathways in urban water sectors

7.1 OVERVIEW

On October 24, 2021, the State Council of China issued the *Action Plan for Carbon Dioxide Peaking Before 2030* (NDRC of China, 2021) to steer towards pathways of reaching carbon emission peak by 2030. This plan sets out ten actions to achieve peak carbon emissions throughout the whole processes and economic and social development, and highlights the importance of reaching targets in an integrated manner. As explicitly stated in this plan, one of the goals of attaining peak carbon emission is the formation of a low-carbon development model for whole society. On June 13, 2022, the State Council issued another guide named *Implementation Plan for Synergizing the Reduction of Pollution and Carbon Emission* (The State Council of China, 2022). This plan clearly defines the contribution of wastewater resource utilization to carbon reduction, encourages wastewater treatment plants to actively implement energy saving and clean energy utilization, promotes the carbon accounting of urban wastewater treatment and resource utilization, optimizes the management of energy consumption, and carbon emission of wastewater treatment facilities. For China's urban water sector, water consumption and wastewater production will continue to grow as the population grows and living standards improve (Huang et al., 2023). In addition, the burden of urban stormwater system is becoming heavier due to increasing extreme rainfall events (Ma et al., 2022). As such, the carbon intensity of the urban water sector should be reduced to offset the continuous increase of handled water to achieve peak carbon emissions. In other words, carbon accounting is not the goal and endpoint, but rather enhancing the synergy of pollution reduction and carbon reduction of water sector is, to promote the sustainable development, to guide the development and evaluation of carbon reduction plan, to assist in the optimization of technology selection, energy saving, and finally carbon reduction. As depicted in Figure 7.1, carbon accounting goes through the whole process of carbon reduction actions in the urban water sector. Carbon accounting can clarify carbon emission sources and contributions, which can in turn guide and target carbon reduction more specifically. Moreover, carbon accounting can help choose optimal technologies by clarifying their carbon footprint.

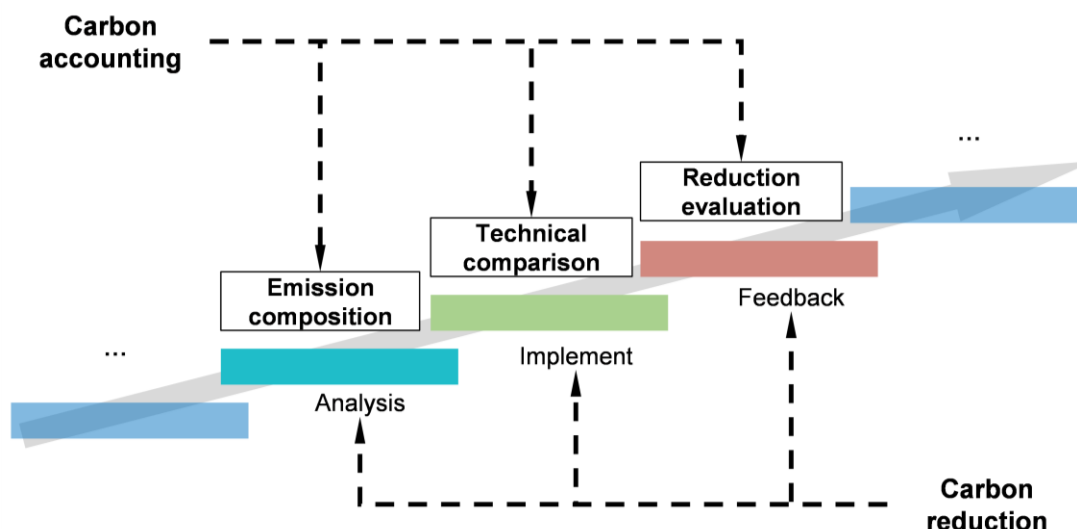


Figure 7.1 Schematic diagram of the interaction between carbon accounting work and carbon emission reduction actions.

From the perspective of the whole life cycle of the urban water sector, carbon emissions are associated with planning and construction and operation and maintenance, as well as asset replacement and demolition. Thus, the carbon accounting and reduction of the urban water sector should not be limited to the operation and maintenance stage but its whole life cycle of *from cradle to grave* (Seyedabadi et al., 2023). The planning and construction and demolition process belongs to the general construction industry, which is one of the traditional high carbon emission industries. Thus, carbon emission from planning and construction is referred to as hidden carbon and should be taken into consideration in the water sector. In terms of the contribution of planning and construction to total carbon emissions of the water sector, hidden carbon is closely related to the operating life of the structure. As shown in Figure 7.2, when the structure life is 20-30 years, the carbon emissions from planning and construction accounted for about 10% of the total carbon emissions of the water supply system while the contribution for wastewater systems accounted for about 3%-5% (NEEP, 2023). In terms of the demolition stage, the proper management and recycling of building waste is generally able to offset the carbon emissions of demolition operations. Overall, within the life cycle of the urban water sector, the operation and maintenance phase is the largest contributor of carbon emissions and should be the key focal point of carbon emission reduction – although the carbon emission of planning and construction should also not be underestimated.

Carbon reduction pathways in water sector

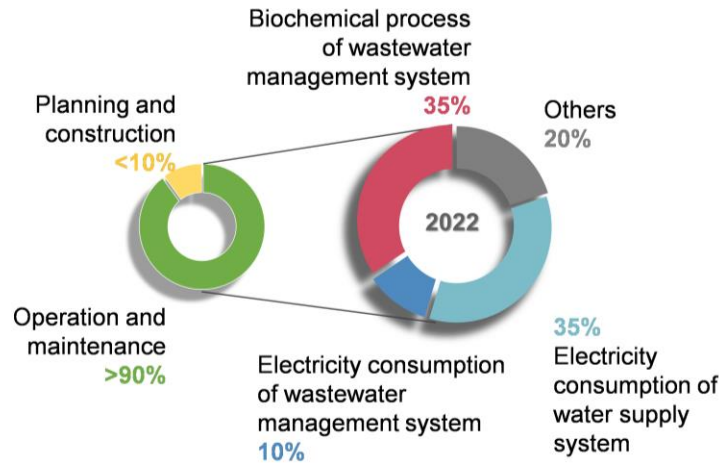


Figure 7.2 Composition of Carbon Emissions from China's Urban Water Affairs System (Related research is extremely limited, and the results vary widely. This Figure is only for reference).

The operation and maintenance phase of the urban water sector accounts for about 6.1% of the world's total electricity consumption (The World Bank, 2019). Within the water sector, the water supply system accounts for about 42% of total electricity consumption while the seawater desalination and reclaimed water treatment accounts for about 26% (Salas, 2023). In addition, wastewater and sludge treatment, building plumbing system, and long-distance water delivery contribute 14%, 13%, and 5%, respectively (Salas, 2023). In terms of the carbon emission of the water sector in China, information is very limited and some of the results available from different studies vary greatly. As such, based on China's *Urban and Rural Construction Statistical Yearbook* (MHURD of China, 2020) and *Urban Drainage Statistical Yearbook* (CUWA, 2018), these *Guidelines* firstly calculates the energy consumption and composition of China's urban water sector. The results show that the annual power consumption (water production and water supply) of China's water supply system (excluding long-distance water transfer) is about 4.75×10^{10} kWh/a, and wastewater treatment facilities (including reclaimed water plants but excluding wastewater drainage system) consumes about 1.69×10^{10} kWh/a (CUWA, 2018). In terms of stormwater management systems, electricity consumption statistics are lacking.

According to previous studies, the major carbon emission activity of the water supply system, water reclamation system, and stormwater system is electricity consumption. By contrast, wastewater management systems present some differences due to the existence of biological processes in drainage systems and wastewater treatment plants, and lead to CH_4 and N_2O emission which are denoted as direct emissions. This part of carbon emissions is about three times that of the power consumption by wastewater systems. By extrapolating to the whole water sector, the percentages of carbon emissions from electricity consumption of water supply systems, electricity consumption of wastewater management system, and biological processes of wastewater management system are about 30%, 10%, and 30%, respectively (Figure 7.2). The sum of these three parts accounts for more than 80% of the total carbon emissions of the whole urban water sector.

Next, the carbon emissions of China's water sector are estimated roughly. A typical carbon emission intensity of the water sector is obtained from a literature review (Huang et al., 2023; Xi et al., 2023). The activity data including the quantity of water consumption and wastewater production of China is referred

to the *Urban Construction Statistical Yearbook* (MHURD of China, 2020). As depicted in Figure 7.3, the total carbon emissions of the water sector in 2022 was about 114.88 million t CO₂-eq, accounting for 0.82% of the total carbon emission of China. Also, wastewater management systems are the largest emission contributor of water sector. It seems that even if the urban water sector achieves carbon neutrality, its contribution to the carbon neutrality of society as a whole will be very limited. However, water as a product is closely related to other industries. Therefore, carbon reduction in the water sector can be of great benefit, helping other industries to reduce their carbon emissions. In addition, climate change has brought new challenges to the operation and management of the urban water sector (Winchell, 2021). Most importantly, the introduction of carbon neutrality to the urban water sector is also conducive to their own sustainable improvement and optimization, and strengthens resilience. Therefore, the urban water sector should take the initiative in incorporating carbon reduction into operations, development, evaluation, etc.

From the perspective of the implementation route of carbon reduction in urban water sectors, alignment with the national *30 and 60 dual carbon goals* can be considered as the most basic time point. To provide a clear picture of the carbon reduction targets that need to be achieved at each stage of the urban water sector, these *Guidelines* have simulated the trends in total carbon emissions of the water sector in China under different scenarios based on literature data (Figure 7.3). Given the continued rise in urbanization levels, population growth, and infrastructure coverage, the amount of water handled by water sectors is bound to increase. The expectations for improved water quality and stringent wastewater discharge standards will also need more electricity and material input. Therefore, based on the baseline value of carbon emissions in 2022, if the water industry does not actively adopt carbon reduction measures, its total carbon emissions will reach 198.53 million t CO₂-eq by 2060, which is nearly twice that of the current level of carbon emissions. Admittedly, carbon reduction will not happen overnight. If carbon peaking is to be achieved by 2030, urban water sector will need to start reducing carbon emissions immediately. By reducing the carbon intensity of the water sector in order to offset the increase in carbon emissions caused by increased water use. It can be estimated that to achieve carbon neutrality target by 2060, continued and increased reductions will be required between 2030 and 2060 to offset the remaining 128.9 Mt CO₂-eq of carbon emissions (Figure 7.3).

From the composition of carbon emissions in China's urban water sectors, it is clear that the focus of carbon reduction should be on electricity consumption in water supply systems, stormwater management systems (lift pumps), wastewater management systems (lift pumps and aeration pumps), and biological processes producing CH₄ and N₂O in wastewater management systems (septic tanks, wastewater collection networks, and wastewater treatment plants) (Sun et al., 2018; Short et al., 2017). In order to reduce carbon emission from electricity consumption, it is necessary to rely on power companies to green their electricity generation. However, urban water sectors should also work hard to reduce electrical use as much as possible to offset the continued increase in water and wastewater volumes. This pathway will not only achieve carbon reduction but also reduce operating costs in the water sector. This is particularly important for stormwater management systems as forced drainage systems require pumping stations and resultant electricity consumption. Thus, optimizing the design of stormwater runoff and reducing deep tunneling are key to reducing the electricity consumption of stormwater systems.

Figure 1 is a line graph showing annual carbon emission trends and projections. The Y-axis represents Annual carbon emission ($\times 10^4$ t CO₂-eq), and the X-axis represents the Year. The graph displays historical data points for 2014, 2022, and 2030, and three projected scenarios from 2030 to 2060.

The historical data points are as follows:

Year	Annual carbon emission ($\times 10^4$ t CO ₂ -eq)
2014	11,488
2022	11,488
2030	12,890

The projected scenarios from 2030 to 2060 are:

- Without carbon emission reduction measures:** This scenario shows a continuous increase in emissions, reaching 19,853 by 2060.
- Maintain carbon peaking levels:** This scenario shows emissions remaining constant at 12,890 from 2030 to 2060.
- With carbon emission reduction measures:** This scenario shows a steady decline in emissions, reaching 0 by 2060.

Based on the principles of carbon reduction, carbon reduction technologies can be internationally classified into three categories – namely, carbon reduction, renewable energy, and carbon sinks (Li et al., 2023). Carbon reduction refers to the reduction of fossil fuel consumption or direct carbon emissions through the optimization or innovation of existing processes and technologies. Renewable energy refers to the replacement of fossil fuels with clean energy to achieve emission reductions. Carbon sinks refers to the absorption and fixation of greenhouse gases in the atmosphere through tree planting and other means. Specifically, for urban water sectors, the complex composition of carbon emissions and carbon emission activities dictate that there are many corresponding technologies that can be used to reduce carbon emissions. These *Guidelines* have further subdivided these three categories into five types of action strategies. As shown in Figure 7.4.

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- **Technology upgrading:** developing new low-energy, low-carbon-emission processes and systems to replace high-energy, high-carbon-emission traditional processes and systems;
- **Low-carbon energy:** tapping and recovering the energy contained within urban water sectors, supplemented using other clean energy sources (wind, solar, organic matter, waste heat, etc.) to reduce fossil fuel consumption;
- **CO₂ sequestration:** sequestering CO₂ from the atmosphere through measures such as afforestation and revegetation.

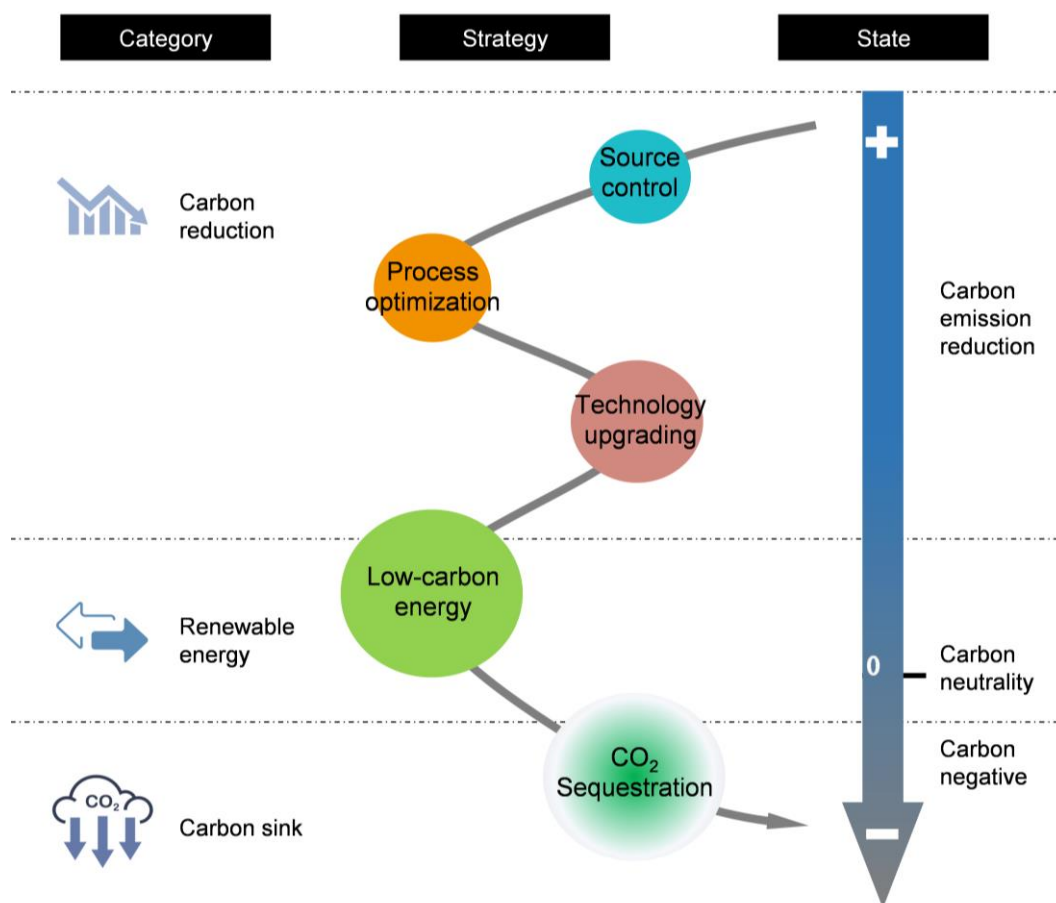


Figure 7.4 Action plan roadmap for carbon emission reduction in urban water sectors.

As summarized in Figure 7.4, source control, process optimization and technology upgrading action strategies belong to the carbon reduction category, while low carbon energy strategies belong to the renewable energy category. Generally, carbon reduction strategies are based on existing process technologies, are quick to implement and are small in scale in terms of engineering and investment, and should be the first choice for developing carbon reduction plans for urban water sectors. However, it is not possible to achieve carbon neutrality by relying on carbon reduction strategies alone, and low-carbon energy is the way to go. The low-carbon energy action strategy focuses on the recovery of energy within the water sector (e.g. chemical and thermal energy) (Meerburg, et al., 2015; Cipolla and Maglionico, 2014). As energy recovery facilities in water sectors (e.g., anaerobic digestion) are currently not common in China, the implementation of this strategy will inevitably involve a huge investment. However, the

carbon reduction potential is huge, as not only will the water system itself become carbon neutral, but it is also expected to export clean energy to society and/or to be used for carbon trading (Hao et al., 2019a). The CO₂ sequestration strategy focuses on plant sink enhancement and carbon sequestration by green facilities. Stormwater management systems involve a wide range of spaces such as green roofs, permeable pavements, bioretention areas, and vegetated filter strips, and are naturally coupled with urban ecosystems (Pelorosso et al., 2017). Green infrastructures are widely adopted in the planning and construction of stormwater management systems and have multiple benefits, enhancing stormwater runoff control and carbon sequestration in ecosystems. Moreover, their low implementation cost can be widely promoted in urban water sectors. Of course, the realization of carbon neutrality in urban water sectors does not depend solely on technological progress, but requires a high level of foresight and unified planning from the management, forming a carbon reduction approach of *policy traction, normative constraints, and technological impetus*.

This chapter firstly analyses the focus of carbon emission reduction and the collaboration between different sub-systems from the urban water sectors as a whole, and then further refines and summarizes the five types of action strategies. Based on the characteristics of carbon emissions at different stages of the life cycle, the emission activities of planning and construction and asset replacement and demolition are analyzed, with an emphasis on the emission reduction pathways in the operation and maintenance phase of water sectors. The technical choices, action strategies, and planning implementation catering for carbon emission reduction of sub-system are analyzed separately, with a view to providing reference for the development of carbon emission reduction plans for urban water sectors in China. From the perspective of carbon reduction, the water reclamation system is very close to the water supply system. Therefore, the two sub-systems are analyzed in the same section.

7.2 SYNERGY OF SUBSYSTEMS' CARBON EMISSION REDUCTION IN URBAN WATER SECTORS

Although the water supply system, wastewater management system, water reclamation system and stormwater system have different operating entities, as an integral part of the urban water sector, these different subsystems are interlinked and affect each other. Therefore, when formulating carbon reduction plans for urban water sectors, full consideration should be given to the synergy between the different subsystems. In other words, the carbon reduction strategies of one subsystem should be linked to the other subsystems. Because of this, the interlink and associated action strategies are summarized in Table 7.1. Water authorities or industry associations should strengthen the links between different subsystem operators and prioritize the promotion of such action strategies in order to achieve more effective carbon reduction with less effort.

Table 7.1 Summary of synergies of action strategies among various subsystems in urban water sector for carbon reduction.

	Water supply system	Wastewater management system	Water reclamation system	Stormwater system
Water supply system		The implementation of water conservation campaigns and enforceable metering will not only reduce the load on the water supply system, but also reduce the load on wastewater collection and treatment facilities so as to achieve carbon reduction	Water conservation, or tiered pricing for different volumes and areas of water use, will not only save water, but also provide a cost incentive for some areas to adopt recycled water and reduce carbon emissions.	Water conservation, or tiered pricing for different volumes and areas of water use, will promote the reuse of stormwater, reduce the load on stormwater drainage systems, and achieve carbon reduction
Wastewater management system	When the effluent from a wastewater treatment plant is discharged upstream of the water source, strict discharge standards can reduce water purification costs but increase energy and chemical consumption for wastewater treatment. The adoption of a separate drainage system can reduce the adverse impact of CSO on water sources, reduce water purification cost, and achieve carbon emissions.		When the effluent of wastewater treatment plant is subsequently reused, the treatment processes should be designed in accordance with the reuse purpose to avoid over-processing and unnecessary energy input.	

Continued

Carbon reduction pathways in water sector

	Water supply system	Wastewater management system	Water reclamation system	Stormwater system
Water reclamation system	In towns where local water resources are scarce, the construction of recycled water systems and the reuse of effluent should be promoted in accordance with local conditions, which can save energy consumption in water supply systems over long distances and achieve carbon emission reduction.	The wastewater treatment processes should be designed in accordance with the reuse purpose to avoid over-processing and unnecessary energy input.		\
Stormwater system	Stormwater reuse can reduce the load on the water supply system and achieve carbon emission reduction	Separate drainage systems or stormwater control facilities are adopted to reduce the amount of stormwater entering the wastewater drainage system, reduce the load of wastewater conveyance and treatment, and achieve carbon emission reduction	\	

7.3 PLANNING AND CONSTRUCTION AND ASSET REPLACEMENT AND DEMOLITION

7.3.1 Overview

From the carbon emission activities of the water sector, we can tell that the selection of water sources in the water supply system will affect the power consumption of water intake facilities. The location of water treatment plants, and their distance from water users will affect the power consumption of the water supply system. In the wastewater management system, the gravity drainage pipe network and the distribution of wastewater treatment plants determines the depth at which pipes are buried which, in turn, affects the number of lift pumps and lifting height, as well as the associated power consumption. In addition, the different wastewater treatment processes and the required effluent discharge standard will also lead to differences in carbon emissions. In a stormwater management system, the adoption of stormwater control facilities or drainage network will result in a difference in carbon emissions as well. From the perspective of the life cycle of the water sectors, the above aspects are determined in the planning and design stage, and are not easy to change and optimize after construction and operation begins. Therefore, the impact of planning and design on the carbon emissions of the water sectors is decisive, as the results cannot be completely reversed through optimization of operation and maintenance (Zhou et al., 2023; Sun et al., 2023). In addition, although construction is not directly controlled by the water sectors, carbon emissions are included in the accounting boundary. As such, planning and construction, as the beginning of the life cycle of the water sector, is also the primary point for implementing carbon reduction.

In general, 97% of carbon emissions in the planning and construction stage come from the use of various building materials such as steel and concrete (Hong et al., 2015). Therefore, the carbon emissions in the construction stage can be reduced by saving building material consumption while maintaining building strength, or using low-carbon building materials. Prefabricated components are usually uniformly designed and produced, and need less energy and raw materials in their manufacture. Green materials and recycled materials are highly recommended for reducing carbon emissions in the construction stage (Zhao et al., 2023).

During building demolition, carbon emissions from the demolition operation account for approximately 70% of total emissions, with the rest coming from the transportation and disposal of demolition waste (Wang et al., 2018). Through the process of sorting, shredding and screening of waste, recycling and reuse can be achieved, which is a pathway to offsetting a certain amount of carbon emissions.

Carbon reduction pathways in water sector

Table 7.2 Summary of action strategies for carbon reduction in planning and construction and asset replacement and demolition.

Stage	Strategy	Authority	Technical recommendation	Implementation notes
Planning and construction	Regime and spatial layout of urban water sector		Selection of water sources; spatial location and design capacity of water treatment plants; pressure zone-based water distribution; regime of wastewater drainage system; treatment capacity, technology selection, and effluent discharge standards of wastewater treatment plants; sludge disposal pathway; stormwater management systems, etc.	Taking wastewater treatment plants as an example, we can see that carbon emission intensity caused by electricity consumption is associated with the design capacity. Generally, the larger the scale, the lower the electricity consumption intensity is. Moreover, different processes also lead to different levels of carbon emission intensity. For example, the SBR process is suitable for small wastewater treatment plants to obtain a low electricity consumption intensity, while the oxidation ditch and AAO processes are more suitable for large-scale wastewater treatment plants.
	High-efficiency equipment	Planning and design department	Choose mechanical equipment with high energy efficiency, long service life, and low failure rate	Reasonable and appropriate selection and operation of mechanical equipment can reduce energy consumption and carbon emissions in future operations, and prevent additional emissions caused by frequent maintenance.
	Pressure management of water distribution system		The water distribution network is separated into pressure zones which are, in turn, separated by means of pressure reducing stations, gate valves and check valves	Appropriate pressure management can avoid pressure waste whereby, through electricity consumption and resultant carbon emission, water leakage can be decreased. The key to implementation pressure management is to obtain a scientific pressure zone. This shows that pressure management can reduce leakage loss by 6.2 m ³ /km, equivalent to reducing 68 t CO ₂ - eq/km emissions.
	Optimization of wastewater		Considering the construction of wastewater drainage and treatment plants in an integrated	The process of adjusting and optimizing drainage system is complex and requires consideration of factors such as urban

Continued

Guidelines for Carbon Accounting and Emission Reduction in the Urban Water Sector

Stage	Strategy	Authority	Technical recommendation	Implementation notes
	drainage system		manner according to local conditions, reducing the amount of wastewater that does not have to enter the treatment plant as far as possible, optimizing the design of drainage systems and reducing the formation of a long-term anaerobic environment in the drainage pipes, e.g. by adopting a separate drainage system, maintaining pipes to reduce infiltration of external water, optimizing the setting of septic tanks	construction and the degree of sophistication of supporting facilities. If a separate drainage system is adopted, initial stormwater runoff pollution should be considered to avoid polluting the receiving water bodies. Overall, optimization of the drainage system requires a long period of exploration and experience accumulation, as well as corresponding top-level design and policy guidance
	New drainage system		Innovation in drainage systems with low carbon footprint solutions, e.g., vacuum drainage systems to reduce the residence time of wastewater in the pipes, reducing the formation of an anaerobic environment and greenhouse gas generation.	The innovation and application of the new drainage system requires top-level design and policy guidance. The vacuum drainage system uses negative pressure suction, and investment and operation energy consumption analysis should be carried out to avoid the transfer of carbon emissions. Typically, vacuum drainage systems can reduce the consumption of flushing water by 40%.
	Adopting stormwater treatment plants and deep tunnels for caution		Large-scale centralized gray facilities of stormwater treatment plants and deep tunnel projects will generate a lot of energy consumption and carbon emissions during construction and operation. Green stormwater management facilities could be alternatives.	Reducing energy input
	Adopting prefabricated		Use of uniformly designed and produced prefabricated components for assembly construction	Prefabricated components have a fixed production process, which is more efficient and saves material and energy

Continued

Carbon reduction pathways in water sector

Stage	Strategy	Authority	Technical recommendation	Implementation notes
	components and prefabricated construction		methods to reduce the amount of on-site construction	consumption in production; the assembly process has standard operating methods, which improves construction efficiency and generally reduces carbon emissions by 3-7 kg CO ₂ -eq/m ² . However, in practice, there are also spatial constraints on factory sites, which may require longer transport distances. Renewable building materials such as wood, and recyclable building materials such as recycled aggregates are green building materials with low carbon emissions. For non-load-bearing structures in buildings, the use of green building materials to replace traditional high-carbon-emitting materials can reduce the carbon emissions of building materials. Long-life building materials should be considered.
	Green building materials	Construction department	Use of construction materials with renewable or recyclable properties, and local building materials to reduce transport emissions	
Asset replacement and demolition	Screen, process and recycle construction waste	Construction department	Construction waste cannot be buried, but should be properly screened and processed into recycled building materials according to actual conditions to offset carbon emissions	The recycling of construction waste can generate more carbon offsets. If it is properly treated, it can generally completely offset the carbon emissions of asset replacement and demolition

7.3.2 Carbon reduction pathways

7.3.2.1 Prefabricated building

A prefabricated building is a building that is manufactured and constructed using prefabrication. It consists of factory-made components or units that are transported and assembled on-site to form the complete building while cast-in-place buildings are poured and constructed on the construction site. In terms of carbon emissions, the main contributors in prefabricated building are the factory-made stage and transportation which are higher than cast-in-place building. However, less energy input is needed in the construction stage for prefabricated building compared to cast-in-place building, which leads to less carbon emissions overall. In short, prefabricated building can reduce carbon emission by 3-7 kg CO₂-eq/m² compared with cast-in-place building (Cai et al., 2023). In addition, there is less construction waste, noise, and dust pollution generated during the construction of prefabricated buildings, thus making them a greener and more environmentally friendly option.

Carbon emissions from the lives of construction workers are unavoidable and are not included in the carbon accounting according to the accounting principles proposed in this *Guidelines*. However, the use of many people instead of machinery can lead to a significant reduction in work efficiency, longer construction periods, and a surge in investment costs. Therefore, it is important to choose the best results for employing construction manpower based on the actual situation and we should not abandon machinery for fear of carbon emissions.

7.3.2.2 Optimization of lighting and ventilation design

Light and ventilation is one of the key design priorities, not only in relation to the energy consumption and carbon emissions of the equipment in operation, but also in relation to the health of the people who work and live there. For water sectors, there are some facilities placed indoors where the light and ventilations system should be paid attention to, so as to avoid too much electricity consumption. For example, doors and windows in the building should be appropriately expanded to enhance the natural lighting capacity of the building, thus reducing the energy consumption and carbon emissions. The direction of the building doors and windows should be reasonably designed to leverage the natural wind flow for indoor ventilation, which could also reduce indoor odor and heat dissipation, thus carbon emission.

7.3.2.3 Optimization of equipment selection

The normal operation of water sectors cannot be achieved without a large amount of mechanical equipment, including pumps, agitators, aeration equipment, etc. (Aghabalaee et al., 2023; Gu et al., 2023). The electricity consumption this mechanical equipment is a carbon emission activity that cannot be ignored in water systems. Improving the efficiency of machinery and equipment, and subsequently reducing electricity consumption, is the main route to achieving a reduction in carbon emissions. For example, the optimum energy efficiency of a pump should keep it operating in the high efficiency, i.e. the water volume should be kept as close to the design volume as possible. However, water volumes in water systems often fluctuate widely, leading to increased electricity consumption. In this regard, variable frequency pumps can dynamically adjust with fluctuations in water volume and can always be kept operating in the high efficiency zone, ensuring the lowest carbon footprint possible. In addition, aeration equipment in wastewater treatment plants is one of the main electricity consuming activities, which is attributed to the lower oxygenation efficiency of the aeration equipment. This, in turn, can be improved by a reasonable choice of aeration units to reduce electricity consumption. In addition, the choice of

machinery and equipment should take full account of the realities of the situation. For effluents with a high grit content, screens and their cleaning equipment that are easy to clear and not easily blocked should be selected. Equipment with a long-life cycle should be selected in order to avoid carbon emission caused by frequent replacement of equipment.

7.3.2.4 Pressure management of water distribution

In water distribution systems, pumps are generally selected in accordance with the highest hourly flow rate, and therefore deviate from the optimum rated working conditions most of the time. In practice, pressure control is often implemented by reducing the valve, which results in wasted energy and reduces its service life. In addition, for some high-rise buildings, secondary pressurization is needed, which cannot leverage the pressure the water distribution and results in pressure waste. Water leakage due to excessive pressure in the distribution network is also a challenge. One of the solutions for pressure management is to divide the service area into several pressure-specific zones which have their own distribution system with different distribution pressures. One study indicated that a 5.6 m head reduction in the inlet pressure of Beijing's water distribution network could reduce leakage by 6.2 m³ per km of pipe, equivalent to a 68 t CO₂-eq/km reduction in emissions. It can be seen that optimizing pumping conditions or pressure zone-based distribution can help to save water and reduce electricity consumption simultaneously.

7.3.2.5 Drainage system upgrade

Conventional sewers collect and convey wastewater to the wastewater treatment plant by gravity flow. Considering the amount of construction and the difficulty of subsequent maintenance, sewers generally control slope and burial depth. In conventional gravity drainage systems, the flow rate of wastewater is slow. For larger urban drainage systems, wastewater at the beginning takes 1 to 2 days to get to the treatment plant. Such a long residence time can lead to an anaerobic environment in the pipes, which in turn leads to the production and emission of large amounts of greenhouse gases. Upgrading and optimizing the drainage system and improving the environment inside the pipes is therefore an effective means of reducing greenhouse gas production.

In this regard, vacuum drainage systems offer significant advantages (Islam, 2017; Miszta-Kruk, 2016). Under negative vacuum pressure, the waste water can be lifted vertically and transported to the trunk sewer or to a wastewater treatment plant. The shallow depth of the vacuum drainage system, the more flexible arrangement of the pipes, and the faster flow rate of the wastewater make it less prone to blockages and leaks, which can greatly reduce the production of greenhouse gases in the pipes. Again, the adoption of separate drainage systems can prevent stormwater and sewage from mixing, avoiding pumping stations in the sewer to pump stormwater, reducing the amount of water entering the treatment plant, to achieve the purpose of reducing the energy consumption of pumping stations and treatment plants. At the same time, it can also prevent CSO pollution.

7.3.2.6 Optimizing the setting of septic tanks according to local conditions

Before the adoption of centralized wastewater treatment systems, septic tanks served as a simple, decentralized wastewater treatment device that degraded organic matter and had a limited role in water environmental protection. However, with centralized treatment becoming prevalent, the disadvantages and hidden dangers of septic tanks are becoming increasingly apparent. Firstly, septic tanks are sources of high carbon emissions. In China, the total annual CH₄ emission from septic tanks is as high as 3×10^7

t CO₂-eq/a, which is equivalent to the direct carbon emissions from wastewater treatment plants. In addition, septic tanks can remove part (30%) of the COD resulting in insufficient carbon sources for biological nutrient removal in wastewater treatment plants. Moreover, septic tanks occupy underground space and pose a safety hazard because of the CH₄ accumulation, and over 95% of septic tanks are likely to leak after 1-2 years of use, not only polluting groundwater and corroding pipes, but also softening the foundations of buildings (Somlai et al., 2019; Withers et al., 2011). It is worth noting that some cities in China have started to plan for the elimination of septic tanks.

7.3.2.7 Carefully consider deep tunnel projects

According to the different objectives of stormwater control, deep tunnels include volume control, stormwater runoff storage, CSO storage, stormwater transfer, etc. Take the analysis of the combined flow system overflow control target of the storage tunnel as an example: when it rains stormwater is stored in the deep tunnel; and then, afterwards, the rain is lifted to the shallow pipe drainage system through the tail-end pumping station, reducing the number of times the gates are opened in the watershed, which can reduce pollution of the combined flow sewage and initial stormwater in the nearby watershed during the rainy season, as well as reducing the appearance of black smelly water bodies in the watershed and improving the quality of the water environment in the watershed, thus further reducing carbon emissions generated by fossil energy consumption (Luo et al., 2021). However, the deep tunnel project is a large project, and its construction boosts carbon emissions by more than 50% compared to shallow surface pipeline systems. Moreover, deep tunnels require the deployment of sewage lift pumping stations, which increase their energy consumption by around 10.40 kWh/10³ m³. Deep tunnels are generally long in distance and prone to siltation in the tunnels, as well as generating significant carbon emissions in terms of operation and maintenance, so other green stormwater facilities should be used in the stormwater system instead where possible.

7.3.3 Carbon replacement pathways

At the end of the life cycle of buildings and structures in water sectors, a large quantity of construction waste will be produced after demolition. Instead of landfilling the waste, recycling and reusing it is highly recommended as a sustainable solution through the reasonable processing. This approach could extend the life cycle of materials and produce carbon credit.

7.4 WATER SUPPLY AND WATER RECLAMATION SYSTEMS

7.4.1 Overview

Based on the composition of carbon emissions of the water supply system, the main contributor in the operation and maintenance stage is electricity consumption. Other sources include chemical and/or various materials input. All these emission activities are classified as indirect emission. Specifically, nearly 100% of the GHG emissions in the long-distance water transmission facilities, water intake facilities, and water distribution system come from electricity consumption of pumps. In terms of water treatment plants and seawater desalination plants, electricity consumption also accounted for 95%, 82%, and 98% GHG emission (Tan et al., 2023; Ma et al., 2022). Therefore, the key to formulating carbon emission reduction plans for water supply systems is to improve management levels and optimize treatment technologies to reduce electricity consumption. A series of available emission reduction pathways and solutions for water supply systems are summarized in Figure 7.5 and Table 7.2.

Carbon reduction pathways in water sector

In the aspect of **source control**, measures such as water conservation by users, strengthening water consumption measurement, gradient pricing, and water source protection can all reduce water demand and thereby the work loading and carbon emissions of water intake facilities, water distribution network, and water treatment. In fact, water reclamation system can reduce the input of external water, which is probably achieved by long-distance water delivery and water intake facilities. Thus, the reuse of wastewater for crop irrigation, green area watering or toilet flushing could decrease electricity consumption.

In terms of **process optimization**, water supply system operators can also take various technical measures to reduce carbon emissions, mainly focusing on improving the operational efficiency of pumps and water use efficiency, including the application of technologies such as network leakage detection, variable frequency speed regulating pumps and filter backwash optimization.

Regarding **technology upgrading**, new water supply systems and efficient treatment technologies should be actively developed, such as pressure management models and low-energy seawater desalination which can significantly reduce energy consumption and carbon emissions.

From the perspective of **low-carbon energy**, the water supply system can also produce renewable energy, including thermal energy extraction and potential electricity generation. However, it should be noted that the implementation of energy recovery from water supply systems should avoid affecting the water quality and water pressure designed for users. The specific action strategies are summarized in Table 7.3.

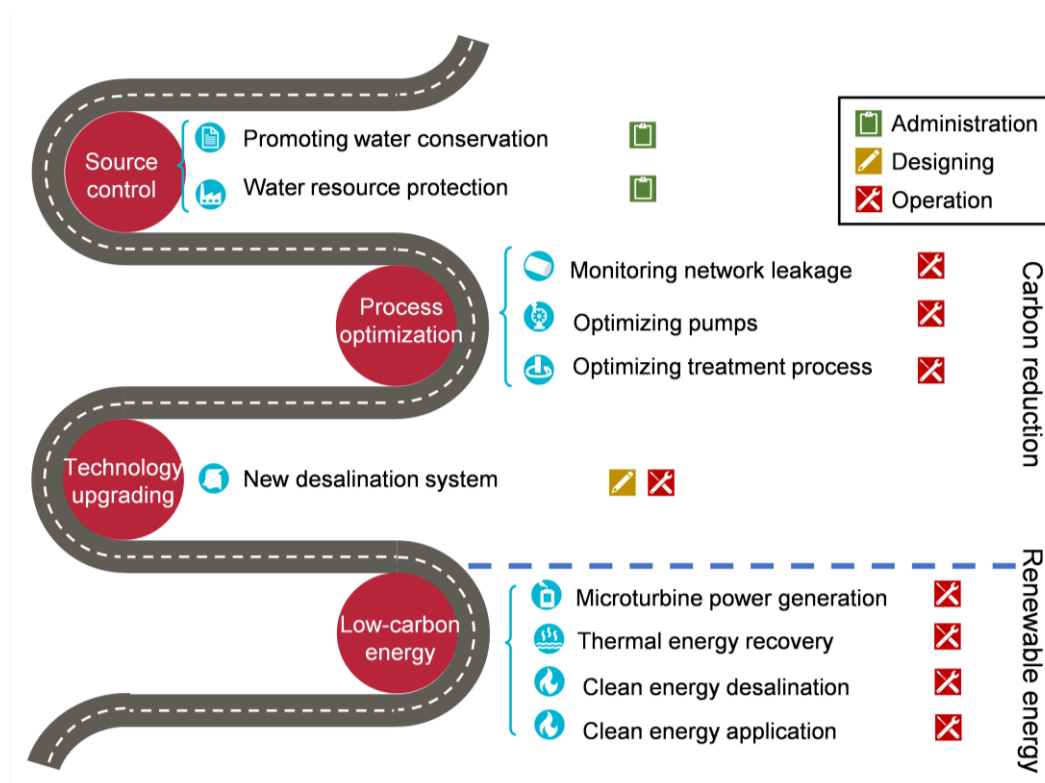


Figure 7.5 Carbon reduction roadmap for water supply system.

Guidelines for Carbon Accounting and Emission Reduction in the Urban Water Sector

Table 7.3 Summary of action strategies for carbon emission reduction in operation and maintenance of water supply system.

Type	Strategy	Authority	Technical recommendation	Implementation notes
Source control	Compulsory water metering/ tiered pricing/water-saving advocacy	Management department	Water metering should be promoted to strengthen water monitoring. Water utilization efficiency should be benchmarked and formulated for various buildings to enhance water-saving along with gradient pricing schemes and advocacy.	The cost of these strategies is low but effective supervision and long-term water-saving advocacy are necessary. It has been shown that mandatory water metering can reduce water consumption by 6 L per person per day (refer to the <i>National Mature and Applicable Water-saving Technology Promotion Catalog</i> of the Ministry of Water Resources)
	Water source protection		By protecting the water source and reducing the input of pollutants, the pollutant removal loading of the water treatment plant can be reduced, thereby reducing the consumption of electricity or chemicals, and achieving carbon emission reduction	Water source protection is a systematic project involving multi-sectoral joint actions, and with the increase of climate change and extreme weather, protection will become more challenging
	Pump optimization		Adoption of VFD pumps	It has been shown that compared with traditional water pumps, VFD pumps can reduce energy consumption and carbon emissions by about 50%
Process optimization	Water leakage detection	Operation department	By installing sensors at specific sites of the pipe, the information of flow rate, pressure, and other indicators of the pipe network can be collected and leveraged to quickly locate abnormal pipe sections	Water leakage detection has been practiced more and more, and it has been demonstrated that about 40% of water leakage in the pipe network can be reduced
	Water treatment process optimization		In water treatment plants and seawater desalination plants, a great deal of work can be done to reduce electricity consumption and carbon emission, such as optimization of backwashing frequency and increasing membrane pretreatment processes	The adjustment and optimization of the process has a lot to do with the quality of the treated water – that is, the optimization depends on the feedback of real-time information
Technology upgrading	New Seawater Desalination System	Design and planning department	Choose new equipment with higher efficiency and less energy consumption	Applicable to water supply systems in coastal cities

Continued

Carbon reduction pathways in water sector

Type	Strategy	Authority	Technical recommendation	Implementation notes
Low carbon energy	Microturbines for electricity generation	Operation department	Adoption of microturbines to recover excessive energy in pipe networks to generate electricity	The recovery of potential energy and the size of head loss should be balanced
	Utilization of thermal energy		Using Heat Exchangers in Pipelines to recover thermal energy to replace fossil fuel energy	Thermal energy extraction should not deteriorate the water quality
	Sea water desalination with renewable energy		Completely use clean energy to realize the desalination process of seawater	Wide applicability without any carbon emissions
	Application of renewable energy		Application of clean energy such as wind energy and solar energy	Make full use of the space of water sectors for new energy power generation to replace fossil energy consumption

7.4.2 Carbon reduction pathways

7.4.2.1 Pipe leak detection

According to relevant standards and regulations of China, the average water loss of urban water supply pipelines should not be higher than 10%. According to the *China Urban and Rural Construction Statistical Yearbook* (MHURD of China, 2020), the state average water loss in 2020 is 13.26%. In some cities, the water loss even exceeds 25%. This not only causes a huge waste of water resources, but also increased energy consumption and carbon emissions in water supply systems. Undoubtedly, the implementation of leak detection will help to monitor, locate, and repair water loss points in a timely manner, whereby carbon emission intensity of the water supply system can be reduced. In fact, the exploration and development of leak detection has always been an important topic in the water industry. For example, the advancement of sensors and wireless transmission technology enables the real-time monitoring of water pressure and flow based on which water loss can be determined and then solved in water distribution systems. In a full-scale project, a leak detection system was built for a water distribution system serving a population of 150,000. The results shown that the system helped to avoid $2.0 \times 10^8 \text{ m}^3$ water leakage and $10^4 \text{ t CO}_2\text{-eq}$ emissions over 12 years' running.

7.4.2.2 New seawater desalination technology

Desalination is an important technological solution to urban water shortages. However, from a technical point of view, the production process, whether distillation or reverse osmosis, often requires large amounts of electricity or fossil fuels such as coal, which results in high levels of carbon emissions (Ayaz et al., 2022). For this reason, a number of new desalination technologies are gradually being developed that consume less energy and have lower total carbon emissions, such as the deep-sea desalination technology and the high-efficiency absorption vapor compression system process (Fasano et al., 2021; Wang et al., 2020). The former is a desalination unit submerged on the seabed, using natural hydrostatic pressure as the “pre-membrane pressure”, and the desalination unit is connected to the coast through a corridor that connects electricity, communication cables and treated desalinated water, reducing carbon emissions by 40% compared to conventional desalination processes. The latter reduces the need for external heating steam by reusing the final steam generated, thereby reducing coal consumption and reducing indirect carbon emissions from operation and maintenance by approximately 73%.

7.4.3 Carbon replacement pathways

7.4.3.1 Water energy recovery

As an energy carrier, water contains many forms of usable energy, including potential energy from pressurization and its own thermal energy. If this energy can be potentially recovered through available technologies, it can feed back into the energy consumption of the water supply system, thereby reducing fossil fuel consumption and accomplishing carbon reduction. The excess energy (water head) of the water in the network can be recovered using microturbine power generation technology, which, based on the current level of operation of the water supply network, can generate up to millions of kWh per year without affecting the effective water head in the network, and can be used as a power source for intelligent monitoring devices in the network system (Amjadi et al., 2020).

In addition, the low and more constant temperature of the water in the network can be used as a cooling source for water source heat pumps for building cooling. Moreover, urban civil buildings are connected to a network of distribution pipes and can guarantee a 24 h supply, ensuring the stability of water as a cooling source. Thus, approaches could produce carbon credit to offset the carbon emission of water sectors (Meggers and Leibundgut, 2011).

7.4.3.2 New seawater desalination technology

Clean energy can also be leveraged to support water supply systems, particularly for seawater desalination. A number of new desalination technologies using clean energy are currently being developed, avoiding a significant portion of the carbon footprint as fossil fuels are not required. For example, large concentrating solar desalination technologies, which boil seawater by building huge glass domes to concentrate light and heat the water. The installation comes with energy storage (over 4 GW of energy storage) for night or cloudy day production (Ahmed et al., 2022).

7.5 WASTEWATER MANAGEMENT SYSTEMS

7.5.1 Overview

Due to the high concentration of organic matters and nitrogen compounds in wastewater, carbon emission of the wastewater management system in the operation and maintenance stage are different from other systems. More specifically, there are considerable amounts of CH₄ and N₂O emissions (direct carbon emissions). According to the above carbon accounting and literature review, the direct carbon emission sources of the wastewater management system include septic tanks, wastewater sewers, combined sewage overflow (CSO), biological wastewater treatment, and facilities for sludge treatment and disposal. It is estimated that the annual total amount of CH₄ produced by urban septic tanks in China is as high as 3×10⁷ t CO₂-eq/a (Hao et al., 2017), which is at the same level as the total carbon emission from wastewater treatment plants (WWTPs), and is necessarily the key goal of developing carbon emission reduction plans for wastewater collection networks. For wastewater treatment plants, the direct carbon emissions caused by wastewater and sludge treatment units account for about 35% to 65% (Gallego-Schmid and Tarpan, 2019), which is related to influent composition and operation parameters. Thus, direct carbon emission is the focus for carbon emission reduction in wastewater treatment plants. Overall, in terms of carbon emission reduction pathways in wastewater management system, the situation can be considered and analyzed from the perspective of four types of action strategies as depicted in Figure 7.6.

In terms of **source control**, reducing the amount of wastewater and the total amount of pollutants entering wastewater sewers and wastewater treatment plants will undoubtedly reduce the energy consumption, material input and direct carbon emissions required for lifting or treating wastewater. Possible action strategies include the implementation of separated drainage systems, source separation technology, etc. Source control also has its own unique aspect, that is, to formulate the discharge standards of wastewater treatment effluent. The discharge standard should be adopted specifically and flexibly according to the condition of the receiving water body instead of one-size-fits-all standard whereby the energy consumption for wastewater treatment can be optimized to achieve carbon emission reduction (Buonocore et al., 2016).

In terms of **process optimization**, a range of measures can still be taken by wastewater management systems, with the potential to reduce carbon emissions. For example, the cross-section and slope of wastewater pipes can be optimized to reduce the formation of anaerobic environment and thus reduce the generation of CH₄. The operation parameters of biological treatment processes can be another hotspot influencing the emissions of CH₄ and N₂O (Hao et al., 2019b). Moreover, aeration optimization combined with water pump optimization can also reduce electricity consumption, thereby reducing indirect carbon emissions.

In terms of **the technology upgrading** strategy, the research and development of new, low-energy, low-carbon, high-efficiency wastewater treatment technology has always been a hotspot in the water sector. In recent years, with the proposal and application of compact and efficient nitrogen removal

processes, it has become possible to achieve carbon emission reduction through the upgrading of new treatment processes.

In terms of **low-carbon energy**, the wastewater management system also has its own advantages as wastewater contains chemical energy as well as huge amounts of thermal energy. Typically, the thermal energy embodied in wastewater accounts for 40% of the total waste heat of a city, and nine times that of the chemical energy in wastewater (Hao et al., 2019a). Relying on anaerobic digestion and water source heat pump technology, the energy in wastewater can be recovered and support the operation of wastewater treatment plants or even export outside. All these practices could help the water sector achieve carbon reduction. Overall, the specific carbon reduction action strategies for wastewater management systems are summarized in Table 7.4.

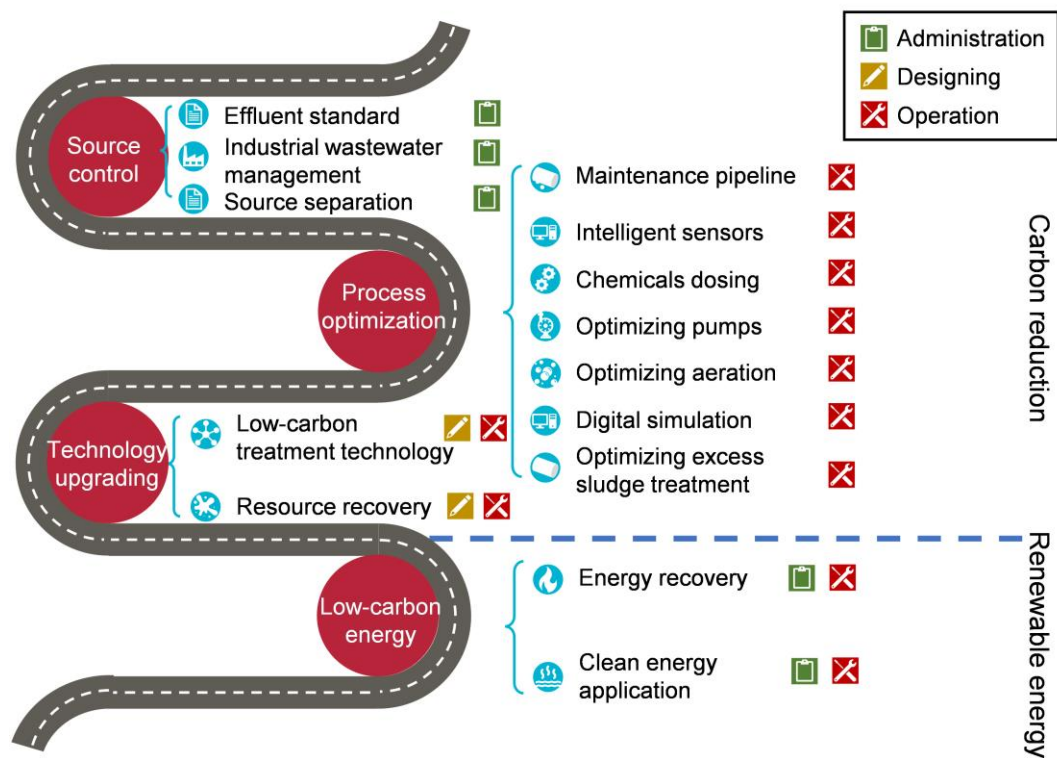


Figure 7.6 Carbon reduction roadmap for wastewater management systems.

Table 7.4 Summary of action strategies for carbon emission reduction in wastewater management system.

Type	Strategy	Authority	Technical recommendation	Implementation notes
Source control	Plant-specific effluent discharge standards		It is not necessary to adopt a one-size-fit-all discharge standard for each WWTP. Instead, plant-specific standard is highly recommended, ones that consider the environmental capacity of the receiving body and effluent reuse purpose.	Unnecessarily stringent effluent discharge standards could lead to more electricity and chemical input in WWTPs and result in more carbon emissions. It is necessary to establish a regular monitoring of the receiving water body quality and adjust the discharge standards accordingly. A low-polluted water body could accept more pollutants and reduce the treatment burden of WWTPs, thereby reducing carbon emissions. It is also worth noting that the improvement of the quality of the receiving body relies on the synergic actions and efforts of other industries or departments.
	Strengthening industrial wastewater management	Management	The quality standard of industrial wastewater discharging into wastewater collection network should be scientifically formulated to avoid input of toxic compounds or wastewater with unbalanced organic to nutrient ratios.	It is necessary to establish regular supervision of the industrial wastewater quality discharged to the collection network.
	Promotion of source separation technology		Black water and urine can be collected, conveyed, and handled separately to recover nutrients, save water demand, and reduce the pollutant input to WWTPs, thereby reducing carbon emission. Moreover, nutrients recovery and reuse could produce carbon credits.	Sanitary appliances are not within the organizational boundary of wastewater management system, but they do benefit WWTPs. Investment analysis is required, and policy & regulations are necessary to enhance the attractiveness of this technology. Typically, with source separation appliances, 47-95 kg CO ₂ -eq/(ca·a) carbon emissions of wastewater treatment plants can be achieved (Hao et al., 2016).
Process optimization	Curbing CH ₄ generation in	Operation department	Efforts should be allocated to avoid formation of anaerobic environment and dead zone in sewers by	Chemical dosing and forced ventilation require costs, energy consumption, and carbon emission analysis in order to avoid

Continued

Guidelines for Carbon Accounting and Emission Reduction in the Urban Water Sector

Type	Strategy	Authority	Technical recommendation	Implementation notes
	wastewater sewers		chemical dosing, flow optimization, and forced ventilation, thereby reducing the production of CH ₄ . Electrical consumption is mainly due to the lifting pump and aeration system. By updating and upgrading equipment, or optimizing the operation, the working efficiency of pumps can be improved and electrical consumption and carbon emissions can be reduced. Specifically, variable frequency pump, high efficiency aeration systems, micro-bubble aeration heads, and forward feedback or post-feedback aeration optimization control technology are recommended.	the trade-off of carbon emission.
	Reduction of power consumption		The operation of biological treatment units can be optimized in terms of parameters, aeration, and other measures to reduce CH ₄ and N ₂ O production. Specifically, mathematical models or digital twins can be applied to guide operational adjustments to minimize CH ₄ and N ₂ O emissions (Mannina et al., 2016).	Investment analysis should be done before adopting technology upgrading or process optimization to increase the economic feasibility. It is worth noting that the potential of these approaches for carbon reduction is highly plant-specific. The optimization of aeration systems should consider direct greenhouse gas (GHG) emissions simultaneously. Typically, by implementing these measures, energy consumption can be reduced by 20% to 50% (Lundin et al., 2000).
	Reduction of direct carbon emissions from wastewater treatment units		Waste sludge should be processed in a timely manner and avoid long-term stacking in the plants. Sludge volume should be reduced as much as possible prior to transportation to landfilling, thereby reducing carbon emission.	This digital technology has been already applied in full-scale projects (Lim et al., 2012), but its application in GHG emission control depends on the development of sensors, control electrical equipment, etc.
	Optimization of excess sludge treatment			Systems should be considered and designed in conjunction with energy recovery from sludge.

Continued

Carbon reduction pathways in water sector

Type	Strategy	Authority	Technical recommendation	Implementation notes
Technology upgrading	Low-carbon wastewater treatment technology	Design planning department &	Some innovative treatment technologies, such as compact nitrogen removal process and/or aerobic granular sludge, can be applied to reduce electricity and chemical consumption, thereby reducing carbon emissions.	It is noteworthy that some novel technologies require a large investment and are probably not suitable for upgrading existing plants. Besides, although some new technologies can reduce power or chemical consumption, they probably produce more direct carbon emissions, such as CH ₄ and N ₂ O.
	Promotion of resource recovery	management department	Resource recovery from wastewater, such as nutrients, cellulose, etc., should be given priority to produce carbon credit (Hao et al., 2022).	Policy and regulations should be formulated to promote resource recovery and recycling. Investment analysis should be done prior to implementation.
Low carbon energy	Strengthening wastewater energy recovery & use of clean energy	Management & operation department	The recovery of chemical energy and thermal energy contained in wastewater can fuel the operation of WWTPs and reduce the consumption of fossil fuel via anaerobic digestion technology (supplemented by thermal hydrolysis pretreatment or high solid concentration) and water source heat pump technology (Hao et al., 2019a). In addition, solar and wind energy can also be leveraged to fuel the operation of WWTPs (Mo and Zhang, 2013).	The investment of sludge anaerobic digestion facilities is huge and needs the subsidy support from government (Ashrafi, et al., 2014). The recovery potential of thermal energy is large, but the recovery points and application scenarios should be balanced and developed. The government should formulate policy orientation to allow the generated CH ₄ and thermal energy to enter the grid. Also, the photovoltaic power generation capacity of a plant is limited.

7.5.2 Carbon reduction pathways

7.5.2.1 Source control

The wastewater treatment facilities in WWTPs (e.g., aeration and mixing) consume a great deal of energy and chemicals to remove/destroy pollutants, which leads to the production of indirect GHG emissions. Indeed, the biochemical processes of pollutants also emit CH_4 and N_2O directly. This means that the WWTPs meet the effluent discharge standard at the cost of increased GHG emissions. Therefore, trying to reduce the pollutant loading of WWTPs could reduce the total carbon emission, which is classified as a source control pathway.

One of the measures that can be taken to reduce the pollutants in wastewater from flowing into WWTPs is source separation. The essence of source separation technology is to collect and process black water and urine separately (Bisinella de Faria, et al., 2015). In this way, the nutrients contained in urine can easily be captured and recycled which means they could then be used in agricultural production. At the same time, source separation reduces the amounts of pollutants, particularly nutrients, entering to WWTPs. As such, electrical and chemical consumption can be minimized, thereby reducing the carbon emission intensity of wastewater treatment. It is estimated that the application of source separation can reduce carbon emissions by about 47-95 kg $\text{CO}_2\text{-eq}/(\text{ca}\cdot\text{a})$ (Hao et al., 2016).

As stated above, WWTPs achieve pollutant removal at the cost of GHG emissions. Stringent effluent discharge standard can reduce the pollutant input to the water bodies and avoid environmental problems of black and odorous water bodies and eutrophication, and an even more stringent discharge standard usually only means more GHG emission. Indeed, for some WWTPs, a one-size-fit-all stringent discharge standard is not necessary probably due to the large environmental capacity. Therefore, a plant-specific discharge standard is recommended to balance effluent quality and carbon emissions. Since China has a large area covering a range of climate zones, wastewater characteristic presents differences. Therefore, a local discharge standard should be scientifically formulated (Mo and Zhang, 2013).

In general, industrial wastewater can be discharged into municipal wastewater sewers after treatment and meeting quality requirements. However, there are always some companies illegally discharging untreated industrial wastewater into sewers. The arbitrary discharge of industrial wastewater will increase the pollutant loading of WWTPs. Moreover, toxic and harmful substances in industrial wastewater will affect biological treatment performance. Thus, it is necessary for the water department to implement strict supervision to reduce unnecessary carbon emission from WWTPs.

7.5.2.2 Digital transformation of wastewater treatment

Relying on the development of digital tools, WWTPs can use a variety of sensors to collect detailed information and use mathematical model and controls to make decisions and optimize the efficiency of wastewater treatment. The core of a digital operation is a scientific, reliable and accurate mathematical model representing the physical assets of treatment units (Cheng et al., 2023). The model is running based on a large number of operational data and can simulate and predict the influence of changing parameters on performance. Then, the model can also propose solutions to optimize the operation of wastewater treatment processes, such as the aeration intensity, flow control, chemical dosage (Therrien et al., 2020). Then, energy and chemical input could be minimized, thereby reducing carbon emissions.

One highlighted application of the digital tools is the accurate aeration in biological wastewater treatment units. Typically, aeration in wastewater treatment accounts for approximately 50% of the total energy consumption in wastewater treatment plants (Sweetapple et al., 2014). If the aeration intensity is insufficient, it will affect the microbial biochemical reaction, thereby impairing the pollutant removal

performance. As such, aeration intensity is usually controlled at a surplus and constant level which is independent of wastewater or pollutant fluctuations. As a result, it will lead to high energy consumption and carbon emissions. As such, digital tools enable optimal control of the aeration process by advanced monitoring of dissolved oxygen (DO), $\text{NH}_4^+ \text{-N}$, and even suspended solids (Haimi et al., 2013). The controller based on mathematical models, can give feedback to the blowers and adjust aeration intensity in real time. As such, electricity consumption can be minimized without exacerbating the treatment performance. It is worth noting that the performance of digital tools in aeration control is highly dependent on the development of instruments, e.g., sensors (Therrien et al., 2020).

Pump optimization with digital tools is also one of the most important approaches to reducing electrical consumption and carbon emissions (Fan et al., 2018). With the operation of pumps, the working efficiency is not always as high as it is designed to be, and may be compromised through long-term use. There are two reasons for this: firstly, components may become damaged after continuous operation, resulting in a gradual decrease and loss of pump efficiency. In addition, pumps do not always work at peak efficiency due to the fluctuations in wastewater flow. As a result, unnecessary electrical consumption leads to further carbon emissions. In terms of solutions, existing pumps could be switched to smart pumping by leveraging digital tools. The emergence of pumps with embedded intelligence is a critical step forward in the evolution of performance management and pump system optimization. Intelligent pump controllers can automatically configure a system with the optimal settings for their application, taking the guesswork out of the process. Systems can be customized with pump protections and for multiple pump operations. Curve control technology relies on pump-specific algorithms, which can accurately predict where a pump operates on its curve. Using speed, torque and power data to know where the pump operates on the curve, sensor less pump controllers can be set up to take actions based on those factors. Overall, there are many case studies of companies providing this service to optimize their pumps, thereby reducing carbon emissions.

7.5.2.3 Compact wastewater treatment processes

Currently, conventional activated sludge is widely applied in WWTPs. However, one of the disadvantages of conventional activated sludge is the low biomass content in the reactors. As a result, the footprint of biological reactors is usually large, which mean that more construction and building materials are needed, which in turn generate greater carbon emissions. In addition, the large volume of the reactors needs more mixers which results in their using more electricity. This shows the necessity of more compact wastewater treatment technologies. Aerobic granular sludge technology is an innovative wastewater treatment technology that provides advanced biological treatment using the unique features of aerobic granular biomass. These technological features translate into a flexible and compact process that offers energy efficiency and significantly lower chemical consumption. The reactor footprint of aerobic granular sludge system is usually only 1/4 that of conventional activated sludge systems (Bengtsson et al., 2019). Moreover, this technology reduces energy consumption by 30% to 50 % (Hamza et al., 2022).

7.5.2.4 Efficient denitrification technology

The biological nitrogen removal process is composed of an aerobic environment and an anoxic environment where oxygen and a carbon source are required, respectively. For wastewater with a low carbon to nitrogen (C/N) ratio, an external carbon source should be supplemented to achieve effluent standard. All these features lead to carbon emission due to electrical and chemical consumption. Since,

this process is also the main contributor to N_2O emission, it should receive the greatest attention. Some novel nitrogen removal processes have been proposed and even applied in full-scale projects, such as partial nitrification-denitrification process and ANAMMOX process. The oxygen demand and carbon sources in these novel processes is reduced compared to the conventional processes, as a result, carbon emissions are minimized. It is worth noting that N_2O emissions in these novel processes need more monitoring to confirm their precise levels (Vasilaki et al., 2019).

7.5.2.5 Resource recovery from wastewater

It is a consensus that wastewater is the carrier of various resources (Hao et al., 2022), such as phosphorus, nitrogen, cellulose, EPS, etc. They can be recovered, and bring certain economic benefits and carbon credits by replacing the mining and production process of the industrial products, and thereby reducing carbon emissions.

Phosphorus recovery from wastewater has been a hotspot, as the conservation of phosphate rock is limited and phosphorus is a necessary element for life and modern agriculture. At present, phosphorus recovery technology can be divided into three categories: i) recovery from the mainstream treatment line, i.e., the supernatant of the anaerobic tank rich in phosphorus with a recovery efficiency of 40%-50% (Jupp et al., 2021); ii) recovery from sludge treatment line, i.e., the side-stream liquid from the anaerobic digestion and/or dewatering units; iii) recovery from the ash after sludge incineration with a recovery efficiency of 90% (Wang et al., 2023).

Another example is the recovery of high-valued alginate-like exopolymer (ALE) from waste sludge. ALE is a mixture of biopolymers extracted from the extracellular polymeric substance matrix of bacterial aggregate (Li et al., 2021). ALE recovered from waste sludge can be used to produce flame retardants, absorbing gels, ink thickeners, gluing agents for fertilizer pellets, corrosion inhibitors, coatings for improving the water resistance of paper, or fire-resistant boards (Li et al., 2022). Overall, many resource recovery technologies have been applied in full-scale projects and should be promoted to reduce the carbon emission of WWTPs.

7.5.3 Carbon replacement pathways

7.5.3.1 Chemical energy recovery

Wastewater contains a large amount of organic matter, which can be recovered and converted into biogas through an anaerobic digestion process, and then converted into heat or electricity through combined heat and power (CHP) (Hao et al., 2020). The electricity and heat recovered can be used within the plants to reduce electricity import and carbon emissions, such as fueling aerators, pumps, and anaerobic digestion. Besides, the surplus energy can also be exported to the grid and generate carbon credits. Currently, a few WWTPs globally have achieved energy neutrality relying on an anaerobic digestion process or co-digestion with food waste (Campos et al., 2016). It is worth noting that the energy recovery potential of anaerobic digestion is highly associated with influent organic content.

In China, anaerobic digestion is not widely used in WWTPs, partly due to the low organic content in influent as well as the high content of inorganic matter in sludge. As a result, biogas production efficiency is low and not attractive. Even if anaerobic digestion is promoted, the capital cost will bring a heavy economic burden to WWTPs (Chai et al., 2015). Incineration of sludge after dewatering has been proposed as an alternative to anaerobic digestion and a more suitable pathway for China. Through incineration, the organic energy contained in the sludge can be more completely released, and converted into electricity and heat via CHP (Hao et al., 2020). Moreover, the ash from incineration could be used for phosphorus recovery (Wang et al., 2023).

7.5.3.2 Thermal energy recovery

In terms of the sources of wastewater, part of them comes from kitchens and bathroom where the hot water is supplied. As such, the wastewater discharged into the drainage pipes has a stable temperature. Given the stable flow, wastewater can be a high-quality heat source which can be utilized by the water source heat pump (WSHP) technology. There are two points where the thermal energy in wastewater can be utilized: drainage pipes and the effluent of WWTPs. Energy extraction at the drainage pipes probably has an adverse impact on the following treatment processes and has corrosive problems. Thus, thermal energy recovery from the effluent of WWTPs is preferable. It should be noted is that the heat extracted from wastewater is a kind of low-grade energy (40-80 °C), which is difficult to use directly for power generation (Hao et al., 2019a). Thus, the heat could be utilized within the plant, for example, for anaerobic digestion warming or sludge drying. Then, the surplus heat could be exported to residents within an economic distance of 3-5 km (Hao et al., 2019a).

In terms of thermal energy recovery potential, Hao et al. (2019a) presented a detailed calculation and comparison. When energy is extracted by 4 °C, the theoretical thermal energy is 4.64 kWh/m³ (Hao et al., 2019a). The achievable thermal energy capacity supplied by WSHP can be calculated through the coefficient of performance (COP, the typical value of which is in the range of 3~5). For heating purposes, the net electrical equivalent is 1.77 kWh/m³ (38% of the theoretical thermal energy, COP = 3.5). When used for cooling, the net electrical equivalent is 1.18 kWh/m³ (25% of the theoretical thermal energy, COP = 4.8) (Hao et al., 2019a). The amount of thermal energy that can be recovered is 6~8 times larger than the amount of chemical energy.

7.6 STORMWATER SYSTEMS

7.6.1 Overview

Carbon emissions of stormwater management systems mainly come from the planning and construction stage. In the operation and maintenance stage, the pumping station in the stormwater sewers also make contributions to carbon emissions. As such, to reduce carbon emissions from stormwater management systems (Figure 7.7), planning and construction should be the first concern, and green infrastructures and green building materials should be given priority. In the operation stage, the carbon emission amount is associated with the amount of stormwater handled by the system, particularly for the pumping stations in stormwater sewers. Thus, source reduction via green infrastructures and/or separated drainage systems should be promoted and gravity flow drainage is preferable. Besides, the integration of the stormwater system and the urban ecosystem has great potential to achieve carbon emission reduction by rainwater recycling, being heat sinks, and CO₂ sinks of vegetation.

In terms of **source control**, reducing the amount of stormwater into the drainage system is a key solution. Priority is given to the leverage of green infrastructures to achieve source reduction. As such, natural storage, infiltration, purification and sustainable water circulation of rainwater can be achieved, which can also effectively achieve multiple goals such as waterlogging mitigation and water environment improvement. In the planning and construction of stormwater systems, giving priority to the use of green facilities and green building materials can greatly reduce carbon emissions.

In terms of **process optimization**, the key is to give priority to gravity flow drainage systems and implement separated drainage systems. The former approach can greatly reduce the electrical consumption of pumping stations, and the latter solution is not only conducive to subsequent rainwater reuse and reduction of CSO pollution, but also conducive to carbon emission reduction in the wastewater management system. Besides, stormwater treatment plants and deep tunnel projects are mainly large-

scale gray facilities, with huge carbon emissions from planning and construction, and operation and maintenance also inevitably lead to carbon emission from the pumping station. Thus, these grey facilities should be the final solution to cater for.

In terms of **technology upgrading**, the key is maximum utilization of rainwater instead of drainage. Rainwater is an important non-traditional water source, as it can not only save water resources, but also effectively reduce the carbon emission of the water supply system. At the same time, the stormwater management system and the urban ecosystem are integrated with each other, and can provide cold source and space resources. Thus, the cooling energy consumption of the built environment can be minimized. Besides, stormwater management systems can also provide large spaces to develop renewable energy sources such as photovoltaics.

In terms of **carbon removal**, green facilities are the key. The use of green facilities in the stormwater management system, such as green roofs, rain gardens, grass ditches, bioretention ponds, etc., can achieve the goal of rainwater runoff control and improve the living environment, as well as having a better carbon sink capacity.

The specific action strategies for stormwater management systems are shown in Table 7.5.

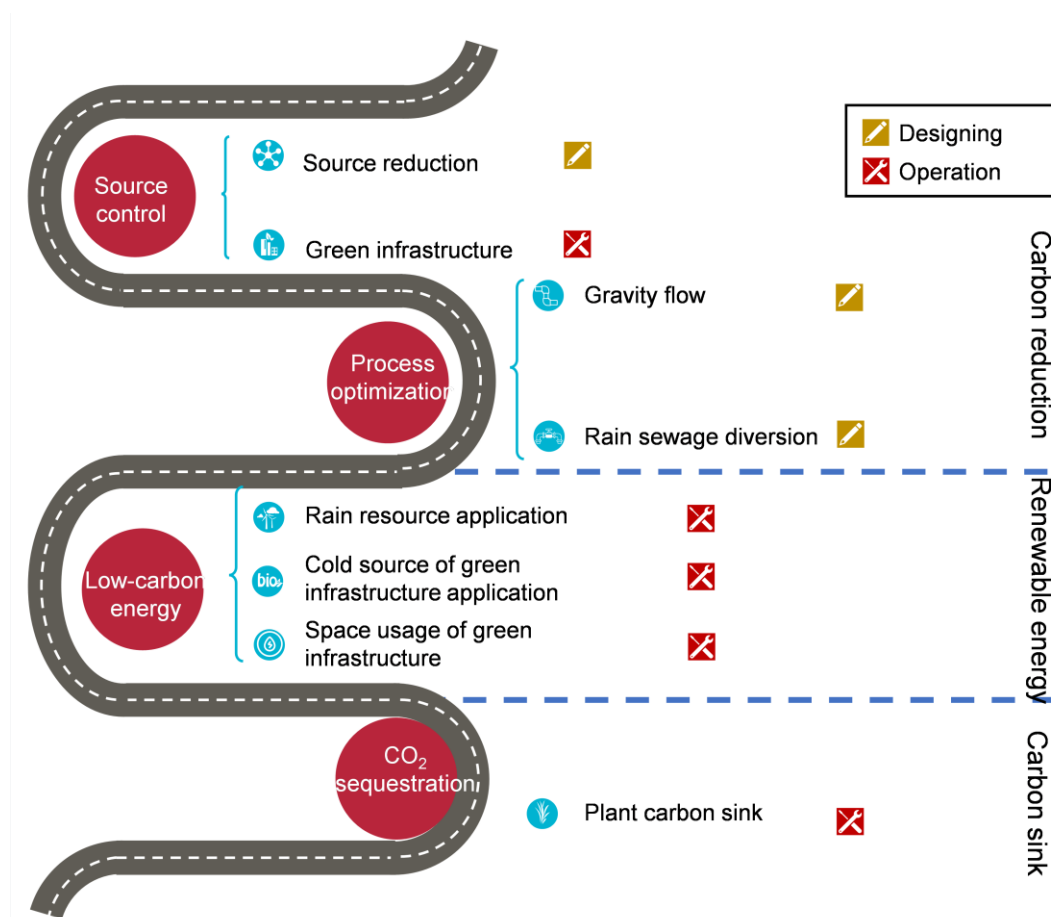


Figure 7.7 Carbon reduction roadmap for stormwater system.

Carbon reduction pathways in water sector

Table 7.5 Summary of action strategies for carbon emission reduction in stormwater management systems.

Type	Strategy	Authority	Technical recommendation	Implementation notes
Source control	Source reduction	Design and planning department	The urbanization and grey building density should be balanced with the natural and ecological land to keep the necessary proportion of impervious areas, and to minimize damage to the original water ecological environment of the city. At the same time, rivers, lakes and ditches should be appropriately excavated to increase the watershed area, and the capacity of rainwater retention, infiltration and purification.	The necessary urban hydrological cycle should be ensured to improve the infiltration, retention and storage, purification, utilization and discharge capacity of runoff.
	Gravity flow	Design and planning department	To adjust and optimize the operation mode of the pumping station according to the water volume, determine the best pumping scheme, use gravity flow, reduce the use of pumping system, and save energy consumption.	To reduce operation energy consumption and cost.
Process management	Rain and wastewater diversion	Design and planning department	To implement separated drainage systems ¹	Water pollution can be controlled, and water loading of WWTPs and stormwater management systems can be reduced.
	Utilization of rainwater resources		Rainwater collection, purification, storage, and reuse for irrigation, toilet flushing, car washing, alternative water sources, etc.	Rainwater harvesting can increase the amount of available water resources.
	Utilization of cold sources for green facilities	Operation department	Green facilities should be given priority to sequester CO ₂ and reduce building energy consumption and save water consumption.	Green facilities can improve the resilience of cities.
	Utilization of green facility space		Green space can be used to develop and build photovoltaic power stations.	Flexible to implement, effective energy saving, carbon emission reduction.

Continued

Guidelines for Carbon Accounting and Emission Reduction in the Urban Water Sector

Type	Strategy	Authority	Technical recommendation	Implementation notes
Plant sink	Plant sink technology		Adopt green facilities in the stormwater management system, such as green roofs, rain gardens, grass ditch, bioretention ponds, etc.	It has a good carbon sink function while realizing the target of stormwater runoff control and improving the living environment.

7.6.2 Carbon reduction pathways

7.6.2.1 Source reduction

The essence of source reduction is to build stormwater control facilities to leverage natural systems to reduce stormwater runoff through decentralized facilities. The reduction of the total amount of rainwater runoff can effectively reduce the carbon emissions of stormwater management systems. Typically, the total annual runoff control rate of 60% can be achieved by source reduction (Lin et al., 2018). With reference to operation energy consumption of the pumping station, power consumption per unit flow for single pump and the double pump running at full speed is 45.50-54.40 kWh/10³ m³ (Su et al., 2022). Thus, in the case of drainage by a pumping station, total rainwater runoff control can reduce CO₂ emissions by 26.57-31.76 kg/10³ m³ during operation and maintenance (Lin et al., 2018).

7.6.2.2 Green infrastructures

The use of green facilities in the planning and construction stage to replace reinforced concrete facilities can significantly reduce GHG emissions. For example, with the same rainwater control rate, the carbon emission intensity of reinforced concrete rainwater storage tanks is 513.32 kg CO₂-eq/m³ (calculated by water storage capacity), while the carbon emission intensity of most green facilities is below 200 kg CO₂-eq/m², or even below 100 kg CO₂-eq/m², with an emission reduction rate of over 80% (Su et al., 2022). Besides, the carbon emission of concrete lined ditch construction is about 50% of that of reinforced concrete pipes, and the carbon emission of grass ditch construction is almost zero (Lin et al., 2018).

Green infrastructure can reduce the use of non-renewable materials, reduce the amount of earthwork in the construction process, and have a longer service life than gray facilities under normal maintenance conditions. At the end of service life, green infrastructure has a small quantity of building waste to be disposed of, which greatly reduces the carbon emissions as well. In addition, because the low infiltration capacity of the gray facilities, the gray facilities often need to be built larger and consume more building materials to achieve the same rainwater control rate. Besides, the plants in the green facilities can act as carbon sinks.

Therefore, in the case of meeting the design needs, it is possible to consider the use of green infrastructure to replace the grey facilities, such as ecological ditches instead of grey open channels.

7.6.2.3 Gravity flow drainage system

The carbon emissions of the stormwater sewer mainly come from the pumping station. Referring to the operation energy consumption of the pumping station, the power consumption per unit flow used by single pump and a double pump in full speed operation is 26.57-31.75 kg CO₂-eq/10³ m³, and carbon emissions account for a relatively large proportion (Campos et al., 2016). In the design of stormwater pipelines, the minimum slope and minimum flow rate should be ensured by considering the local groundwater level, the bury depth of the pipeline, and the distance from the discharge point. Overall, gravity flow drainage systems should be given priority with no or less lifting pump stations.

7.6.2.4 Separated drainage system

The combined drainage system is a system that mixes domestic wastewater, industrial wastewater and rainwater in the same canal. The separated drainage system discharges domestic wastewater, industrial wastewater and rainwater in two or more independent pipes. In a scenario of a combined drainage system, rainwater and domestic wastewater are mixed into the same pipelines. When they enter the green facilities such as rainwater wetlands, the treatment loading of the facilities increases due to the high

concentration of pollutants and poor water quality, and the microorganisms and more organic matter in the mixed water also increase the carbon emissions of rainwater wetlands. This problem will be avoided if a separated drainage system is adopted to ensure less pollutants concentration and relatively clean water quality. In addition, rainwater collected by separated drainage systems can easily be reused on-site after simple sedimentation and disinfection. Moreover, the combined drainage system conveys more wastewater to WWTPs, which will lead to more electrical consumption and carbon emission in WWTPs. Besides, a separated drainage system can reduce CSO pollution and 70%-80% of water pollution (Campos et al., 2016).

7.6.3 Carbon replacement pathways

7.6.3.1 Utilization of rainwater resources

Rainwater has the characteristics of slight pollution, neutral pH, little salt content, and very low water hardness, so there is no need to soften it. Collected rainwater can be used for non-drinking purposes after simple treatment, and can be directly used for watering, flushing, car washing, etc. Since the rainwater does not enter the urban rainwater pipe network, it can also reduce loading of the urban flood control drainage and treatment system at the same time. In areas of severe water shortage, rainwater can be used to produce drinking water. The production of drinking water from rainwater is very economical, and the operating cost is estimated to be 1.5-2.5 CNY/m³ (Du et al., 2019).

7.6.3.2 Utilization of cold sources for green facilities

Green infrastructure can not only effectively respond to climate change through its own resilience, but can also use its own characteristics to enhance the resilience of cities in the face of climate change. Green facilities can greatly increase the green area by ecological retention facilities, sunken green spaces, and green roofs. The improvement of green land coverage can significantly improve urban climate environments. First, plants can absorb a large amount of heat through transpiration. Green vegetation itself has the function of carbon fixation. It uses photosynthesis to absorb CO₂, thereby reducing the concentration of CO₂ in the atmosphere, inhibiting the greenhouse effect, and releasing O₂, which results in cooling effect. Secondly, the specific heat capacity of water in green space is considerable, and can control the rapid rise of temperature. Finally, plants can retain dust in the atmosphere, reduce solar radiation heat absorption, and further play a role in reducing the heat island effect. Studies have shown that the use of green infrastructure will reduce the ambient temperature by 7-8 °C.

By integrating with the environment, green facilities can effectively reduce the urban heat island effect, thereby reducing building cooling energy consumption. At the same time, it has a positive impact on the lives and health of the residents. The mitigation of the heat island effect provides residents with a more comfortable living environment and better indoor air circulation, which reduces the incidence of diseases caused by the heat island effect. This, in turn, results in a 30% reduction in neurological morbidity (Su et al., 2022). The use time of the refrigeration system in buildings is greatly reduced, saving a lot of power consumption and indirectly reducing CO₂ emissions. Taking the green roof as an example, through the transpiration of plants and large areas of shading, thus avoiding direct sunlight, it can significantly reduce the temperature of the roof surface and the surrounding ambient air, thereby reducing the energy consumption of the building in which it is located. Studies show that the simple green roof greening system can save 6.31 kWh/m² air conditioning of energy consumption and reduce CO₂ emissions by 10.00 kg CO₂-eq/(m²·a) per year (Su et al., 2022).

7.6.3.3 Utilization of green facility space

Solar energy is a clean and renewable energy suitable for local and nearby development and utilization. The large space of green facilities can be leveraged to install photovoltaic panel to produce electricity and achieve carbon reduction. For example, green facilities such as green roofs can take advantage of their space to install distributed photovoltaic power generation facilities. Rainwater gardens at the bottom of the plane can make full use of its sunny side to install small area photovoltaic power generation facilities, while the ecological environment of green facilities can also keep the temperature of photovoltaic modules in a lower range to help improve photovoltaic power generation, thus providing more green energy for the rainwater system. After the use of photovoltaic facilities to generate electricity, the energy used by green facilities will shift from traditional energy to low-carbon energy. For example, photovoltaic power generation facilities can use solar power to replace traditional conventional energy used on lift pumps in storm sewer systems. In addition, photovoltaic power generation facilities can also be applied to the automated rainwater irrigation devices for landscape rainwater control facilities such as rain gardens or the power supply of public lighting systems such as ground lights.

7.6.4 Carbon removal pathways

Carbon sequestration refers to the long-term storage of atmospheric CO₂ as organic matter in plants and soils. The area covered by green vegetation is the main area to produce carbon sinks. These are different from other water systems, in that the green facilities in the stormwater management system can not only achieve the goal of controlling the source of rainwater runoff and improve the quality of the living environment, but also have different degrees of carbon sink potential in the green space formed by the combination of facilities. In addition, the green space of the rainwater system relates to other green areas in the city to form an urban green network combining point-line-surface, which provides green support for the carbon neutrality of the water system.

Ecosystems with high biodiversity can absorb and store more carbon than those with low biodiversity, and green spaces composed of green facilities have significant differences in their carbon sink capacity during their life cycle. Studies have shown (Table 7.6) that the carbon sequestration capacity of the bioretention facilities, the vegetation filter zone and the rainwater garden over a 30-year life cycle is about 44.2 ± 35.8 kg CO₂-eq/m², 2.3 kg CO₂-eq/m², and 75.5 ± 68.4 kg CO₂-eq/m², respectively (Zhou and Li, 2009). The carbon sequestration capacity of a green roof with a 40-year life cycle is about 58.4 ± 24.7 kg CO₂-eq/m² (Su et al., 2022). Among the many green facilities, vegetation buffer strips, bioretention facilities, rain gardens, and green roofs demonstrate significant carbon sequestration. Therefore, according to planned regional conditions, the proportion of green facilities with significant carbon sequestration such as green roofs and rain gardens can be appropriately increased. According to regional environmental conditions, plants with high carbon sequestration coefficients can be selected as the main vegetation in green spaces. For the specific carbon sink potential of different facilities, field measurements are needed to obtain more accurate data. In practical applications, various factors should be considered comprehensively, and construction should be planned and constructed reasonably according to local conditions.

Table 7.6 Carbon sequestration amount of typical green facilities for rainwater control

Green facility	Bioretention facility	Vegetation filter strip	Rain garden	Green roof
Life cycle (a)	30	30	30	40
Carbon fixation capacity (kg CO ₂ -eq/m ²)	44.2 ± 35.8	8.57 ± 2.3	75.5 ± 68.4	58.4 ± 24.7

In the actual planning and construction applications, factors such as geographical location, climatic conditions, plant diversity, and growth status should be considered carefully, and the spatial layout should be optimized according to local conditions, so as to improve the urban green space system and achieve considerable “carbon recovery” benefits.

Chapter 8

Data acquisition and management

8.1 OVERVIEW

The urban water sectors encompass various types of system modules and correspondingly involve multiple carbon emission activities. The required data types and quantities for carbon emission accounting differ for different subsystems and carbon emission activities. The data required for carbon emission accounting of urban water sectors mainly consists of two categories, i.e., activity data and emission factors. To enhance the accuracy and representativeness of the data and ensure the accuracy of its accounting results, this chapter summarizes and standardizes data types and acquisitions during the planning and construction, operation and maintenance, and asset replacement and demolition stages of the urban water sectors. For data that needs on-site collection or detection, the existing technical standards or specifications are recommended and preferred. In the absence of corresponding technical specifications, this chapter provides literature references to guide data collection. It is noteworthy that throughout the entire process of carbon emission accounting, the reporting entity should strengthen the management of carbon emission data collection and acquisition, including:

- Establish carbon emission accounting and reporting regulations, including responsible institutions and personnel, workflow and content, work cycles and timeframes, etc. Appoint dedicated personnel to be responsible for carbon emission accounting and reporting.
- Establish the inventory of carbon emission activities, and identify the key and more complicated activities on which special attention about data and emission factors should be placed.
- Assess existing monitoring conditions, continuously improve the monitoring capabilities, and formulate corresponding monitoring plans of both activity data and emission factors, etc.
- Establish data recording and reporting regulation and regularly manage and maintain measuring instruments, equipment or online monitoring devices.
- Establish an internal review system for carbon emission reporting. Conduct regular cross-verification of carbon emission data, identify potential risks of data errors, and propose corresponding solutions.

8.2 ACTIVITY DATA ACQUISITION

The carbon emission accounting of the water supply system mainly involves the acquisition of electrical and material consumption data, which are summarized in Table 8.1. For the wastewater management system, in addition to the electricity and material consumption, it also involves data associated with direct carbon emissions due to biochemical processes. Thus, the data requirements for carbon emission accounting in wastewater management systems are more complex, as summarized in Table 8.2. For water reclamation systems and stormwater management systems, the required activity data for carbon emission accounting are shown in Table 8.3 and Table 8.4, respectively.

Guidelines for Carbon Accounting and Emission Reduction in the Urban Water Sector

Table 8.1 Summary of the source and acquisition of data required for carbon emission accounting of water supply system.

Stage	Activity data	Associated accounting equation	Data requirements	Sources, reference specifications or methods
Planning and construction ¹	Types and amounts of fossil fuel consumption	(4-1)	Monitoring the entire construction period ²	Obtained from the construction company's ledger
	Types and amounts of machine shifts ³	(4-2)	Monitoring the entire construction period	Obtained from the construction company's ledger
	and electricity consumption	(4-3)	Monitoring the entire construction period	Obtained from the construction company's ledger
	Types and consumption of construction materials	(4-4)	Monitoring the entire construction period	Obtained from the construction company's ledger
	Amounts of materials transported and corresponding transportation type and distance	(4-5)	Monitoring the entire construction period	Obtained from the construction company's ledger
Operation and maintenance	Electricity consumption	(5-2)	Continuous monitoring for the inventory year ⁴	Operation ledger or electricity bill receipt
	Types and consumption of materials and/or chemicals	(5-3)		
	Amounts of materials transported and corresponding transportation type and distance	(5-4)	Continuous monitoring in the inventory year	Operation ledger or the procurement contract
		(5-5)		
	Water volume treated		Continuous monitoring in the inventory year	Monitoring according to <i>Technical specifications for water resources (Water volume) monitoring</i> (DB37/T 3858-2020)
	Turbidity, color of raw water		Daily monitoring in the	Monitoring according to <i>Technical specifications for water</i>

Continued

Stage	Activity data	Associated accounting equation	Data requirements	Sources, reference specifications or methods
Asset replacement and demolition	Suspended solid content of sludge		inventory year	<i>resources (Water volume) monitoring</i> (DB37/T 3858-2020)
	Types and amounts of fossil fuel consumption	(6-1)	Monitoring the entire demolition period	Obtained from the construction company's ledger
	Types and quantity of machine shifts	(6-2)	Monitoring the entire demolition period	Obtained from the construction company's ledger
	Power consumption	(6-3)	Monitoring the entire demolition period	Obtained from the construction company's ledger
	Amounts of building waste transported and corresponding transportation type and distance	(6-4)	Monitoring the entire demolition period	Obtained from the construction company's ledger
	Carbon offset generated from recovered materials	(6-5)	Monitoring the entire demolition period	Obtained from the construction company's ledger

1. When it is difficult to obtain the data of planning and construction activities, the design scale or total construction investment can be used to make a rough estimation in combination with the chart in Section 4.5.
2. The construction period refers to the whole duration of the planning and construction project from the start of the formal project to the time when it is fully put into service.
3. Non-essential data collected. When it is difficult to obtain the exact types and consumption of fossil fuels, this data can be used as an alternative method.
4. The electricity consumption in the inventory year refers to the level after operation getting stable.

Guidelines for Carbon Accounting and Emission Reduction in the Urban Water Sector

Table 8.2 Summary of the source and acquisition of data required for carbon emission accounting of wastewater management system.

Stage	Activity data	Associated accounting equation	Data requirements	Sources, reference specifications or methods
Planning and construction ⁵	Types and amounts of fossil fuel consumption	(4-1)	Monitoring the entire construction period ¹	Obtained from the construction company's ledger
	Types and quantity of machine shifts ²	(4-2)	Monitoring the entire construction period	Obtained from the construction company's ledger
	Electricity consumption	(4-3)	Monitoring the entire construction period	Obtained from the construction company's ledger
	Types and amounts of construction material consumption	(4.4-1)	Monitoring the entire construction period	Obtained from the construction company's ledger
	Amounts of materials transported and corresponding transportation type and distance	(4.5-1)	Monitoring the entire construction period	Obtained from the construction company's ledger
Operation and maintenance	Electricity consumption	(5-2)	Monitoring for the inventory year	Operation ledger/can be obtained by reading the electricity meter, taking the difference between the electricity meter readings at the end of the year (December 31st) and the beginning of the year (January 1st). It can also be based on the settlement voucher from the electricity supply department
	Types and amounts of material and chemical consumption	(5-3)	Monitoring for the inventory year	Operation ledger/can be obtained according to the statistical records of the

Continued

Data acquisition and management

Stage	Activity data	Associated accounting equation	Data requirements	Sources, reference specifications or methods
	Amounts of materials transported and corresponding transportation type and distance	(5-4)		water supply system operating units or the purchase contract from the supply enterprises
	Sewage sewer facilities			
	Influent volume and COD concentration of septic tank	(5-10) (5-12)	Monitoring all septic tanks within the organizational boundary	4 times per month for the inventory year ⁴ /24 hours equal proportion mixed sampling/ <i>Technical specifications for wastewater monitoring</i> (HJ 91.1-2019)
	The population served by the septic tank ³	(5-13)	-	Design data
	Diameter, slope and length of wastewater sewers and the conveyed wastewater temperature and volume	(5-14)	Monitoring all wastewater sewers within the organizational boundary	Design data/monitoring 4 times per month for the inventory year// <i>Technical specifications for wastewater monitoring</i> ” (HJ 91.1-2019), “ <i>Methods for the examination of municipal sewage</i> (CJ/T 51-2018)
	The average organic matter and TN concentration of wastewater in sewers	(5-15~16) (5-18)		<i>Technical specifications for wastewater monitoring</i> (HJ 91.1-2019)/ <i>Methods for the examination of municipal sewage</i> (CJ/T 51-2018)
	The water volume and COD/TN concentration in CSO event	(5-17) (5-20)	Monitoring each CSO event in the inventory year within the organizational boundary	
	The working head and efficiency of wastewater lifting pump	(5-3)	-	Design data
	Wastewater treatment plant			

Continued

Guidelines for Carbon Accounting and Emission Reduction in the Urban Water Sector

Stage	Activity data	Associated accounting equation	Data requirements	Sources, reference specifications or methods
	Volume of wastewater handled	(5-31 ~34) (5-35~45)		Monitoring
	BOD ₅ /TN concentration in influent and effluent	(5-21~30)	Average monitoring data in the inventory year	<i>Methods for the examination of municipal sewage</i> (CJ/T 51-2018)
	HRT, water temperature, MLVSS concentration of biological treatment units	(5-21~23)		
	Production of waste sludge	(5-35~36) (5-40~41)		
	VSS concentration of sludge before and after anaerobic digestion	(5-35) (5-40)		<i>Determination method for municipal sludge in wastewater treatment plant</i> (CJ/T 221-2005)
	Volume of biogas production	(5-36) (5-41)		
	Dry weight of sludge	(5-37~39) (5-40~43)	Average monitoring data in the inventory year	<i>Soil-Determination of organic carbon-Potassium dichromate oxidation spectrophotometric method</i> (HJ 615-2011)
	TOC of waste sludge	(5-37~39) (5-42)		<i>Soil-Determination of organic carbon-Combustion oxidation- titration method</i> (HJ 658-2013)
				<i>Soil-Determination of organic carbon-Combustion oxidation nondispersive</i>

Continued

Data acquisition and management

Stage	Activity data	Associated accounting equation	Data requirements	Sources, reference specifications or methods
				<i>infrared absorption method</i> (HJ 695-2014)
	DOC of waste sludge	(5-37~39) (5-42)		-
	The amounts of energy exported (if applicable)	(5-31~32)	Monitoring for the inventory year	
	The amounts of resources exported (if applicable)	(5-33)	Monitoring for the inventory year	Operation ledger
	Types and amounts of fossil fuel consumption	(6-1)	Monitoring the entire demolition period	Obtained from the construction company's ledger
	Types and quantity of machine shifts	(6-2)	Monitoring the entire demolition period	Obtained from the construction company's ledger
Asset replacement and demolition	Electricity consumption	(6-3)	Monitoring the entire demolition period	Obtained from the construction company's ledger
	Amounts of building waste transported and corresponding transportation type and distance	(6-4)	Monitoring the entire demolition period	Obtained from the construction company's ledger
	The amounts of building waste recycled	(6-5)	Monitoring the entire construction period	Obtained from the construction company's ledger

1. The construction period refers to the whole duration of the planning and construction project from the start of the formal project to the time when it is fully put into service.
2. Non-essential data collected. When it is difficult to obtain the exact types and consumption of fossil fuels, this data can be used as an alternative method.
3. Non-essential data. When the data of septic tank influent volume and water quality are not available, this data can be used as an alternative method.
4. The electricity consumption in the inventory year refers to the level after operation getting stable.
5. When it is difficult to obtain the data of planning and construction activities, the design scale or total construction investment can be used to make a rough estimation in combination with the chart in Section 4.5.

Table 8.3 Summary of the source and acquisition of data required for carbon emission accounting of water reclamation system.

Stage	Activity data	Associated accounting equation	Data requirements and recommendations	Sources, reference specifications or methods
Planning and construction ⁵	Types and amounts of fossil fuel consumption	(4-1)	Monitoring the entire construction period	Obtained from the construction company's ledger
	Types and quantity of machine shifts ²	(4-2)	Monitoring the entire construction period	Obtained from the construction company's ledger
	Electricity consumption	(4-3)	Monitoring the entire construction period	Obtained from the construction company's ledger
	Types and amounts of construction material consumption	(4-4)	Monitoring the entire construction period	Obtained from the construction company's ledger
	Amounts of materials transported and corresponding transportation type and distance	(4-5)	Monitoring the entire construction period	Obtained from the construction company's ledger
Operation and maintenance	Electricity consumption	(5-2)	Continuous monitoring for the inventory year	Operation ledger/can be obtained by reading the electricity meter, taking the difference between the electricity meter readings at the end of the year (December 31st) and the beginning of the year (January 1st). It can also be based on the settlement voucher from the electricity supply department Operation ledger/can be obtained according to the statistical records of the water supply system operating units or the purchase contract from the supply enterprises
		(5-3)		
	Types and amounts of material and chemical consumption	(5-4)	Continuous monitoring for the inventory year	
	Amounts of materials and chemicals transported and	(5-5)	Continuous monitoring for the inventory year	Operation ledger/can be obtained according to the statistical records of the water supply system

Continued

Data acquisition and management

Stage	Activity data	Associated accounting equation	Data requirements and recommendations	Sources, reference specifications or methods
Asset replacement and demolition	corresponding transportation type and distance			operating units or the purchase contract from the supply enterprises
	Water volume treated		Continuous monitoring for the inventory year	Monitoring/Technical specifications for water resources (Water volume) monitoring (DB37/T 3858-2020)
	Types and amounts of fossil fuel consumption	(6-1)	Monitoring the entire construction period	Obtained from the construction company's ledger
	Types and quantity of machine shifts	(6-2)	Monitoring the entire construction period	Obtained from the construction company's ledger
	Electricity consumption	(6-3)	Monitoring the entire construction period	Obtained from the construction company's ledger
	Amounts of building waste transported and corresponding transportation type and distance	(6-4)	Monitoring the entire construction period	Obtained from the construction company's ledger
	Carbon offset generated from recovered materials	(6-5)	Monitoring the entire construction period	Obtained from the construction company's ledger

1. The construction period refers to the whole duration of the planning and construction project from the start of the formal project to the time when it is fully put into service.
2. Non-essential data collected. When it is difficult to obtain the exact types and consumption of fossil fuels, this data can be used as an alternative method.
3. The electricity consumption in the inventory year refers to the level after operation getting stable.
4. When it is difficult to obtain the data of planning and construction activities, the design scale or total construction investment can be used to make a rough estimation in combination with the chart in Section 4.5.

Table 8.4 Summary of statistical and monitoring data required for carbon emission accounting of rainwater management sectors.

Stage	Activity data	Associated accounting equation	Data requirements and recommendations	Sources, reference specifications or methods
Planning and construction ⁵	Types and amounts of fossil fuel consumption	(4-1)	Monitoring the entire construction period	Obtained from the construction company's ledger
	Types and quantity of machine shifts ²	(4-2)	Monitoring the entire construction period	Obtained from the construction company's ledger
	Electricity consumption	(4-3)	Monitoring the entire construction period	Obtained from the construction company's ledger
	Types and amounts of construction material consumption	(4-4)	Monitoring the entire construction period	Obtained from the construction company's ledger
	Amounts of materials transported and corresponding transportation type and distance	(4-5)	Monitoring the entire construction period	Obtained from the construction company's ledger
	Types and amounts of fossil fuel consumption	(5-1) (5-46~48)	Monitoring the entire construction period	Obtained from the construction company's ledger
Operation and maintenance	Electricity consumption	(5-2 ~3) (5-49~51)	Continuous monitoring for 1 calendar year ³	Operation ledger can be obtained by reading the electricity meter, taking the difference between the electricity meter readings at the end of the year (December 31st) and the beginning of the year (January 1st). It can also be based on the settlement voucher from the electricity supply department

Continued

Data acquisition and management

Stage	Activity data	Associated accounting equation	Data requirements and recommendations	Sources, reference specifications or methods
Asset replacement and demolition	Vegetation covering area of green facilities	(5.6.2-7)	The whole operation and maintenance stage	Obtained by querying the vegetation carbon sequestration coefficient
	Rainwater controlling facilities			
	Influent COD/TN concentration entering to wetland	(5-53~55)	Continuous monitoring of COD, TN and other data in the rainwater entering the wetland for the inventory year	Obtained from the construction company's ledger
	Types and amounts of fossil fuel consumption	(6-1)	Monitoring the entire demolition period	Obtained from the construction company's ledger
	Types and quantity of machine shifts	(6-2)	Monitoring the entire demolition period	Obtained from the construction company's ledger
	Electricity consumption	(6-3)	Monitoring the entire demolition period	Obtained from the construction company's ledger
	Amounts of building waste transported and corresponding transportation type and distance	(6-4)	Monitoring the entire demolition period	Obtained from the construction company's ledger
	The amounts of recovered and exported materials	(6-5)	Monitoring the entire demolition period	Obtained from the construction company's ledger

1. The construction period refers to the whole duration of the planning and construction project from the start of the formal project to the time when it is fully put into service.
2. Non-essential data collected. When it is difficult to obtain the exact types and consumption of fossil fuels, this data can be used as an alternative method.
3. The electricity consumption in the inventory year refers to the level after operation getting stable.
4. When it is difficult to obtain the data of planning and construction activities, the design scale or total construction investment can be used to make a rough estimation in combination with the chart in Section 4.5.

8.3 EMISSION FACTORS ACQUISITION

8.3.1 General emission factors

The emission factors required for carbon accounting of urban water sectors can be roughly divided into two categories, namely, indirect emission factors and direct emission factors. The indirect emission factors are mainly used to account for emission activities, including electricity consumption and material consumption. They depend on the optimization and progress of social energy structures and industrial levels, respectively, and are not controlled by urban water sectors. They are mainly calculated and updated by other sectors and are suitable for each system of urban water sector. Therefore, they are also called general emission factors of which the values, sources and updating methods used in this *Guidelines* are summarized in Table 8.5.

Table 8.5 Summary of general emission factors and their sources.

Systems	Targeted activities	Sources	Links	Notes
General		<i>The People's Republic of China national greenhouse gas inventory</i> (NDRC of China, 2007)		
	Fossil fuels	<i>Provincial greenhouse gas inventory guidelines (trial)</i> <i>China energy statistical yearbook 2011</i> (National Bureau of Statistics, 2011)	Appendix B.1	
	Electricity	<i>Ministry of Ecology and Environment of the People's Republic of China, 2020</i> <i>Standard for building carbon emission calculation</i> (GB/T 51366-2019) (MHURD of China, 2019)	Appendix B.2	Official websites
	Building materials	<i>Standard for building carbon emission calculation</i> (GB/T 51366-2019) (MHURD of China, 2019)	Appendix B.3	
	Transportation	<i>Standard for building carbon emission calculation</i> (GB/T 51366-2019) (MHURD of China, 2019) Winnipeg. (2012) Emission factors in kg CO ₂ -equivalent per unit.	Appendix B.4	Data lacks updates and may not match the current situation well
	Chemicals	Alexis A. (2021) Carbon Footprint of Finnish Wastewater Treatment Plants. Finland: Aalto University. Johnston A H and Karanfil T. (2013). Calculating the	Appendix B.5	
Continued				

Systems	Targeted activities	Sources	Links	Notes
Water supply system	Carbon sequestration by vegetation	greenhouse gas emissions of water utilities. Journal-American Water Works Association, 105(7), E363-E371.	Appendix B.8	-
		Incopa. (2014) Life Cycle Analysis of Leading Coagulants: Executive Summary. Karlsruhe: IPCC. (2006) Emission factor database. CLCD. (2011) Chinese Life Cycle Database. Cui Pengfei. (2019) Research on staged emission reduction strategy of construction industry based on measurement of life cycle carbon emissions. Taiyuan University of Technology.		
	Construction of water treatment plants	Prepared by analyzing 12 water treatment plant projects with various treatment capacity and processes.	Appendix C.1	
	Construction of water distribution networks	Prepared by analyzing 18 water distribution network projects with various dimeters and materials.	Appendix C.2	
	Construction of wastewater sewer	Prepared by analyzing 32 wastewater sewer projects with various diameter and materials.	Appendix C.3	-
Waste-water management system	Construction of wastewater treatment plants	Prepared by analyzing 20 wastewater treatment plant projects with various treatment capacity, processes, and effluent standards.	Appendix C.4	
Storm-water system	Construction of stormwater system		Appendix C.5	
	Construction information			

8.3.2 CH₄ and N₂O emission factors in wastewater management systems

Some of the wastewater management facilities produce and emit CH₄ and N₂O due to the biochemical processes of organic and nitrogen compounds in wastewater. In general, the accounting methods of CH₄ and N₂O emissions are mostly based on emission factors which are determined in advance and also closely related to the process type, management and operation level, and regional climate. Currently, there are no uniform values for CH₄ and N₂O emission factors globally due to their fluctuations and uncertainty. IPCC recommended typical values for CH₄ and N₂O emission factors in their report by reviewing literature which is usually adopted in carbon accounting practices. Besides, a few countries have also formulated their own country-specific emission factors. For example, New Zealand published a carbon accounting guideline for wastewater treatment in 2021 in which the emission factors of CH₄ and N₂O were modified. In addition, Australia, Denmark, and the United Kingdom also have their own specific emission factors. In view of the fact that monitoring works and information about CH₄ and N₂O emission factors in wastewater management systems in China are limited, this *Guidelines* presents updated emission factors by referring to literature based on IPCC's original data, which are summarized in Table 8.6. In addition, this *Guidelines* summarizes the monitoring methods for determining emission factors in wastewater management systems. With sufficient local monitoring results, country-specific emission factors for China can be obtained.

Table 8.6 Summary of sources of CH₄ and N₂O emission factors in wastewater management system.

Modules	Facilities	Sources	Links	Notes
Sewage sewer facilities	Septic tank	CH ₄	International Water Association <i>Evaluation of Greenhouse Gas Emissions from Septic Systems</i> <i>2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories</i> (IPCC,2019) <i>2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories</i> (IPCC,2019)	On-site testing results are the most accurate, and local monitoring work samples are limited.
	Waste-water drainage	CH ₄	<i>Temperature and Organic Loading Dependency of Methane and Carbon Dioxide Emission Rates of a Full-Scale Anaerobic Waste Stabilization Pond</i> DOI:10.1016/0043-1354(94)00251-2 <i>IPCC Guidelines for National Greenhouse Gas Inventories</i> <i>Municipal Sewer Networks as Sources of Nitrous Oxide, Methane and Hydrogen Sulphide Emissions: A Review and Case Studies</i> DOI:10.1016/j.jece.2015.07.006.	
		CH ₄	<i>2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories</i> (IPCC,2019)	
		N ₂ O	<i>2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories</i> (IPCC,2019)	
	CSO	CH ₄	Table 5.4	
		N ₂ O	Table 5.5	

Continued

Modules	Facilities	Sources	Links	Notes
Wastewater treatment plants	Waste-water treatment plants	Two types of emission factors, international and domestic, are provided respectively, which are derived from the collation of literature data. This guideline is based on the <i>2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories</i> correcting 14 pieces of data, extending 18 pieces of data to update the emission factors	Appendix B.6	actual monitoring to determine the emission factors
		Two types of emission factors, international and domestic, are provided respectively, which are derived from the collation of literature data. This guideline is based on the <i>2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories</i> correcting 30 pieces of data, extending 46 pieces of data to update the emission factors		CH ₄ and N ₂ O emissions monitoring works in actual wastewater treatment plants are rare, and there is a significant variation in emissions between different wastewater treatment plants. When there are enough samples, the emission factors that are more suitable for China's wastewater treatment plants can be sorted out. The monitoring and determination methods are summarized in the article.
	Sludge disposal	<i>2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories</i>	Section 5.3.3	On-site testing results are the most accurate, and local monitoring work samples are limited.

It is worth noting that, with regard to the calculation of CH_4 emissions from wastewater treatment plants, this *Guidelines* modified the equations recommended by the IPCC to match current emission factors Equation (5.25). In other words, the activity data of Equation (5.25) is influent BOD_5 instead of BOD_5 removed. However, it should be noted that the accounting equations provided by IPCC is more scientific. If the corresponding emission factor is available, IPCC's equation is recommended and accurate. Therefore, the monitoring and determination of CH_4 emission factors in wastewater treatment plants should be consistent with the equations adopted. In terms of the calculation of N_2O emissions, there are two accounting methods available in which the activity data is influent TN and TN removal, respectively. These two methods have their own advantages and disadvantages, as discussed in detail by de Hass et al. (2022). In addition, studies shown that N_2O emission intensity is associated with the TN removal efficiency. Therefore, a dynamic emission factor by considering this correlation is suggested as a focus of future research. Overall, on-site monitoring should be recommended and should be promoted in order to ensure the accuracy of carbon accounting and/or formulate a more representative local emission factor. If the monitoring results are used to obtain emission factor, some suggestions are listed to guide the monitoring practices.

- The monitoring should be conducted in full-scale centralized WWTPs that primarily receive municipal wastewater or domestic wastewater.
- $\text{CH}_4/\text{N}_2\text{O}$ emission monitoring sites should cover all facilities that may generate $\text{CH}_4/\text{N}_2\text{O}$ in the WWTPs, including pumping stations, grit chambers, sedimentation tanks, biological tanks, sludge storage tanks, sludge thickening and dewatering rooms, mud cake stacking areas (if sludge is transported out of the plant timely, it may not be considered), etc., but does not include anaerobic digestion tanks.
- The monitoring activity should last for an inventory year.

Currently, the most commonly used methods for monitoring $\text{CH}_4/\text{N}_2\text{O}$ emissions in WWTPs include the flux chamber method and the tracer gas dispersion method. The principles, equipment, and implementation procedures of these two methods can be found in published articles. In addition, for WWTPs or treatment units with enclosed covers, it is recommended that offgas volume and $\text{CH}_4/\text{N}_2\text{O}$ concentration should be monitored to determine emission intensity. This *Guidelines* also summarizes the requirements and applicable scenario of various methods in Table 8.7, and the calculation of $\text{CH}_4/\text{N}_2\text{O}$ emission factors can be done using Equation (8.1) to (8.8).

Table 8.7 Summary of monitoring requirements of various methods for CH₄ and N₂O emission factors in WWTPs.

Methods	Scenario	Gas monitoring Data	Water quality data	Notes
Flux chamber method	Non-aerated facility	Flux chamber volume V, coverage area A, facility area At, target gas concentration in a certain time interval in the box C	Influent and effluent volume Q, BOD ₅ and TN concentration, outbound sludge volume W, VSS concentration	The most commonly used method, different facility units can determine the number of monitoring sites according to the dissolved oxygen concentration or ORP
		Gas flux M, coverage area A, target gas concentration in the box after monitoring C	Influent and effluent volume Q, BOD ₅ and TN concentration, outbound sludge volume W, VSS concentration	
	Aeration facility	Continuous measurement of tracer gas and target gas concentrations at the downwind outlet Ctr/Ctg, background concentration (tracer gas and target gas in the atmosphere) Ctr-b/Ctg-b, tracer gas release rate Qtr	Influent and effluent volume Q, BOD ₅ and TN concentration, outbound sludge volume W, VSS concentration	
Tracer gas dispersion method	\			Commonly uses acetylene as tracer gas
Closed wastewater treatment plants	\	Tail gas emission M, target gas concentration in tail gas C	Influent and effluent volume Q, BOD ₅ and TN concentration, outbound sludge volume W, VSS concentration	\

$$E_g = \sum (V/A) \rho (dC/dt) \cdot A_t \quad (8.1)$$

$$E_g = \sum (M \rho C / A) \cdot A_t \quad (8.2)$$

$$E_g = Q_{tr} \cdot \frac{\int (C_{tg} - C_{tg-b}) dx \cdot MW_{tg}}{\int (C_{tr} - C_{tr-b}) dx \cdot MW_{tr}} \quad (8.3)$$

$$E_g = M \cdot C \quad (8.4)$$

$$E_{CH_4-1} = E_g / (Q \cdot BOD_{in}) \quad (8.5)$$

$$E_{CH_4-2} = E_g / (Q \cdot BOD_{in} - Q \cdot BOD_{eff} - W \cdot VSS \cdot K_{BOD}) \quad (8.6)$$

$$E_{N_2O-1} = E_g / (Q \cdot TN_{in}) \quad (8.7)$$

$$E_{N_2O-2} = E_g / (Q \cdot TN_{in} - Q \cdot TN_{eff} - W \cdot VSS \cdot K_N) \quad (8.8)$$

Where:

- E_g — The target gas emission of the wastewater treatment plant for a certain period of time
- E_{CH_4-1}/E_{N_2O-1} — Emission factors based on influent BOD₅ or TN loading (* applicable to this guideline)
- E_{CH_4-2}/E_{N_2O-2} — Emission factors based on BOD₅ or TN removal load
- ρ — The density of the target gas at the corresponding temperature
- MW_{tg}/MW_{tr} — Relative molecular mass of target gas and tracer gas
- K_{BOD} — Equivalent of sludge
- K_N — N equivalent of sludge

Chapter 9

Result interpretation and reporting

9.1 RESULT INTERPRETATION

The purpose of carbon emission accounting is to serve as a tool for guiding the planning, implementation, and achievement of carbon emission reduction for urban water sectors. There are two time points in the life cycle of urban water sector to fulfill carbon emission reduction, i.e., the planning and construction stage and operation and maintenance stage. For the existing facilities of urban water sectors under service, in the operation and maintenance stage is the focus. As stated above, water companies, water sector associations, and water authorities have their specific roles in promoting the carbon accounting and carbon emission reduction of urban water sector. To make better use of the carbon accounting results, this *Guidelines* briefly provide some tips for interpreting and analyzing the results, facilitating the formulation of policy and regulations, guiding the role of different entities in carbon emission reduction (Figure 9.1). Overall, water company, water sector association, and water authority should take their exclusive role as summarized:

- **Water company:** Based on the accounting results, water company analyzes and identifies the contribution of each emission activity and the key activities, then formulate the plan and the cluster of strategies for carbon reduction. Regular quantification of the carbon emission is to provide feedback on the efficiency and adjustments of emission reduction measures.
- **Water sector association:** Water sector association has the advantages of links with more water companies and accounting results. Thus, they could guide water companies to implement carbon reduction in a more systematic pathway by establish a benchmark evaluation system and a sector average carbon emission level. Besides, they can also collect and compile a workbook of efficient reduction technologies and roadmaps to provide direction for water companies to carry out carbon emission reduction practices.
- **Water authority:** Water authorities mainly take the responsibility of policy and regulation establishment to help water companies implement carbon reduction more actively, scientifically, and economically. For example, water authorities should actively discuss with other government sectors to promote the acceptance of resources from wastewater which will undoubtedly promote the recovery enthusiasm of water companies. Besides, water authorities can also promote the water sectors in be included in carbon trading market, thereby enhancing the motivation of water companies to implement carbon reduction. Additionally, relevant tax reduction and subsidy policies should be formulated to guide and support the transformation of low-carbon development.

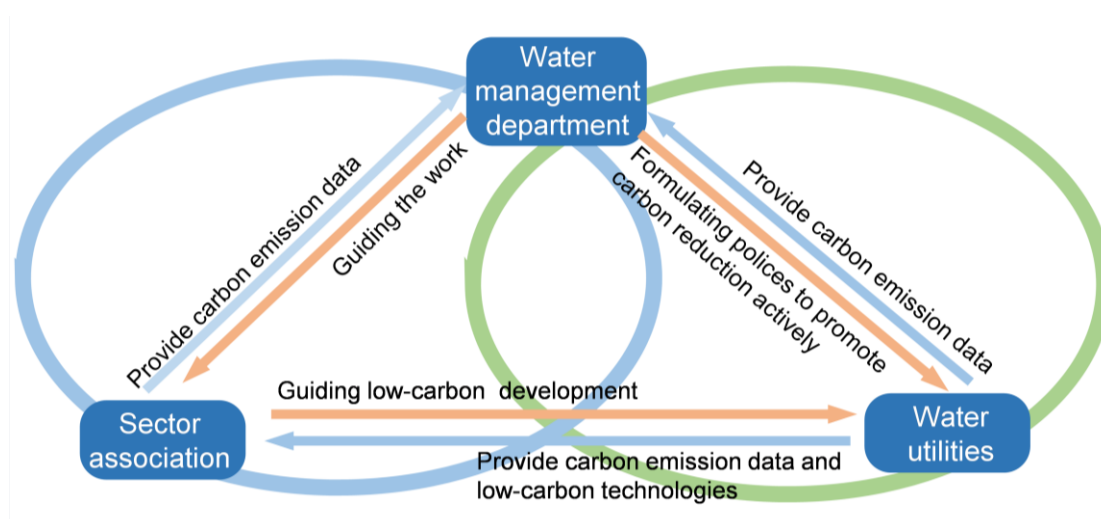


Figure 9.1 The roles and relationship among different entities within the urban water sectors in the analysis of carbon emission accounting results.

Based on the results of carbon emission accounting results of urban water sectors, this section puts forward several feasible analysis ideas and methods for reference, aiming at condensing and sublimating the key links hidden under the accounting results, and assisting entities to formulate effective carbon emission reduction strategies, helping promote the novel carbon emission reduction technologies and policies.

9.1.1 Key information analyzed

The analysis of carbon accounting results that can be conducted according to Table 9.1.

Table 9.1 Summary of the key information extracted from the accounting results.

Entity	Analysis information	Data	Tools
Water company	Establish the baseline		Pareto diagram/Characteristic diagram
	Identify key carbon emission activities	Carbon emission accounting results	Pareto diagram
	Predict the future trend		Pareto diagram
	Summarize the low-carbon operation parameters		Scatter diagram
	Carbon emission comparison	Carbon emission accounting results from various projects	Histogram

Continued

Entity	Analysis information	Data	Tools
Water sector association	Carbon emission intensity and benchmarking	Carbon emission accounting results of water companies	-
Water authority	Relationship between carbon emission and policy	Accounting results of all or representative operating enterprises and industry associations in the jurisdiction or accounting results of similar cities	-

9.1.2 Analyzing methodologies

9.1.2.1 Pareto diagram

The Pareto diagram (Figure 9.2) aims to identify the most significant carbon emission activities to address the issue most effectively. By ranking the carbon emissions (intensity) of all activities, the emission activities are classified into major activities, minor activities, and general activities. The major activities are the primary targets on which carbon reduction measures should be prioritized. It is easier and more effective to reduce the carbon emissions (intensity) focusing on major activities instead of the minor or general factors at the very beginning. By analyzing the results of the Pareto diagram, emission reduction strategies can be developed. The steps to use the Pareto diagram are as follows:

- Collate the required data of carbon accounting results.
- Arrange carbon emissions (intensity) data in descending order based on emission intensity and calculate the cumulative emission ratio.
- Plot the diagram with emission activities on the horizontal axis, carbon emissions (intensity) on the left vertical axis, and cumulative emissions (intensity) ratio on the right vertical axis.
- Divide the emission activities in the 0%~80% range based on the cumulative ratio into major activities. Divide the emission activities in the 80%~90% range into minor activities. Divide the emission activities in the 90%~100% range into general activities.

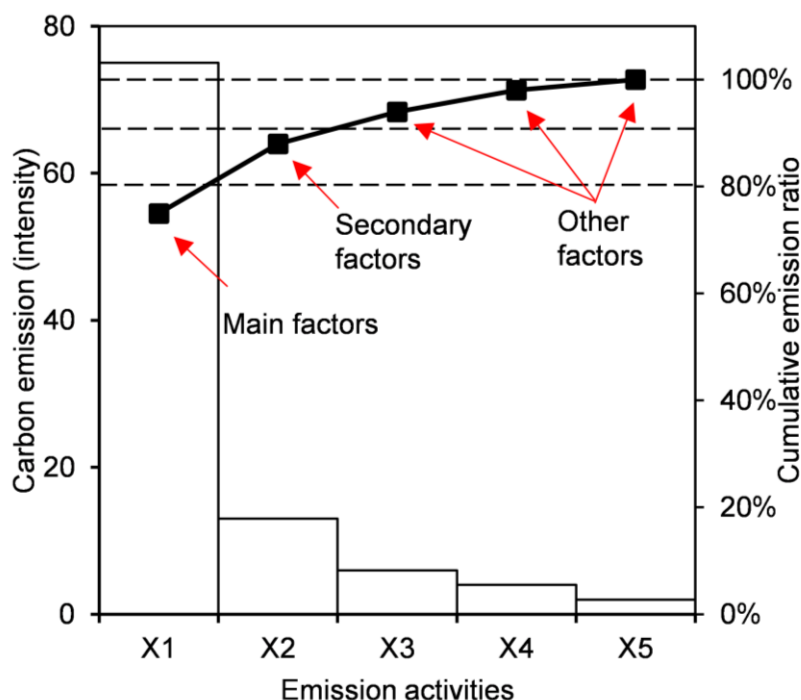


Figure 9.2 Pareto diagram indicating carbon emissions (intensity).

9.1.2.2 Characteristic diagram

The characteristic diagram, also known as the fishbone diagram (Figure 9.3), is a method to analyze and find out the key reasons for a certain problem. The characteristic factor diagram can be used to find out the major influencing factors affecting the target emission activities through analysis, so as to accurately adopt corresponding emission reduction technologies and measures. The steps to use the characteristic diagram are as follows:

- Taking the key factors influencing the targeted carbon emission as the main reason and “main bone” which is represented by a horizontal line.
- Dissect and list the second-level factors that may affect the main reason, and draw the corresponding “middle bone” on the “main bone”.
- Dissect and list the third-level factors that may affect the second-level factors, and draw the corresponding “small bones” on the “middle bone”. If there are further subfactors, it can be deduced in this way until the analysis reaches the level where corresponding emission reduction measures can be taken.
- **According** to the importance of influencing factors at each level, the factors that are considered to have a significant impact on carbon emissions (intensity) are marked.

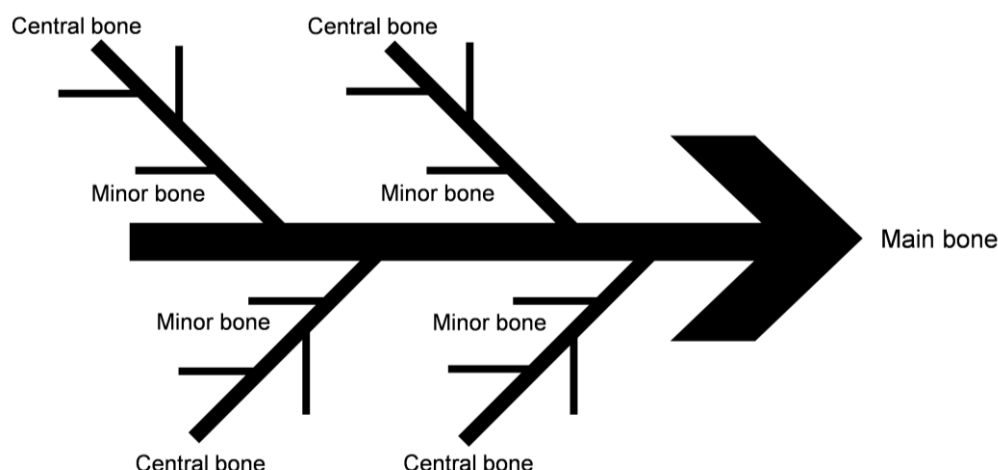


Figure 9.3 Primary and secondary fishbone diagram of carbon emissions.

9.1.2.3 Histogram

For units using the same treatment process and operating mode, the carbon emission levels cannot be the same, but always fluctuate within a certain range. The histogram can be used to show its statistical regularity, present the design and operation level distribution, and even estimate the carbon emission level of the larger group (Figure 9.4). The steps to use the histogram are as follows:

- Collate the carbon emissions (intensity) data of many facilities (number is n) using the same treatment process and/or operating mode.
- Identify the facility with the maximum (Max) and minimum (Min) emissions (intensity).
- Calculate the number of groups: $k = \sqrt{n}$, the class interval: $h = (\max - \min)/k$ (the result is taken as an integer).
- Divide the carbon emission intensity data into groups based on interval. The first group will include data ranging from $\min \sim \min + h$, the second group will include data ranging from $\min + k \sim \min + 2k$, and so on, until the last group includes data ranging from $\max - h \sim \max$. All the data should be assigned to their respective groups.
- Calculate the numbers of data in each group.
- Draw the histogram with each carbon emissions (intensity) data group as the horizontal axis and the number of data in each group as the vertical axis.
- **Analyze** the shape of the histogram. When the graph shows a peak shape to the left, it indicates that the overall operation level of the process is relatively high and the carbon emissions is low. When the graph appears to be peak-shaped to the right, it is reversed.
- **Perform** histogram analysis on multiple treatment processes and operating modes, comparing the carbon emission levels and technological maturity between different processes. Based on this analysis, provide recommendations for promoting and adopting lower-carbon operating processes.

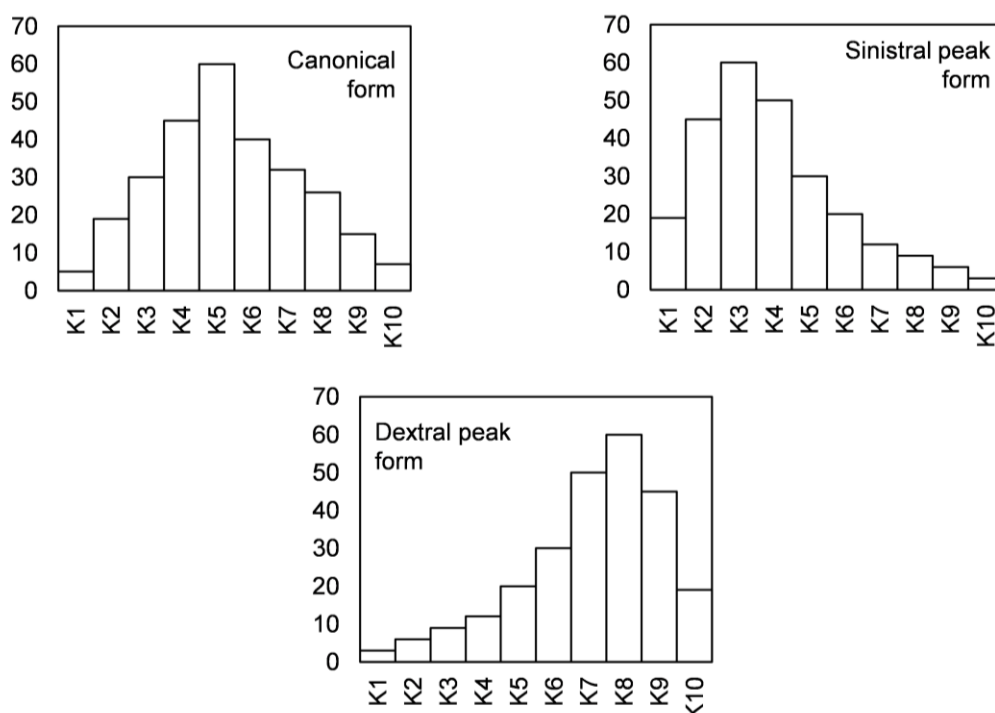


Figure 9.4 Histogram indicating carbon emissions (intensity).

9.1.2.4 Correlation scatter diagram

During the operation of the urban water sectors, there may be some unknown correlations between some influencing factors and carbon emissions (intensity). The correlative relationship can be displayed through the scatter diagram analysis process (Figure 9.5), so as to predict and find out the most suitable low-carbon operating parameters. The steps to use the scatter diagram are as follows:

- **Identify** influencing factors related to carbon emissions (intensity).
- Collate data on carbon emissions (intensity) and influencing factors.
- **Take** the data of influencing factors as the horizontal axis and the carbon emissions (intensity) as the vertical axis to draw each data point.
- **Analyze** the correlogram to find the correlations in it.

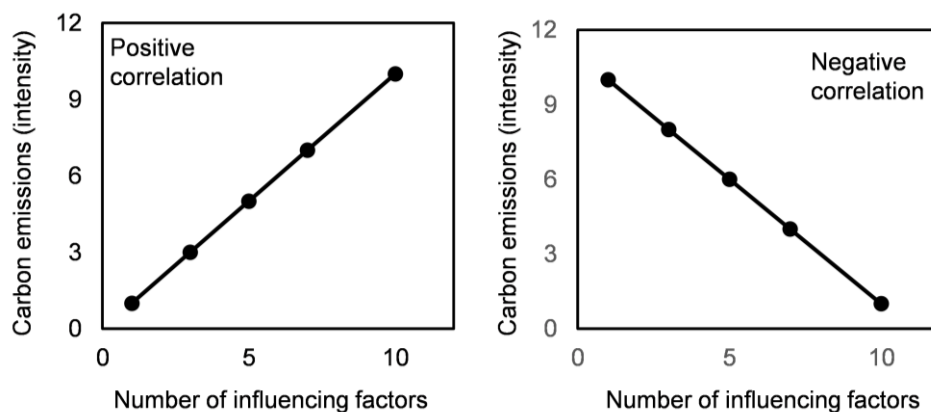


Figure 9.5 Scatter diagram indicating carbon emissions (intensity).

9.2 REPORTING PROTOCOL

The report format should refer to Appendix G, and the reporting content should include the following information.

9.2.1 Information of reporting entity

The reporting entity should be the relevant enterprise or unit responsible for operating the accounting object. The report should include the name and the nature of the entity, inventory year, industry sector, unified social credit code, legal representative, accounting manager, and contact information.

Detailed information should be provided on the process flow, division of accounting units and emission sources. Tables or graphs can be used if necessary.

9.2.2 Data sources

The reporting entity should report the various data source required for the accounting process according to the identification of the emission activities. For the data of indirect emission activities, efforts should be made to differentiate the carbon emission generated by various activities and functions, so as to enhance the comparability of the accounting results. The data sources or credentials, acquisition or monitoring methods, recording frequency and other data should be specified.

9.2.3 Emission factors

The reporting entity should report the emission factors or other necessary calculation parameters corresponding to each dataset, and explain their data sources, reference sources, relevant assumptions and reasons, etc.

9.2.4 Carbon emissions

The reporting entity should describe key information such as accounting boundaries and emission activities. Then, the emissions of each activities should be reported, along with the carbon emissions of electricity or heat consumed by each activities/function, the carbon emissions of materials consumed by each activities/function, the amount of output carbon compensation, etc. Finally, the total carbon emissions during the accounting period is presented.

Appendix A

Terms and symbols

Greenhouse gas

The naturally or anthropogenic gases in the atmosphere that absorb and re-emit infrared radiation. According to the categorization of IPCC ([IPCC, 2006](#)), there are 7 major greenhouse gases: carbon dioxide (CO₂), methane (CH₄), nitrous oxide (N₂O), hydrofluorocarbons (HFCs), perfluorocarbons (PFCs), sulfur hexafluoride (SF₆) and nitrogen trifluoride (NF₃). CO₂, CH₄ and N₂O are primarily considered in this guideline.

Carbon emission from urban water sector

The total greenhouse gas emissions, expressed in CO₂ equivalent, resulting from various activities related to urban water sector (including water supply system, wastewater management system, water reclamation system and stormwater system) in planning & construction, operation & maintenance, and asset replacement & demolition.

Reporting boundary

To avoid duplications or omissions, the identified and determined collection of all carbon emission activities associated with the accounting system which is the focus.

Emission activity

All units and processes directly or indirectly associated with the accounting system that result in greenhouse gas emissions, such as electric energy consumption, chemicals consumption and biological wastewater treatment.

Carbon emission intensity

The total carbon emissions generated by the accounting entity during the production of 1 unit of product.

Direct carbon emission

The GHG emission generated from the sources owned or controlled by the organization.

Indirect carbon emission

The GHG emissions that is a consequence of an organization's operations and activities, but that arises from GHG sources that are not owned or controlled by the organization.

Global warming potential (GWP)

Index, based on radiative properties of GHGs, relative to that of CO₂.

Carbon dioxide equivalent

A common unit for comparing the radiative forcing of a GHG to that CO₂, which can be written as CO₂-eq.

Activity data

Quantitative measure of activity that results in a GHG emission or GHG removal.

Carbon emission factor

The coefficient represents the emissions of greenhouse gases associated with activities.

Carbon reduction

The activities in which enterprises reduce the intensity of greenhouse gas emissions, and thus decrease the total carbon emissions, through technological adjustments, process optimization, and other means.

Renewable alternatives to carbon

The activity in which enterprises reduce their total carbon emissions during the production process by replacing fossil fuels with clean energy.

Carbon sink

The resources or energy produced in the production process of the enterprise can be exported externally, i.e., creating the corresponding carbon sink.

Emission point

The locations and processes within water sector facilities and structures that contribute to greenhouse gas emissions.

Asset reset & demolition

The process of recycling, renewing, or repurposing water sector facilities or structures to reintroduce them for use when they reach the end of their lifecycle.

Gray infrastructure

Conventional municipal infrastructure constructed by humans, such as dams, pipelines, channels, etc. Its basic function is to realize the discharge, transfer and treatment of water and pollutants in cities.

Green infrastructure

Municipal infrastructure constructed utilizing natural species and the environment. Its basic function is to regulate the water quality and quantity in the cities, and enhance the resilience of cities against extreme events.

Terms and symbols

Unit symbol

a	---	year	L	---	liter
°C	---	Celsius	m	---	meter
ca	---	person	mg	---	milligram
d	---	day	mm	---	millimeter
g	---	gram	mol	---	mole
GJ	---	giga joule	MWh	---	megawatt-hour
hm ²	---	hectare	m ²	---	square meter
J	---	joule	m ³	---	cubic meter
kg	---	kilogram	s	---	second
km	---	kilometer	t	---	tonne
kWh	---	kilowatt-hour			

Appendix B

Emission factors

B.1 FOSSIL FUEL EMISSION FACTORS

Table B.1 Summary of emission factors and carbon oxidation rates by different fossil fuel.

Fossil fuel	Emission factor		Carbon oxidation rate
	kg CO ₂ /GJ	kg CO ₂ /kg (solid and liquid fuel)/kg CO ₂ /m ³ (gaseous fuel)	
Anthracite	98.08	1.97	97.3%
General bituminous coal	93.11	1.86	97.0%
Lignite	98.56	2.06	96%
Washed coal	89.41	2.45	96%
Other coal washing	89.41	0.78	96%
Coal products	110.88	1.17	90%
Coke	100.25	2.85	93%
Coke oven gas	49.37	0.00089	99%
Other gas	44.29	0.00017	99%
Crude	72.23	3.02	98%
Fuel oil	75.82	3.17	98%
Gasoline	67.91	2.92	98%
Diesel fuel	72.59	3.10	98%
Jet kerosene	70.07	3.03	98%
General kerosene	70.43	3.03	98%
Liquefied petroleum gas	61.81	3.10	98%
Refinery Dry Gas	65.40	3.04	98%
Naphtha	71.87	3.26	98%
Petroleum coke	98.82	4.14	98%
Other oil	71.87	3.26	98%
Natural gas	55.54	0.0022	99%

1. The data of anthracite and general bituminous coal come from *Research on China's Greenhouse Gas Inventory* (NDRC of China, 2007) and *Guidelines for Provincial Greenhouse Gas Inventory (Trial)* (NDRC of China, 2011).

2. The data of washed clean coal, other washed coal, and other gas come from *China Energy Statistical Yearbook 2011* ([National Bureau of Statistics, 2011](#)), the data of petroleum coke and other oil products come from *Energy Utilization Status of Ten Thousand Enterprises*.

B.2 ELECTRICITY EMISSION FACTORS OF CHINA

Table B.2 Summary of electricity emission factors of China by regions.

Region	Emission factor (kg CO ₂ -eq/kWh)
North China	0.9419
Northeast	1.0826
East China	0.7921
Central China	0.8587
Northwest	0.8922
South China	0.8042

B.3 BUILDING MATERIAL EMISSION FACTORS

Table B.3 Summary of emission factors by building material.

Material	Carbon emission factor
Pebble	11.29 kg CO ₂ /m ³
Planting soil	0.024 kg CO ₂ /t
Clay soil	2.69 kg CO ₂ /t
Sandy soil	2.51 kg CO ₂ /t
Gravel	6.05 kg CO ₂ /t
Herbs	0.024 kg CO ₂ /t
Shale rock	5.08 kg CO ₂ /t
Gravel	2.18 kg CO ₂ /t
Boulder	2.18 kg CO ₂ /t
Rocks	2.18 kg CO ₂ /t
Graded crushed stone	52.8 kg CO ₂ /m ³
Middle sand	2.51 kg CO ₂ /t
Gabion	11.36 kg CO ₂ /t
Shale solid brick	292 kg CO ₂ /m ²
Sand-free concrete permeable brick	336 kg CO ₂ /m ³
Sand	2.51 kg CO ₂ /kg
Sintered standard brick	134 kg CO ₂ /m ³
Grass brick	336 kg CO ₂ /m ³
Permeable brick	2.21 kg CO ₂ /m ³
Wood pile	144.5 kg CO ₂ /m ³

Emission factor

Material	Carbon emission factor
1:2 cement mortar	531.52 kg CO ₂ /m ³
1:2.5 cement mortar	158.75 kg CO ₂ /m ³
1:3 cement mortar	393.65 kg CO ₂ /m ³
1:6 cement mortar	140.16 kg CO ₂ /m ³
Cement	735 kg CO ₂ /t
C15 Concrete	186.43 kg CO ₂ /m ³
C20 Concrete	239.19 kg CO ₂ /m ³
C25 Concrete	289.44 kg CO ₂ /m ³
C30 Concrete	295 kg CO ₂ /m ³
Fine stone concrete	355 kg CO ₂ /m ³
Rebar	2340 kg CO ₂ /t
Seamless steel pipe	3150 kg CO ₂ /t
FRP	2170 kg CO ₂ /t
Masonry Mortar DM5.0-HR	228.03 kg CO ₂ /m ³
Masonry Mortar DM10-HR	315.39 kg CO ₂ /m ³
UPVC perforated collection tube	7.93 kg/kg
Plant fiber blanket	144.5 kg CO ₂ /kg
Bentonite waterproof blanket	1.46 kg CO ₂ /kg
HDPE geomembrane	10.28 kg CO ₂ /kg
Composite waterproof membrane	4.46 kg CO ₂ /kg
SBS coil	2.37 kg CO ₂ /m ²
Permeable geotextile	10280 kg CO ₂ /t
HDPE composite film	2620 kg CO ₂ /t
SBS modified bitumen waterproof membrane	2.37 kg CO ₂ /m ²
Polyester non-woven fabric	10.28 kg CO ₂ /kg
PVC row (storage) water board	1765 kg CO ₂ /t
Polyethylene geomembrane	2620 kg CO ₂ /t
PP Polypropylene	1939 kg CO ₂ /t

B.4 EMISSION FACTORS FOR VARIOUS MODES OF TRANSPORTATION

Table B.4 Summary of emission factors for transportation by load.

Mode of transportation	Emission factor (kg CO ₂ -eq/(t·km))
Gasoline truck (2 t)	0.334
Gasoline truck (8 t)	0.115
Gasoline truck (10 t)	0.104
Gasoline truck (18 t)	0.104
Diesel truck (load 2 t)	0.286
Diesel truck (8 t load)	0.179

Diesel truck (load 10 t)	0.162
Diesel truck (load 18 t)	0.129
Diesel truck (load 30 t)	0.078
Diesel truck (load 46 t)	0.057
Electric locomotive	0.010
Diesel locomotive	0.011
Railway (on average in China)	0.010
Tanker transportation (load 2 000 t)	0.019
Dry bulk carrier (capacity 2 500 t)	0.015
Container ship (200 TEU load)	0.012

B.5 EMISSION FACTORS OF VARIOUS CHEMICALS

Table B.5 Summary of emission factor by chemicals.

Chemical	Emission factor (kg CO ₂ -eq/kg)	Chemical	Emission factor (kg CO ₂ -eq/kg)
	8.01		0.66
Ozone	14.7		0.462
	11.36 ± 3.35¹	Methanol	0.375
Acetic acid	0.852		0.985
Sodium acetate	0.623		0.61 ± 0.27
	0.14	Diammonium hydrogen phosphate	0.018
Sulfuric acid	0.182		0.029
	0.16 ± 0.02	Ferrous sulfate	0.038
Hydrochloric acid	1.20		0.03 ± 0.00
	0.6		0.18
	0.455		0.395
PAC	0.537	Ferric chloride	0.077
	0.53 ± 0.06		0.395
	0.92		0.26 ± 0.14
Sodium hypochlorite	1.065		1.08
	0.99 ± 0.07	Liquid chlorine	0.78
	0.41		0.93 ± 0.15
Oxygen	0.226		0.3
	0.32 ± 0.09		1.26
	0.59	Lime	1.74
	1.84		1.10 ± 0.60
Sodium carbonate	0.415	Polyacrylamide	1.48
	0.95 ± 0.63	Aluminum sulfate	0.16

1. The data in bold in the table is the average emission factor.

B.6 CH₄ EMISSION FACTORS FOR CENTRALIZED WASTEWATER TREATMENT

In the wastewater treatment plants that receiving domestic wastewater, a large quantity of CH₄ and N₂O is produced due to the biological processes. This part of carbon emission is denoted as direct emission of the water sector and cannot be excluded out of the boundary. By far, the accounting methods for these two gases includes emission factor method and model-based method (Hui *et al.*, 2022; Kati *et al.*, 2018). Thereinto, emission factor method is the one most widely applied in which the key is to obtain a specific emission factor. In the 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories (IPCC, 2019) released, IPCC provided integrated emission factors for CH₄ and N₂O accounting which are widely convincing and generally cited for CH₄ and N₂O quantification. Equation (B.1) and (B.3) are the equations provided in the IPCC's guidelines.

$$CH_4 \text{ Emission}_j = (TOW_j - S_j) \times EF_j - R_j \quad (B.1)$$

$$S_j = S_{mass} \cdot K_{rem} \cdot 1\,000 \quad (B.2)$$

$$EF_j = B_0 \cdot MCF_j \quad (B.3)$$

Where:

$CH_4 \text{ Emission}_j$	—	CH ₄ emissions from wastewater treatment in inventory year, kg CH ₄ /a
TOW_j	—	Influent organics of wastewater treatment plant in inventory year, kg BOD ₅ /a
S_j	—	Organic matter removed from wastewater treatment in the form of sludge in inventory year, kg BOD ₅ /a. See Equation (B.2)
EF_j	—	CH ₄ emission factor, kg CH ₄ /kg BOD ₅ . See Equation (B.3)
B_0	—	Maximum CH ₄ production capacity, kg CH ₄ /kg BOD ₅ . Default is 0.6
MCF_j	—	CH ₄ correction factor. See Table B.6
S_{mass}	—	Amount of raw sludge removed as dry mass, t/a
K_{rem}	—	Sludge factor, kg BOD ₅ /kg sludge. See Table B.7
R_j	—	Amount of CH ₄ recovered from treatment in inventory year, kg CH ₄ /a. Default is 0

Table B.6 Default MCF values by treatment process.

Treatment process	MCF
Centralized, aerobic treatment plant	0.03
Anaerobic reactor (e.g., UASB)	0.8
Anaerobic shallow lagoon and facultative lagoons	0.2
Anaerobic deep lagoon	0.8

Table B.7 Removal of organic component from wastewater as sludge (K_{rem}).

Treatment type	K_{rem} (kg BOD ₅ /kg dry sludge)	
	Defaults	Range
Mechanical treatment plants	0.5	0.4~0.6
Aerobic wastewater treatment plants without separate primary treatment	1.16	1.0~1.5

Aerobic treatment plants with primary treatment	0.8	0.65~0.95
Aerobic treatment plants with primary treatment and anaerobic sludge digestion	1.0	0.8~1.2

The emission factors for CH₄ accounting provided by IPCC is based on the data from 14 wastewater treatment plants by literature review. As summarized in Table (B.8), a collection of 7 scientific studies were selected according to the criteria of being full-scale wastewater treatment plant and providing key information. Overall, an MCF of 0.03 was calculated for aerobic wastewater treatment plants and the corresponding emission factor of CH₄ is 0.018 kg CH₄/kg BOD. According to Equation (B.1), the activity data is the organic matter which is obtained by excluding that in the form of sludge. Thus, the emission factor should be calculated to match the equation. However, by going through the 7 scientific articles, it shows that the emission factors of some studies were calculated without subtracting the BOD discharged in the form of sludge. In other words, the emission factor provided by IPCC does not match the equation provided. Besides, some errors of citing data were found in IPCC's guidelines. Therefore, this guideline optimizes the equation for accounting for CH₄ emission. Meanwhile, the guideline corrects and updates the emission factor by including more latest studies (Table (B.8)). The updated CH₄ emission factor of urban wastewater treatment plants is 0.0121 kg CH₄/kg BOD_{-influent}. Besides, a local emission factor of 0.0036 kg CH₄/kg BOD_{-influent} is calculated for China specifically. To distinguish the differences in CH₄ emissions by different treatment processes, emission factors for AAO, AO, SBR, oxidation ditch (OD), and aerobic treatment process are calculated respectively (Figure (B.1))

It is worth noting that the equation provided by IPC is more scientific because the production of CH₄ can only come from the mineralized portion of BOD. Therefore, the on-situ monitoring of CH₄ emissions in full-scale wastewater treatment plants should be promoted and encouraged to obtain more accurate CH₄ emission factor.

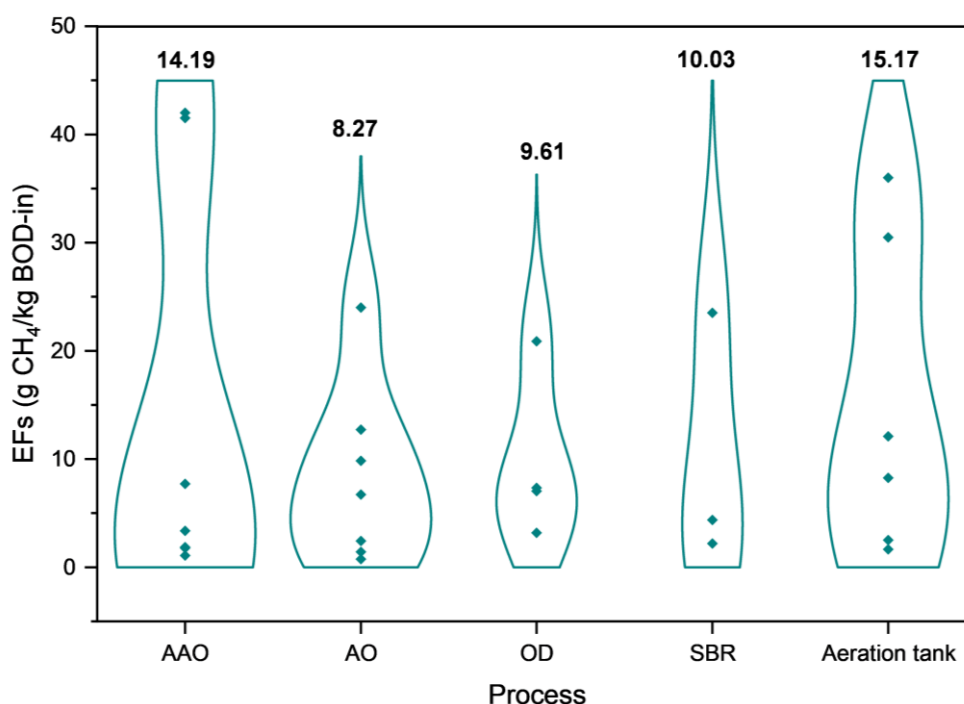


Figure B.1 CH₄ emission factors of different wastewater treatment processes (numbers in the figure are average values).

Emission factor

Table B.8 CH₄ emission factor based on measurement in full-scale wastewater treatment plants ¹.

Type of treatment process	Reference	MCF	EF (kg CH ₄ /kg BOD-influent)	EF (kg CH ₄ /kg COD-influent)
Studies referred by IPCC				
Activated sludge (Stickney)	Bellucci <i>et al.</i> , 2010	0.017	0.0121	\
Activated sludge (Northside)	Bellucci <i>et al.</i> , 2010	0.004	0.00251	\
Activated sludge (Egan)	Bellucci <i>et al.</i> , 2010	0.014	0.00826	\
Activated sludge	Czepiel <i>et al.</i> , 1993	0.013	0.00165	\
Biological nutrient removal (Kralingseveer)	Daelman <i>et al.</i> , 2013	0.03	0.024	0.01
Biological nutrient removal (Kortenoord)	Daelman <i>et al.</i> , 2013	0.02	0.01272	0.0053
Biological nutrient removal (Papendrecht)	Daelman <i>et al.</i> , 2013	0.04	0.02088	0.0087
Sequencing batch reactor (Holbæk)	Delre <i>et al.</i> , 2017	0.038	0.02352	0.0098
Activated sludge (Källby)	Delre <i>et al.</i> , 2017	0.048	0.03048	0.0127
Biological nutrient removal (Lundtofte)	Delre <i>et al.</i> , 2017	0.014	0.0096	0.004
Biological nutrient removal (Lynetten)	Delre <i>et al.</i> , 2017	0.015	0.00672	0.0028
Activated sludge	Kozak <i>et al.</i> , 2009	0.09	0.0124	\
Five sections of bardenpho	Kyung <i>et al.</i> , 2015	0.07	0.042	\
Anaerobic/Anoxic/Aerobic (A ² O) process	<u>Wang <i>et al.</i>, 2011a</u>	<u>0.003</u>	<u>0.00186</u>	<u>0.00078</u>
Studies further included in this guideline				
MBBR process ³	Delre <i>et al.</i> , 2017		0.04152	0.0173
Push stream	Ribera-Guardia <i>et al.</i> , 2019		0.000984	0.0041
A ² O	Hwang <i>et al.</i> , 2016		0.0077	\
Aerated activated sludge	Noyola <i>et al.</i> , 2018		0.036	\
Anaerobic/aeration tank	Noyola <i>et al.</i> , 2018		0.048	\
Oxidation ditch	Masuda <i>et al.</i> , 2018		0.0032	\
AO	Masuda <i>et al.</i> , 2018		0.0024	\
AO	Masuda <i>et al.</i> , 2018		0.00075	\
<u>Oxidation ditch</u>	<u>Ren <i>et al.</i>, 2013</u>		<u>0.00704</u>	\
<u>A²O</u>	<u>Ren <i>et al.</i>, 2013</u>		<u>0.0177</u>	\
<u>inverted A²O</u>	<u>Ren <i>et al.</i>, 2013</u>		<u>0.0014</u>	\
<u>Oxidation ditch</u>	<u>Yan <i>et al.</i>, 2019</u>		<u>0.0007344</u>	<u>0.00306</u>
<u>A²O</u>	<u>Yan <i>et al.</i>, 2019</u>		<u>0.000336</u>	<u>0.0014</u>
<u>inverted A²O</u>	<u>Yan <i>et al.</i>, 2019</u>		<u>0.001848</u>	<u>0.00077</u>

Type of treatment process	Reference	MCF	EF (kg CH ₄ /kg BOD-influent)	EF (kg CH ₄ /kg COD-influent)
<i>A²O</i>	<i>Liu et al., 2014</i>		<i>0.0011</i>	<i>0.00045</i>
<i>SBR</i>	<i>Liu et al., 2014</i>		<i>0.0022</i>	<i>0.000914</i>
<i>AO</i>	<i>Bao et al., 2016</i>		<i>0.001416</i>	<i>0.00059</i>
<i>SBR</i>	<i>Bao et al., 2016</i>		<i>0.004368</i>	<i>0.00182</i>
Mean (International) ²			0.0121 ± 0.0140	\
<u>Mean (domestic) ³</u>			0.0036 ± 0.0050	\

1. The italic value is calculated based the original data and the underlined value is based on measurements in China.

2. This emission factor is calculated based on all the values in the table.

3. This emission factor is calculated based on the measurements in China.

B.7 N₂O EMISSION FACTORS FOR CENTRALIZED WASTEWATER TREATMENT

N₂O is another potent GHG of which the global warming potential is three hundred times of that of CO₂. Due to the existence of nitrogen compounds in wastewater, N₂O is produced in the biological nitrogen removal processes and should be accounted in sewage system. At present, the quantification of N₂O emission also widely adopts the emission factor method (Equation (5.28)) in which the accuracy and representativeness of emission factor is vital important. Currently, the most accepted emission factor of N₂O for wastewater treatment plants is provided by IPCC. Similar to CH₄ emission factor, emission results were collected from 13 scientific literature as summarized in Table (B.9). By correlating the N₂O emission and influent nitrogen load (Figure (B.2)), the slope is the integrated emission factor, i.e., 1.6 % (0.016 kg N₂O-N/kg N_{-influent}).

Emission factor

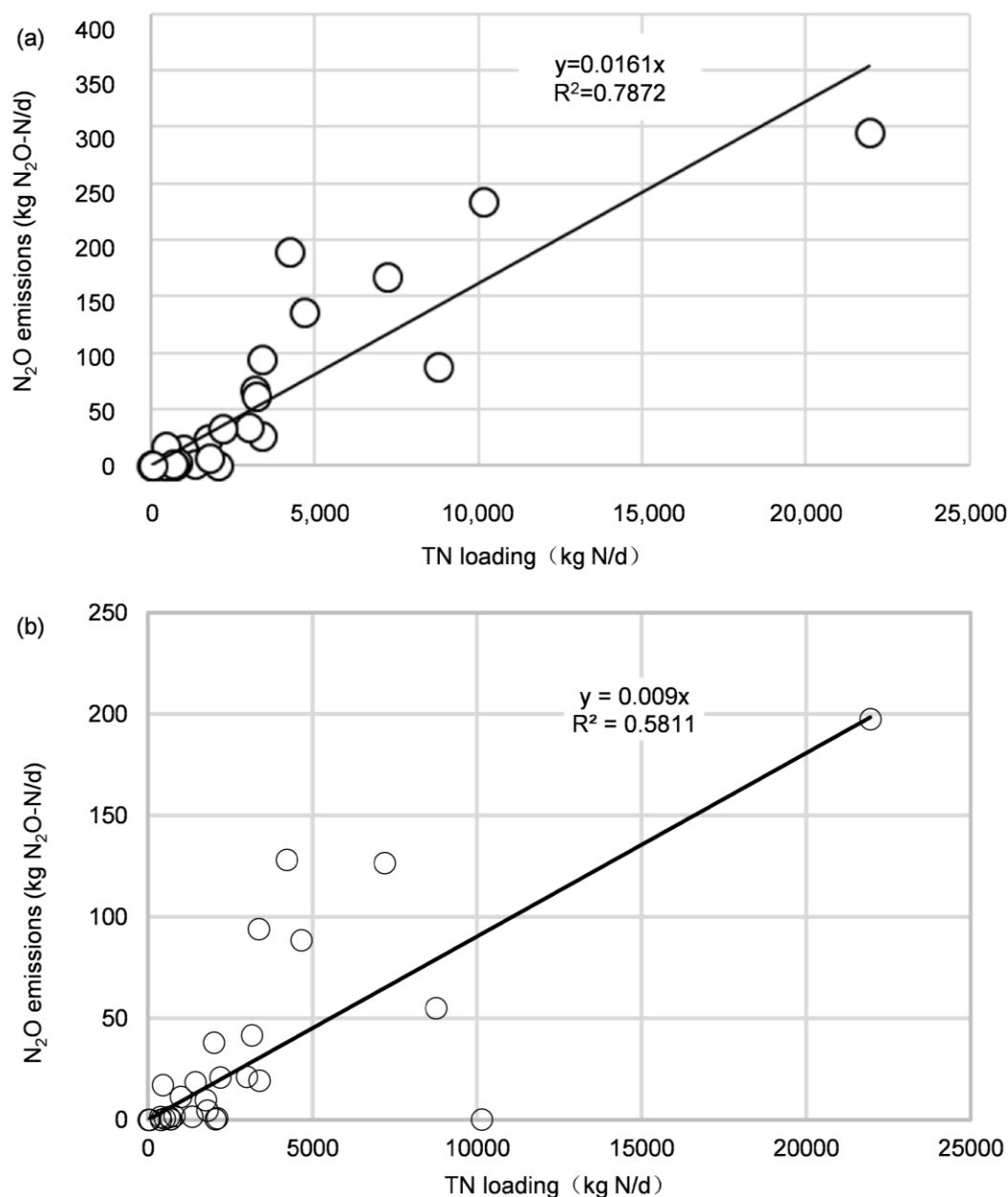


Figure B.2 Correlation between influent total nitrogen loading and N₂O emissions (a. data from IPCC; b. corrected data from IPCC).

Noteworthy is that some errors were found in the data processing for formulating N₂O emission factor by IPCC. Thus, for the first step, the data cited in Figure (B.2(a)) was double-checked and corrected to obtain an updated emission factor of N₂O which are summarized in Table (B.9). By correlating the revised data, a new emission factor of 0.9% (0.009 kg N₂O-N/kg N_{in}fluent) is obtained (Figure (B.2b)) which is significantly smaller than the value calculated by the IPCC. Then, this guideline did another literature review with a focus on studies after 2018, and 16 scientific papers were identified and summarized in Table (B.9). Then, an updated N₂O emission factor of 0.93 % (0.0093 kg N₂O-N/kg N_{in}fluent) was obtained with all the data in Table (B.9). Besides, this guideline also provides specific N₂O emission factors for different treatment processes (Figure (B.4)).

In addition, this guideline exclusively analyzes the N₂O emission factor of wastewater treatment plants in China, as shown in Figure (B.3b). An emission factor of 1.06 % (0.0106 kg N₂O-N/kg N_{-influent}) is obtained referring to 6 studies. However, the correlation coefficient was low, which indicated that the on-site monitoring of N₂O in full-scale wastewater treatment plants in China is limited and should be encouraged.

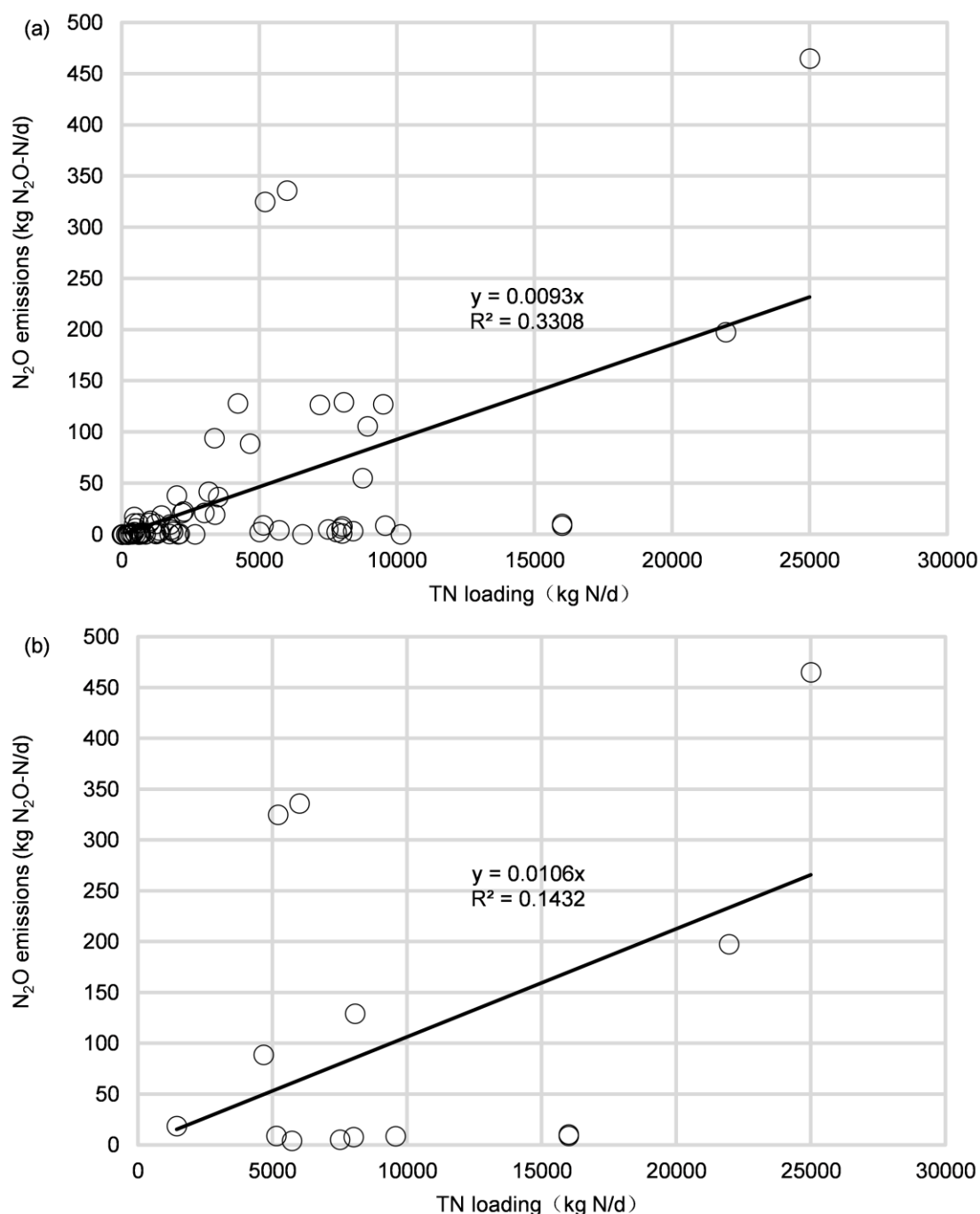


Figure B.3 Correlation between influent total nitrogen loading and N₂O emissions (a. based on global-wide case studies; b. based on China specific case studies).

Emission factor

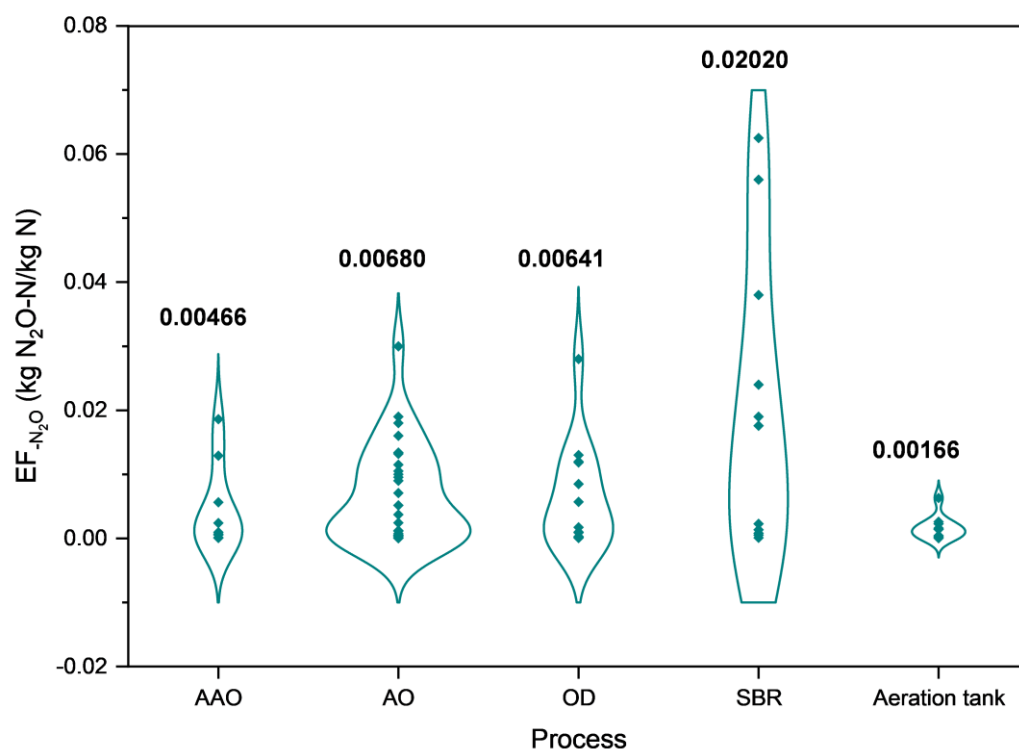


Figure B.4 N₂O emission factors of different wastewater treatment processes (numbers in the figure are average values).

Table B.9 N₂O emission factor based on measurement in full-scale wastewater treatment plants.

Type of treatment process	Category	Reference	Original data in IPCC guideline	Corrected data in the present guideline
			kg N ₂ O-N/kg N _{-influent}	
Studies referred to by IPCC				
AO	BNR	Daelman <i>et al.</i> , 2015	0.028	0.028
AO	BNR	Foley <i>et al.</i> , 2010	0.021	0.0132 ¹
AO	BNR	Foley <i>et al.</i> , 2010	0.045	0.03
A ² O	BNR	Foley <i>et al.</i> , 2010	0.013	0.00562
SBR	BNR	Foley <i>et al.</i> , 2010	0.023	0.0176
Oxidation ditch	BNR	Foley <i>et al.</i> , 2010	0.008	0.00571
EA	BNR	Foley <i>et al.</i> , 2010	0.015	0.0115
Intermittent aeration	BNR	Kimochi <i>et al.</i> , 1998	0.0005	0.0005
A ² O	BNR	Wang <i>et al.</i> , 2016	0.013	0.0129
Conventional activated sludge	BNR	Aboobakar <i>et al.</i> , 2013	0.00036	0.00036

Type of treatment process	Category	Reference	Original data in IPCC guideline	Corrected data in the present guideline
			kg N ₂ O-N/kg N-influent	
AO	BNR	Rodriguez-Caballero <i>et al.</i> , 2014	0.12	0.00116
Oxidation ditch	BNR	Masuda <i>et al.</i> , 2018	0.00016	0.00014 ²
AO	BNR	Masuda <i>et al.</i> , 2018	0.0013	0.0012
AO	BNR	Masuda <i>et al.</i> , 2018	0.0049	0.0037
Biological nitrogen and phosphorus removal	BNR	Ahn <i>et al.</i> , 2010	0.00019	0.000127 ³
Bardenpho	BNR	Ahn <i>et al.</i> , 2010	0.0036	0.00242
Biological nitrogen and phosphorus removal by multi-point water inlet	BNR	Ahn <i>et al.</i> , 2010	0.011	0.00706
MLE	BNR	Ahn <i>et al.</i> , 2010	0.0007	0.000445
MLE	BNR	Ahn <i>et al.</i> , 2010	0.0006	0.000382
Oxidation ditch	BNR	Ahn <i>et al.</i> , 2010	0.0003	0.000191
Biological nitrogen and phosphorus removal by multi-point water inlet	BNR	Ahn <i>et al.</i> , 2010	0.015	0.00954
Multi-point water inlet plug flow reactor	BNR	Ni <i>et al.</i> , 2015	0.019	0.019
SBR	BNR	Bao <i>et al.</i> , 2016	0.029	0.019
SBR	BNR	Rodriguez-Caballero <i>et al.</i> , 2015	0.038	0.038
Push stream	Non-BNR	Ahn <i>et al.</i> , 2010	0.004	0.00226
Push stream	Non-BNR	Ahn <i>et al.</i> , 2010	0.0062	0.00258
Multi-point water inlet aeration tank	Non-BNR	Ahn <i>et al.</i> , 2010	0.0018	0.0063
Push stream	Non-BNR	Masuda <i>et al.</i> , 2015	0.023	0.00002299

Continued

Emission factor

Type of treatment process	Category	Reference	Original data in IPCC guideline	Corrected data in the present guideline
			kg N ₂ O-N/kg N-influent	
AO	Non-BNR	Bao <i>et al.</i> , 2016	0.013	0.009
Intermittent aeration	Non-BNR	de Mello <i>et al.</i> , 2013	0.0016	0.0016
Studies further included in this <i>Guidelines</i>				
Intermittent aeration	BNR	Kimochi <i>et al.</i> , 1998		0.0008
Intermittent aeration	BNR	Kimochi <i>et al.</i> , 1998		0.0001
Aerated biological pool	BNR	Brotto <i>et al.</i> , 2010		0.0014
A ² O	BNR	Wang <i>et al.</i> , 2011b		0.00068
SBR	BNR	Delre <i>et al.</i> , 2017		0.00133
Biological nitrogen and phosphorus removal	BNR	Delre <i>et al.</i> , 2017		0.0024
Conventional activated sludge	BNR	Delre <i>et al.</i> , 2017		0.0004
MBBR	BNR	Delre <i>et al.</i> , 2017		0.00277
Bardenpho	BNR	Delre <i>et al.</i> , 2017		0.00074
Aerated biological pool	Non-BNR	Czepiel <i>et al.</i> , 1995		0.00022
Conventional activated sludge	BNR	Gruber <i>et al.</i> , 2020		0.018
Intermittent aeration	BNR	Gruber <i>et al.</i> , 2020		0.01
SBR	BNR	Gruber <i>et al.</i> , 2020		0.024
Conventional activated sludge	BNR	Tumendelger <i>et al.</i> , 2019		0.00008
MLE	BNR	Tumendelger <i>et al.</i> , 2019		0.00001
Intermittent aeration	BNR	Valkova <i>et al.</i> , 2021		0.001
SBR	BNR	Valkova <i>et al.</i> , 2021		0.00228
Intermittent aeration	BNR	Valkova <i>et al.</i> , 2021		0.00007
Multistage	BNR	Valkova <i>et al.</i> , 2021		0.01198

Type of treatment process	Category	Reference	Original data in IPCC guideline	Corrected data in the present guideline
			kg N ₂ O-N/kg N-influent	
activated sludge				
Conventional activated sludge	BNR	Valkova <i>et al.</i> , 2021		0.01184
Push stream	BNR	Valkova <i>et al.</i> , 2021		0.000295
CSTR	BNR	Valkova <i>et al.</i> , 2021		0.00005
Conventional activated sludge	BNR	Valkova <i>et al.</i> , 2021		0.00516
Conventional activated sludge	BNR	Valkova <i>et al.</i> , 2021		0.0085
Conventional activated sludge	BNR	Valkova <i>et al.</i> , 2021		0.013
AO	BNR	Blomberg <i>et al.</i> , 2018		0.0134
SBR	BNR	Sun <i>et al.</i> , 2013a		0.056
AO	BNR	Sun <i>et al.</i> , 2017		0.016
A ² O	BNR	Sun <i>et al.</i> , 2013b		0.0186
SBR	BNR	Sun <i>et al.</i> , 2013b		0.0625
A ² O	BNR	Vieira <i>et al.</i> , 2019		0.000055
Push stream	Non-BNR	Vieira <i>et al.</i> , 2019		0.00143
MLE	BNR	Vieira <i>et al.</i> , 2019		0.000025
MLE	BNR	Vieira <i>et al.</i> , 2019		0.000089
MLE	BNR	Vieira <i>et al.</i> , 2019		0.00051
Short-course digestion-anammox	BNR	Castro-Barros <i>et al.</i> , 2015		0.02
Oxidation ditch	BNR	Yan <i>et al.</i> , 2014		0.00173
Inverted A ² O	BNR	Yan <i>et al.</i> , 2014		0.00064
A ² O	BNR	Yan <i>et al.</i> , 2014		0.00055
Oxidation ditch	BNR	Ren <i>et al.</i> , 2013		0.000899
A ² O	BNR	Ren <i>et al.</i> , 2013		0.000969
Inverted A ² O	BNR	Ren <i>et al.</i> , 2013		0.000712
AO	BNR	Chen <i>et al.</i> , 2019		0.0105
AO	BNR	Bellandi <i>et al.</i> , 2018		0.00012
Conventional activated sludge	Non-BNR	Bellandi <i>et al.</i> , 2018		0.0004
UCT	BNR	Bellandi <i>et al.</i> , 2018		0.0005
Aerobic granular sludge	BNR	van Dijk <i>et al.</i> , 2021		0.0033

B.8 CARBON SEQUESTRATION CAPACITY OF DIFFERENT VEGETATION

Table B.10 Carbon sequestration capacity by different vegetation.

Vegetation types	Sequestration factor [kg CO ₂ -eq/(m ² a)]
Mixed planting of trees, shrubs, flowers and plants	27.5
Mixed planting of large and small trees	22.5
Deciduous tree	20.2
Deciduous small trees, conifers, thin-leaved trees	13.4
About 1.25 m high densely planted shrubs	10.3
High about 0.85 m densely planted shrubs	8.2
High about 0.55 m densely planted shrubs	5.2
High 1m wild grass	1.2
High 0.25 m low stem weeds	0.4
Artificial turf	0.0

Appendix C

Construction inventory and integrated emission factor formulation

C.1 CONSTRUCTION INVENTORY OF WATER SUPPLY SYSTEM

Table C.1 Summary of construction inventory and emission intensity of water treatment plants.

Treatment capacity	Materials	Consumption	Emission intensity of energy consumption (t CO ₂ -eq/10 000 m ³)	Emissions intensity of material consumption (t CO ₂ -eq/10 000 m ³)	Emissions intensity of transportation ¹ (t CO ₂ -eq/10 000 m ³)	Integrated emissions factor (t CO ₂ -eq/10 000 m ³)	Capital-based emissions factor (t CO ₂ -eq/10 000 yuan)
Conventional treatment							
Small-scale plant (< 50 000 m ³ /d)	Cement	607.680	t/10 000 m ³	-	446.645	0.693	447.338
	Wood	144.000	m ³ /10 000 m ³	-	20.808	0.164	20.972

Continued

Guidelines for Carbon Accounting and Emission Reduction in the Urban Water Sector

Treatment capacity	Materials	Consumption	Emission intensity of energy consumption (t CO ₂ -eq/10 000 m ³)	Emissions intensity of material consumption (t CO ₂ -eq/10 000 m ³)	Emissions intensity of transportation ¹ (t CO ₂ -eq/10 000 m ³)	Integrated emissions factor (t CO ₂ -eq/10 000 m ³)	Capital-based emissions factor (t CO ₂ -eq/10 000 yuan)
Medium-scale plant (50 000 m ³ /d~100 000 m ³ /d)	Rebar	194.418 t/10 000 m ³	-	454.938	0.222	455.160	
	Sand	313.600 m ³ /10 000 m ³	-	1.102	0.358	1.459	
	Gravel	2 352.000 m ³ /m ³	-	7.691	2.681	10.372	
	Steel pipe and fittings	38.020 t/10 000 m ³	-	96.191	0.043	96.234	
	Diesel fuel	25.212 t/10 000 m ³	76.392	-	0.029	76.421	
	Electricity	17.394 kWh/m ³	101.564	-	-	101.564	
	Total		177.956	1 027.375	4.189	1 209.520	
	Cement	540.956 t/10 000 m ³	-	397.603	0.617	398.219	
	Wood	99.000 m ³ /10 000 m ³	-	14.306	0.113	14.418	1.018
	Rebar	173.266 t/10 000 m ³	-	405.442	0.198	405.640	

Continued

Construction inventory and integrated emission factor formulation

Treatment capacity	Materials	Consumption	Emission intensity of energy consumption (t CO ₂ -eq/10 000 m ³)	Emissions intensity of material consumption (t CO ₂ -eq/10 000 m ³)	Emissions intensity of transportation ¹ (t CO ₂ -eq/10 000 m ³)	Integrated emissions factor (t CO ₂ -eq/10 000 m ³)	Capital-based emissions factor (t CO ₂ -eq/10 000 yuan)
Large-scale plant (≥ 100 000 m ³ /d)	Sand	137.900 m ³ /10 000 m ³	-	0.048	0.157	0.206	0.996
	Gravel	1 969.000 m ³ /m ³	-	6.439	2.245	8.683	
	Steel pipe and fittings	33.207 t/10 000 m ³	-	84.014	0.038	84.052	
	Diesel fuel	22.604 t/10 000 m ³	68.490	-	0.026	68.516	
	Electricity	15.594 kWh/m ³	91.053	-	-	91.053	
	Total		159.543	907.852	3.393	1 070.787	
	Cement	416.907 t/10 000 m ³	-	306.426	0.475	306.902	
	Wood	76.125 m ³ /10 000 m ³	-	8.7163	0.087	8.803	
	Rebar	132.920 t/10 000 m ³	-	311.034	0.152	311.185	
	Sand	648.925 m ³ /10 000 m ³	-	2.280	0.740	3.020	
	Gravel	1 611.250 m ³ /m ³	-	5.269	1.837	7.106	

Continued

Guidelines for Carbon Accounting and Emission Reduction in the Urban Water Sector

Treatment capacity	Materials	Consumption	Emission intensity of energy consumption (t CO ₂ -eq/10 000 m ³)	Emissions intensity of material consumption (t CO ₂ -eq/10 000 m ³)	Emissions intensity of transportation ¹ (t CO ₂ -eq/10 000 m ³)	Integrated emissions factor (t CO ₂ -eq/10 000 m ³)	Capital-based emissions factor (t CO ₂ -eq/10 000 yuan)
	Steel pipe and fittings	27.151	t/10 000 m ³	-	68.693	0.031	68.724
	Diesel fuel	18.040	t/10 000 m ³	54.661	-	0.021	54.682
	Electricity	12.445	kWh/m ³	72.666	-	-	72.666
	Total			127.327	702.4183	3.342	833.087
Pretreatment + conventional treatment							
Small-scale plant (< 50 000 m ³ /d)	Cement	743.748	t/10 000 m ³	-	546.655	0.848	547.503
	Wood	137.000	m ³ /10 000 m ³	-	19.797	0.156	19.953
	Rebar	233.642	t/10 000 m ³	-	546.722	0.266	546.989
	Sand	378.600	m ³ /10 000 m ³	-	1.330	0.432	1.762
	Gravel	2 627.000	m ³ /m ³	-	8.590	2.995	11.585
	Steel pipe and fittings	38.390	t/10 000 m ³	-	97.127	0.044	97.170

Continued

Construction inventory and integrated emission factor formulation

Treatment capacity	Materials	Consumption	Emission intensity of energy consumption (t CO ₂ -eq/10 000 m ³)	Emissions intensity of material consumption (t CO ₂ -eq/10 000 m ³)	Emissions intensity of transportation ¹ (t CO ₂ -eq/10 000 m ³)	Integrated emissions factor (t CO ₂ -eq/10 000 m ³)	Capital-based emissions factor (t CO ₂ -eq/10 000 yuan)
Medium-scale plant (50 000 m ³ /d~100 000 m ³ /d)	Diesel fuel	31.298	t/10 000 m ³	94.833	-	0.036	94.869
	Electricity	21.592	kWh/m ³	126.076	-	-	126.076
	Total			220.909	1 220.221	4.776	1 445.906
	Cement	672.116	t/10 000 m ³	-	494.005	0.766	494.771
	Wood	127	m ³ /10 000 m ³	-	18.352	0.145	18.496
	Rebar	211.176	t/10 000 m ³	-	494.152	0.241	494.393
	Sand	171	m ³ /10 000 m ³	-	0.060	0.195	0.255
	Gravel	2 443	m ³ /m ³	-	7.989	2.785	10.774
	Steel pipe and fittings	33.839	t/10 000 m ³	-	85.613	0.039	85.651
	Diesel fuel	27.821	t/10 000 m ³	84.298	0.032	84.329	
	Electricity	19.192	kWh/m ³	112.062	-	-	112.062

Continued

Guidelines for Carbon Accounting and Emission Reduction in the Urban Water Sector

Treatment capacity	Materials	Consumption	Emission intensity of energy consumption (t CO ₂ -eq/10 000 m ³)	Emissions intensity of material consumption (t CO ₂ -eq/10 000 m ³)	Emissions intensity of transportation ¹ (t CO ₂ -eq/10 000 m ³)	Integrated emissions factor (t CO ₂ -eq/10 000 m ³)	Capital-based emissions factor (t CO ₂ -eq/10 000 yuan)
Large-scale plant (≥ 100 000 m ³ /d)	Total		196.36	1 100.171	4.202	1 300.732	
	Cement	510.978 t/10 000 m ³	-	375.568	0.583	376.151	
	Wood	95.875 m ³ /10 000 m ³	-	10.978	0.109	11.087	
	Rebar	165.329 t/10 000 m ³	-	386.870	0.188	387.058	
	Sand	804.075 m ³ /10 000 m ³	-	2.826	0.917	3.742	1.006
	Gravel	1 795.250 m ³ /m ³	-	5.870	2.047	7.917	
	Steel pipe and fittings	28.029 t/10 000 m ³	-	70.914	0.032	70.946	
	Diesel fuel	21.844 t/10 000 m ³	66.186	-	0.025	66.211	
	Electricity	15.069 kWh/m ³	87.988	-	-	87.988	
	Total		154.174	853.026	3.900	1 011.100	
Pretreatment + conventional treatment + advanced treatment							

Continued

Construction inventory and integrated emission factor formulation

Treatment capacity	Materials	Consumption	Emission intensity of energy consumption (t CO ₂ -eq/10 000 m ³)	Emissions intensity of material consumption (t CO ₂ -eq/10 000 m ³)	Emissions intensity of transportation ¹ (t CO ₂ -eq/10 000 m ³)	Integrated emissions factor (t CO ₂ -eq/10 000 m ³)	Capital-based emissions factor (t CO ₂ -eq/10 000 yuan)
Small-scale plant (< 50 000 m ³ /d)	Cement	1 014.540	t/10 000 m ³	-	745.687	1.157	746.843
	Wood	174.000	m ³ /10 000 m ³	-	25.143	0.198	25.341
	Rebar	311.392	t/10 000 m ³	-	728.657	0.355	729.012
	Sand	463.000	m ³ /10 000 m ³	-	1.627	0.528	2.155
	Gravel	3 432.000	m ³ /m ³	-	11.223	3.912	15.135
	Steel pipe and fittings	51.268	t/10 000 m ³	-	129.708	0.058	129.766
	Diesel fuel	41.732	t/10 000 m ³	126.448	-	0.048	126.496
	Electricity	28.788	kWh/m ³	168.093	-	-	168.093
Medium-scale plant	Total			294.541	1 642.045	6.256	1 942.842
	Cement	934.611	t/10 000 m ³	-	686.939	1.065	688.005

Continued

Guidelines for Carbon Accounting and Emission Reduction in the Urban Water Sector

Treatment capacity	Materials	Consumption	Emission intensity of energy consumption (t CO ₂ -eq/10 000 m ³)	Emissions intensity of material consumption (t CO ₂ -eq/10 000 m ³)	Emissions intensity of transportation ¹ (t CO ₂ -eq/10 000 m ³)	Integrated emissions factor (t CO ₂ -eq/10 000 m ³)	Capital-based emissions factor (t CO ₂ -eq/10 000 yuan)
(50 000 m ³ /d~100 000 m ³ /d)	Wood	153.000 m ³ /10 000 m ³	-	22.109	0.174	22.283	
	Rebar	284.550 t/10 000 m ³	-	665.847	0.324	666.171	
	Sand	211.300 m ³ /10 000 m ³	-	0.074	0.241	0.315	
	Gravel	3 116.000 m ³ /m ³	-	10.189	3.552	13.742	
	Steel pipe and fittings	46.940 t/10 000 m ³	-	118.758	0.054	118.812	
	Diesel fuel	39.123 t/10 000 m ³	118.543	-	0.045	118.587	
	Electricity	26.989 kWh/m ³	157.589	-	-	157.589	
	Total		276.132	1 503.916	5.455	1 785.503	
Large-scale plant (≥ 100 000 m ³ /d)	Cement	754.789 t/10 000 m ³	-	554.770	0.860	555.630	0.993
	Wood	121.250 m ³ /10 000 m ³	-	13.883	0.138	14.021	

Continued

Construction inventory and integrated emission factor formulation

Treatment capacity	Materials	Consumption	Emission intensity of energy consumption (t CO ₂ -eq/10 000 m ³)	Emissions intensity of material consumption (t CO ₂ -eq/10 000 m ³)	Emissions intensity of transportation ¹ (t CO ₂ -eq/10 000 m ³)	Integrated emissions factor (t CO ₂ -eq/10 000 m ³)	Capital-based emissions factor (t CO ₂ -eq/10 000 yuan)
	Rebar	227.828	t/10 000 m ³	-	533.118	0.260	533.377
	Sand	1 050.213	m ³ /10 000 m ³	-	3.690	1.197	4.888
	Gravel	2 498.625	m ³ /m ³	-	8.171	2.848	11.019
	Steel pipe and fittings	38.247	t/10 000 m ³	-	96.765	0.044	96.808
	Diesel fuel	31.407	t/10 000 m ³	95.164	-	0.036	95.199
	Electricity	21.666	kWh/m ³	126.510	-	-	126.510
	Total			221.674	1 210.397	5.383	1 437.453
Water distribution pump station							
Small-scale (< 50 000 m ³ /d)	Cement	193.636	t/10 000 m ³	-	142.322	0.221	142.543
	Wood	36.000	m ³ /10 000 m ³	-	5.202	0.041	5.243
	Rebar	61.918	t/10 000 m ³	-	144.888	0.071	144.959

Continued

Guidelines for Carbon Accounting and Emission Reduction in the Urban Water Sector

Treatment capacity	Materials	Consumption	Emission intensity of energy consumption (t CO ₂ -eq/10 000 m ³)	Emissions intensity of material consumption (t CO ₂ -eq/10 000 m ³)	Emissions intensity of transportation ¹ (t CO ₂ -eq/10 000 m ³)	Integrated emissions factor (t CO ₂ -eq/10 000 m ³)	Capital-based emissions factor (t CO ₂ -eq/10 000 yuan)
Medium-scale (50 000 m ³ /d~100 000 m ³ /d)	Sand	96.800 m ³ /10 000 m ³	-	0.340	0.110	0.451	0.975
	Gravel	727.000 m ³ /m ³	-	2.377	0.829	3.206	
	Steel pipe and fittings	4.372 t/10 000 m ³	-	11.061	0.014	11.075	
	Diesel fuel	12.024 t/10 000 m ³	36.433	-	0.009	36.442	
	Electricity	7.824 kWh/m ³	45.684	-	-	45.684	
	Total		82.117	306.191	1.294	389.602	
	Cement	174.017 t/10 000 m ³	-	127.902	2.180	130.083	
	Wood	31.000 m ³ /10 000 m ³	-	4.480	0.198	4.678	
	Rebar	55.475 t/10 000 m ³	-	129.812	0.035	129.847	
	Sand	44.400 m ³ /10 000 m ³	-	0.016	0.063	0.079	
	Gravel	634.000 m ³ /m ³	-	2.073	0.051	2.124	

Continued

Construction inventory and integrated emission factor formulation

Treatment capacity	Materials	Consumption		Emission intensity of energy consumption (t CO ₂ -eq/10 000 m ³)	Emissions intensity of material consumption (t CO ₂ -eq/10 000 m ³)	Emissions intensity of transportation ¹ (t CO ₂ -eq/10 000 m ³)	Integrated emissions factor (t CO ₂ -eq/10 000 m ³)	Capital-based emissions factor (t CO ₂ -eq/10 000 yuan)
Large-scale (≥ 100 000 m ³ /d)	Steel pipe and fittings	10.305	t/10 000 m ³	-	26.072	0.723	26.794	
	Diesel fuel	6.955	t/10 000 m ³	21.074	-	0.012	21.085	
	Electricity	4.798	kWh/m ³	28.016	-	0.008	28.023	
	Total			49.090	290.355	-	339.443	
	Cement	131.333	-	t/10 000 m ³	96.530	0.150	96.679	
	Wood	24.750	-	m ³ /10 000 m ³	2.834	0.028	2.862	
	Rebar	41.927	-	t/10 000 m ³	98.109	0.048	98.157	
	Sand	212.063	-	m ³ /10 000 m ³	0.745	0.242	0.987	
	Gravel	520.625	-	m ³ /m ³	1.702	0.594	2.296	0.989
	Steel pipe and fittings	3.186	-	t/10 000 m ³	8.061	0.010	8.071	
	Diesel fuel	8.368	25.354	t/10 000 m ³	-	0.006	25.360	
	Electricity	5.542	32.362	kWh/m ³	-	-	32.362	
	Total		57.716		207.981	1.077	266.774	

1. Calculated with 20 kilometers transported by heavy-duty diesel trucks (with a load capacity of 46 tons).

Continued

C.2 CONSTRUCTION INVENTORY OF WATER DISTRIBUTION NETWORKS

Table C.2 Summary of construction inventory and emission intensity of water distribution networks.

Pipes	Diameter	Emission intensity of material consumption (t CO ₂ -eq/km)	Emission intensity of construction operation (per meter buried depth) (t CO ₂ - eq/(km·m))	Emission intensity of transportation ¹ (t CO ₂ -eq/km) (20 km)	Integrated emission factor (t CO ₂ -eq/km)	Capital-based emission factor (t CO ₂ -eq/10 000 yuan)
Ductile iron pipe	DN 300	103.28	2.47	0.05	105.07	1.33
	DN 400	153.44	2.70	0.08	155.41	1.54
	DN 500	211.58	3.27	0.11	213.98	1.60
	DN 600	278.16	3.39	0.14	280.67	2.12
	DN 700	353.40	3.62	0.18	356.11	2.42
	DN 800	437.76	4.12	0.22	440.87	2.91
	DN 900	528.96	4.35	0.26	532.27	3.13
	DN 1000	627.00	4.58	0.31	630.52	3.40
Steel pipe	DN 1200	852.04	5.02	0.43	855.97	4.05
	DN 300	179.53	2.47	0.08	181.34	1.88
	DN 400	226.65	2.68	0.11	228.63	1.99
	DN 500	284.80	3.25	0.13	287.21	1.90
	DN 600	342.87	3.40	0.16	345.41	2.40
	DN 700	400.71	3.59	0.19	403.41	2.68
	DN 800	458.78	4.07	0.22	461.85	2.98
	DN 900	516.62	4.29	0.24	519.86	2.91
	DN 1 000	574.70	4.50	0.27	578.11	2.98
	DN 1 200	690.61	4.93	0.32	694.38	3.16

1. Calculated with 20 kilometers transported by heavy-duty diesel trucks (with a load capacity of 46 tons).

C.3 CONSTRUCTION INVENTORY OF WASTEWATER SEWER

Table C.3 Summary of construction inventory and emission intensity of wastewater sewer.

Pipes	Diameter	Emission intensity of material consumption (t CO ₂ -eq/km)	Emission intensity of construction operation (per meter buried depth) (t CO ₂ -eq/(km·m))	Emissions intensity of transportation ¹ (t CO ₂ -eq/km)	Integrated emission factor ² (t CO ₂ -eq/km)	Capital-based emission factor (t CO ₂ -eq/10 000 yuan)
Reinforced concrete pipe	dn 600 ³	41.3	15.5	0.29	52.4	3.1
	dn 800	70.9	21.8	0.50	86.7	4.3
	dn 1 000	106.2	28.3	0.72	126.7	5.2
	dn 1 200	152.1	36.6	1.03	178.8	6.1
	dn 1 400	234.5	39.2	1.59	263.5	8.1
	dn 1 600	309.5	49.7	2.10	346.4	8.8
	dn 1 800	373.5	55.6	2.43	414.9	9.2
	dn 2 000	491.5	60.0	2.99	536.5	10.7
HDPE pipe	DN 600 ⁴	172.6	18.2	0.002	17.9	0.5
	DN 800	292.2	18.5	0.003	20.8	0.5
	DN 1 000	455.9	19.2	0.005	24.4	0.5
	DN 1 200	547.8	21.0	0.007	29.9	0.5
	DN 1 400	596.7	21.6	0.009	36.1	0.5
	DN 1 600	656.4	21.9	0.013	44.1	0.5
	DN 1 800	716.1	22.2	0.015	49.6	0.5
	DN 2 000	835.4	22.5	0.017	55.1	0.5

1. Calculated with 20 kilometers transported by heavy-duty diesel trucks (with a load capacity of 46 tons).

Continued

2. Construction emission intensity in the total emission intensity is calculated with a depth of cover of 0.7 m (according to “GB 50014-2021 (2021 Edition) Standard for design of outdoor wastewater engineering” the minimum burial depth of the pipe top under the roadway should be 0.7 m).
3. The diameter of concrete pipes and other pipes is represented by the inner diameter of the pipe, denoted as “dn”.
4. The pipe diameter is represented by the nominal diameter “DN”, which is the average of the outer diameter and inner diameter of the pipe.

C.4 CONSTRUCTION INVENTORY OF WASTEWATER TREATMENT PLANTS

Table C.4 Summary of construction inventory and emission intensity of wastewater treatment plants.

Treatment capacity	Materials	Consumption	Emission intensity of energy consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of material consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of transportation ¹ (t CO ₂ -eq/10 000 m ³)	Integrated emission factor (t CO ₂ -eq/10 000 m ³)	Capital-based emission factor (t CO ₂ -eq/10 000 yuan)
Wastewater treatment plants (excluding sludge digestion) with Class B discharge standards							
Small-scale plant (< 10 000 m ³ /d)	Cement	1 953.2	t/10 000 m ³	-	1 435.6	2.2	1 437.8
	Wood	247.0	m ³ /10 000 m ³	-	0.0	0.3	
	Rebar	421.2	t/10 000 m ³	-	985.6	0.5	986.1
	Sand	4 771.0	m ³ /10 000 m ³	-	16.8	5.4	22.2
	Gravel	7 787.0	m ³ /m ³	-	24.6	8.9	33.5

Continued

Construction inventory and integrated emission factor formulation

Treatment capacity	Materials	Consumption	Emission intensity of energy consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of material consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of transportation ¹ (t CO ₂ -eq/10 000 m ³)	Integrated emission factor (t CO ₂ -eq/10 000 m ³)	Capital-based emission factor (t CO ₂ -eq/10 000 yuan)
Medium-scale plant (10 000 m ³ /d~100 000 m ³ /d)	Steel pipe and fittings	89.6 t/10 000 m ³	-	226.7	0.1	226.8	1.3
	Diesel fuel	79.7 t/10 000 m ³	247.1	0.0	0.1	247.2	
	Electricity	550.0 MWh/10 000 m ³	321.1	0.0	-	321.1	
	Total		568.2	2 689.3	17.5	3 275.0	
	Cement	1176.5 t/10 000 m ³	-	864.7	1.3	866.1	
	Wood	150.0 m ³ /10 000 m ³	-	0.0	0.2	0.2	
	Rebar	317.1 t/10 000 m ³	-	742.0	0.4	742.4	
	Sand	3 375.0 m ³ /10 000 m ³	-	11.9	3.8	15.7	
	Gravel	4 690.0 m ³ /m ³	-	14.8	5.3	20.2	

Continued

Guidelines for Carbon Accounting and Emission Reduction in the Urban Water Sector

Treatment capacity	Materials	Consumption	Emission intensity of energy consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of material consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of transportation ¹ (t CO ₂ -eq/10 000 m ³)	Integrated emission factor (t CO ₂ -eq/10 000 m ³)	Capital-based emission factor (t CO ₂ -eq/10 000 yuan)
Large-scale plant (≥ 100 000 m ³ /d)	Steel pipe and fittings	54.0 t/10 000 m ³	-	136.6	0.1	136.7	1.1
	Diesel fuel	64.3 t/10 000 m ³	199.3	-	0.1	199.4	
	Electricity	444.0 MWh/10 000 m ³	259.3	-	-	259.3	
	Total		458.6	1 770.1	11.2	2 239.9	
	Cement	951.0 t/10 000 m ³	-	699.0	1.1	700.1	
	Wood	120.0 m ³ /10 000 m ³	-	0.0	0.1	0.2	
	Rebar	241.4 t/10 000 m ³	-	564.9	0.3	565.2	
	Sand	2 663.3 m ³ /10 000 m ³	-	9.4	3.0	12.4	
	Gravel	3 800 m ³ /m ³	-	12.0	4.3	16.3	

Continued

Construction inventory and integrated emission factor formulation

Treatment capacity	Materials	Consumption	Emission intensity of energy consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of material consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of transportation ¹ (t CO ₂ -eq/10 000 m ³)	Integrated emission factor (t CO ₂ -eq/10 000 m ³)	Capital-based emission factor (t CO ₂ -eq/10 000 yuan)
	Steel pipe and fittings	43.7 t/10 000 m ³	-	110.6	0.0	110.6	
	Diesel fuel	55.0 t/10 000 m ³	170.5	-	0.1	170.6	
	Electricity	380.0 MWh/10 000 m ³	0.0	-	-	0.0	
	Total		170.5	1 395.8	9.0	1 575.3	
Wastewater treatment plants (including sludge digestion) with Class B discharge standards							
	Cement	2 519.6 t/10 000 m ³	-	1 851.9	2.9	1 854.8	
Small factory (< 10 000 m ³ /d)	Wood	319.0 m ³ /10 000 m ³	-	0.1	0.4	0.4	
	Rebar	543.4 t/10 000 m ³	-	1 271.6	0.6	1 272.2	1.3
	Sand	6 155.0 m ³ /10 000 m ³	-	21.6	7.0	28.6	
	Gravel	10 045.0 m ³ /m ³	-	31.8	11.5	43.2	

Continued

Guidelines for Carbon Accounting and Emission Reduction in the Urban Water Sector

Treatment capacity	Materials	Consumption	Emission intensity of energy consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of material consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of transportation ¹ (t CO ₂ -eq/10 000 m ³)	Integrated emission factor (t CO ₂ -eq/10 000 m ³)	Capital-based emission factor (t CO ₂ -eq/10 000 yuan)
Medium-scale plant (10 000 m ³ /d~100 000 m ³ /d)	Steel pipe and fittings	115.6 t/10 000 m ³	-	292.5	0.1	292.6	1.2
	Diesel fuel	103.0 t/10 000 m ³	319.3	-	0.1	319.4	
	Electricity	711.0 MWh/10 000 m ³	415.2	-	-	415.2	
	Total		734.5	3 469.4	22.6	4 226.4	
	Cement	1 446.1 t/10 000 m ³	-	1 062.9	1.6	1 064.5	
	Wood	184.6 m ³ /10 000 m ³	-	0.0	0.2	0.2	
	Rebar	311.9 t/10 000 m ³	-	729.8	0.4	730.2	
	Sand	3 534.0 m ³ /10 000 m ³	-	12.4	4.0	16.4	
	Gravel	5 765.0 m ³ /m ³	-	18.2	6.6	24.8	

Continued

Construction inventory and integrated emission factor formulation

Treatment capacity	Materials	Consumption	Emission intensity of energy consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of material consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of transportation ¹ (t CO ₂ -eq/10 000 m ³)	Integrated emission factor (t CO ₂ -eq/10 000 m ³)	Capital-based emission factor (t CO ₂ -eq/10 000 yuan)
Large-scale plant (≥ 100 000 m ³ /d)	Steel pipe and fittings	66.4 t/10 000 m ³	-	168.0	0.1	168.1	1.2
	Diesel fuel	75.7 t/10 000 m ³	234.7	-	0.1	234.8	
	Electricity	522.0 MWh/10 000 m ³	304.8	-	-	304.8	
	Total		539.5	1 991.4	13.0	25 43.8	
	Cement	1 131.4 t/10 000 m ³	-	831.6	1.3	832.9	
	Wood	140.0 m ³ /10 000 m ³	-	0.0	0.2	0.2	
	Rebar	243.2 t/10 000 m ³	-	569.1	0.3	569.4	
	Sand	2 766.5 m ³ /10 000 m ³	-	9.7	3.2	12.9	
	Gravel	4 515.5 m ³ /m ³	-	14.3	5.1	19.4	

Continued

Guidelines for Carbon Accounting and Emission Reduction in the Urban Water Sector

Treatment capacity	Materials	Consumption	Emission intensity of energy consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of material consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of transportation ¹ (t CO ₂ -eq/10 000 m ³)	Integrated emission factor (t CO ₂ -eq/10 000 m ³)	Capital-based emission factor (t CO ₂ -eq/10 000 yuan)
	Steel pipe and fittings	51.9 t/10 000 m ³	-	131.3	0.1	131.4	
	Diesel fuel	64.1 t/10 000 m ³	198.7	-	0.1	198.8	
	Electricity	442.0 MWh/10 000 m ³	258.1	-	-	258.1	
	Total		456.8	1 556.0	10.2	2 022.9	
	Wastewater treatment plants (excluding sludge digestion) with Class A discharge standards						
Small-scale plant (< 10 000 m ³ /d)	Cement	2 832.1 t/10 000 m ³	-	2 081.6	3.2	2 084.8	1.3
	Wood	358.0 m ³ /10 000 m ³	-	0.1	0.4	0.5	
	Rebar	610.8 t/10 000 m ³	-	1 429.3	0.7	1 430.0	
	Sand	6 918.0 m ³ /10 000 m ³	-	24.3	7.9	32.2	
	Gravel	1 1291.0 m ³ /m ³	-	35.7	12.9	48.6	

Continued

Construction inventory and integrated emission factor formulation

Treatment capacity	Materials	Consumption	Emission intensity of energy consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of material consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of transportation ¹ (t CO ₂ -eq/10 000 m ³)	Integrated emission factor (t CO ₂ -eq/10 000 m ³)	Capital-based emission factor (t CO ₂ -eq/10 000 yuan)
Medium-scale plant (10 000 m ³ /d~100 000 m ³ /d)	Steel pipe and fittings	130.0 t/10 000 m ³	-	328.9	0.1	329.0	1.2
	Diesel fuel	115.6 t/10 000 m ³	358.4	-	0.1	358.5	
	Electricity	797.0 MWh/10 000 m ³	465.4	-	-	465.4	
	Total		823.7	3 899.8	25.4	4 748.9	
	Cement	1 658.8 t/10 000 m ³	-	1 219.2	1.9	1 221.1	
	Wood	211.5 m ³ /10 000 m ³	-	0.0	0.2	0.3	
	Rebar	357.7 t/10 000 m ³	-	837.0	0.4	837.4	
	Sand	4 053.8 m ³ /10 000 m ³	-	14.2	4.6	18.9	
	Gravel	6 612.9 m ³ /m ³	-	20.9	7.5	28.4	

Continued

Guidelines for Carbon Accounting and Emission Reduction in the Urban Water Sector

Treatment capacity	Materials	Consumption	Emission intensity of energy consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of material consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of transportation ¹ (t CO ₂ -eq/10 000 m ³)	Integrated emission factor (t CO ₂ -eq/10 000 m ³)	Capital-based emission factor (t CO ₂ -eq/10 000 yuan)
Large-scale plant (≥ 100 000 m ³ /d)	Steel pipe and fittings	76.1 t/10 000 m ³	-	192.5	0.1	192.6	1.2
	Diesel fuel	87.1 t/10 000 m ³	270.0	-	0.1	270.1	
	Electricity	601.0 MWh/10 000 m ³	350.9	-	-	350.9	
	Total		620.9	2 284.0	14.9	2 919.8	
	Cement	1 282.5 t/10 000 m ³	-	942.6	1.5	944.1	
	Wood	158.5 m ³ /10 000 m ³	-	0.0	0.2	0.2	
	Rebar	275.7 t/10 000 m ³	-	645.1	0.3	645.5	
	Sand	3 136.0 m ³ /10 000 m ³	-	11.0	3.6	14.6	
	Gravel	5 119.0 m ³ /m ³	-	16.2	5.8	22.0	

Continued

Construction inventory and integrated emission factor formulation

Treatment capacity	Materials	Consumption	Emission intensity of energy consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of material consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of transportation ¹ (t CO ₂ -eq/10 000 m ³)	Integrated emission factor (t CO ₂ -eq/10 000 m ³)	Capital-based emission factor (t CO ₂ -eq/10 000 yuan)
	Steel pipe and fittings	58.9 t/10 000 m ³	-	149.0	0.1	149.1	
	Diesel fuel	72.6 t/10 000 m ³	225.1	-	0.1	225.1	
	Electricity	501.0 MWh/10 000 m ³	292.5	-	-	292.5	
	Total		517.6	1 764.0	11.5	2 293.1	
Wastewater treatment plants (including sludge digestion) with Class A discharge standards							
	Cement	3 379.0 t/10 000 m ³	-	2 483.6	3.9	2 487.4	
Small-scale plant (< 10 000 m ³ /d)	Wood	427.0 m ³ /10 000 m ³	-	0.1	0.5	0.6	
	Rebar	728.7 t/10 000 m ³	-	1 705.2	0.8	1 706.0	1.3
	Sand	8 254.0 m ³ /10 000 m ³	-	29.0	9.4	38.4	
	Gravel	13 472.0 m ³ /m ³	-	42.6	15.4	57.9	

Continued

Guidelines for Carbon Accounting and Emission Reduction in the Urban Water Sector

Treatment capacity	Materials	Consumption	Emission intensity of energy consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of material consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of transportation ¹ (t CO ₂ -eq/10 000 m ³)	Integrated emission factor (t CO ₂ -eq/10 000 m ³)	Capital-based emission factor (t CO ₂ -eq/10 000 yuan)
Medium-scale plant (10 000 m ³ /d~100 000 m ³ /d)	Steel pipe and fittings	153.9 t/10 000 m ³	-	389.4	0.2	389.5	0.9
	Diesel fuel	137.2 t/10 000 m ³	425.3	-	0.2	425.5	
	Electricity	946.0 MWh/10 000 m ³	552.4	-	-	552.4	
	Total		977.7	4 649.8	30.3	5 657.7	
	Cement	1 419.0 t/10 000 m ³	-	1 043.0	1.6	1 044.6	
	Wood	181.9 m ³ /10 000 m ³	-	0.0	0.2	0.2	
	Rebar	306.0 t/10 000 m ³	-	716.0	0.3	716.4	
	Sand	3 468.6 m ³ /10 000 m ³	-	12.2	4.0	16.1	
	Gravel	5 655.7 m ³ /m ³	-	17.9	6.4	24.3	

Continued

Construction inventory and integrated emission factor formulation

Treatment capacity	Materials	Consumption	Emission intensity of energy consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of material consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of transportation ¹ (t CO ₂ -eq/10 000 m ³)	Integrated emission factor (t CO ₂ -eq/10 000 m ³)	Capital-based emission factor (t CO ₂ -eq/10 000 yuan)
Large-scale plant (≥ 100 000 m ³ /d)	Steel pipe and fittings	65.1 t/10 000 m ³	-	164.7	0.1	164.8	1.2
	Diesel fuel	75.3 t/10 000 m ³	233.4	-	0.1	233.5	
	Electricity	519.0 MWh/10 000 m ³	303.0	-	-	303.0	
	Total		536.4	1 953.8	12.7	2 503.0	
	Cement	1 538.6 t/10 000 m ³	-	1 130.9	1.8	1 132.6	
	Wood	190.5 m ³ /10 000 m ³	-	0.0	0.2	0.3	
	Rebar	330.7 t/10 000 m ³	-	773.8	0.4	774.2	
	Sand	3 763.0 m ³ /10 000 m ³	-	13.2	4.3	17.5	
	Gravel	6 141.5 m ³ /m ³	-	19.4	7.0	26.4	

Continued

Guidelines for Carbon Accounting and Emission Reduction in the Urban Water Sector

Treatment capacity	Materials	Consumption	Emission intensity of energy consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of material consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of transportation ¹ (t CO ₂ -eq/10 000 m ³)	Integrated emission factor (t CO ₂ -eq/10 000 m ³)	Capital-based emission factor (t CO ₂ -eq/10 000 yuan)
	Steel pipe and fittings	70.6 t/10 000 m ³	-	178.6	0.1	178.7	
	Diesel fuel	87.1 t/10 000 m ³	270.0	-	0.1	270.1	
	Electricity	601.0 MWh/10 000 m ³	350.9	-	-	350.9	
	Total		620.9	2 116.0	13.8	2 750.8	
Upgrading and transforming of wastewater treatment plants (excluding sludge digestion) to Class A discharge standards							
	Cement	957.1 t/10 000 m ³	-	703.5	1.1	704.6	
Small-scale plant (< 10 000 m ³ /d)	Wood	121.0 m ³ /10 000 m ³	-	0.0	0.1	0.2	
	Rebar	206.4 t/10 000 m ³	-	483.0	0.2	483.2	1.3
	Sand	2 338.0 m ³ /10 000 m ³	-	8.2	2.7	10.9	
	Gravel	3 816.0 m ³ /m ³	-	12.1	4.4	16.4	

Continued

Construction inventory and integrated emission factor formulation

Treatment capacity	Materials	Consumption	Emission intensity of energy consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of material consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of transportation ¹ (t CO ₂ -eq/10 000 m ³)	Integrated emission factor (t CO ₂ -eq/10 000 m ³)	Capital-based emission factor (t CO ₂ -eq/10 000 yuan)
Medium-scale plant (10 000 m ³ /d~100 000 m ³ /d)	Steel pipe and fittings	43.9 t/10 000 m ³	-	111.1	0.1	111.1	
	Diesel fuel	38.7 t/10 000 m ³	120.0	-	0.0	120.0	
	Electricity	267.0 MWh/10 000 m ³	155.9	-	-	155.9	
	Total		275.9	1 317.8	8.6	1 602.3	
	Cement	589.2 t/10 000 m ³	-	433.1	0.7	433.7	1.2
	Wood	75.1 m ³ /10 000 m ³	-	0.0	0.1	0.1	
	Rebar	127.1 t/10 000 m ³	-	297.4	0.1	297.6	
	Sand	1 439.9 m ³ /10 000 m ³	-	5.1	1.6	6.7	
	Gravel	2 348.9 m ³ /m ³	-	7.4	2.7	10.1	

Continued

Guidelines for Carbon Accounting and Emission Reduction in the Urban Water Sector

Treatment capacity	Materials	Consumption	Emission intensity of energy consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of material consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of transportation ¹ (t CO ₂ -eq/10 000 m ³)	Integrated emission factor (t CO ₂ -eq/10 000 m ³)	Capital-based emission factor (t CO ₂ -eq/10 000 yuan)
Large-scale plant (≥ 100 000 m ³ /d)	Steel pipe and fittings	27.0 t/10 000 m ³	-	68.3	0.0	68.3	1.2
	Diesel fuel	31.0 t/10 000 m ³	96.1	-	0.0	96.1	
	Electricity	214.0 MWh/10 000 m ³	125.0	-	-	125.0	
	Total		221.1	811.3	5.3	1 037.6	
	Cement	444.7 t/10 000 m ³	-	326.9	0.5	327.4	
	Wood	55.0 m ³ /10 000 m ³	-	0.0	0.1	0.1	
	Rebar	95.6 t/10 000 m ³	-	223.7	0.1	223.8	
	Sand	1 087.5 m ³ /10 000 m ³	-	3.8	1.2	5.1	
	Gravel	1 775.0 m ³ /m ³	-	5.6	2.0	7.6	

Continued

Construction inventory and integrated emission factor formulation

Treatment capacity	Materials	Consumption	Emission intensity of energy consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of material consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of transportation ¹ (t CO ₂ -eq/10 000 m ³)	Integrated emission factor (t CO ₂ -eq/10 000 m ³)	Capital-based emission factor (t CO ₂ -eq/10 000 yuan)
	Steel pipe and fittings	20.4 t/10 000 m ³	-	51.6	0.0	51.6	
	Diesel fuel	25.0 t/10 000 m ³	77.5	-	0.0	77.5	
	Electricity	173.0 MWh/10 000 m ³	101.0	-	-	101.0	
	Total		178.5	611.6	4.0	794.1	
Upgrading and transforming of wastewater treatment plants (excluding sludge digestion) to pseudo Class IV discharge standards							
	Cement	1445.3 t/10 000 m ³	-	1 062.3	1.6	1 063.9	1.3
Small-scale plant (< 10 000 m ³ /d)	Wood	183.0 m ³ /10 000 m ³	-	0.0	0.2	0.2	
	Rebar	311.7 t/10 000 m ³	-	729.4	0.4	729.7	
	Sand	3 531.0 m ³ /10 000 m ³	-	12.4	4.0	16.4	
	Gravel	5 762.0 m ³ /m ³	-	18.2	6.6	24.8	

Continued

Guidelines for Carbon Accounting and Emission Reduction in the Urban Water Sector

Treatment capacity	Materials	Consumption	Emission intensity of energy consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of material consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of transportation ¹ (t CO ₂ -eq/10 000 m ³)	Integrated emission factor (t CO ₂ -eq/10 000 m ³)	Capital-based emission factor (t CO ₂ -eq/10 000 yuan)
Medium-scale plant (10 000 m ³ /d~100 000 m ³ /d)	Steel pipe and fittings	66.3 t/10 000 m ³	-	167.7	0.1	167.8	1.0
	Diesel fuel	58.6 t/10 000 m ³	181.7	-	0.1	181.7	
	Electricity	405.0 MWh/10 000 m ³	236.5	-	-	236.5	
	Total		418.1	1 990.1	12.9	2 421.2	
	Cement	884.3 t/10 000 m ³	-	650.0	1.0	651.0	
	Wood	112.7 m ³ /10 000 m ³	-	0.0	0.1	0.1	
	Rebar	190.7 t/10 000 m ³	-	446.2	0.2	446.5	
	Sand	2 160.9 m ³ /10 000 m ³	-	7.6	2.5	10.1	
	Gravel	3 525.5 m ³ /m ³	-	11.1	4.0	15.2	

Continued

Construction inventory and integrated emission factor formulation

Treatment capacity	Materials	Consumption	Emission intensity of energy consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of material consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of transportation ¹ (t CO ₂ -eq/10 000 m ³)	Integrated emission factor (t CO ₂ -eq/10 000 m ³)	Capital-based emission factor (t CO ₂ -eq/10 000 yuan)
Large-scale plant (≥ 100 000 m ³ /d)	Steel pipe and fittings	40.6 t/10 000 m ³	-	102.7	0.0	102.8	1.0
	Diesel fuel	46.4 t/10 000 m ³	143.8	-	0.1	143.9	
	Electricity	320.0 MWh/10 000 m ³	0.0	-	-	0.0	
	Total		143.9	1 217.7	7.9	1 369.5	
	Cement	675.9 t/10 000 m ³	-	496.8	0.8	497.6	
	Wood	83.5 m ³ /10 000 m ³	-	0.0	0.1	0.1	
	Rebar	145.3 t/10 000 m ³	-	340.0	0.2	340.2	
	Sand	1 653.0 m ³ /10 000 m ³	-	5.8	1.9	7.7	
	Gravel	2 698.0 m ³ /m ³	-	8.5	3.1	11.6	

Continued

Treatment capacity	Materials	Consumption	Emission intensity of energy consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of material consumption (t CO ₂ -eq/10 000 m ³)	Emission intensity of transportation ¹ (t CO ₂ -eq/10 000 m ³)	Integrated emission factor (t CO ₂ -eq/10 000 m ³)	Capital-based emission factor (t CO ₂ -eq/10 000 yuan)
	Steel pipe and fittings	31.0 t/10 000 m ³	-	78.4	0.0	78.5	
	Diesel fuel	38.4 t/10 000 m ³	119.0	-	0.0	119.1	
	Electricity	265.0 MWh/10 000 m ³	0.0	-	-	0.0	
	Total		119.1	929.6	6.1	1 054.7	

1. Calculated with 20 kilometers transported by heavy-duty diesel trucks (with a load capacity of 46 tons).

C.5 CONSTRUCTION INVENTORY OF STORMWATER SYSTEM

Table C.5 Summary of construction inventory and emission intensity of stormwater system.

Category	Facility	Emission intensity of material consumption	Emission intensity of construction operation	Emission intensity of transportation	Integrated emission factor	Unit
	PE tube	0.664	0.161	11.93 3	1 2.75 8	kg CO ₂ -eq/m

Continued

Construction inventory and integrated emission factor formulation

Category	Facility			Emission intensity of material consumption	Emission intensity of construction operation	Emission intensity of transportation	Integrated emission factor	Unit
Stormwater pipeline facilities	Pipeline structures	Seepage pipe		20.307	0.396	8.044	28.747	kg CO ₂ -eq/m ³
		Seepage channel		77.765	0.397	8.416	86.578	kg CO ₂ -eq/m ³
		Diversion channel		133.618	0.402	6.262	140.282	kg CO ₂ -eq/m
	Other transfer facilities	Transfer type						
		dry grass swale		4.880	0.409	3.141	8.430	kg CO ₂ -eq/m
			Grass Permeable swales					
		dry grass swale			17.220	0.657	5.709	19.757
			Wet grass swale		25.367	0.778	11.746	36.651
		Permeabl e pavemen t (taking permeabl e cement concrete as an example)		Sidewalk	17.679	0.194	17.68 0	37.891
			Driving load ≤5 t	39.937	0.334	39.94 0	80.211	kg CO ₂ -eq/m ²
Stormwater controlling facilities	Osmotic facilities	Driving load 5~8 t	51.828	0.373	51.83 0	104.031	kg CO ₂ -eq/m ²	
		Driving load 8~13 t	59.755	0.420	59.75 0	119.925	kg CO ₂ -eq/m ²	
		Bioretention zone	11.684	0.1 30	3. 666	15.480	kg CO ₂ -eq/m ²	

Guidelines for Carbon Accounting and Emission Reduction in the Urban Water Sector

Category	Facility	Emission intensity of material consumption	Emission intensity of construction operation	Emission intensity of transportation	Integrated emission factor	Unit
Collection and utilization facilities	Rain garden (Take the purification rain garden as an example)	35.114	0.662	8.074	43.850	kg CO ₂ -eq/m ²
	Simple ecological tree pool	151.666	0.593	13.684	165.943	kg CO ₂ -eq/m ²
	Purifying ecological tree pool	142.982	0.816	12.039	155.837	kg CO ₂ -eq/m ²
	High flower bed (Take the lingering high flower bed as an example)	184.010	0.520	45.082	229.612	kg CO ₂ -eq/m ²
	Sunken green space	10.608	0.207	2.281	13.096	kg CO ₂ -eq/m ²
	Simple green roof	77.865	4.372	3.351	85.588	kg CO ₂ -eq/m ²
	Garden type green roof	113.289	71.445	12.675	197.409	kg CO ₂ -eq/m ²
	Infiltration pond	7.440	0.408	3.458	11.084	kg CO ₂ -eq/m ²
	Seepage well	183.234	9.150	35.197	227.580	kg CO ₂ -eq/m ³
	Reservoir (calculated according to water storage capacity)	506.635	1.971	16.786	525.392	kg CO ₂ -eq/m ³
	Rainwater tank (PE)	73.982	1.932	2.346	78.260	kg CO ₂ -eq/m ³

Construction inventory and integrated emission factor formulation

Category	Facility	Emission intensity of material consumption	Emission intensity of construction operation	Emission intensity of transportation	Integrated emission factor	Unit
Detention and retention facilities	Rainwater tank (fiberglass)	62.623	1.932	2.450	67.005	kg CO ₂ -eq/m ³
	Wet pond	0.490	0.518	7.688	8.696	kg CO ₂ -eq/m ³
	Reinforced concrete					
	rainwater storage tank (calculated according to water storage capacity)	493.937	2.319	17.064	513.320	kg CO ₂ -eq/m ³
	Rainwater storage tank (module)	261.618	2.334	11.530	275.482	kg CO ₂ -eq/m ³
	Vegetation buffer zone			Reference planting ditch		
	Pile revetment	117.814	2.333	14.890	135.037	kg CO ₂ -eq/m
	Gabion revetment	197.778	6.645	86.72	291.142	kg CO ₂ -eq/m
	Grass brick revetment	408.908	1.955	41.572	452.435	kg CO ₂ -eq/m
	Stone revetment	144.639	2.299	16.939	163.877	kg CO ₂ -eq/m
Interception and purification facilities	Ecological block revetment	48.754	2.587	26.025	77.366	kg CO ₂ -eq/m
	Rain wetland			Reference wet pond		
	Sedimentation tank	325.897	Manual construction, ignore	11.368	337.265	kg CO ₂ -eq/m ³

Appendix D

Carbon accounting: case studies

D.1 CARBON EMISSION OF WATER SUPPLY SYSTEM

A water treatment plant in Beijing was designed with a total capacity of 500,000 m³/d, which would be constructed in three phases. Currently, the first phase project with a capacity of 100,000 m³/d has already been put into operation. The raw water is taken from a nearby reservoir at a relative low elevation of 40 m. The turbidity is below 15 NTU and the suspended solids (SS) concentration is 10 mg/L on average. A combination of four 350S75A pumps, each with a working efficiency of 70%, was designed for the water abstraction with three pumps operating and one serving as a backup. The water is then transported to the water treatment plant and stored in the tanks. The first phase of the project includes following process: water abstraction - coagulation - flocculation -sedimentation -filtration - adsorption - disinfection - water distribution. The coagulant used is polyaluminum chloride (PAC), with a dosage of 20 mg/L. The disinfection chemical is provided by the on-site chlorine dioxide generator, with a dosage of 1 mg/L. The monthly electricity consumption is 15,000 kWh. The procurement of chemicals is sourced within a 20 km radius of the project site. The waste sludge produced by the water treatment is 4,000 m³/d. And lime is added to adjust the pH, with a dosage based on 10% of SS. The sludge from water treatment is exported and transported to landfill site with 20 km away for landfilling disposal after conditioning (pre-treatment) and dewatering. All transportation uses medium-sized diesel trucks with a load of 8 t. Water pumps (3 operating and 1 backup) are working to distribute the water to users at an average relative elevation of 20 m.

D.1.1 Reporting boundaries

The reporting boundary includes all the carbon emissions generated by water abstraction stations, treatment, and distribution. The present carbon accounting only focuses on the operation and maintenance stage. The main carbon emission activities of the water supply system include:

- Indirect carbon emissions from electricity: Indirect carbon emissions caused by the electricity consumption by the operation of water treatment equipment such as water abstraction and distribution pumps operation and stirring.
- Indirect carbon emissions from materials: Mainly refers to the chemicals used in the water treatment and their transportation. Indirect carbon emissions are generated by the PAC for coagulation, the chlorine dioxide agent for disinfection, and the lime for the sludge conditioning. And the indirect carbon emissions from the transportation of chemicals are caused by use of medium-sized diesel trucks.
- Indirect carbon emissions from sludge disposal: Indirect carbon emissions from transportation of sludge for disposal.

D.1.2 Accounting method

The accounting methods for the emission activities are referred to this *Guidelines*, and the equation index are summarized in Table D.1.

Table D.1 Summary of accounting equations.

Activities		Accounting equations	Data
Electricity consumption	Water intake stations	Section 5.2: Equation (5.2~3)	Electricity consumption (kWh/d)
		Section 5.3: Equation (5.6)	Actual lifting head (m)
	Water treatment	Section 5.2: Equation (5.2)	Parameters of pumps such as type/efficiency (%)
			Electricity consumption (kWh/d)
Chemicals consumption	Distribution	Section 5.2: Equation (5.2~3)	Electricity consumption (kWh/d)
			Actual lifting head (m)
	Chemicals consumption	Section 5.2: Equation (5.4)	Parameters of pumps such as type/efficiency (%)
			Chemicals consumption (kg/d)
	Transportation	Section 5.2: Equation (5.5)	Transportation load (t), Distance (km)

D.1.3 Data acquisition and collection

According to the equations summarized in Table D.1, the activity data required are summarized in Table D.2.

Table D.2 Summary of the activity data.

Index	Value	Index	Value
Water intake head	40 m	Chemical transport distance	20 km
Pump efficiency	70%	Sludge transportation distance	15 km
Water treatment electricity consumption	500 kWh/d	Mode of transport	Medium-sized diesel truck (8 t load)
PAC consumption	2,000 kg/d	Water head	20 m
CIO ₂ consumption	100 kg/d	Sludge production	4.1 t/d
Lime	100 kg/d		

D.1.4 Accounting results

The accounting results are shown in Table D.3.

Table D.3 Summary of emission accounting results of the water supply system.

Activities		Emission intensity	Annual total emissions	Proportion
		(kg CO ₂ -eq/m ³)	(t CO ₂ -eq)	
Electricity consumption	Water intake station	0.147	5,365.5	56.7%
	Water treatment	0.005	182.5	1.9%
Material consumption	Distribution	0.074	2,701	28.5%
	Chemicals	0.033	1,204.5	12.8%
Sludge disposal	Transportation	7.59×10^{-5}	2.78	0.03%
	Transportation	1.47×10^{-4}	5.37	0.057%
Total		0.26	9,462	-

According to Table D.3, the carbon emissions from the electricity consumption of pump stations and the use of chemicals account for 98% of the total carbon emission amount in the water sector. Among them, the carbon emissions caused by water abstraction and distribution pumping stations are the most significant contributors. Therefore, to reduce the carbon emission from the operation and maintenance in water supply system, it is necessary to significantly reduce the electricity consumption of pump stations and chemicals dosage. In this project, the water abstraction and distribution pump stations all use the same type of pumps, which is overly standardized and may result in pressure waste. It is recommended to use a combination of multiple variable frequency pumps to better match the water usage conditions of users. Additionally, pressure zoning and water distribution can be implemented. To reduce the chemicals consumption of water treatment, the recovery of sludge coagulant and establishment real-time dynamic chemical dosing through hydraulic modelling can be considered.

D.2 CARBON EMISSION OF SEAWATER DESALINATION PLANT

A seawater desalination plant is located at Tianjin City, with a capacity of 3,000 m³/d. The plant employs a high-temperature brine-recirculation long-tube multi-stage flash (MSF) seawater desalination facilities, which produces 18 kg freshwater per 4,681.8 J heat consumed. Anthracite is utilized to provide thermal energy required, and the total daily electricity consumption is 1,000 kWh.

D.2.1 Reporting boundaries

The organizational boundary of the seawater desalination covers entire processes from water inflow to outflow within the plant area. The reporting boundary includes energy and resource consumption, and carbon emission associated with material transportation from other enterprises. The emission accounting focused on the operation and maintenance stage. The main GHG emission activities of the desalination plant include:

- Direct emission from desalination: This seawater desalination plant uses MSF technology. In the process of desalination, due to the combustion of fossil fuels, fossil source CO₂ is produced, resulting in direct emissions.
- Indirect emissions from resource and energy consumption: The seawater desalination plant consumes electricity and thus indirect emissions.

D.2.2 Accounting method

According to the *Guidelines*, the carbon emissions calculation methods are summarized in Table D.4.

Table D.4 Summary of accounting equations.

Activities		Accounting equations	Data
Carbon emissions from desalination plants	Fossil CO ₂	Section 5.3.4 Equation (5.8)	Thermal energy consumed by desalinating unit volume of seawater (GJ/m ³)
			Proportion of each type of fuel (%)
	Electricity consumption	Section 5.2.2 Equation (5.2)	Daily electricity consumption (kWh/d)
			Daily treated water yield (m ³ /d)

D.2.3 Data acquisition and collection

According to the equations summarized in Table D.4, the activity data required are summarized in Table D.5.

Table D.5 Summary of the activity data.

Index	Value
Heat consumed by desalinate a unit volume of sea water	2.6×10 ⁶ GJ/m ³
Emission factor of the i-th fuel	98.08 kg CO ₂ -eq/GJ
Proportion of i-th fuel	100%
A total of n types of fossil fuels used	1 type
Daily electricity consumption	1,000 kWh/d
Emission factor of electricity in the region	0.9419 kg CO ₂ -eq/kWh
Daily treated water yield	3,000 m ³ /d

D.2.4 Accounting results

The accounting results are shown in Table D.6.

Table D.6 Summary of accounting results of the seawater desalination plant.

Activity	Emission intensity	Annual total emissions	Proportion
	(kg CO ₂ -eq/m ³)	(10,000 t CO ₂ -eq)	
Fossil CO ₂	2.55 × 10 ⁻²	2.79	7.6%
Electricity consumption	0.31 kg CO ₂ -eq/m ³	33.9	92.4%
Total	0.3355 kg CO ₂ -eq/m ³	36.74	-

D.3 CARBON EMISSION OF WASTEWATER TREATMENT PLANTS

A wastewater treatment plant is located at Shenzhen City, Guangdong Province, with a treatment capacity of 125,000 m³/d. The main source of wastewater is domestic wastewater. The plant employs the AAO process, and produces about 150 t of excess sludge (with a moisture content of 80%) every day. The sludge is transported to a disposal site 40 km away for centralized incineration treatment.

D.3.1 Reporting boundaries

The organizational boundary of the wastewater treatment plant should cover the entire process from the influent to the effluent, including all wastewater and excess sludge treatment structures. The present accounting only focuses on the operation and maintenance stage. The main GHG emission activities of the wastewater treatment plant include:

- Direct emissions from wastewater sludge treatment processes: GHG is produced from the microbial biochemical reactions degrading the pollutants in the wastewater.
- Indirect emissions from excess sludge disposal: The treatment plant adopts incineration for the dispose of excess sludge in which fossil CO₂ and N₂O will be produced.
- Indirect emissions from resource and energy consumption: The treatment plant consumes electricity for operation, resulting in generating indirect emissions. Polyaluminum chloride was used for phosphorus removal and polyacrylamide was used as coagulant, resulting in generating indirect emissions. Additionally, the transportation of excess sludge and chemicals between the plant and other enterprises, resulting in generating indirect emissions.

D.3.2 Accounting method

The accounting methods for the emission activities are referred to this *Guidelines*, and the equation index are summarized in Table D.7.

Table D.7 Summary of accounting equations

Activities		Accounting equations	Data
Direct emissions from wastewater treatment	Fossil CO ₂	Section 5.4.3: Equation (5.21~23)	BOD ₅ concentration of influent and effluent (mg/L)
			TN concentration of influent and effluent (mg/L)
			Sludge retention time (SRT) (d)
			Hydraulic retention time (HRT) (d)
			Mixed liquid volatile suspended solid concentration (mg/L)
Indirect emissions from sludge disposal	Fossil CO ₂	Section 5.4.3: Equation (5-39)	Water temperature (°C)
			BOD ₅ concentration of influent (mg/L)
			TN concentration of influent (mg/L)
			Excess sludge dry weight (kg/d)
			Excess sludge carbon content (%)
			Wastewater flow (m ³ /d)

Activities		Accounting equations	Data
Indirect emissions from resource and energy consumption	N ₂ O	Section 5.4.3: Equation (5.45)	Excess sludge dry weight (kg/d) Wastewater flow (m ³ /d)
	Electricity consumption	Section 5.2.2: Equation (5.2)	Electric consumption (kWh/d) Wastewater flow (m ³ /d)
	Chemicals consumption	Section 5.2.3: Equation (5.4)	Chemicals consumption (kg/d) Wastewater flow (m ³ /d)
	Transportation	Section 5.2.4: Equation (5.5)	Transportation volume (kg) and distance (km)
			Wastewater flow (m ³ /d)

D.3.3 Data collection and acquisition

According to the equations summarized in Table D.7, the activity data required are summarized in Table D.8.

Table D.8 Summary of activity data.

Index	Value	Index	Value
Wastewater flow	125,000 m ³ /d	Excess sludge dry weight	30,000 kg/d
BOD ₅ concentration of influent	97.13 mg/L	BOD ₅ concentration of effluent	0.73 mg/L
Fossil C ratio	1%	TN concentration of effluent	15 mg/L
TN concentration of influent	37.1 mg/L	MLVSS concentration	3,074 mg/L
Water temperature	23.78 °C	SRT	23 d
HRT	0.42 d	Electricity consumption	42,318 kWh /d
C content of excess sludge	1.2%	PAC Consumption	144,80 kg/d
Polyacrylamide Consumption	85 kg/d		

D.3.4 Accounting results and analysis

The accounting results are shown in Table D.9.

Table D.9 Summary of accounting results of the wastewater treatment plant.

Activity		Emission intensity (kg CO ₂ -eq/m ³)	Annual total emissions (10,000 t CO ₂ -eq)	Proportion
Direct emissions from wastewater treatment	Fossil CO ₂	0.035	0.157	5.08%
	CH ₄	0.010	0.045	1.46%
	N ₂ O	0.104	0.48	15.53%

Continued

Activity		Emission intensity (kg CO ₂ -eq/m ³)	Annual total emissions (10,000 t CO ₂ -eq)	Proportion
Indirect emissions from sludge disposal	Fossil CO ₂	0.0006	0.003	0.10%
	N ₂ O	0.063	0.29	9.39%
Indirect emissions from resource and energy consumption	Electricity consumption	0.272	1.24	40.13%
	Chemicals PAC	0.188	0.86	27.83%
	Consumption polyacryl amide	0.001	0.005	0.16%
	Transportation	0.003	0.01	0.33%
Total		0.6716	3.09	-

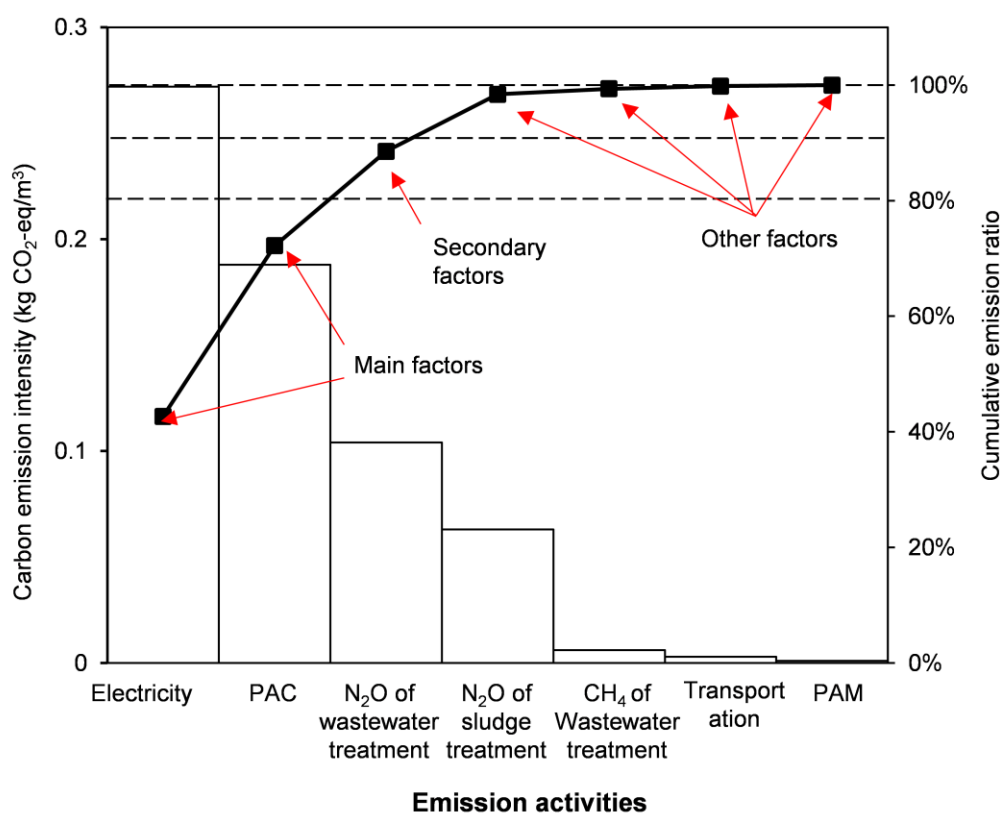


Figure D.1 Arrangement chart of accounting results of the case sewage treatment plant.

The emission of fossil CO₂ from the wastewater treatment process is determined by the composition of influent, and is beyond the control of wastewater treatment plant. The emission profiles of other carbon emission activities are shown in Figure D.1. The main factors contributing to carbon emissions from the wastewater treatment plant are: (1) indirect carbon emissions from electricity consumption; (2) indirect emissions from polyaluminum chloride consumed during the treatment process. Secondary factor is direct emission of N₂O generated in wastewater treatment. Other factors are: (a) N₂O emissions from sludge incineration; (b) CH₄ emissions from wastewater treatment; (c) fossil CO₂ emissions from sludge

incineration; (d) carbon emissions from transportation; (e) indirect emissions from consumption of polyacrylamide.

Therefore, in order to reduce the carbon emission amount and intensity of the case wastewater treatment plant, efforts should primarily be focused on reducing the carbon emission intensity of the main factors, followed by reducing the carbon emission intensity of the secondary factor, and finally try to reduce the carbon emission intensity of other factors.

D.3.5 Develop a strategy to reduce emissions

Based on the carbon emission reduction pathways of the wastewater treatment plant in section 7.5 of this *Guidelines*, the following emission reduction strategies are proposed for the case wastewater treatment plant.

D.3.5.1 Carbon reduction pathways of reducing electricity consumption

Indirect emissions from electricity consumption is the main factor of this wastewater treatment plant. Therefore, optimizing the energy efficiency of mechanical equipment and reducing electricity consumption are important strategies for emission reduction. The wastewater treatment plant should conduct a detailed analysis of the proportion of electricity consumption, and subsequently optimize and update accordingly. For the aeration process, the following measures can be taken: 1) adopting new equipment and techniques to improve aeration efficiency, such as replacing with new high-efficiency equipment, using microbubble aeration, etc. 2) employing pre-feedback or post-feedback aeration optimization control technology to adjust the aeration rates promptly to avoid the excessive aeration and electricity losses. 3) combining the biological models and automatic control technology to precisely control the aeration flow rate, thereby maximizing the electricity efficiency of the aeration system.

For water pump units, the following measures can be taken: 1) maintaining worn-out or corroded equipment in time, and upgrading to new high-efficiency equipment. 2) optimizing the operation schemes of water pump units, including their operating status and timing, to avoid inefficient or ineffective operation; 3) adding frequency conversion functionality to the water pump units, enabling electricity reduction during low flow rate to prevent idling and wasting electricity.

In addition, the wastewater treatment process of the plant can also be modified by adopting efficient and low-carbon processes, such as compact wastewater treatment process, high-efficiency denitrification process, etc., to reduce the electricity consumption.

D.3.5.2 Carbon reduction pathway of reducing chemicals consumption

The poor wastewater treatment performance requires additional dosing of polyaluminum chloride for chemical phosphorus removal, which is another main factor for the carbon emissions. Based on this, the plant should improve wastewater treatment, enhance biological treatment capacity, and reduce the consumption of polyaluminum chloride to effectively reduce its carbon emission intensity.

D.3.5.3 Renewable energy pathways

In addition to the carbon reduction strategy, the plant can also adopt the renewable energy pathways to offset part of its carbon emissions by recycling resources and energy. For example, the ashes of sludge incineration can be recycled and processed into phosphate fertilizer products to replace ore phosphate fertilizers. The plant can also use water source heat pump technology to recover the residual heat in wastewater and provide cooling or heating for surrounding areas, replacing traditional fire heating, thereby reducing carbon emissions.

D.4 CARBON EMISSION OF WATER RECLAMATION PLANT

A water reclamation plant located at Beijing serves as the upgrading project of a wastewater treatment plant. Half of the effluent which is 100,000 m³/d of the wastewater treatment plant is used as the source of the reclaimed water. The effluent is composed of BOD₅= 8 mg/L, COD= 30 mg/L, TN= 7 mg/L, TP= 0.1 mg/L, ammonia nitrogen= 4 mg/L, SS= 6 mg/L, pH= 7.5, respectively. The treatment line involves micro-flocculation, media filtration, disinfection, water pumping station for non-portable utilization. The sludge produced in reclamation plant is handled with the sludge from wastewater treatment plant. The coagulant used is PAC with a dosage of 6 mg/L. The disinfection is made by the on-site chlorine dioxide generator, with a dosage of 1 mg/L. The monthly electricity consumption of reclaimed water treatment is 10,000 kWh. The procurement of chemicals is sourced within a 20 km radius of the project site. All transportation uses medium-sized diesel trucks with a load of 8 t. Centrifugal water pumps (3 operating and 1 backup) are used to supply to users located at relative elevation of 18 m.

D.4.1 Reporting boundaries

The organizational boundary of the water reclamation plant covers the treatment facilities and distribution pumping stations. The emission accounting focused on the operation and maintenance stage. The main GHG emission activities of the water reclamation plant include:

- Indirect carbon emissions from electricity consumed by the operation of water pumping stations and treatment facilities.
- Indirect carbon emissions from materials and chemical consumption, as well as their associated transportation, including PAC for coagulation, the chlorine dioxide agent for disinfection.

D.4.2 Accounting method

The accounting methods for the emission activities are referred to this Guidelines, and the equations index are summarized in Table D.10.

Table D.10 Summary of accounting equations.

Activities		Accounting equations	Data
Electricity consumption	Water purification	Section 5.2: Equation (5.2~3)	Electricity consumption (kWh)
	Distribution	Section 5.2: Equation (5.2~3)	Electricity consumption (kWh)
			Actual lifting head (m) Parameters of pumps such as type/efficiency (%)
Chemicals consumption	Chemicals consumption	Section 5.2: Equation (5.4)	Chemicals consumption (kg/d)
	Transportation	Section 5.2: Equation (5.5)	Transportation load (t), Distance (km)

D.4.3 Data acquisition and collection

According to the equations summarized in Table D.10, the activity data required are summarized in Table D.11.

Table D.11 Summary of the activity data.

Index	Value	Index	Value
Electricity consumption	333.34 kWh/d	Mode of transport	Medium-sized diesel truck (8 t load)
PAC consumption	600 kg/d	Water head	18 m
ClO ₂ consumption	100 kg/d	Pump efficiency	70%
Chemicals transport distance	20 km		

D.4.4 Accounting results

The accounting results are shown in Table D.12.

Table D.12 Summary of emission accounting results of the water reclamation plant.

Activity		Emission intensity (kg CO ₂ -eq/m ³)	Annual total emissions (t CO ₂ -eq)	Proportion
Electricity consumption	Water purification	0.003	109.5	3.8%
	Distribution	0.066	2,409	83.52%
Material consumption	Chemicals	0.01	365	12.65%
	Transportation	2.5×10^{-5}	0.9	0.03%
Total		0.08 kg	2,884.4	-

According to Table D.12, the carbon emissions from the electricity consumption of the distribution and the use of chemicals account for 96.17% of the total carbon emission amount in the water reclamation plant. Therefore, to reduce the carbon emission from the operation and maintenance in reclaimed water sector, it is necessary to significantly reduce the electricity consumption of distribution and chemicals dosage, for example, adopting a combination of multiple variable frequency pumps to better match the water usage conditions of users.

D.5 CARBON EMISSION OF STORMWATER SYSTEM

A central urban area is taken as the research case with a catchment area of 660 km², and a comprehensive runoff coefficient of 0.7. The annual runoff volume in the study area is 4.90×10^8 m³. Based on field monitoring data and drainage statistical yearbook data, the carbon emissions from the operation and maintenance of the stormwater system only refers to the conventional stormwater control facilities (i.e., the drainage pipe network).

D.5.1 Reporting boundary

The reporting boundary covers from the stormwater pumping station to the natural water bodies, including all carbon emissions generated by the water lifting process. The present carbon accounting only focuses on the operation and maintenance stage. The main carbon emission activities include:

- Direct carbon emissions from fossil fuels for pump systems for rapid stormwater removal during emergency urban flooding, resulting in the carbon emissions from the combustion of fossil fuel fuels.
- Indirect carbon emissions from electricity: Indirect carbon emissions from the electricity consumption in the operation of the water pumps.

D.5.2 Accounting method

The accounting methods for the emission activities are referred to this *Guidelines*, and the equation index are summarized in Table D.13.

Table D.13 Summary of accounting equations.

Activities	Accounting equations	Data
Fossil fuels consumption	Section 5.6: Equation (5.46)	Actual lifting head (m)
	Section 5.6: Equation (5.47)	Parameters of pumps such as type/efficiency (%)
	Section 5.6: Equation (5.48)	Parameters of pipe such as diameter(mm), length (m), flow velocity (m/s) and so on
Electricity consumption	Section 5.6: Equation (5.49)	Electricity consumption (kWh/a)
	Section 5.2: Equation (5.2~3)	Actual lifting head, head loss (m), Parameters of pumps such as type/efficiency (%)

D.5.3 Data acquisition and collection

According to the equations summarized in Table D.14, the activity data required are summarized in Table D.14.

Table D.14 Summary of the activity data.

Index	Value	Index	Value
Lifting head	8 m	Acceleration of gravity	9.8 m/s ²
Oil pump system efficiency	60%	Rainwater density	1,000 kg/m ³
Electric pump system efficiency	60%	Pipe length	5 km

D.5.4 Accounting results

According to the calculation method provided in this *Guidelines*, the calculation and statistical results are shown below.

D.5.4.1 Carbon emission from stormwater pressure sewers

A separated drainage system was applied in the area, and lifting pumps are required to transfer the stormwater of which the head loss of pipeline network is calculated as follows.

$$H_{\text{loss}} = 0.00124 \times \frac{v^2}{d^{1.33}} \times L = 0.00124 \times \frac{9}{0.3^{1.33}} \times 5000 = 277 \text{ m}$$

Where:

- v — the flow velocity (m/s)
 d — the diameter (m) of the pipelines

L — the length (m) of the pipelines

Considering only the treatment capacity of stormwater pipelines, the head loss of pipeline network can be calculated to be 277 m.

D.5.4.1.1 Oil pump system

If the pump system in the study area uses diesel with an emission factor of 72.59 t CO₂/TJ (Appendix B.1), the carbon emission is calculated to be 2.58×10^5 t CO₂.

$$W_{rl} = \sum_{i=1}^n \frac{\rho g (H_{net} + H_{loss})}{\eta_i} \times Q = \frac{1000 \times 9.8 \times (8 + 277)}{0.27} \times 4.90 \times 10^8 \times 0.7 = 3548.1 \text{ TJ}$$

$$CE_{rl} = W_{rl} \times EF_{rl} = 3548.1 \times 72.59 = 2.58 \times 10^5 \text{ t CO}_2$$

D.5.4.1.2 Electric pump system

If electric pumps are adopted, the carbon emission is calculated to be 2.57×10^5 t CO₂.

$$E_d = \sum_{i=1}^n \frac{\rho g (H_{net} + H_{loss})}{3.6 \times 10^6 \times \eta_i} \times Q = \frac{1000 \times 9.8 \times (8 + 277)}{3.6 \times 10^6 \times 0.6} \times 4.90 \times 10^8 \times 0.7 = 4.435 \times 10^8 \text{ kWh}$$

$$CE_d = E_d \times EF_d = 4.435 \times 10^8 \times 0.5810 = 2.57 \times 10^5 \text{ t CO}_2$$

D.5.4.2 Carbon emission from stormwater gravity sewers

Based on stormwater pressure sewers using electric pump system, without considering changes in flow velocity, pipe diameter, pipe length, pump efficiency, etc., only a part of the pump system was redesigned and converted to a gravity flow system, as shown in Table D.15.

Table D.15 Carbon emissions of stormwater sewers under different gravity flow and pressure flow ratio scenarios.

Scene	Gravity flow/%	Pressure flow/%	Pump station capacity /10 ⁵ m ³	Carbon emissions/ 10,000t CO ₂
Conventional drainage (full pump drainage)	0	100	3,430	25.95
Scenario one	10	90	3,087	23.35
Scenario two	20	80	2,744	20.76
Scenario Three	30	70	2,401	18.16
Scenario Four	40	60	2,058	15.57
Scenario five	50	50	1,715	12.97
Scenario six	60	40	1,372	10.38
Scenario seven	70	30	1,029	7.78
Scenario eight	80	20	686	5.19
Scenario nine	90	10	343	2.59
Scenario ten (full gravity flow)	100	0	0	0

D.5.4.3 Carbon emissions of stormwater sewers with source reduction

As stated above, the source reduction of stormwater is relying on the reconstruction of the low-impact development facilities and the decrease of urban runoff. It prioritizes the application of natural drainage systems, and construction of ecological drainage facilities, fully utilizing urban green spaces, roads, water systems, and other elements for the absorption, retention, and attenuation of stormwater. It is designed that the total annual runoff control rate in the study area is 85%, and the integrated runoff coefficient is 0.15 with a flow velocity of 2 m/s. With other data unchanged, the annual runoff in the study area reduced to $7.35 \times 10^7 \text{ m}^3$ and the head loss is then reduced.

$$H_{\text{loss}} = 0.00124 \times \frac{v^2}{d^{1.33}} \times L = 0.00124 \times \frac{4}{0.3^{1.33}} \times 5000 = 124 \text{ m}$$

Based on calculations, the head loss along the pipeline network is 124 m. The carbon emissions of the oil and the electric pump system are calculated as follows.

D.5.4.3.1 Oil pump system

$$W_{\text{rl}} = \sum_{i=1}^n \frac{\rho g (H_{\text{net}} + H_{\text{loss}})}{\eta_i} \times Q = \frac{1000 \times 9.8 \times (8 + 124)}{0.27} \times 7.35 \times 10^7 = 352.1 \text{ TJ}$$

$$CE_{\text{rl}} = W_{\text{rl}} \times EF_{\text{rl}} = 352.1 \times 72.59 = 2.56 \times 10^4 \text{ tCO}_2$$

D.5.4.3.2 Electric pump system

$$E_{\text{d}} = \sum_{i=1}^n \frac{\rho g (H_{\text{net}} + H_{\text{loss}})}{3.6 \times 10^6 \times \eta_i} \times Q = \frac{1000 \times 9.8 \times (8 + 124)}{3.6 \times 10^6 \times 0.6} \times 7.35 \times 10^7 = 4.4 \times 10^7 \text{ kWh}$$

$$CE_{\text{d}} = E_{\text{d}} \times EF_{\text{d}} = 4.4 \times 10^7 \times 0.5810 = 2.56 \times 10^4 \text{ t CO}_2$$

Table D.16 Comparison of carbon emissions from pressure sewers and sewers with source reduction facilities.

Scenario	Integrated runoff coefficient	Diameter /m	speed	Annual runoff /m ³	Oil pumps /t CO ₂	Electric pumps /t CO ₂
Pressure sewers	0.7	0.3	3 m/s	4.90 × 10 ⁸	2.57 × 10 ⁵	2.59 × 10 ⁵
Pressure sewers with source reduction	0.15	0.3	2 m/s	7.35 × 10 ⁷	2.56 × 10 ⁴	2.56 × 10 ⁴

D.6 Accounting cases for water sector association

A local water sector association consists of 10 wastewater treatment companies, with a total treatment capacity of 150,000 m³/d. According to the procedures in Section 3.3.3, the association organized and guided member companies to carry out carbon emission accounting work.

D.6.1 Accounting plan

The water sector association formulated and unified the accounting plan as follows.

- Accounting boundary: The organizational boundary should cover the entire processes from the influent to the effluent, including all wastewater and sludge treatment facilities. The carbon accounting only includes operation and maintenance stage. The main GHG emission activities include: (I) direct emissions from wastewater treatment, (II) direct emissions from sludge treatment, (III) indirect emissions from resource and energy consumption, and transportation.
- Accounting methods: The equations of Section 5.3 are referred in this *Guidelines*.
- Data acquisition and emission factors: According to the provisions of Chapter 8, member enterprises have agreed on the same activity data monitoring method and selected the same emission factors.

D.6.2 Accounting implementation

According to the accounting plan, the association takes the roles as follows.

- Supervising the data acquisition work to ensure the quality of activity data.
- Providing guidance for the process of the accounting work in each member enterprise to ensure the accuracy of the accounting results.
- Summarizing the accounting results in Table D.17.

Table D.17 Summary of the members' carbon emission accounting results.

Number	Treatment process	Treatment capacity (m ³ /d)	Carbon intensity (kg CO ₂ -eq/m ³)	
			Wastewater treatment	Sludge treatment
1	AAO	10,000	0.575	0.206
2	AAO	20,000	0.597	0.181
3	OD	20,000	0.638	0.111
4	SBR	5,000	0.687	0.114
5	AAO	30,000	0.564	0.209
6	Multi-segment AO	10,000	0.763	0.225
7	OD	10,000	0.652	0.121
8	SBR	5,000	0.560	0.189
9	SBR	10,000	0.552	0.196
10	AAO	30,000	0.542	0.211

Based on the accounting results, the average carbon emission intensity of wastewater treatment is calculated as follows (Equation D.1).

$$\overline{CES} = \frac{\sum_{i=1}^{10} (CES_i \times Q_i)}{\sum_{i=1}^{10} Q_i} \quad (D.1)$$

Where: $\overline{CES}$ — The average carbon emission intensity of association, kg CO₂-eq/m³

- CES_i — The carbon emission intensity of i -th wastewater treatment plant, kg CO₂-eq/m³
- Q_i — The daily water treatment volume of i -th wastewater treatment plant, m³/d

The average carbon emission intensity of the association member companies is about 0.78 kg CO₂-eq/m³.

D.6.3 Formulating report

A report is prepared and submitted to the relevant management department.

D.7 Carbon emission of stormwater treatment

The carbon emissions of stormwater treatment in a city or region can be estimated by the following equation (Equation D.2).

$$CE_{CL} = 5693.8 \times P \times S \times \frac{n \times [0.1a + (1-a)k] + (1-n) \times [0.1(1-\varphi) + k \times \varphi]}{3.6 \times 10^6 \eta} \times (H_{net} + H_{loss}) \quad (D.2)$$

- Where:
- CE_{CL} — Carbon emissions from stormwater treatment, kg CO₂-eq/a
 - P — Average annual precipitation, m
 - n — Sponge built-up area ratio
 - S — Area of built-up, m²
 - a — Total stormwater runoff control ratio
 - k — The proportion of runoff handled by pressure sewer in the area
 - φ — The integrated stormwater runoff coefficients in undeveloped areas. See *Technical Guidelines for Sponge City Construction*
 - H_{net} — Elevation difference between the start and end points of sewer, m
 - H_{loss} — Head loss along the sewer, m, See Equation (5.48)
 - η — Pumps efficiency, %

The targeted area is 1,400 km² with 21% coverage by sponge facilities to achieve 85% total runoff control rate. The proportion of the pressure sewers is $k = 0.2$. The average annual precipitation is 560 mm. The elevation difference between the starting and ending points is 8 m. The operating efficiency of the pump station is 0.9. The stormwater runoff coefficient of the traditional undeveloped area φ is 0.7. The head loss along the way is calculated to be about 74.7 m. The carbon emission of the stormwater treatment operation is calculated as follows.

$$CE_{CL} = 5693.8 \times P \times S \times (H_{net} + H_{loss}) \times \frac{n \times [0.1a + (1-a)k] + (1-n) \times [0.1(1-\varphi) + k \times \varphi]}{3.6 \times 10^6 \eta} = 5693.8 \times 1400 \times 560 \times (8 + 74.7) \times \frac{0.21 \times [0.1 \times 0.85 + 0.15 \times 0.2] + 0.79 \times [0.1(1-0.7) + 0.2 \times 0.7]}{3.6 \times 10^3 \times 0.9} = 2.08 \times 10^4 \text{ t CO}_2\text{-eq}$$

D.8 Carbon emission of the water sector in a city

A city is located at the North China Plain, with a permanent population of 21 million. The city's current water resource reservation is about 4.011 billion m³/a, of which surface water (excluding evaporative seepage) reservation is 665 million m³/a, river inflow water is 661 million m³/a, long-distance water conveyance is 934 million m³/a, and the groundwater storage is 1.751 billion m³/a. The water management department of the city organized a carbon emission accounting work regarding the whole water sector.

D.8.1 Guiding company's accounting plan

The water management department of the city formulates the accounting plan for water association and companies.

D.8.1.1 Determining the accounting boundary:

- Water supply system: water abstraction facilities, water treatment plants, and distribution
- Water reclamation system: water treatment, and distribution;
- Wastewater treatment plants: wastewater and sludge treatment facilities;
- Stormwater system: pumping station and source control facilities.

D.8.1.2 Unifying accounting methods

The accounting methods and equations refer to Chapter 5 in this *Guidelines*.

D.8.1.3 Supervising data acquisition

The data acquisition is according to Chapter 8 of this *Guidelines*.

D.8.1.4 Summarizing the accounting results

The accounting results should be summarized and analyzed.

D.8.2 Carbon accounting implementation

Based on the accounting results from water sector associations and companies, the water management department conducted accounting work in terms of the whole water sector.

D.8.2.1 Analyzing water balance

The basic operation information of water sector is summarized below.

- Water supply system: There are currently 130 water treatment plants in the city, with an annual production and distribution (including leakage loss) capacity of 4.06 billion m³/a. Among this amount, 1.70 billion m³/a is allocated for domestic use, 1.74 billion m³/a for ecological use, 300 million m³/a for industrial use, and 320 million m³/a for agricultural use. The water source structure includes 0.85 billion m³/a of surface water, 1.35 billion m³/a of groundwater, 1.20 billion m³/a of reclaimed water, and 0.66 billion m³/a of long-distance water conveyance.
- Wastewater management system: The city has an annual wastewater discharge of 2 billion m³/a. There is a total length of 10,000 km of concrete sewage pipelines in the city, with a minimum flow rate of 0.6 m/s. Currently, there are 70 wastewater treatment plants with a combined capacity

of 7 million m³/d, achieving a wastewater treatment ratio of 97%.

- Water reclamation system: There are 30 water reclamation plants in the city, which use the effluent of wastewater treatment plants as water sources. These plants have a capacity to produce 1.20 billion m³/a of reclaimed water for industry, landscape or agriculture.
- Stormwater system: The average annual precipitation in the city is 585 mm, with an average precipitation of 488 mm during the flood season (June to September) and 97 mm during the non-flood season.

The water balance of the city is shown in Figure D.2.

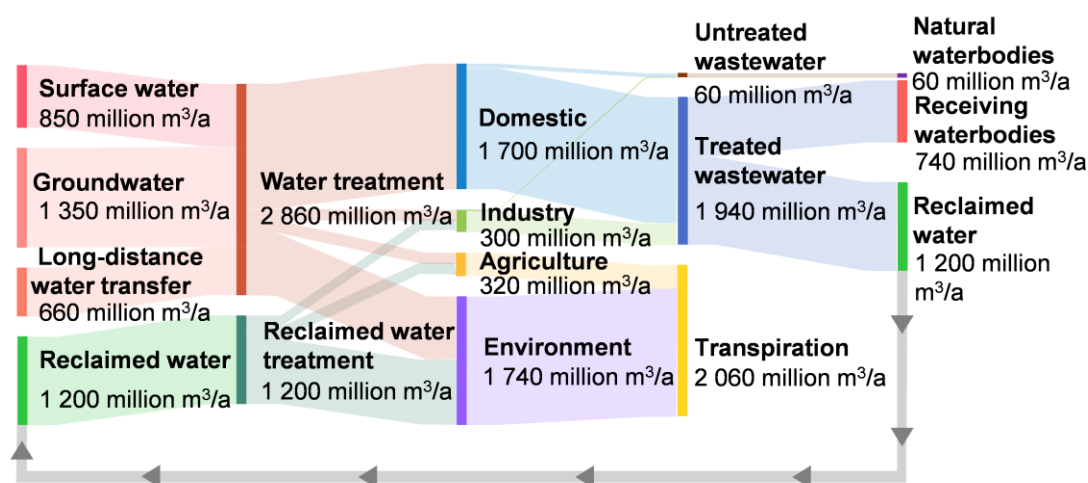


Figure D.2 Water resources in a city's water affairs system.

D.8.2.2 Determination of average emission intensity

Based on the carbon emission accounting results of companies, appropriate average carbon emission intensities are shown in Table D.18.

Table D.18 Accounting parameters and sources of the case city

Emission sites		Average carbon intensity (kg CO ₂ -eq/m ³)	Source
Water supply sectors	Surface water intaking	0.147	Average of intaking calculation results of 20 water treatment plants
	Groundwater intaking	1.159	Average of intaking calculation results of 10 water treatment plants
	Long-distance water conveyance	5.93	Accounting results of 1 water conservancy project
	Water treatment	0.038	Average of accounting results of 30 water treatment plants

Continued

Emission sites		Average carbon intensity (kg CO ₂ -eq/m ³)	Source
Wastewater sectors	Distribution	0.074	Average of accounting results of 5 water supply enterprises
	Gravity flow sewer	0.72	Average of accounting results of 1-km sewage
	Untreated wastewater pollution	0.427	Estimated value based on wastewater water quality
	Wastewater and sludge treatment	0.8	Average of accounting results of 20 wastewater treatment plants
	Anaerobic digestion recovery	-0.32	Average of accounting results of 20 wastewater treatment plants
Reclaimed water sectors	Reclaimed water treatment	0.08	Average of the accounting results of 10 reclaimed water plants
Rainwater sectors	Rainwater treatment	0.145	Accounting results 1,000-km ² sponge built-up area

D.8.2.3 Carbon emission of the whole water sector

According to the water balance and emission intensity, the carbon emission of the whole water sectors is calculated, and the results are shown in Table D.19.

Table D.19 Summary of carbon emissions of the whole water sector.

System		Average carbon intensity (kg CO ₂ -eq/m ³)	Amount of water (100 million m ³ /a)	Carbon emission (10,000 t CO ₂)
Water supply system	Surface water abstraction	0.147	8.5	12.50
	Groundwater abstraction	1.159	13.5	156.47
	Long-distance water conveyance	5.93	6.6	391.38
	Water treatment plant	0.038	28.6	10.87
	Distribution	0.074	40.6	30.04
Wastewater management system	gravity flow sewer	0.72	20	147.02
	Untreated wastewater pollution	0.427	0.59	2.52
	Wastewater and sludge treatment	0.8	19.41	155.28
	Energy recovery via anaerobic digestion	-0.32	10	-32.00

Continued

System		Average carbon intensity (kg CO ₂ -eq/m ³)	Amount of water (100 million m ³ /a)	Carbon emission (10,000 t CO ₂)
Water reclamation system	Water reclamation treatment	0.08	1 2	9.60 _
Stormwater system	Stormwater treatment	0.145	8.23	11.93
Total		-	-	895.61

The carbon emission accounting results of the city's water sectors show that the total carbon emission of the water supply system is 6,012,600 t CO₂-eq. The total carbon emissions of the wastewater management system, water reclamation system, and stormwater system are 2,728,200 t CO₂-eq, 96,000 t CO₂-eq, and 119,300 t CO₂-eq, respectively. The total carbon emission of the whole water sector is 8,956,100 t CO₂-eq with a per capita carbon emission of about 426 kg CO₂-eq.

D.8.3 Emission reduction pathway analysis

Based on the analysis of the total amount and composition of carbon emissions of the city's water sector, the water management department proposed some carbon emission reduction pathways as summarized below.

D.8.3.1 Water saving

The annual per capita water resources in this city is approximately 191 m³, indicating a relatively scarce water resources situation. According to the *Standard for Urban Domestic Water Consumption*, the minimum water consumption for residents' daily life should be above 100 L/(ca·d), and the reasonable water requirement for improving living conditions should be above 150 L/(ca·d). Currently, the city's total residential domestic water consumption is 1.70 billion m³/a, which is about 221.8 L/(ca·d), far exceeding the above-mentioned water consumption quotas. It indicates that there is significant potential for water saving in the city. By saving residents' domestic water consumption, the operation burden of the water supply system can be reduced, leading to a decrease in the total carbon emissions. Additionally, optimizing the structure of water sources and prioritizing the reduction of high carbon-intensity water resource can effectively reduce carbon emissions. Under scenario of different water-saving policies, the carbon emissions of water supply and water reclamation system are estimated in accordance with Chapter 5, and the results are shown in Figure D.3.

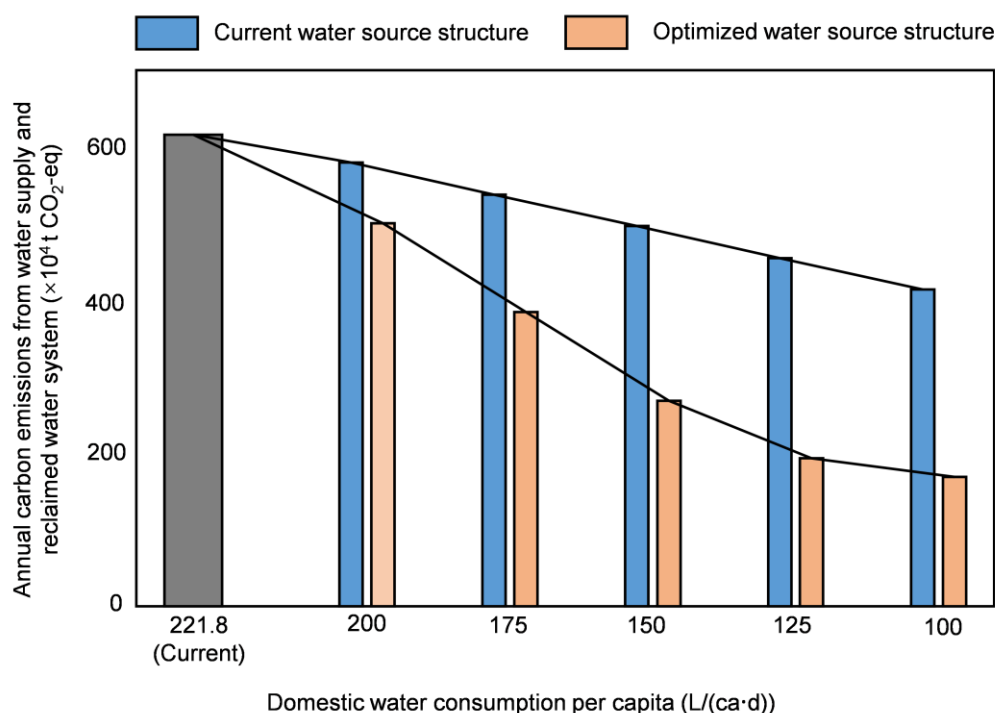


Figure D.3 Changes in carbon emissions of water supply and water reclamation systems under different water-saving scenarios

According to Figure D.3, a water quota of 150 L/(ca·d) is more reasonable and can not only effectively reduce the total carbon emissions of water supply and reclaimed water system, but also meet the needs of residents to improve living conditions.

D.8.3.2 Development of non-traditional water sources

For a long time, the local surface water and ground water resources of the city has been overexploited and exceeded the sustainable limits. The current surface water resource reservation is 665 million m³, which is far lower than the long-term average reservation of 1.772 billion m³. The groundwater resource reservation is 1.751 billion m³, and decreased by 5.2 billion m³ compared with that in last century, with a water level decline by 10 m. If no solutions are proposed, water resources will become an important constraint on the urban development. Currently, the city has adopted long-distance water conveyance and wastewater reclamation to obtain and supplement local water resources.

D.8.3.2.1 Wastewater reclamation

Wastewater reclamation refers to the advanced treatment of the secondary effluent produced by the wastewater treatment plants, which undergoes further treatment either on-site or off-site to enhance the water quality for recycling purpose (Yi *et al.*, 2011). Generally, reclaimed water is not recommended for portable use, but it can be utilized for industrial, ecological, and agricultural purposes (Chen *et al.*, 2013). Currently, the recovery ratio of recycled water in the city is approximately 62%. Increasing the recovery and utilization of reclaimed water can alleviate the problem of water shortage in the city. At the same time, it can also optimize the structure of water supply sources, reduce the use of high carbon-intensity water sources, and effectively decrease carbon emissions. Based on different scenarios of reclaimed water reuse, the carbon emissions of water supply and water reclamation system are estimated according to Chapter 5, and the results are shown in Figure D.4. It can be observed that a wastewater recovery ratio

of 90% yields a more cost-effective and efficient carbon emission reduction, making it more conducive to achieving sustainability goals.

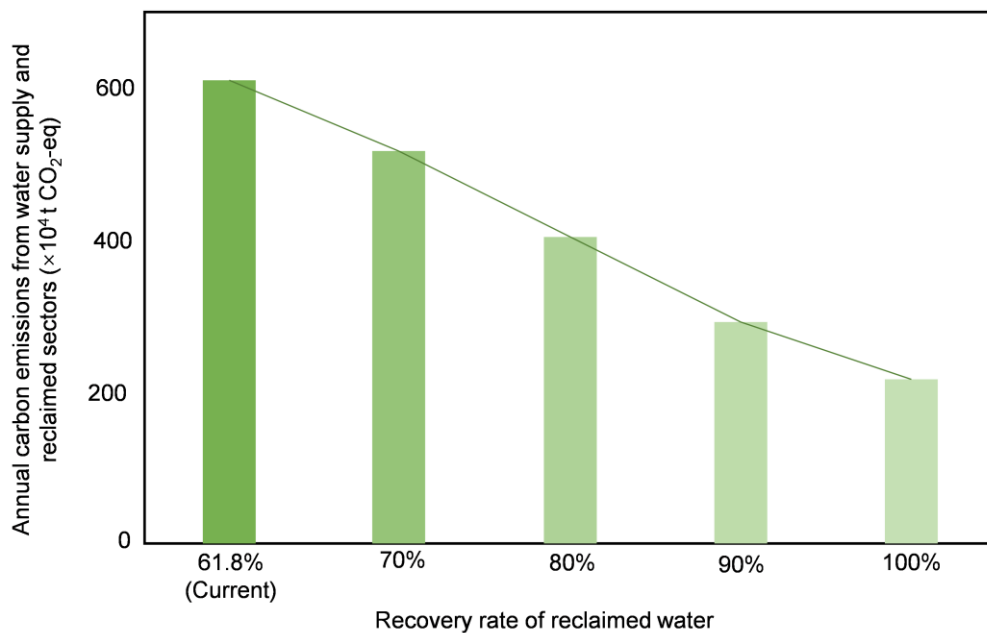


Figure D.4 Changes in total carbon emissions of water supply and reclaimed water systems under different reclaimed water recovery rates in a city.

D.8.3.2.2 Rainwater reuse

Rainwater reuse refers to the collection and treatment of rainwater for toilet flushing, green irrigation, road flushing, landscape water (Leong *et al.*, 2017), etc. It can effectively solve the current severe problems of water resource shortage and water environmental pollution. Currently, the city's annual rainwater utilization is about $3 \times 10^8 \text{ m}^3$, and the rainwater reuse rate is about 36.45%. By reusing part of the rainwater, the drainage pressure of the water supply system and the stormwater system can be reduced, leading to a decrease in carbon emissions. Based on different scenarios of rainwater reuse, the carbon emissions of stormwater system are estimated according to Chapter 5, and the results are shown in Figure D.5. With the increase of the city's rainwater reuse rate, the carbon emission reduction also increases.

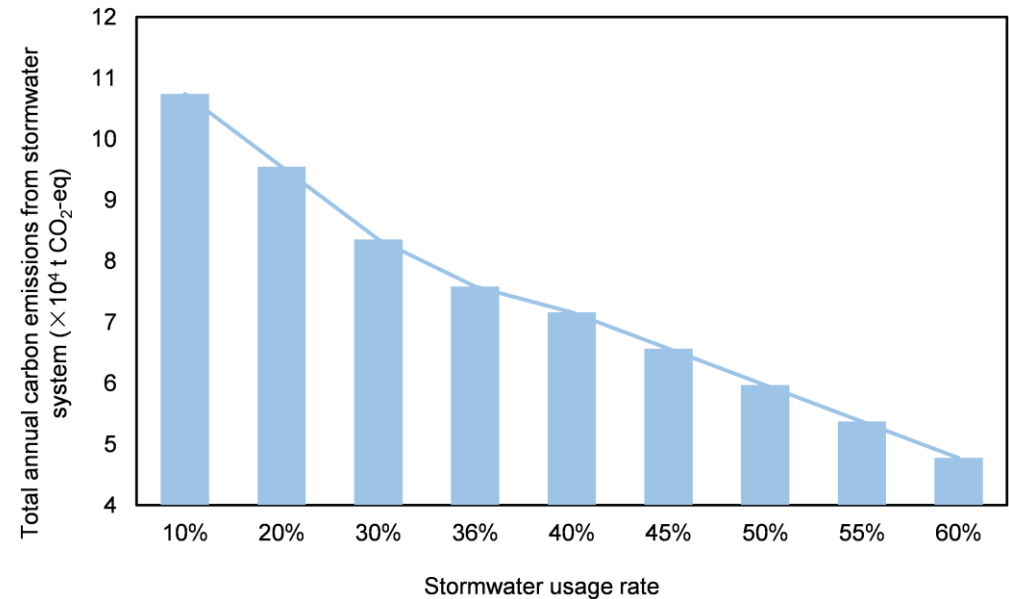


Figure D.5 Changes in carbon emissions of rainwater systems under different rainwater reuse rates in a city.

D.8.3.2.3 Seawater desalination

The city is located at an area approximately 300 km far away from the nearest coastline, which provides access to seawater resources for desalination and supplementing freshwater resources. The carbon emission intensity of seawater desalination is approximately $0.336 \text{ kg CO}_2\text{-eq/m}^3$. Currently, the city has not yet utilized seawater desalination as a source of fresh water. By using seawater desalination, local water resource shortage can be alleviated. At the same time, it can also optimize the structure of water supply sources, reduce the use of high carbon-intensity water sources, and effectively decrease carbon emissions. Based on different scenarios of utilizing seawater desalination to provide water source, the carbon emissions of the water supply and water reclamation system can be estimated, and the results are shown in Figure D.6.

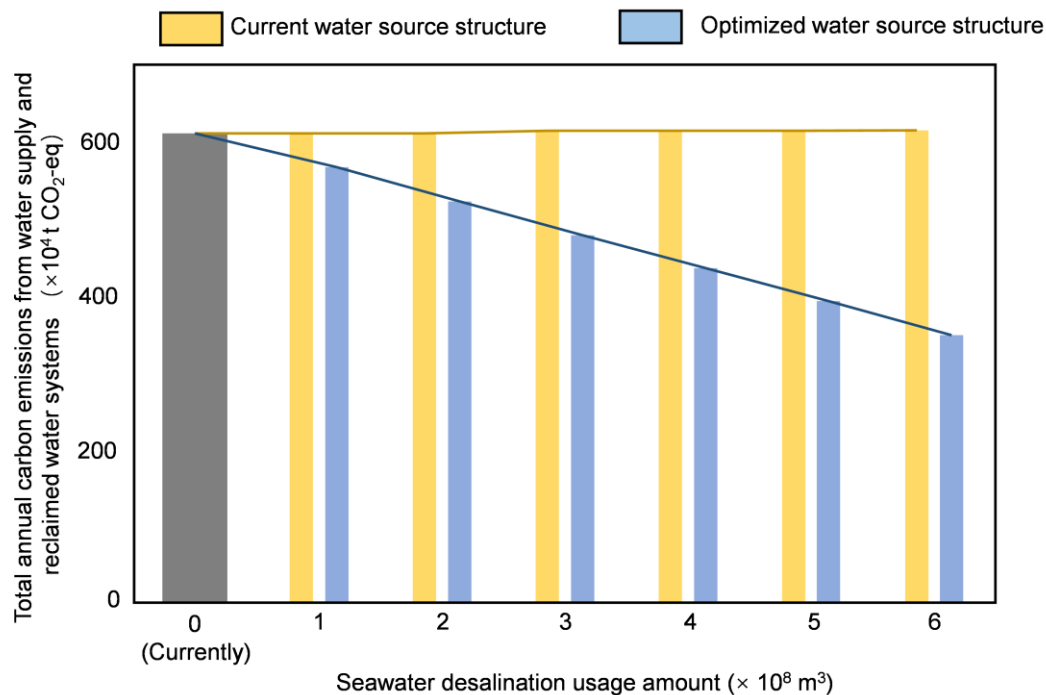


Figure D.6 Changes in carbon emissions of water supply and reclaimed water systems under different desalination water consumption in a city.

The carbon emission intensity of seawater desalination and distribution is slightly higher than that of general surface water intake and treatment, especially when the city is far away from coastline (Martinez-Mate *et al.*, 2018). Therefore, considering the specific geographical conditions of a city, seawater desalination should be adopted appropriately, and its use should be targeted towards reducing the carbon intensity of higher-emission water sources, leading to an optimization of the water sources structure.

D.8.3.3 Energy recovery from wastewater

Domestic Wastewater is rich in energy such as chemical and thermal energy the recovery of which can effectively offset the carbon emissions generated in the process of wastewater and sludge treatment (Hao *et al.*, 2015).

D.8.3.3.1 Anaerobic digestion

In the wastewater treatment plant, approximately 50% of the waste sludge is treated with anaerobic digestion, while the rest is directly disposed of in landfills. By increasing the anaerobic digestion rate and recovering energy through digestion, etc., it is possible to offset a part of the carbon emissions of wastewater treatment, and the results are shown in Figure D.7.

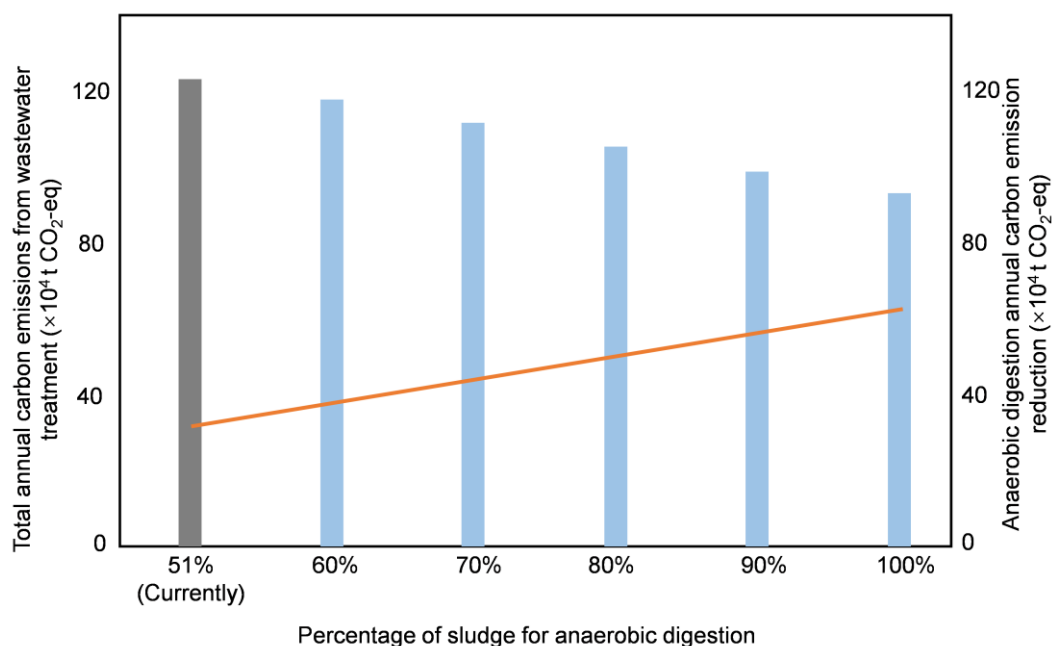


Figure D.7 Changes in total carbon emissions of sewage treatment plants under different anaerobic digestion rates in a city.

D.8.3.3.2 Recovery of thermal energy

In the city, the average outdoor temperature during winter (January to February next year) is 0 °C, and the indoor heating standard is set at 18 °C. Currently, natural gas is widely used for heating, with an average consumption of approximately 6.6 m³/(m²·d). However, wastewater thermal energy has not been used for heating purpose. The average temperature of the effluent treated by the city's wastewater treatment plants during winter is about 14 °C. By recovering the thermal energy from wastewater, it can provide heating for nearby buildings, thereby offsetting the carbon emissions from wastewater treatment. Based on other engineering case for wastewater heat recovery, the emission reduction of 1.76 kg CO₂-eq/m³ of wastewater can be achieved. Thus, under the scenario of different wastewater temperature heat energy recovery ratios for heating during winter, the carbon emissions are shown in Figure D.8:

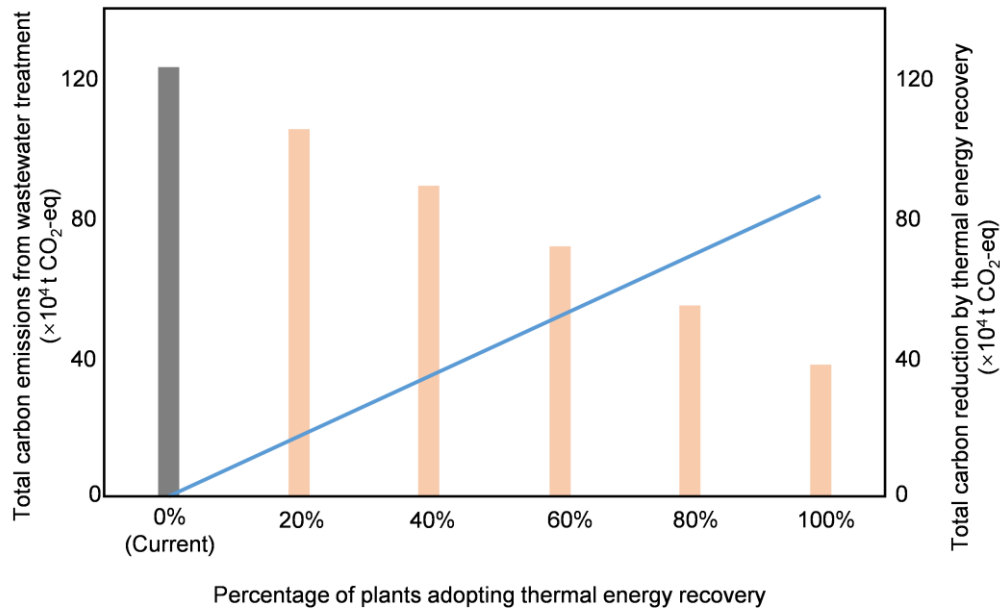


Figure D.8 Changes in carbon emissions of sewage treatment plants under different heating ratios of recycled sewage heat in a city in winter.

It can be observed that by utilizing wastewater thermal energy for heating during winter, it is possible to offset up to nearly 70 % of the total annual carbon emissions, demonstrating its significant emission reduction potential. Furthermore, the wastewater heat energy can also be used for cooling during summer. Therefore, it can further offset the carbon emissions generated by wastewater treatment to achieve a "carbon neutral" or even a "carbon negative" state (Hao *et al.*, 2019a).

D.8.3.4 Carbon sink from green stormwater space

Both plants and soil in the green space can sequester carbon dioxide (Lessmann *et al.*, 2022). Assuming a total area of 100 m² for the green facility, the unit carbon sequestration rate ranges from approximately 1.60 to 2.23 kg CO₂-eq/(m²·a). Therefore, the carbon sequestration total of the sponge city's green facility is approximately 160.2 to 222.6 kg CO₂-eq/(100 m²·a).

Appendix E

Carbon emission reduction potential of typical technologies

E.1 EMISSION REDUCTION POTENTIAL OF WASTEWATER THERMAL ENERGY RECOVERY

A wastewater treatment plant is located at Shanghai, with a treatment capacity of 28,800 m³/d. It adopts water source heat pump technology with the effluent as the heat source. The extracted and recovered heat energy is provided nearby residents heating during winter and cooling during summer for an area covering 1,800 m². The system can replace traditional coal-fired or electricity-consuming heating and cooling mode, resulting in a reduction in carbon emissions. The heating operation takes place from December to next February, and the cooling operation occurs from July to September. The specific operation mode and parameters are shown in Figure E.1.

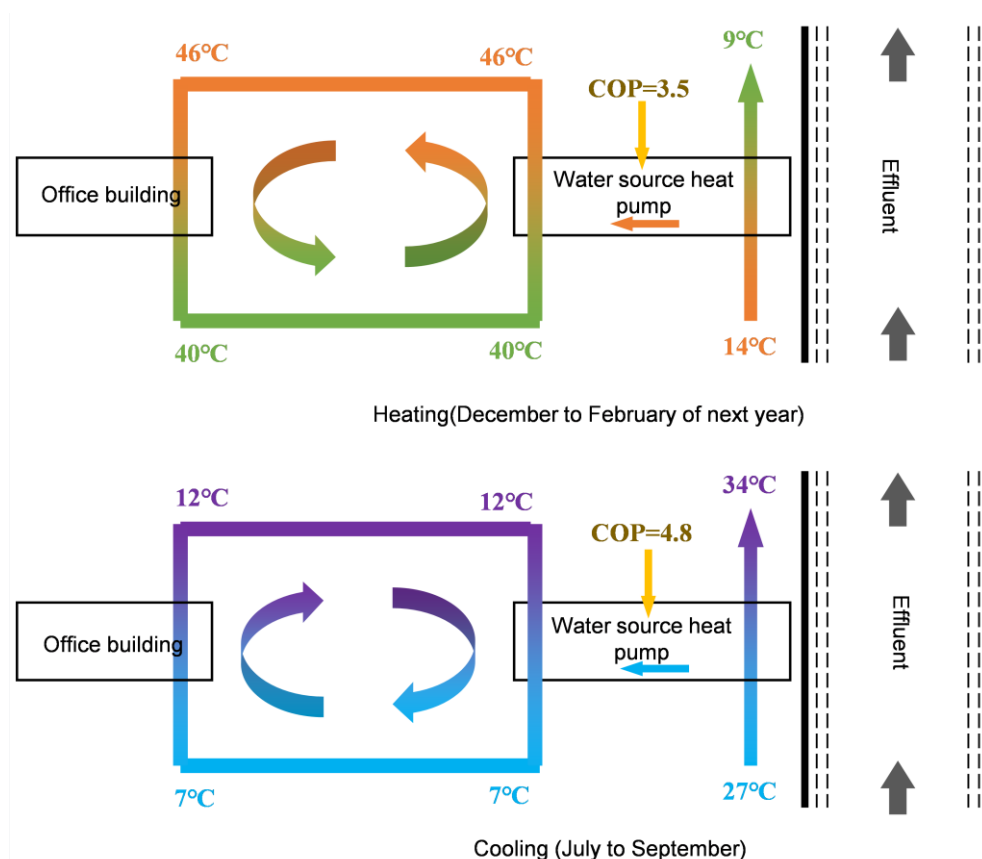


Figure E.1 Operation parameters of waste temperature heat energy recovery in the case sewage treatment plant.

The theoretical extraction potential of wastewater thermal energy in the case plant using a water source heat pump is calculated as follows.

$$A = Q \times c \times \Delta T \quad (E.1)$$

$$A_H = A + \frac{A}{COP_H - 1} \quad (E.2)$$

$$A_C = A - \frac{A}{COP_C + 1} \quad (E.3)$$

$$M_{CO_2,H} = A \times \left(1 + \frac{\delta - \alpha}{\alpha \cdot COP_H}\right) \times EF \times 10^{-6} \quad (E.4)$$

$$M_{CO_2,C} = \frac{A}{COP_{C,A}} - \frac{A}{COP_C} / 3600 \times EF_D \quad (E.5)$$

Where,

A	—	Heat energy extracted from wastewater, kJ
Q	—	Wastewater flow, m ³
c	—	Specific heat capacity of water, 4.2×10^3 kJ/m ³
ΔT	—	Extract heat energy temperature difference, °C
A_H	—	The theoretical heat that can be obtained when heat energy is extracted from wastewater for heating, kJ
COP_H	—	The energy efficiency ratio of the water source heat pump during heating is usually 1.77~10.63, generally 3.5
A_C	—	The cooling capacity that can be obtained when extracting heat energy to wastewater for cooling, kJ
COP_C	—	The energy efficiency ratio of the water source heat pump during cooling is usually 2.23~5.35, generally 4.8
$M_{CO_2,H}$	—	The carbon emission reduction amount from external heating, kg CO ₂ -eq
α	—	Energy efficiency of coal-fired heating, 55%~70%, generally 60 %
δ	—	Energy efficiency of coal-fired thermal power generation, the average level is 33%
EF	—	Coal combustion emission factor, 9 6.10 kg CO ₂ -eq/GJ
$M_{CO_2,C}$	—	The carbon emission reduction amount from external cooling, kg CO ₂ -eq
$COP_{C,A}$	—	Energy efficiency ratio of air source heat pump refrigeration, 3.4
EF_D	—	Emission factor of electricity, kg CO ₂ -eq/kWh
3600	—	1 kWh = 3,600 kJ

The calculations show that in the case wastewater treatment plant, the theoretical heat energy extraction in winter is 604.8 GJ/d, resulting in a carbon emission reduction of 50.65 t CO₂/d. The emission reduction intensity is 1.76 kg CO₂-eq/m³ of wastewater, leading to a total reduction of 4,558.4 t CO₂-eq during winter. In summer, the wastewater could accept 846.72 GJ/d energy to cool the office, resulting in a carbon emission reduction of 13.43 t CO₂-eq/d. The carbon emission intensity is 0.47 kg CO₂-eq/m³ of wastewater, yielding a carbon emission reduction of 1,208.6 t CO₂-eq during summer. In full-scale project, due to heat loss in pipeline transmission and multiple heat pump conversions in the energy distribution of heating/cooling areas, the actual annual energy recovery is 1 5,582 GJ. The annual carbon reduction is about 1,109 t CO₂-eq, reducing the coal consumption of 423.28 t.

It is evident that the thermal energy recovery potential of wastewater for heating is higher than that for cooling (Bruno *et al.*, 2009). It can be explained by the current heating and cooling methods. On the one hand, there is a significant temperature difference between the indoor and outdoor environment in winter, and the efficiency of air source heat pumps is relatively low. Therefore, coal burning or direct conversion of electricity is generally used to obtain heat energy. In summer, the temperature difference between indoor and outdoor environment is relatively small, and the energy efficiency of air source heat pump for cooling is higher. Consequently, extracting thermal energy for heating purpose yields more carbon emission reduction.

China generated a total of 57.14 billion tons of domestic wastewater in 2020. Based on the above calculation, if water source heat pumps are widely implemented throughout society, 3.0×10^8 GJ of heat energy can be generated, equivalent to a reduction of about 25.1416 million t CO₂-eq. And during the cooling period (calculated as 3 months), 4.2×10^8 GJ cooling energy can be generated, equivalent to a reduction of about 67.1395 million t CO₂-eq.

In terms of economic benefits, wastewater source heat pumps also have advantages. Take a wastewater treatment plant in Qingdao as an example, the plant provides heating to the 40,000 m² of surrounding residential buildings during winter heating period, which lasts about 140 days each year. The plant has a treatment capacity of 200,000 m³/d, and the average wastewater temperature in winter is 13 °C. After heat exchange by the heat pump, the hot water with an average temperature of 46 °C is obtained, which is then delivered to the end of the users. The initial investment for the entire wastewater source heat pump system, including the equipment, pump stations, and pipeline network, is 14.8 million yuan. In comparison, the traditional coal-fired boiler system cost 11.6 million yuan for initial investment. However, during the operation for heating, the actual annual operating cost of the wastewater source heat pump is about 420,000 yuan, while the actual annual operating cost of the coal-fired system is about 1.3 million yuan. It is evident that although the initial investment of the wastewater source heat pump is high, the operating cost is low. After about 4 years, the investment of heat pump system can be recovered and higher income can be obtained thereafter.

Noteworthy is that there are many challenges in the implementation process of the wastewater source heat pump project, such as higher initial investment cost, which poses difficulties in fundraising and obtaining loans for economic activities. There is a lack of policy support for connecting the wastewater source heat pump to the municipal heating pipe network in China, resulting in a lack of proper utilization of recovered heat energy. Therefore, the wastewater source heat pump project still needs to overcome these challenges.

E.2 EMISSION REDUCTION POTENTIAL OF SOLAR POWER GENERATION

In 2013, the State Council of China issued *Several Opinions of the State Council on Promoting the Healthy Development of the Photovoltaic Industry* (State Council of China, 2013), which clarified the action plan to vigorously develop the distributed photovoltaic power generation market. The plan aimed to fully develop the surface space of the various buildings and structures in different industries and support the installation of small-scale distributed photovoltaics power system. With the adoption of economic subsidies, the installed capacity of distributed photovoltaic power generation in China has greatly increased. For wastewater treatment plants, there are multiple structures with a large space. And they are generally located at the suburb area without high-rise buildings, with good sunlight condition, which makes wastewater treatment plants good conditions for installing photovoltaic equipment. Since the pioneering installation of photovoltaic panel in wastewater treatment plants in 2014, a total of 22

plants (20 wastewater treatment plants and 2 water treatment plants) have completed the installation of photovoltaic panels and started to self-generate electricity. To clearly demonstrate the potential of photovoltaic power generation technology in carbon emission reduction of urban water sectors, this section presents an analysis of the current full-scale projects in China.

As shown in Table E.1, the treatment capacity of wastewater treatment plants adopting photovoltaic power generation is between 0.6 and 1.7 million t/d. The installed photovoltaic panels are distributed on the headspace of secondary sedimentation tanks and the biological tanks, with an annual power generation capacity between 6 and 22 million kWh. Since the footprints of the case wastewater treatment plants are not available, considering the correlation between the treatment capacity and the footprint, the annual solar power generation is normalized to the treatment capacity. The results show that the potential electricity generation per unit of treated water in China is approximately 21.31 kWh/m³ (Figure E.2). It can be applied for determination of the photovoltaic power generation projects in other wastewater treatment plants. Combining with the electricity consumption data of the case wastewater treatment plants, the photovoltaic power generation can cover about 4% to 37% of the electricity consumption (Figure E.3), with an average of approximately 15%. In terms of the carbon emission reduction, photovoltaic power generation can reduce carbon emissions by an average of about 9.7% by saving external electricity consumption (base emission including electricity consumption of wastewater treatment plants and direct CH₄ and CO₂ emissions, excluding chemicals consumption, fossil carbon emissions and transportation emissions, Figure E.4). Among these, Wangxiaoying Wastewater Treatment Plant can achieve a carbon emissions reduction of about 25.8% through photovoltaic power generation, while some individual wastewater treatment plants can only reduce carbon emissions by 1.7%. In term of return on investment (the magnitude of the annual return on investment), the application of photovoltaic technology in all wastewater treatment plants can reduce electricity and costs, and can achieve an investment return of approximately 1.3% to 7.9%.

In addition, the last two cases in Table E.1 represent the application of photovoltaic panels in water treatment plants, which can contribute to about 17% to 19% of energy consumption. It is important to emphasize that, as shown in Figure E.2, the electricity generation intensity (electricity generation per unit of water) of wastewater treatment plants with installed photovoltaic panels is related to the capacity of wastewater treatment plants. Figure E.2 shows that the solar power generation intensity of relatively small-scale wastewater treatment plants is significantly lower than the average level. Considering that the electricity consumption intensity of small-scale wastewater treatment plants is often higher than the average, the application of photovoltaic panels should consider the relationship between emission reduction and investment.

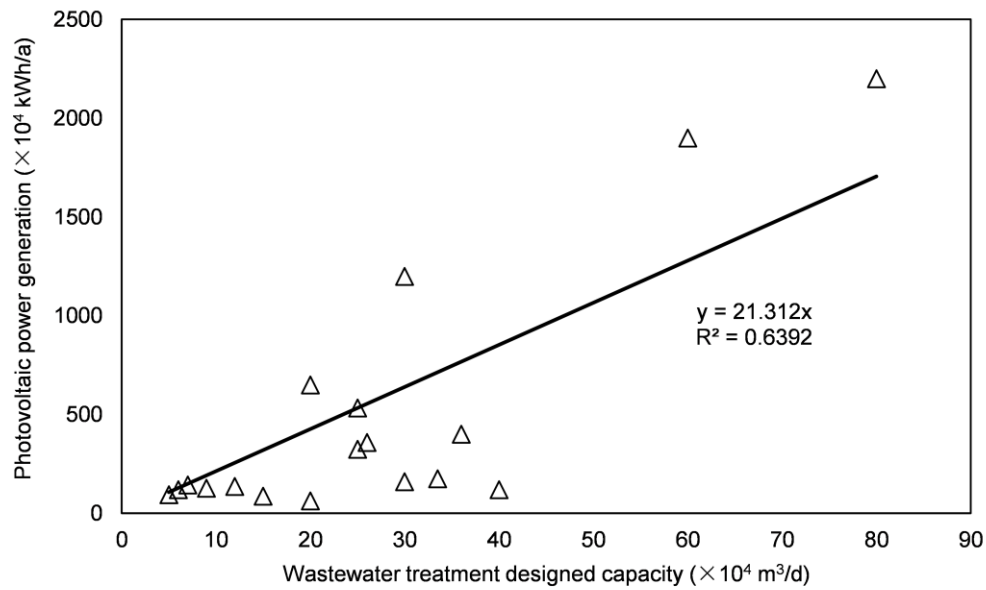


Figure E.2 Correlation between power generation and sewage treatment capacity of photovoltaic power generation equipment that can be installed in sewage treatment plants (data are obtained from cases 1 to 18 in Table E.1).

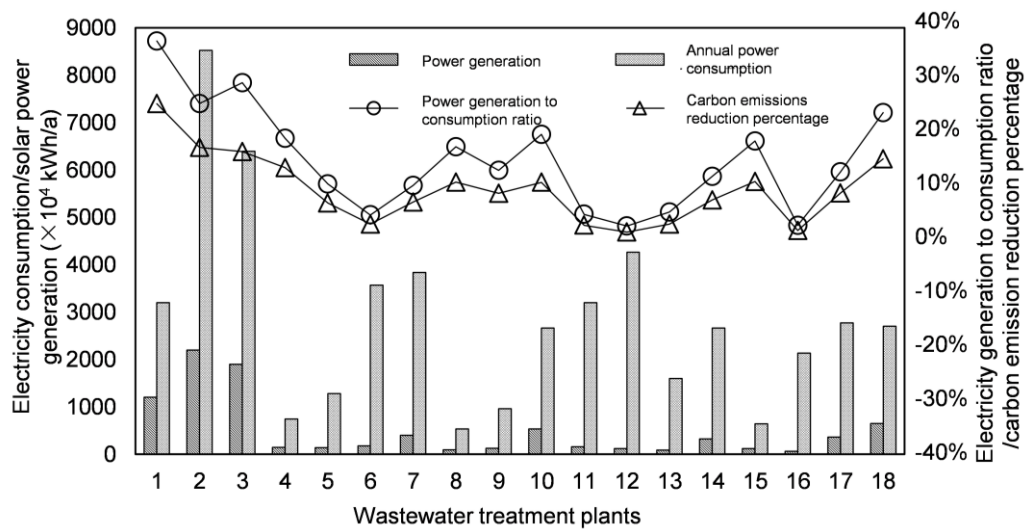


Figure E.3 Annual photovoltaic power generation, total power consumption, back-feeding power ratio and carbon emission reduction ratio of the case sewage treatment plant.

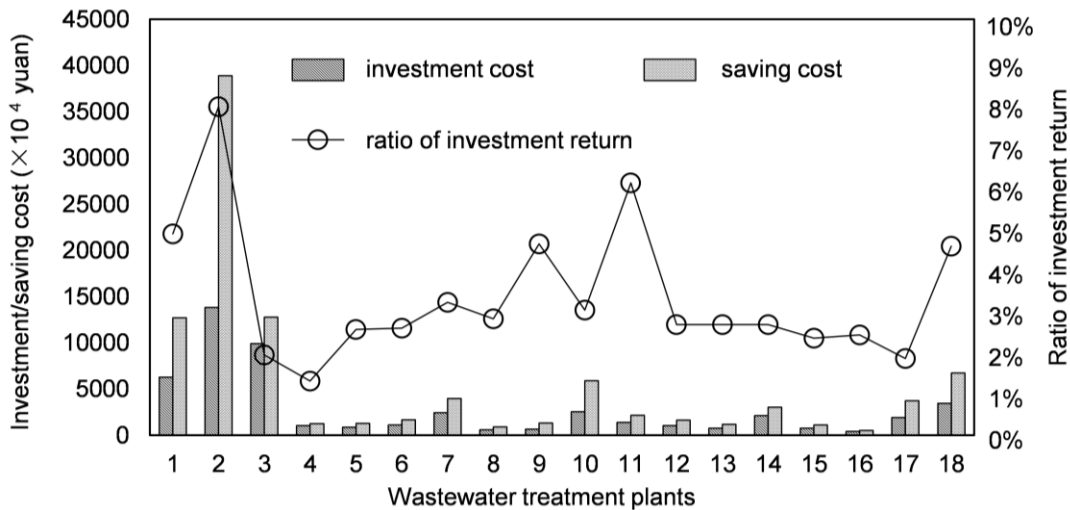


Figure E.4 Investment cost, cost saving and investment return ratio of photovoltaic power generation equipment in sewage treatment plant.

Table E.1 Photovoltaic power generation case of wastewater treatment plant.

Serial number	Design treatment capacity (10,000 m ³ /d)	Area of photovoltaic panel (10,000 m ²)	Electricity generation (10,000 kWh/a)	Electricity consumption (10,000 kWh/a)	Investment cost ¹ (10,000 yuan)	Cost savings ² (10,000 yuan)	Carbon emission reduction (10,000 t CO ₂ -eq)	Total carbon emissions ³ (10,000 t CO ₂ -eq)	Carbon emission reduction potential (%)
1	30	11	1,200.00	3,197.40	6,285.6	12,686.40	237,630.00	922,611.41	25.76
2	80	18.17	2,200.00	8,526.40	13,809.4632	38,880.54	472,285.00	2,691,874.27	17.54
3	60	15	1,900.00	6,394.80	9,894	12,763.50	407,882.50	2,433,636.95	16.76
4	7	2.26	143.82	746.06	1,047.6	1,245.25	30,874.56	224,287.74	13.76
5	12	1	136.73	1,278.96	878.82	1,300.99	29,352.51	412,473.25	7.12
6	33.5	\	174.92	3,570.43	1,112.4	1,653.08	34,638.53	1,100,128.66	3.15
7	36	2.42	400.00	3,836.88	2,421.12	3,955.88	85,870.00	1,173,975.53	7.31
8	5	\	94.24	532.90	582	894.74	18,661.88	168,613.21	11.07
9	9	1	127.00	959.22	672.21	1,317.88	25,149.18	281,091.04	8.95
10	25	\	532.90	2,664.50	2,554.98	5,896.81	105,527.52	957,579.10	11.02
11	30	\	160.14	3,197.40	1,396.8	2,164.31	31,711.72	1,067,142.38	2.97
12	40	\	120.10	4,263.20	1,047.6	1,623.12	23,782.80	1,422,856.51	1.67
13	15	\	86.75	1,598.70	756.6	1,172.50	17,178.67	539,572.47	3.18
14	25	\	323.53	2,664.50	2,121.39	3,036.48	69,453.80	903,468.30	7.69
15	6	\	119.84	639.48	774.06	1,120.61	23,731.31	212,931.62	11.14
16	20	\	62.25	2,131.60	407.4	533.04	13,363.51	677,797.52	1.97
17	26	3.28	359.14	2,771.08	1,908.96	3,718.76	71,118.69	790,592.56	8.99
18	20	5.94	649.86	2,700.00	3,454.17	6,729.13	128,688.52	833,643.56	15.44
19	170	\	1,284.00	18,118.60	7,886.1	12,529.50	254,264.10	\	\
20	15	\	85.70	1,598.70	0	1,409.76	20,180.21	\	\
21	20	\	411.41	2,117.00	2,677.2	3,538.17	88,319.44	\	\
22	100	4.4	511.10	3,000.00	2,677.2	5,331.737	101,210.58	\	\

1 Considering cost and operation and maintenance investment (operation and maintenance investment is calculated as 20% of cost investment);

2 Considering the government's subsidies;

3 Considering the carbon emissions of electricity consumption of wastewater treatment plants, and the total direct carbon emissions of CH₄ and N₂O in wastewater treatment plants;

4 1~20 cases are wastewater treatment plants, 21~22 cases are water supply treatment plants.

E.3 COMPARISON OF CARBON EMISSIONS BETWEEN VACUUM DRAINAGE SYSTEM AND GRAVITY DRAINAGE SYSTEM

A small village is located at Northwest of China, with an area of approximately 86 hm² and a population of 1,137 people. The village is situated near important cultural relics sites. To protect cultural relics, the excavation depth of pipes is limited to no more than 30 cm. If the traditional gravity drainage system is used, the pipe slope can only follow the natural road slope and adopt the minimum gradient, resulting in poor hydraulic conditions in the pipe and increased changes of blockage and carbon emission. As an alternative solution, the vacuum drainage system is not restricted by the terrain and does not require a high excavation depth, making it suitable for areas with cultural heritage preservation (Milsom *et al.*, 1979). Finally, the village adopts a vacuum drainage system.

Table E.2 Summary of parameters of two drainage systems.

Project	Value	Project	Value
Serving population	600 people	Pipeline length	1000 m
Wastewater flow	350 m ³ /d	TN	30 mg/L
Wastewater COD	200 mg/L		
Gravity drainage system			
Diameter	DN 200	Slope	0.003
Flow rate	0.65 m/s	Hydraulic retention time	26 minutes
Daily electricity consumption	1.5 kWh/d		
Vacuum drainage system			
Diameter	DN 150	Flow rate	4 m/s
Hydraulic retention time	4.2 min	Daily electricity consumption	3 kWh/d

The CH₄ emission intensity of the vacuum drainage system is according to Equation (E.6).

$$CES_{CH_4} = \gamma \times \left[\frac{A}{V} \right] \times HRT \times 28 \quad (E.6)$$

Where

- CES_{CH_4} — CH₄ emission intensity of vacuum drainage system, kg CO₂-eq/m³
- γ — Specific CH₄ release rate, kg/m²·h, default is 5.24×10^{-5}
- HRT — The residence time of wastewater in the sewer, h
- 28 — Global warming potential of CH₄, default is 28 kg CO₂-eq/kg CH₄

According to the calculation method provided in this *Guidelines*, the carbon emissions generated from the two drainage systems implemented in the village were calculated and summarized in Table E.3.

Table E.3 Comparison of carbon emission accounting results of two drainage systems.

Emission activities		Accounting Methods	Emission intensity	Daily emissions	Proportion
			(kg CO ₂ - eq/m ³)	(kg CO ₂ - eq/d)	
Gravity Drainage System					
Direct discharge	Fossil CO ₂	Equation (5-10~11)	3.8×10^{-5}	0.01	0.22%
	CH ₄	Equation (5-14~16)	0.007	2.43	39.91%
	N ₂ O	Equation (5-18~19)	0.0066	2.29	37.63%
Indirect emissions	Electricity consumption	Equation (5-2)	0.0039	1.35	22.24%
Total		-	0.017	6.09	-
Vacuum Drainage System					
Direct discharge	Fossil CO ₂	Equation (5-10~11)	6.2×10^{-6}	0.01	0.05%
	CH ₄	Equation (E-6)	0.0027	0.94	21.94%
	N ₂ O	Equation (5-18~19)	0.0019	0.66	15.44%
Indirect emissions	Electricity consumption	Equation (5-2)	0.0077	2.67	62.57%
Total		-	0.0123	4.2 8	-

Based on the calculation results, it can be observed that the main source of carbon emissions from the gravity drainage system is direct emissions (77.76%), and the main source of carbon emissions from the vacuum drainage system is indirect emissions from electricity consumption (62.57%). Despite the higher electricity consumption of the vacuum drainage system compared to the gravity drainage system, the direct carbon emissions the greenhouse gas are significantly reduced. Therefore, the carbon emissions of the vacuum drainage system are lower than those of the traditional gravity drainage system, making it a viable carbon emission reduction technology for wastewater collection system.

In addition to the carbon reduction potential, the vacuum drainage systems also have several advantages in various aspects. Comprehensive evaluation and ratings were conducted for both gravity drainage system and vacuum drainage systems in terms of regional planning, technology, environment, social factors and economic investment, respectively, as follows.

- In terms of regional regulations, considerations are primarily given to the topographical structure, land use conditions, and transportation issues, etc. Additionally, priority is placed on the preservation of archaeological discoveries.
- In terms of technical aspects, considerations are primarily given to flexibility in the construction phase, system performance (considering possible changes in the future), process complexity, professional knowledge requirements, and operational safety.
- In terms of the environmental impacts, considerations are given to the geological conditions of the area, and the compatibility with the local ecosystem.
- In terms of the social dimension, considerations include public acceptance, public health concerns, and user engagement during system operation.
- In terms of the economic standard, considerations include installation costs, operation costs and maintenance costs.

The evaluation results are shown in Table E.4. The vacuum drainage system is better than that of the gravity drainage system for the following reasons: i) the area has a flat terrain, which is suitable for vacuum conveyance. b) the costs of the vacuum drainage system are much lower than those of traditional gravity drainage system. c) there are still many ancient cultural relics that have not been excavated in the area, requiring flexible handling during excavating work. Overall, the vacuum drainage system is suitable for a complex underground situation without the need for deep excavation.

Table E.4 Comprehensive evaluation results of gravity and vacuum drainage system.

Standard		Gravity drainage system	Vacuum drainage system
Non-economic criteria (in score)	Regional Planning	26.03	31.35
	Technology	22.08	25.42
	Environment	24.1	30.31
	Society	16.6	22.7
	total score	88.81	109.78
Economic Standard (in Euros)	Installation cost	5.38×10^6	3.66×10^6
	Operating costs	1.65×10^5	2.92×10^4
	Maintenance cost	7.41×10^4	3.88×10^4
Comprehensive score		0.79	1.00

Appendix F

Reporting protocol

X X Carbon emissions accounting report

Reporting entity (seal):

Reporting year:

Prepared date: Year month day

Reporting utility information

Utility name					
Industry		Industry code		Organization code	
Address					
Legal representative		Telephone		Fax	
Mail address				Post code	
Branch leaders		Telephone		Fax	
Carbon emission management department					
Director		Telephone		Fax	
E-mail					
Liaison		Telephone		Fax	
E-mail					
Mail address				Post code	
The main product or service					
Accounting and reporting boundary	New or expanded facilities (compared to the previous year):				
	Reduced or downsized facilities (compared to previous year):				

Carbon emissions report

Scope (I PCC (2019))	Type (ISO 14064-1: 2018)	Carbon emission activities	Carbon emissions (t CO ₂ -eq)	Total annual carbon emissions (tCO ₂ -eq)
Direct greenhouse gas emissions from activities within the management boundary of the accounting entity	Direct greenhouse gas emissions			
		...	...	
	Carbon sink			
		...	...	
Indirect GHG emissions due to electricity, steam, heating/cooling sources	Indirect Greenhouse Gas Emissions - Electricity and Heat Consumption			
		...	...	
Other indirect greenhouse gas emissions caused by the activities of the accounting subject but outside the management boundary of the accounting subject, such as purchased consumables, waste disposal	Indirect Greenhouse Gas Emissions - Transport			
	Indirect Greenhouse Gas Emissions - Material Inputs and Services			
		...	...	
	Indirect Greenhouse Gas Emissions - Asset and By-Product Disposal ¹			
		...	...	
	Indirect Greenhouse Gas Emissions - Other			
		...	...	
Total:				
Carbon emissions intensity (tCO ₂ -eq/m ³):				

Wastewater treatment plant - direct emission reporting

Month	Water volume/m³	Wastewater treatment					Sludge treatment (treatment method:)						
		Water quality index		Emission factor		Greenhouse gas emissions		Sludge index		Emission factor		Greenhouse gas emissions	
		BOD ₅		Fossil source CO ₂ ratio		Fossil source CO ₂		Total sludge		Sludge fossil source CO ₂ ratio		Fossil source CO ₂	
		TN		N ₂ O		N ₂ O		Sludge TOC		N ₂ O		N ₂ O	
		TOC		CH ₄		CH ₄				CH ₄		CH ₄	
						Total						Total	
		Water quality index		Emission factor		Greenhouse gas emissions		Sludge index		Emission factor		Greenhouse gas emissions	
		BOD ₅		Fossil source CO ₂ ratio		Fossil source CO ₂		Total sludge		Sludge fossil source CO ₂ Ratio		Fossil source CO ₂	
		TN		N ₂ O		N ₂ O		Sludge TOC		N ₂ O		N ₂ O	
		TOC		CH ₄		CH ₄				CH ₄		CH ₄	
						Total						Total	

Material inventory inducing indirect emission

A S/N	B Material type	C Annual consumption (t, ten thousand m ³ , MWh)	D Emission factor (kg CO ₂ /t)	E. Indirect emissions (t CO ₂)

References

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